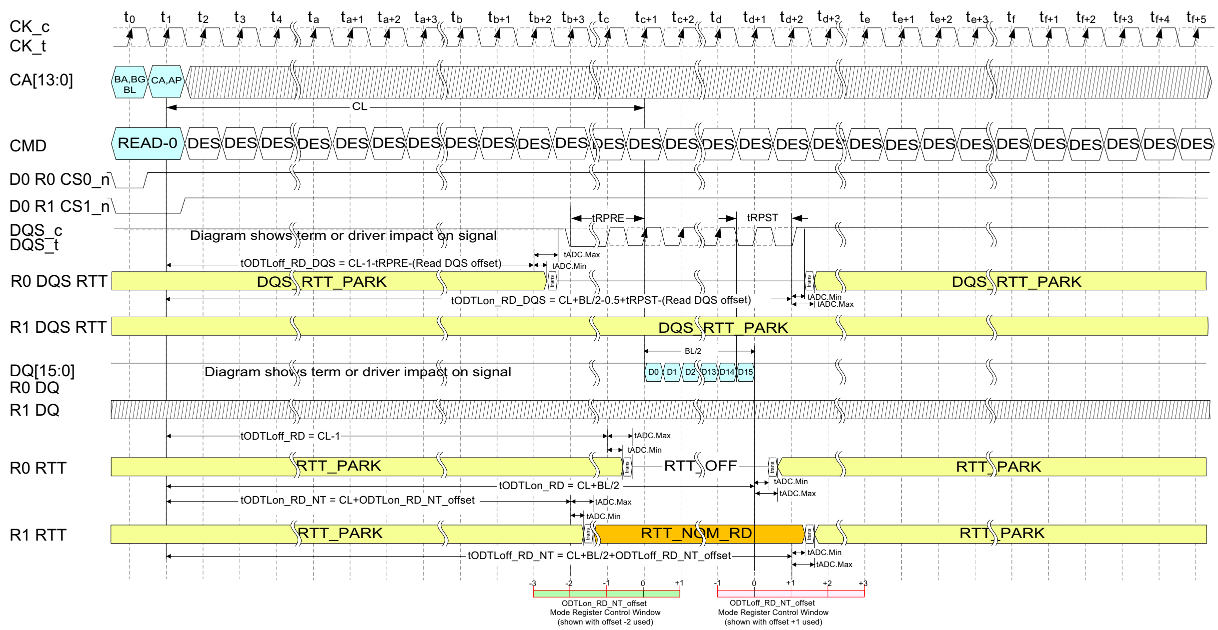
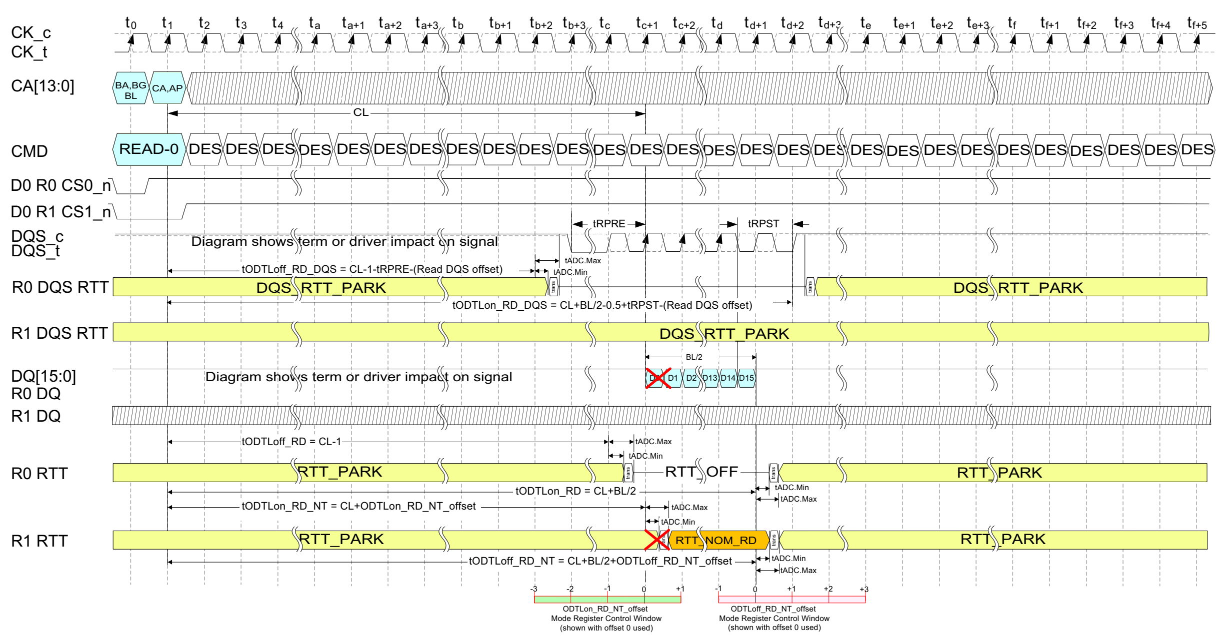
5.3.3 ODT Timing Diagrams (cont'd)



NOTES:
1. The entire range of ODTL control is now shown for simplicity.
2. Example shown with NT_ODT overlapping normal read disable by 1tCK on both sides (ODTLon_RD_NT_offset = -2 & ODTLoff_RD_NT_offset = +1)

Figure 203 — Example of Burst Read Operation ODT Latencies and Control Diagrams



NOTES:
1. The entire range of ODTL control is now shown for simplicity.
2. Since the ODTLon_RD_NT_Offset was left at zero offset and tADC still had to be considered, the NT RTT turned on too late for the non-target device. tADC is not instantaneous.
3. Since the tODTLoff_RD_NT is referenced from the CL, it is not affected by the offset used for the 'on' time and would turn off 1 clock earlier than the read disable RTT if programed to zero offset. tODTLon_RD_NT and tODTLoff_RD_NT are independently set and calculated from CL.

Figure 204 — Example of Burst Read Operation with ODTLon_RD_NT_offset Set Incorrectly

5.4 On-Die Termination for CA, CS, CK_t, CK_c

The DDR5 DRAM includes ODT (On-Die Termination) termination resistance for CK_t, CK_c, CS and CA signals.

The ODT feature is designed to improve signal integrity of the memory channel by allowing the DRAM controller to turn on and off termination resistance for any target DRAM devices via MR setting.

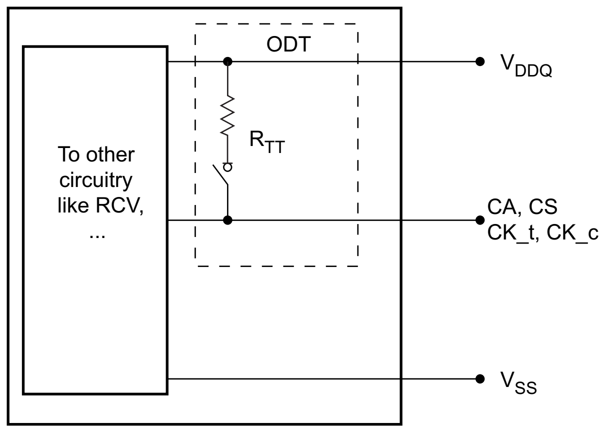


Figure 205 — A Simple Functional Representation of the DRAM CA ODT Feature

The ODT termination resistance during power up will be set to the default values based on MR32 and MR33. The ODT resistance values can be configured by those same registers.

On-Die Termination effective resistance RTT is define by MRS bits. ODT is applied to CK_t, CK_c, CS, and CA pins

$$RTT = \frac{VDDQ\text{-}Vout}{|\ I\ out\ |}$$

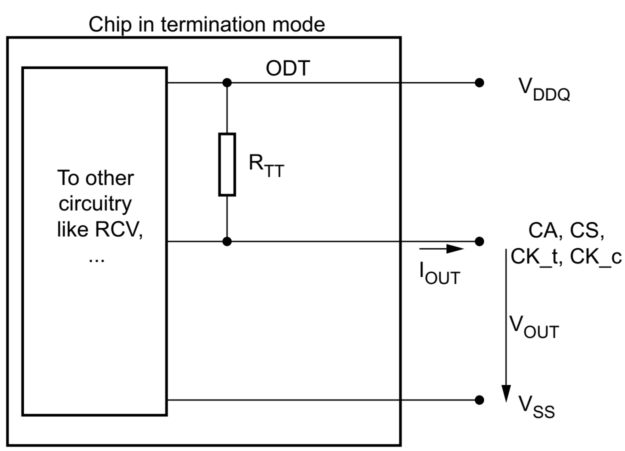


Figure 206 — A Functional Representation of the On-die Termination

5.4.1 Supported On-Die Termination Values

On-die termination effective Rtt values supported are 480, 240,120, 80, 60, and 40 ohms

Table 190 — ODT Electrical Characteristics RZQ=240 Ω +/-1%
Entire Temperature Operation Range; after Proper ZQ Calibration; VDD=VDDQ

MR	RTT	Vout	Min	Nom	Max	Unit	Note
MR32 for CK & CS MR33 for CA	480 Ω	V_{OLdc}= 0.5* VDDQ	0.7	1	1.4	R_{ZQ}*2	1, 2, 4
		V_{OMdc}= 0.8* VDDQ	0.7	1	1.3	R_{ZQ}*2	1, 2, 4
		V_{OHdc}= 0.95* VDDQ	0.6	1	1.3	R_{ZQ}*2	1, 2, 4
MR32 for CK & CS MR33 for CA	240 Ω	V_{OLdc}= 0.5* VDDQ	0.9	1	1.25	R_{ZQ}	1, 2, 4
		V_{OMdc}= 0.8* VDDQ	0.9	1	1.1	R_{ZQ}	1, 2, 4
		V_{OHdc}= 0.95* VDDQ	0.8	1	1.1	R_{ZQ}	1, 2, 4
MR32 for CK & CS MR33 for CA	120 Ω	V_{OLdc}= 0.5* VDD	0.9	1	1.25	R_{ZQ}/2	1, 2, 4, 5
		V_{OMdc}= 0.8* VDD	0.9	1	1.1	R_{ZQ}/2	1, 2, 4, 5
		V_{OHdc}= 0.95* VDD	0.8	1	1.1	R_{ZQ}/2	1, 2, 4, 5
MR32 for CK & CS MR33 for CA	80 Ω	V_{OLdc}= 0.5* VDDQ	0.9	1	1.25	R_{ZQ}/3	1, 2, 4
		V_{OMdc}= 0.8* VDDQ	0.9	1	1.1	R_{ZQ}/3	1, 2, 4
		V_{OHdc}= 0.95* VDDQ	0.8	1	1.1	R_{ZQ}/3	1, 2, 4
MR32 for CK & CS MR33 for CA	60 Ω	V_{OLdc}= 0.5* VDDQ	0.9	1	1.25	R_{ZQ}/4	1, 2, 4
		V_{OMdc}= 0.8* VDDQ	0.9	1	1.1	R_{ZQ}/4	1, 2, 4
		V_{OHdc}= 0.95* VDDQ	0.8	1	1.1	R_{ZQ}/4	1, 2, 4
MR32 for CK & CS MR33 for CA	40 Ω	V_{OLdc}= 0.5* VDDQ	0.9	1	1.25	R_{ZQ}/6	1, 2, 4
		V_{OMdc}= 0.8* VDDQ	0.9	1	1.1	R_{ZQ}/6	1, 2, 4
		V_{OHdc}= 0.95* VDDQ	0.8	1	1.1	R_{ZQ}/6	1, 2, 4
Mismatch CA-CA within Device		0.8* VDDQ	0		10	%	1, 2, 3, 4

NOTE 1 The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see “ZQ Calibration Commands” section.

NOTE 2 Pull-up ODT resistors are recommended to be calibrated at 0.8*VDDQ. Other calibration schemes may be used to achieve the linearity spec shown above, e.g., calibration at 0.5*VDDQ and 0.95*VDDQ.

NOTE 3 CA to CA mismatch within device variation for a given component including CS, CK_t and CK_c (characterized)

$$\text{CA-CA Mismatch in a Device} = \left(\frac{RTTMax - RTTMin}{RTTNOM}\right) \times 100$$

NOTE 4 Without ZQ calibration ODT effective RTT values have an increased tolerance of +/- 30%

NOTE 5 CK ODT RZQ/2 (120Ω), CS ODT RZQ/2 (120Ω), and CA ODT RZQ/2 (120 Ω) are only supported at >5200 MT/s data rates.

5.5 On-Die Termination for Loopback Signals

The DDR5 DRAM includes ODT (On-Die Termination) termination resistance for the Loopback signals LBDQS and LBDQ. The ODT feature is designed to improve signal integrity of the memory channel by allowing the DRAM controller to turn on and off termination resistance for any target DRAM devices via MR setting.

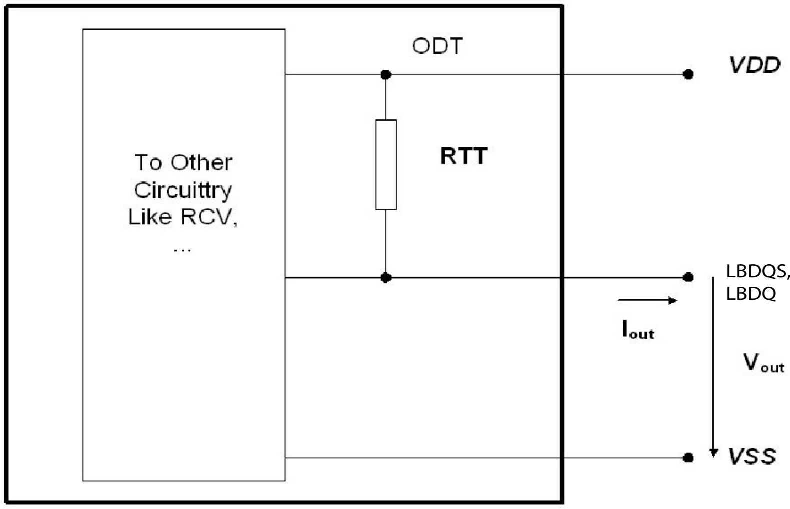


Figure 207 — Functional Representation of Loopback ODT

The ODT termination resistance during power up will be set to the default RTT_OFF values based on MR36:OP[2:0] definition. The ODT resistance values can be configured by MR26:OP[2:0].

On-Die Termination effective resistance RTT is defined by MR36:OP[2:0] bits = 48 ohms.

ODT is applied to Loopback signals LBDQS and LBDQ. On die termination effective Rtt values supported for the Loopback pins is 48 ohms.

$$RTT = \frac{VDDQ - Vout}{|\,Iout\,|}$$

Table 191 — ODT Electrical Characteristics RZQ=240 Ω +/-1%
Entire Temperature Operation Range; after Proper ZQ Calibration; VDD=VDDQ

RTT	Vout	Min	Nom	Max	Unit	NOTE
48 ohms	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/5	1, 2
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/5	1, 2
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/5	1, 2
Mismatch LBDQS - LBDQ within device	VOMdc = 0.8* VDDQ	0		8	%	1, 2, 3

NOTE 1 The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see “ZQ Calibration Commands” section.

NOTE 2 Pull-up ODT resistors are recommended to be calibrated at 0.8*VDDQ. Other calibration schemes may be used to achieve the linearity spec shown above, e.g. calibration at 0.5*VDDQ and 0.95*VDDQ.

NOTE 3 Loopback ODT mismatch within device variation for a given component including LBDQS and LBDQ

$$\textit{LBDQS-LBDQ Mismatch in a Device} = \left(\frac{RTTMax - RTTMin}{RTTNOM}\right) \times 100$$

5.6 On-Die Termination Timing Definitions

5.6.1 Test Load for ODT Timings

The reference load for On-Die Termination (ODT) timings is defined in Figure 208:

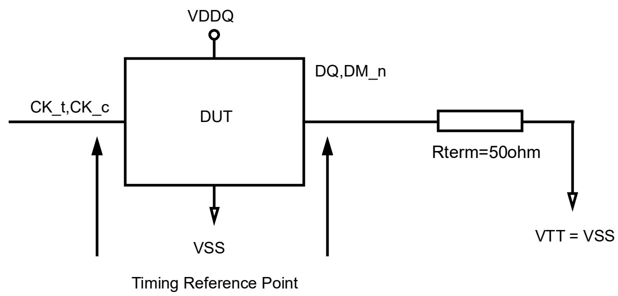


Figure 208 — ODT Timing Reference Load

5.6.2 tADC Measurement Method

Table 192 — tADC Measurement Timing Definitions

Measured Parameter	Begin Point Definition	End Point Definition	Figure
tADC for Target DRAM Write	End of tODTLoff_WR at CK_t/CK_c differential crossing point	Extrapolated point at VRTT_WR	Figure 209
	End of tODTLon_WR at CK_t/CK_c differential crossing point	Extrapolated point at VSS	
tADC for Non Target DRAM Write	End of tODTLoff_WR_NT at CK_t/CK_c differential crossing point	Extrapolated point at VRTT_NOM_WR	Figure 210
	End of tODTLon_WR_NT at CK_t/CK_c differential crossing point	Extrapolated point at VSS	
tADC for Non Target DRAM Read	End of tODTLoff_RD_NT at CK_t/CK_c differential crossing point	Extrapolated point at VRTT_NOM_RD	Figure 211
	End of tODTLon_RD_NT at CK_t/CK_c differential crossing point	Extrapolated point at VSS	

Table 193 — Reference Setting for ODT Timing Measurement

Measured Parameter	RTT_PARK	RTT_WR	RTT_NOM_WR	RTT_NOM_RD	Vsw1	Vsw2	Figure	Note
tADC for Target DRAM Write	RTT_OFF	RZQ/7	-	-	0.20V	0.40V	208	1, 4
tADC for Non Target DRAM Write	RTT_OFF	-	RZQ/7	-	0.20V	0.40V	209	2, 4
tADC for Non Target DRAM Read	RTT_OFF	-	-	RZQ/7	0.20V	0.40V	210	3, 4

NOTE 1 MR34 OP[2:0]=000B (RTT_PARK Setting); MR34 OP[5:3]=111B (RTT_WR Setting)

NOTE 2 MR34 OP[2:0]=000B (RTT_PARK Setting); MR35 OP[2:0]=111B (RTT_NOM_WR Setting)

NOTE 3 MR34 OP[2:0]=000B (RTT_PARK Setting); MR35 OP[5:3]=111B (RTT_NOM_RD Setting)

NOTE 4 tADC measurement method applies to DQs/DM only. It does not apply to DQS_t, DQS_c, TDQS_t and TDQS_c since there is no transition from DQS_RTT_PARK.

5.6.2 tADC Measurement Method (cont'd)

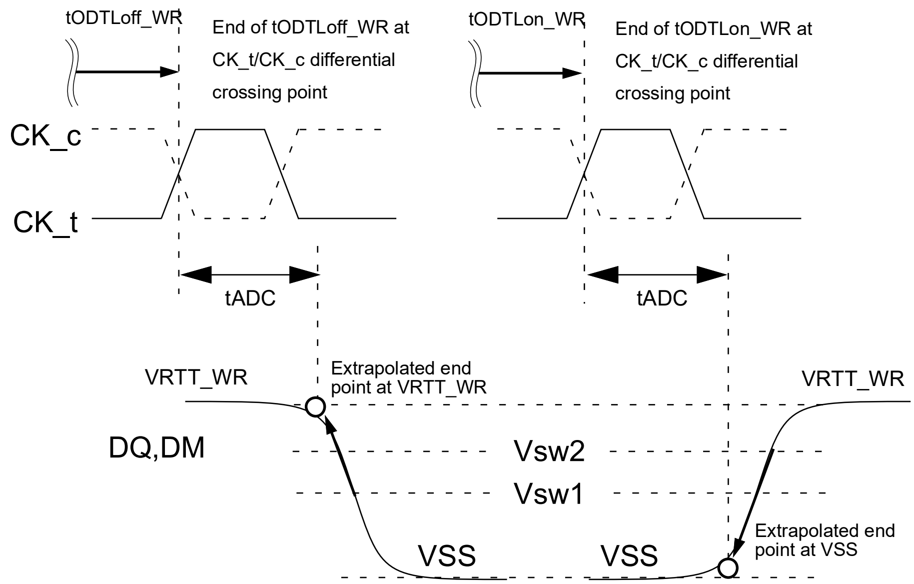


Figure 209 — tADC Measurement Method Target DRAM Write

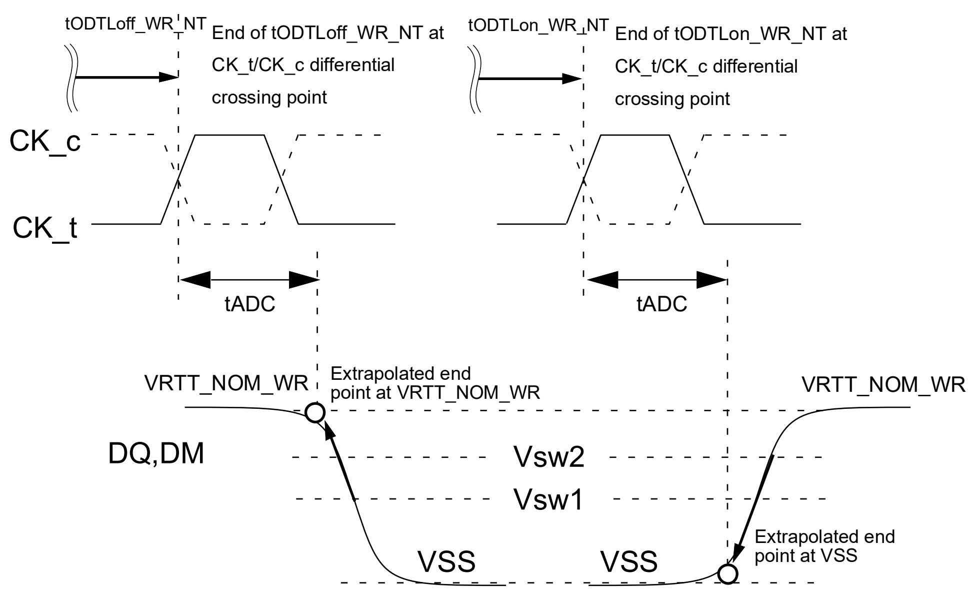


Figure 210 — tADC Measurement Method Non-Target DRAM Write

5.6.2 tADC Measurement Method (cont'd)

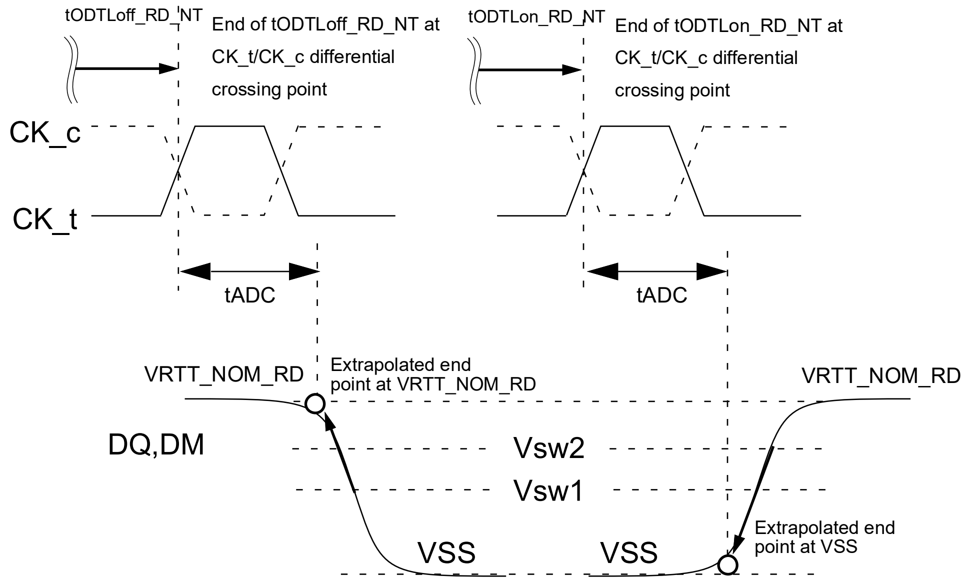


Figure 211 — tADC Measurement Method Non-Target DRAM Read

6 AC and DC Operating Conditions

6.1 Absolute Maximum Ratings

Table 194 — Absolute Maximum DC Ratings

Symbol	Parameter	Rating	Units	NOTE
VDD	Voltage on VDD pin relative to Vss	-0.3 ~ 1.4	V	1, 3
VDDQ	Voltage on VDDQ pin relative to Vss	-0.3 ~ 1.4	V	1, 3
VPP	Voltage on VPP pin relative to Vss	-0.3 ~ 2.1	V	4
V_{IN}, V_{OUT}	Voltage on any pin relative to Vss	-0.3 ~ 1.4	V	1, 3, 5
T_{STG}	Storage Temperature	-55 to +100	°C	1, 2

NOTE 1 Stresses greater than those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress rating only and functional operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect reliability

NOTE 2 Storage Temperature is the case surface temperature on the center/top side of the DRAM. For the measurement conditions, please refer to JESD51-2 standard.

NOTE 3 VDD and VDDQ must be within 300 mV of each other at all times. When VDD and VDDQ are less than 500 mV

NOTE 4 VPP must be equal or greater than VDD/VDDQ at all times.

NOTE 5 Overshoot area above 1.5 V is specified in Section 8.3.4, Section 8.3.5, and Section 8.3.6.

6.2 DC Operating Conditions

Table 195 — DC Operating Conditions

Symbol	Parameter	Low Freq Voltage Spec Freq: DC to 2 MHz				Z(f) [4] Spec Freq: 2 MHz to 10 MHz		Z(f) [4] Spec Freq: 20 MHz		Notes
		Min.	Typ.	Max.	Unit	Zmax	Unit	Zmax	Unit	
VDD	Device Supply Voltage	1.067 (-3%)	1.1	1.166 (+6%)	V	10	mOhm	20	mOhm	1, 2, 3
VDDQ	Supply Voltage for I/O	1.067 (-3%)	1.1	1.166 (+6%)	V	10	mOhm	20	mOhm	1, 2, 3
VPP	Core Power Voltage	1.746 (-3%)	1.8	1.908 (+6%)	V	10	mOhm	20	mOhm	3

NOTE 1 VDD must be within 66 mv of VDDQ

NOTE 2 AC parameters are measured with VDD and VDDQ tied together.

NOTE 3 This includes all voltage noise from DC to 2 MHz at the DRAM package ball.

NOTE 4 Z(f) is defined for all pins per voltage domain. Z(f) does not include the DRAM package and silicon die.

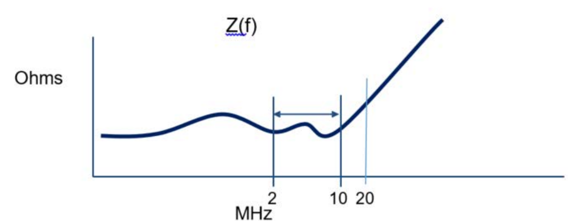


Figure 212 — Zprofile/Z(f) of the System at the DRAM Package Solder Ball (without DRAM Component)

6.2 DC Operating Conditions (cont'd)

A simplified electrical system load model for Z(F) with the general frequency response is shown in Figure 213. The resistance and inductance can be scaled to generalize the spec response to the DRAM pin.

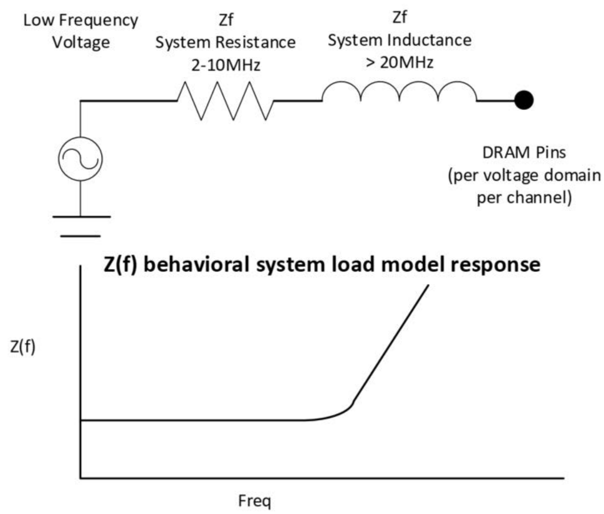


Figure 213 — Simplified Z(f) Electrical Model and Frequency Response of PDN at the DRAM Pin without the DRAM Component

6.3 DRAM Component Operating Temperature Range

Table 196 — DC Operating Temperature Range

Symbol	Parameter	Temperature Range (Unit: °C)		Grade	Notes
		Min	Max		
Toper_normal	Normal Operating Temperature	0	85	NT	1, 2, 3, 4
Toper_extended	Extended Operating Temperature	0	95	XT	1, 2, 3, 4, 5

NOTE 1 All operating temperature symbols, ranges, acronyms are referred from JESD402-1.

NOTE 2 Operating Temperature is the case surface temperature on the center/top side of the DRAM. For the measurement conditions, please refer to JESD51-2 standard.

NOTE 3 All DDR5 SDRDAMs are required to operate in NT and XT temperature ranges.

NOTE 4 When operating above 85 °C, the host shall provide appropriate Refresh mode controls associated with the increased temperature range. The full description of these settings are defined in Table 68 in section 4.13.5

NOTE 5 Operating Temperature for 3DS needs to be derated by the number of DRAM dies as: $[T_{OPER} - (2.5\ °C \times \log_2 N)]$, where N is the number of the stacked dies.

7 AC and DC Global Definitions

7.1 Bit Error Rate

7.1.1 Introduction

This section provides an overview of the Bit Error Rate (BER) and the desired Statistical Level of Confidence.

7.1.2 General Equation

$$n = \left(\frac{1}{BER}\right)\left[-\ln(1 - SLC) + \ln\left(\sum_{k=0}^{N} \frac{(n \cdot BER)^k}{k!}\right)\right]$$

Where:

n = number of bits in a trial
SLC = statistical level of confidence
BER = Bit Error Rate
k = intermediate number of specific errors found in trial
N = number of errors recorded during trial

If no, errors are assumed in a given test period, the second term drops out and the equation becomes:

$$n = \left(\frac{1}{BER}\right)[-\ln(1 - SLC)]$$

JEDEC recommends testing to 99.5% confidence levels; however, one may choose a number that is viable for their own manufacturing levels. To determine how many bits of data should be sent (again, assuming zero errors, or N=0), using BER=E^{-9} and confidence level SLC=99.5%, the result is n=(1/BER)(-ln(1-0.995) =5.298x10^{9}.

Results for commonly used confidence levels of 99.5% down to 70% are shown in Table 197.

Table 197 — Estimated Number of Transmitted Bits (n) for the Confidence Level of 70% to 99.5%

Number Errors	n = -ln(1-SLC)/BER							
	99.5%	99%	95%	90%	85%	80%	75%	70%
0	5.298/BER	4.61/BER	2.99/BER	2.3/BER	1.90/BER	1.61/BER	1.39/BER	1.20/BER

7.1.3 Minimum Bit Error Rate (BER) Requirements

Table 198 specifies the UIavg and Bit Error Rate requirements over which certain receiver and transmitter timing and voltage specifications need to be validated assuming a 99.5% confidence level at BER=E^{-9}.

Table 198 — Minimum BER Requirements for Rx/Tx Timing and Voltage Tests

Parameter	Symbol	DDR5-3200 to DDR5-8800			Unit	Notes
		Min	Nom	Max		
Average UI	UI_{AVG}	0.999* nominal	1000/f	1.001* nominal	ps	1
Number of UI (min)	$N_{Min\_UI\_Validation}$	5.3x109	-	-	UI	2
Bit Error Rate	BER_{Lane}	-	-	E-16	Events	3, 4, 5

NOTE 1 Average UI size, “f” is data rate

NOTE 2 # of UI over which certain Rx/Tx timing and voltage specifications need to be validated assuming a 99.5% confidence level at BER=E^{-9}.

NOTE 3 This is a system parameter. It is the raw bit error rate for every lane before any logical PHY or link layer based correction. It may not be possible to have a validation methodology for this parameter for a standalone transmitter or standalone receiver, therefore, this parameter has to be validated in selected systems using a suitable methodology as deemed by the platform.

NOTE 4 Bit Error Rate per lane. This is a raw bit error rate before any correction. This parameter is primarily used to determine electrical margins during electrical analysis and measurements that are located between two interconnected devices.

NOTE 5 This is the minimum BER requirements for testing timing and voltage parameters listed in Input Clock Jitter, Rx DQS & DQ Voltage Sensitivity, Rx DQS Jitter Sensitivity, Rx DQ Stressed Eye, Tx DQS Jitter, Tx DQ Jitter, and Tx DQ Stressed EH/EW specifications.

7.2 Unit Interval and Jitter Definitions

This section describes the Unit Interval (UI) and UI Jitter definitions associated with the jitter parameters specified in the Input Clock Jitter, Rx Stressed Eye, Tx DQS Jitter, and Tx DQ Jitter sections of this specification.

7.2.1 Unit Interval (UI)

The definition of Unit Interval (UI) is a minimum time interval between condition changes of a signal. DDR-based signals are referenced to the differential crossing point of CK_t and CK_c. 2UI=1tCK for DDR-based signals (for example, DQ, DQS).

The UI definitions shown in Figure 214 and Figure 215 are for DDR-based signals. The times at which the differential crossing points of the clock occur are defined at t1, t2, …, tn-1, tn. The UI at index “n” is defined as an arbitrary time in steady state, where n=0 is chosen as the starting crossing point.

$$UI_n = t_n - t_{n-1}$$

Figure 214 — UI Definition in Terms of Adjacent Edge Timings

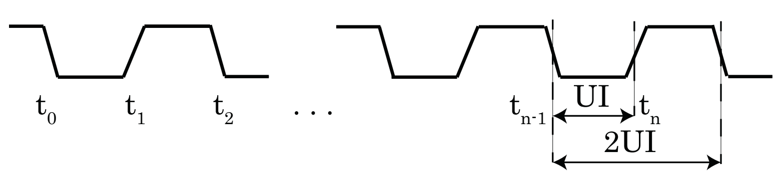


Figure 215 — UI Definition Using Clock Waveforms

7.2.2 UI Jitter Definition

If a number of UI edges are computed or measured at times t1, t2, …, tn-1, tn,..., tK, where K is the maximum number of samples, then the UI jitter at any instance “n” is defined in Figure 216, where T = the ideal UI size.

$$UI(jit)_n = (t_n - t_{n-1}) - T\,, \quad \mathrm{n=1,2,3,...,K}$$

Figure 216 — UI Jitter for “nth” UI Definition (in Terms of Ideal UI)

In a large sample with random Gaussian-like jitter (therefore very close to symmetric distribution), the average of all UI sizes usually turns out to be very close to the ideal UI size.

The equation described in Figure 216 assumes starting from an instant steady state, where n=0 is chosen as the starting point. 1 UI = one bit, which means 2 UI = one full cycle or time period of the forwarded strobe. Example: For 6.4 GT/s signaling, the forwarded strobe frequency is 3.2 GHz, or 1 UI = 156.25 ps.

Deterministic jitter is analyzed in terms of the peak-to-peak value and in terms of specific frequency components present in the jitter, isolating the causes for each frequency. Random jitter is unbounded and analyzed in terms of statistical distribution to convert to a bit error rate (BER) for the link.

7.2.3 UI-UI Jitter Definition

UI-UI (read as “UI to UI”) jitter is defined to be the jitter between two consecutive UI as shown in Figure 217.

$$\Delta UI_n = UI_n - UI_{n-1} \qquad \mathrm{n=2,3,...,K}$$

Figure 217 — UI-UI Jitter Definitions

7.2.4 Accumulated Jitter (Over “N” UI)

Accumulated jitter is defined as the jitter accumulated over any consecutive “N” UI as shown in Figure 218.

$$T^{N}_{acc} = \sum_{p=m}^{m+N-1} (UI_p - \overline{UI}) \qquad \mathrm{m=1,2,...,K\text{-}N}$$

Figure 218 — Definition of Accumulated Jitter (over “N” UI)

where $\overline{UI}$ is defined in the equation shown in Figure 219.

$$\overline{UI} = \frac{\sum_{p=1}^{K} UI_p}{K} \qquad \mathrm{p=1,2,...,N,...,K}$$

Figure 219 — Definition of $\overline{UI}$

8 AC and DC Input Measurement Levels

8.1 Overshoot and Undershoot Specifications for CAC - No Ballot

8.2 CA Rx Voltage and Timings

Note: The following draft assumes internal CA VREF. If the VREF is external, the specs will be modified accordingly.

The command and address (CA) including CS input receiver compliance mask for voltage and timing is shown in Figure 220. All CA signals apply the same compliance mask and CS signal applies its own compliance mask independently. All signals applied to the compliance mask operate in single data rate mode.

The CA input receiver mask for voltage and timing is shown in the figure below is applied across all CA pins. The receiver mask (Rx Mask) defines the area that the input signal must not encroach in order for the DRAM input receiver to be expected to be able to successfully capture a valid input signal; it is not the valid data-eye.

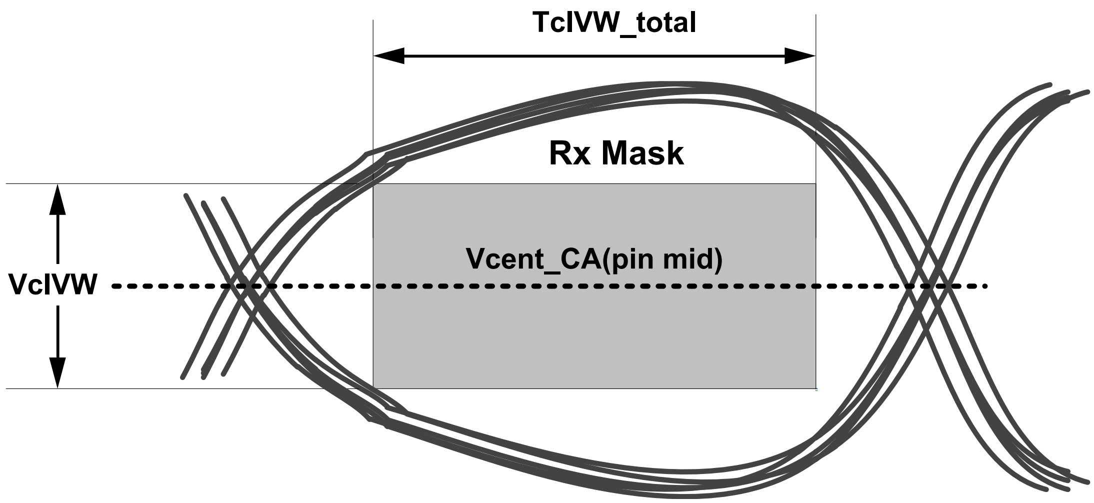


Figure 220 — CA Receiver (Rx) Mask

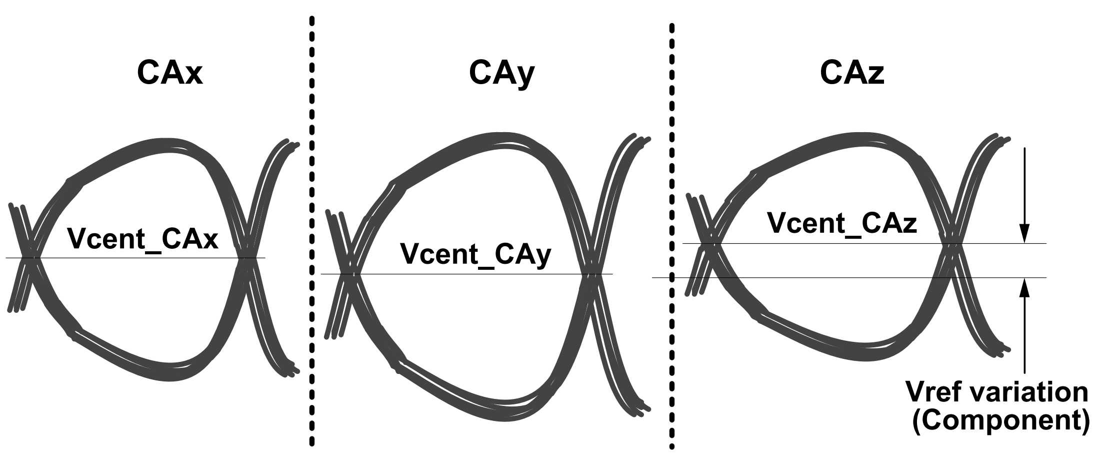


Figure 221 — Across Pin V_{REF}CA Voltage Variation

Vcent_CA(pin mid) is defined as the midpoint between the largest Vcent_CA voltage level and the smallest Vcent_CA voltage level across all CA and CS pins for a given DRAM component. Each CA Vcent level is defined by the center, i.e., widest opening, of the cumulative data input eye as depicted in Figure 221. This clarifies that any DRAM component level variation must be accounted for within the DRAM CA Rx mask. The component level V_{REF} will be set by the system to account for Ron and ODT settings.

8.2 CA Rx Voltage and Timings (cont'd)

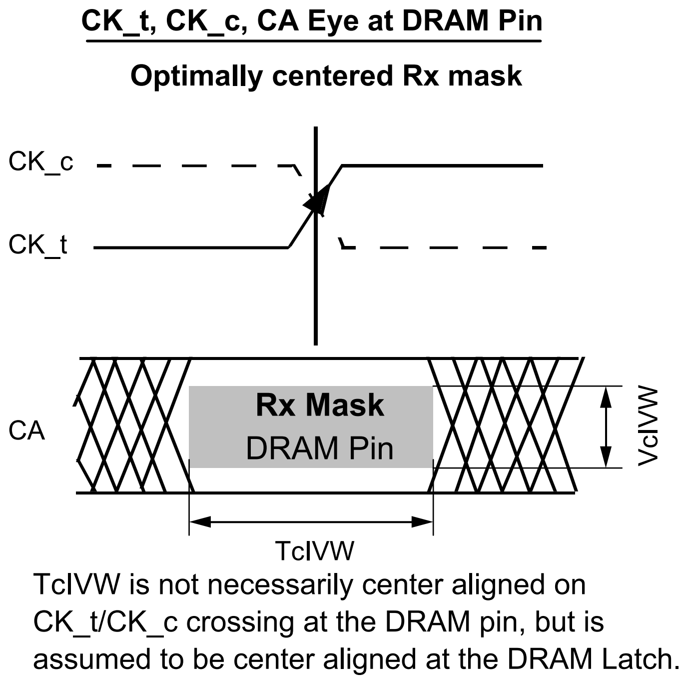
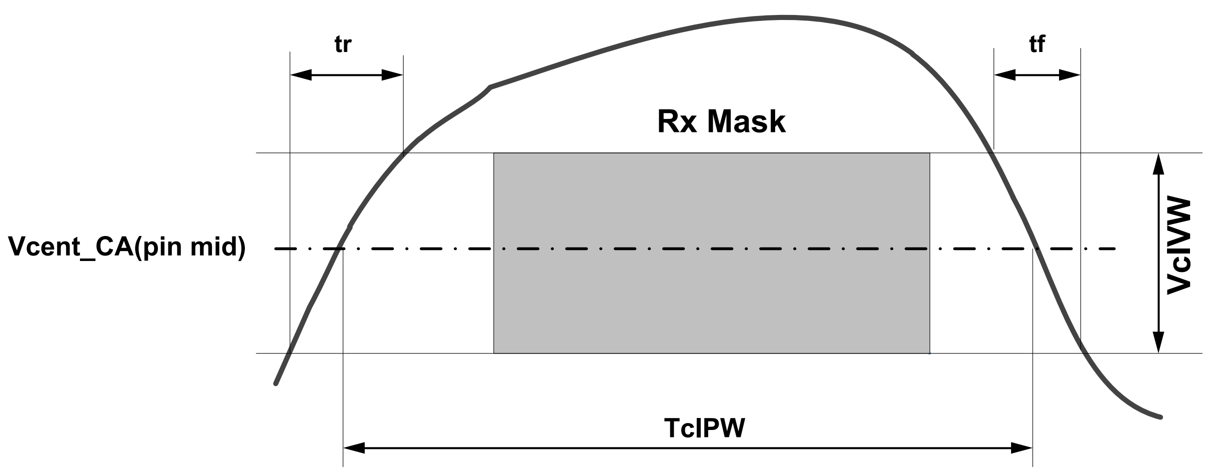


Figure 222 — CA Timings at the DRAM Pins



Note
1. SRIN_cIVW=VcIVW_Total/(tr or tf), signal must be monotonic within tr and tf range.

Figure 223 — CA TcIPW and SRIN_cIVW Definition (for Each Input Pulse)

8.2 CA Rx Voltage and Timings (cont'd)

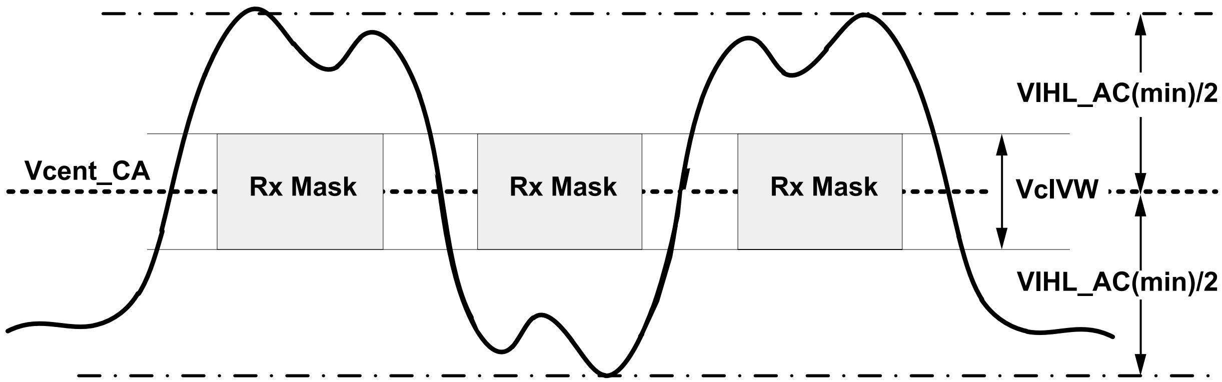


Figure 224 — CA VIHL_AC Definition (for Each Input Pulse)

Table 199 — DRAM CA, CS Parametric Values for DDR5-3200 to 4800

Parameter	Symbol	x4, x8, and x16 Devices										Unit	Notes
		DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800			
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Rx Mask voltage - p-p	VciVW	140	-	140	-	140	-	130	-	130	-	mV	1, 2, 4
Rx Timing Window	TcIVW	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	tCK	1, 2, 3, 4
CA Input Pulse Amplitude	VIHL_AC	160		160		160		150		150		mV	7
CA Input Pulse Width	TcIPW	0.58		0.58		0.58		0.58		0.58		tCK	5
Input Slew Rate over VcIVW	SRIN_cIVW	1	7	1	7	1	7	1	7	1	7	V/ns	6

NOTE 1 CA Rx mask voltage and timing parameters at the pin including voltage and temperature drift.
NOTE 2 Rx mask voltage VcIVW total(max) must be centered around Vcent_CA(pin mid).
NOTE 3 Rx differential CA to CK jitter total timing window at the VcIVW voltage levels.
NOTE 4 Defined over the CA internal V_{REF} range. The Rx mask at the pin must be within the internal V_{REF} CA range irrespective of the input signal common mode.
NOTE 5 CA only minimum input pulse width defined at the Vcent_CA(pin mid).
NOTE 6 Input slew rate over VcIVW Mask centered at Vcent_CA(pin mid).
NOTE 7 VIHL_AC does not have to be met when no transitions are occurring.

8.2 CA Rx Voltage and Timings (cont'd)

Table 200 — DRAM CA, CS Parametric Values for DDR5-5200 to 5600

Parameter	Symbol	x4 and x8 Devices												x16 Device		Unit	Notes
		Option 3				Option 2				Option 1							
		10 ps <= Tpkg_Delay_CA <=29 ps				29 ps < Tpkg_Delay_CA <=32 ps				32 ps < Tpkg_Delay_CA <=35 ps							
		DDR5-5200		DDR5-5600		DDR5-5200		DDR5-5600		DDR5-5200		DDR5-5600		DDR5-5200 to 5600			
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Rx Mask voltage - p-p	VciVW	110	-	110	-	95 (Cl max =0.55 pF) 100 (Cl max =0.5 pF)	-	95 (Cl max =0.55 pF) 100 (Cl max =0.5 pF)	-	80 (Cl max =0.55 pF) 85 (Cl max =0.5 pF)	-	80 (Cl max =0.55 pF) 85 (Cl max =0.5 pF)	-	110	-	mV	1, 2, 4
Rx Timing Window	TcIVW	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	UI*	1, 2, 3, 4, 8
CA Input Pulse Amplitude	VIHL_AC	125	-	125	-	115	-	115	-	110	-	110	-	125	-	mV	7
CA Input Pulse Width	TcIPW	0.58		0.58		0.58		0.58		0.58		0.58		0.58		UI*	5, 8
Input Slew Rate over VcIVW	SRIN_cIVW	1	7	1	7	1	7	1	7	1	7	1	7	1	7	V/ns	6

NOTE 1 CA Rx mask voltage and timing parameters at the pin including voltage and temperature drift.
NOTE 2 Rx mask voltage VcIVW total(max) must be centered around Vcent_CA(pin mid).
NOTE 3 Rx differential CA to CK jitter total timing window at the VcIVW voltage levels.
NOTE 4 Defined over the CA internal V_{REF} range. The Rx mask at the pin must be within the internal V_{REF} CA range irrespective of the input signal common mode.
NOTE 5 CA only minimum input pulse width defined at the Vcent_CA(pin mid).
NOTE 6 Input slew rate over VcIVW Mask centered at Vcent_CA(pin mid).
NOTE 7 VIHL_AC does not have to be met when no transitions are occurring.
NOTE 8 * UI=tck(avg)min

Table 201 — DRAM CA, CS Parametric Values for DDR5-6000 to 6400

Parameter	Symbol	x4 and x8 Devices												x16 Device		Unit	Notes
		Option 3				Option 2				Option 1							
		10 ps <= Tpkg_Delay_CA <=29 ps				29 ps < Tpkg_Delay_CA <=32 ps				32 ps < Tpkg_Delay_CA <=35 ps							
		DDR5-6000		DDR5-6400		DDR5-6000		DDR5-6400		DDR5-6000		DDR5-6400		DDR5-6000 to 6400			
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Rx Mask voltage - p-p	VciVW	95	-	95	-	85	-	85	-	80	-	80	-	95	-	mV	1, 2, 4
Rx Timing Window	TcIVW	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	UI*	1, 2, 3, 4, 8
CA Input Pulse Amplitude	VIHL_AC	115	-	115	-	105	-	105	-	100	-	100	-	115	-	mV	7
CA Input Pulse Width	TcIPW	0.58		0.58		0.58		0.58		0.58		0.58		0.58		UI*	5, 8
Input Slew Rate over VcIVW	SRIN_cIVW	1	7	1	7	1	7	1	7	1	7	1	7	1	7	V/ns	6

NOTE 1 CA Rx mask voltage and timing parameters at the pin including voltage and temperature drift.
NOTE 2 Rx mask voltage VcIVW total(max) must be centered around Vcent_CA(pin mid).
NOTE 3 Rx differential CA to CK jitter total timing window at the VcIVW voltage levels.
NOTE 4 Defined over the CA internal V_{REF} range. The Rx mask at the pin must be within the internal V_{REF} CA range irrespective of the input signal common mode.
NOTE 5 CA only minimum input pulse width defined at the Vcent_CA(pin mid).
NOTE 6 Input slew rate over VcIVW Mask centered at Vcent_CA(pin mid).
NOTE 7 VIHL_AC does not have to be met when no transitions are occurring.
NOTE 8 * UI=tck(avg)min

8.2 CA Rx Voltage and Timings (cont'd)

Table 202 — DRAM CA, CS Parametric Values for DDR5-6800 to 7200

Parameter	Symbol	x4 and x8 Devices												x16 Device		Unit	Notes
		Option 3				Option 2				Option 1							
		10 ps <= Tpkg_Delay_CA <=29 ps				29 ps < Tpkg_Delay_CA <=32 ps				32 ps < Tpkg_Delay_CA <=35 ps							
		DDR5-6800		DDR5-7200		DDR5-6800		DDR5-7200		DDR5-6800		DDR5-7200		DDR5-6800 to 7200			
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Rx Mask voltage - p-p	VciVW	90	-	90	-	80	-	80	-	50	-	50	-	90	-	mV	1, 2, 4
Rx Timing Window	TcIVW	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	UI*	1, 2, 3, 4, 8
CA Input Pulse Amplitude	VIHL_AC	110	-	110	-	100	-	100	-	70	-	70	-	110	-	mV	7
CA Input Pulse Width	TcIPW	0.58		0.58		0.58		0.58		0.58		0.58		0.58		UI*	5, 8
Input Slew Rate over VcIVW	SRIN_cIVW	1	7	1	7	1	7	1	7	1	7	1	7	1	7	V/ns	6

NOTE 1 CA Rx mask voltage and timing parameters at the pin including voltage and temperature drift.
NOTE 2 Rx mask voltage VcIVW total(max) must be centered around Vcent_CA(pin mid).
NOTE 3 Rx differential CA to CK jitter total timing window at the VcIVW voltage levels.
NOTE 4 Defined over the CA internal V_{REF} range. The Rx mask at the pin must be within the internal V_{REF} CA range irrespective of the input signal common mode.
NOTE 5 CA only minimum input pulse width defined at the Vcent_CA(pin mid).
NOTE 6 Input slew rate over VcIVW Mask centered at Vcent_CA(pin mid).
NOTE 7 VIHL_AC does not have to be met when no transitions are occurring.
NOTE 8 * UI=tck(avg)min

Table 203 — DRAM CA, CS Parametric Values for DDR5-7600 to 8000

Parameter	Symbol	x4 and x8 Devices												x16 Device		Unit	Notes
		Option 3				Option 2				Option 1							
		10 ps <= Tpkg_Delay_CA <=29 ps				29 ps < Tpkg_Delay_CA <=32 ps				32ps < Tpkg_Delay_CA <=35 ps							
		DDR5-7600		DDR5-8000		DDR5-7600		DDR5-8000		DDR5-7600		DDR5-8000		DDR5-7600 to 8000			
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Rx Mask voltage - p-p	VciVW	85	-	85	-	75	-	75	-	45	-	45	-	85	-	mV	1, 2, 4
Rx Timing Window	TcIVW	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	UI*	1, 2, 3, 4, 8
CA Input Pulse Amplitude	VIHL_AC	105	-	105	-	95	-	95	-	65	-	65	-	105	-	mV	7
CA Input Pulse Width	TcIPW	0.58		0.58		0.58		0.58		0.58		0.58		0.58		UI*	5, 8
Input Slew Rate over VcIVW	SRIN_cIVW	1	7	1	7	1	7	1	7	1	7	1	7	1	7	V/ns	6

NOTE 1 CA Rx mask voltage and timing parameters at the pin including voltage and temperature drift.
NOTE 2 Rx mask voltage VcIVW total(max) must be centered around Vcent_CA(pin mid).
NOTE 3 Rx differential CA to CK jitter total timing window at the VcIVW voltage levels.
NOTE 4 Defined over the CA internal V_{REF} range. The Rx mask at the pin must be within the internal V_{REF} CA range irrespective of the input signal common mode.
NOTE 5 CA only minimum input pulse width defined at the Vcent_CA(pin mid).
NOTE 6 Input slew rate over VcIVW Mask centered at Vcent_CA(pin mid).
NOTE 7 VIHL_AC does not have to be met when no transitions are occurring.
NOTE 8 * UI=tck(avg)min

8.2 CA Rx Voltage and Timings (cont'd)

Table 204 — DRAM CA, CS Parametric Values for DDR5-8400 to 8800

Parameter	Symbol	x4 and x8 Devices												x16 Device		Unit	Notes
		Option 3				Option 2				Option 1							
		10 ps <= Tpkg_Delay_CA <=29 ps				29 ps < Tpkg_Delay_CA <=32 ps				32 ps < Tpkg_Delay_CA <=35 ps							
		DDR5-8400		DDR5-8800		DDR5-8400		DDR5-8800		DDR5-8400		DDR5-8800		DDR5-8400 to 8800			
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Rx Mask voltage - p-p	VciVW	80	-	80	-	70	-	70	-	40	-	40	-	80	-	mV	1, 2, 4
Rx Timing Window	TcIVW	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	UI*	1, 2, 3, 4, 8
CA Input Pulse Amplitude	VIHL_AC	100	-	100	-	90	-	90	-	60	-	60	-	100	-	mV	7
CA Input Pulse Width	TcIPW	0.58		0.58		0.58		0.58		0.58		0.58		0.58		UI*	5, 8
Input Slew Rate over VcIVW	SRIN_cIVW	1	7	1	7	1	7	1	7	1	7	1	7	1	7	V/ns	6

NOTE 1 CA Rx mask voltage and timing parameters at the pin including voltage and temperature drift.
NOTE 2 Rx mask voltage VcIVW total(max) must be centered around Vcent_CA(pin mid).
NOTE 3 Rx differential CA to CK jitter total timing window at the VcIVW voltage levels.
NOTE 4 Defined over the CA internal V_{REF} range. The Rx mask at the pin must be within the internal V_{REF} CA range irrespective of the input signal common mode.
NOTE 5 CA only minimum input pulse width defined at the Vcent_CA(pin mid).
NOTE 6 Input slew rate over VcIVW Mask centered at Vcent_CA(pin mid).
NOTE 7 VIHL_AC does not have to be met when no transitions are occurring.
NOTE 8 * UI=tck(avg)min

8.3 Input Clock Jitter Specification

8.3.1 Overview

The clock is being driven to the DRAM either by the RCD for L/RDIMM modules, or by the host for U/SODIMM modules (Figure 225).

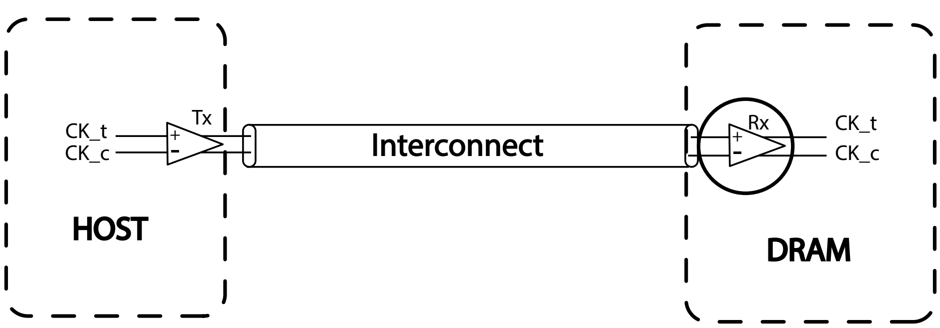


Figure 225 — HOST Driving Clock Signals to the DRAM

8.3.2 Specification for DRAM Input Clock Jitter

The Random Jitter (Rj) specified is a random jitter meeting a Gaussian distribution. The Deterministic Jitter (Dj) specified is bounded. Input clock violating the min/max jitter values may result in malfunction of the DDR5 SDRAM device.

Table 205 — DRAM Input Clock Jitter Specifications for DDR5-3200 to 4400

[BUJ=Bounded Uncorrelated Jitter; DCD=Duty Cycle Distortion; Dj=Deterministic Jitter; Rj=Random Jitter; Tj=Total jitter; pp=Peak-to-Peak]

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
DRAM Reference clock frequency	tCK	0.9999 * f0	1.0001 * f0	0.9999 * f0	1.0001 * f0	0.9999 * f0	1.0001 * f0	0.9999 * f0	1.0001 * f0	0.9999 * f0	1.0001 * f0	MHz	1, 11
Duty Cycle Error	tCK_Duty_UI_Error	-	0.05	-	0.05	-	0.05	-	0.05		0.05	UI	1, 4, 11
Rj RMS value of 1-UI Jitter	tCK_1UI_Rj_NoBUJ	-	0.0037	-	0.0037	-	0.0037	-	0.0037	-	0.0037	UI (RMS)	3, 5, 11
Dj pp value of 1-UI Jitter	tCK_1UI_Dj_NoBUJ	-	0.030	-	0.030	-	0.030	-	0.030	-	0.030	UI	3, 6, 11
Tj value of 1-UI Jitter	tCK_1UI_Tj_NoBUJ	-	0.090	-	0.090	-	0.090	-	0.090	-	0.090	UI	3, 6, 11
Rj RMS value of N-UI Jitter, where N=2,3	tCK_NUI_Rj_NoBUJ, where N=2,3	-	0.0040	-	0.0040	-	0.0040	-	0.0040	-	0.0040	UI (RMS)	3, 7, 11
Dj pp value of N-UI Jitter, where N=2,3	tCK_NUI_Dj_NoBUJ, where N=2,3	-	0.074	-	0.074	-	0.074	-	0.074	-	0.074	UI	3, 7, 11
Tj value of N-UI Jitter, where N=2,3	tCK_NUI_Tj_NoBUJ, where N=2,3	-	0.140	-	0.140	-	0.140	-	0.140	-	0.140	UI	3, 8, 11
Rj RMS value of N-UI Jitter, where N=4,5,6,...,30	tCK_NUI_Rj_NoBUJ, where N=4,5,6,...,30	-	-	-	-	-	-	-	-	-	-	UI (RMS)	3, 9, 11, 12
Dj pp value of N-UI Jitter, N=4,5,6,...,30	tCK_NUI_Dj_NoBUJ, where N=4,5,6,...,30	-	-	-	-	-	-	-	-	-	-	UI	3, 10, 11, 12
Tj value of N-UI Jitter, N=4,5,6,...,30	tCK_NUI_Tj_NoBUJ, where N=4,5,6,...,30	-	-	-	-	-	-	-	-	-	-	UI	3, 10, 11, 12

NOTE 1 f0 = Data Rate/2, example: if data rate is 3200 MT/s, then f0=1600

NOTE 2 Rise and fall time slopes (V / nsec) are measured between +100 mV and -100 mV of the differential output of reference clock

NOTE 3 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining Tx lanes send patterns to the corresponding Rx receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 4 Duty Cycle Error defined as absolute difference between average value of all UI with that of average of odd UI, which in magnitude would equal absolute difference between average of all UI and average of all even UI.

NOTE 5 Rj RMS value of 1-UI jitter without BUJ, but on-die system-like noise present. This extraction is to be done after software correction of DCD

NOTE 6 Dj pp value of 1-UI jitter (after software assisted DCC). Without BUJ, but on-die system like noise present. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE 7 Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 1 < N < 4. This extraction is to be done after software correction of DCD

NOTE 8 Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 1 < N < 4. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE 9 Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 3 < N < 31. This extraction is to be done after software correction of DCD

NOTE 10 Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 3 < N < 31. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE 11 The validation methodology for these parameters will be covered in future ballots.

NOTE 12 If the clock meets total jitter Tj at BER of $1E^{-16}$, then meeting the individual Rj and Dj components of the spec can be considered optional. Tj is defined as Dj + 16.2*Rj for BER of $1E^{-16}$

8.3.2 Specification for DRAM Input Clock Jitter (cont'd)

Table 206 — DRAM Input Clock Jitter Specifications for DDR5-5200 to 6400

[BUJ=Bounded Uncorrelated Jitter; DCD=Duty Cycle Distortion; Dj=Deterministic Jitter; Rj=Random Jitter; Tj=Total jitter; pp=Peak-to-Peak]

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
DRAM Reference clock frequency	tCK	0.9999* f0	1.0001* f0	0.9999* f0	1.0001* f0	0.9999* f0	1.0001* f0	0.9999* f0	1.0001* f0	MHz	1, 11
Duty Cycle Error	tCK_Duty_UI_Error	-	0.05	-	0.05	-	0.05	-	0.05	UI	1, 4, 11
Rj RMS value of 1-UI Jitter	tCK_1UI_Rj_NoBUJ	-	0.0037	-	0.0037	-	0.0037	-	0.0037	UI (RMS)	3, 5, 11
Dj pp value of 1-UI Jitter	tCK_1UI_Dj_NoBUJ	-	0.030	-	0.030	-	0.030	-	0.030	UI	3, 6, 11
Tj value of 1-UI Jitter	tCK_1UI_Tj_NoBUJ	-	0.090	-	0.090	-	0.090	-	0.090	UI	3, 6, 11
Rj RMS value of N-UI Jitter, where N=2,3,4,5	tCK_NUI_Rj_NoBUJ, where N=2,3,4,5	-	0.0040	-	0.0040	-	0.0040	-	0.0040	UI (RMS)	3, 7, 11
Dj pp value of N-UI Jitter, where N=2,3,4,5	tCK_NUI_Dj_NoBUJ, where N=2,3,4,5	-	0.074	-	0.074	-	0.074	-	0.074	UI	3, 7, 11
Tj value of N-UI Jitter, where N=2,3,4,5	tCK_NUI_Tj_NoBUJ, where N=2,3,4,5	-	0.140	-	0.140	-	0.140	-	0.140	UI	3, 8, 11
Rj RMS value of N-UI Jitter, where N=6,...,30	tCK_NUI_Rj_NoBUJ, where N=6,...,30	-	-	-	-	-	-	-	-	UI (RMS)	3, 9, 11, 12
Dj pp value of N-UI Jitter, N=6,...,30	tCK_NUI_Dj_NoBUJ, where N=6,...,30	-	-	-	-	-	-	-	-	UI	3, 10, 11, 12
Tj value of N-UI Jitter, N=6,...,30	tCK_NUI_Tj_NoBUJ, where N=6,...,30	-	-	-	-	-	-	-	-	UI	3, 10, 11, 12

NOTE 1 f0 = Data Rate/2, example: if data rate is 3200 MT/s, then f0=1600

NOTE 2 Rise and fall time slopes (V / nsec) are measured between +100 mV and -100 mV of the differential output of reference clock

NOTE 3 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining Tx lanes send patterns to the corresponding Rx receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 4 Duty Cycle Error defined as absolute difference between average value of all UI with that of average of odd UI, which in magnitude would equal absolute difference between average of all UI and average of all even UI.

NOTE 5 Rj RMS value of 1-UI jitter without BUJ, but on-die system-like noise present. This extraction is to be done after software correction of DCD

NOTE 6 Dj pp value of 1-UI jitter (after software assisted DCC). Without BUJ, but on-die system like noise present. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE 7 Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $1 < N < 4$. This extraction is to be done after software correction of DCD

NOTE 8 Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $1 < N < 4$. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE 9 Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $3 < N < 31$. This extraction is to be done after software correction of DCD

NOTE 10 Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $3 < N < 31$. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE 11 The validation methodology for these parameters will be covered in future ballots.

NOTE 12 If the clock meets total jitter Tj at BER of $1E^{-16}$, then meeting the individual Rj and Dj components of the spec can be considered optional. Tj is defined as Dj + 16.2*Rj for BER of $1E^{-16}$

8.3.2 Specification for DRAM Input Clock Jitter (cont'd)

Table 207 — DRAM Input Clock Jitter Specifications for DDR5-6800 to 8400

[BUJ=Bounded Uncorrelated Jitter; DCD=Duty Cycle Distortion; Dj=Deterministic Jitter; Rj=Random Jitter; Tj=Total jitter; pp=Peak-to-Peak]

Parameter	Symbol	DDR5-6800		DDR5-7200		DDR5-7600-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max		
DRAM Reference clock frequency	tCK	0.9999* f0	1.0001* f0	0.9999* f0	1.0001* f0	0.9999* f0	1.0001* f0	MHz	1, 11
Duty Cycle Error	tCK_Duty_UI_Error	-	0.05	-	0.05	-	0.05	UI	1, 4, 11
Rj RMS value of 1-UI Jitter	tCK_1UI_Rj_NoBUJ	-	0.003145	-	0.003145	-	0.003145	UI (RMS)	3, 5, 11
Dj pp value of 1-UI Jitter	tCK_1UI_Dj_NoBUJ	-	0.0255	-	0.0255	-	0.0255	UI	3, 6, 11
Tj value of 1-UI Jitter	tCK_1UI_Tj_NoBUJ	-	0.0765	-	0.0765	-	0.0765	UI	3, 6, 11
Rj RMS value of N-UI Jitter, where N=2,3,4,5	tCK_NUI_Rj_NoBUJ, where N=2,3,4,5	-	0.0034	-	0.0034	-	0.0034	UI (RMS)	3, 7, 11
Dj pp value of N-UI Jitter, where N=2,3,4,5	tCK_NUI_Dj_NoBUJ, where N=2,3,4,5	-	0.0629	-	0.0629	-	0.0629	UI	3, 7, 11
Tj value of N-UI Jitter, where N=2,3,4,5	tCK_NUI_Tj_NoBUJ, where N=2,3,4,5	-	0.119	-	0.119	-	0.119	UI	3, 8, 11
Rj RMS value of N-UI Jitter, where N=6,...,30	tCK_NUI_Rj_NoBUJ, where N=6,...,30	-	-	-	-	-	-	UI (RMS)	3, 9, 11, 12
Dj pp value of N-UI Jitter, N=6,...,30	tCK_NUI_Dj_NoBUJ, where N=6,...,30	-	-	-	-	-	-	UI	3,1 0, 11, 12
Tj value of N-UI Jitter, N=6,...,30	tCK_NUI_Tj_NoBUJ, where N=6,...,30	-	-	-	-	-	-	UI	3, 10, 11, 12

NOTE 1 f0 = Data Rate/2, example: if data rate is 3200 MT/s, then f0=1600

NOTE 2 Rise and fall time slopes (V / nsec) are measured between +100 mV and -100 mV of the differential output of reference clock

NOTE 3 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining Tx lanes send patterns to the corresponding Rx receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 4 Duty Cycle Error defined as absolute difference between average value of all UI with that of average of odd UI, which in magnitude would equal absolute difference between average of all UI and average of all even UI.

NOTE 5 Rj RMS value of 1-UI jitter without BUJ, but on-die system-like noise present. This extraction is to be done after software correction of DCD

NOTE 6 Dj pp value of 1-UI jitter (after software assisted DCC). Without BUJ, but on-die system like noise present. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE 7 Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $1 < N < 4$. This extraction is to be done after software correction of DCD

NOTE 8 Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $1 < N < 4$. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE 9 Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $3 < N < 31$. This extraction is to be done after software correction of DCD

NOTE 10 Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $3 < N < 31$. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE 11 The validation methodology for these parameters will be covered in future ballots.

NOTE 12 If the clock meets total jitter Tj at BER of $1E^{-16}$, then meeting the individual Rj and Dj components of the spec can be considered optional. Tj is defined as Dj + 16.2*Rj for BER of $1E^{-16}$

8.4 Differential Input Clock (CK_t, CK_c) Cross Point Voltage (VIX)

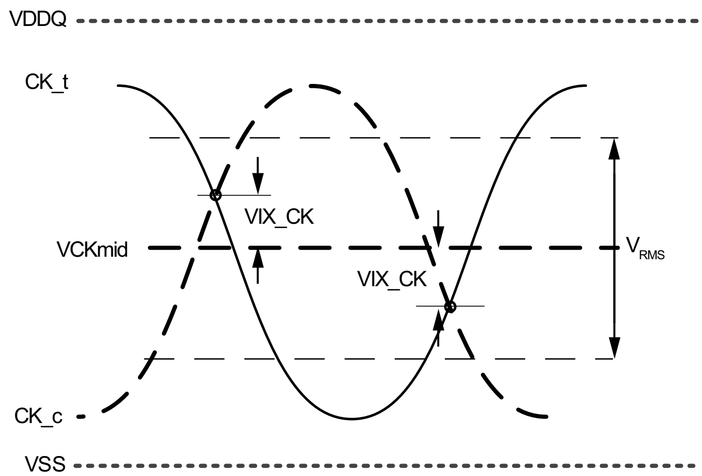


Figure 226 — VIX Definition (CK)

Table 208 — Crosspoint Voltage (VIX) for Differential Input Clock

Parameter	Symbol	DDR5-3200 - 8800		Unit	Notes
		Min	Max		
Clock differential input crosspoint voltage ratio	VIX_CK_Ratio	-	50	%	1, 2, 3

NOTE 1 The VIX_CK voltage is referenced to VCKmid(mean) = (CK_t voltage + CK_c voltage) /2, where the mean is over 8 UI

NOTE 2 VIX_CK_Ratio = (|VIX_CK| / |V_{RMS}|)*100%, where V_{RMS} = RMS(CK_t voltage - CK_c voltage)

NOTE 3 Only applies when both CK_t and CK_c are transitioning

8.5 Differential Input Clock Voltage Sensitivity

The differential input clock voltage sensitivity test provides the methodology for testing the receiver's sensitivity to clock by varying input voltage in the absence of Inter-Symbol Interference (ISI), jitter (Rj, Dj, DCD) and crosstalk noise. This specifies the Rx voltage sensitivity requirement. The system input swing to the DRAM must be larger than the DRAM Rx at the specified BER

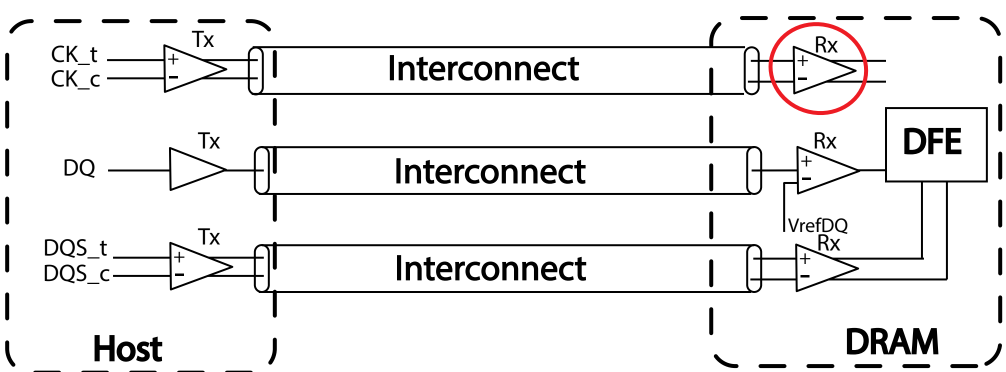


Figure 227 — Example of DDR5 Memory Interconnect

8.5.1 Differential Input Clock Voltage Sensitivity Parameter

Differential input clock (CK_t, CK_c) VRx_CK is defined and measured as shown in Table 209 through Table 211. The clock receiver must pass the minimum BER requirements for DDR5.

Table 209 — Differential Input Clock Voltage Sensitivity Parameter for DDR5-3200 to 4800

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Input Clock Voltage Sensitivity (differential pp)	VRx_CK	-	200	-	200	-	180	-	180	-	160	mV	1, 2

NOTE 1 Refer to the minimum BER requirements for DDR5
NOTE 2 The validation methodology for this parameter will be covered in future ballot(s)

Table 210 — Differential Input Clock Voltage Sensitivity Parameter for DDR5-5200 to 6400

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
Input Clock Voltage Sensitivity (differential pp)	VRx_CK	-	140	-	120	-	100 (*120)	-	100 (*120)	mV	1, 2, 3

NOTE 1 Refer to the minimum BER requirements for DDR5
NOTE 2 The validation methodology for this parameter will be covered in future ballot(s)
NOTE 3 * indicates that it's supported if the CK buffer is present on the module.

Table 211 — Differential Input Clock Voltage Sensitivity Parameter for DDR5-6800 to 8400

Parameter	Symbol	DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Input Clock Voltage Sensitivity (differential pp)	VRx_CK	-	100 (*120)	-	100 (*120)	-	100 (*120)	-	100 (*120)	-	100 (*120)	-	100 (*120)	mV	1, 2

NOTE 1 Refer to the minimum BER requirements for DDR5
NOTE 2 The validation methodology for this parameter will be covered in future ballot(s)
NOTE 3 * indicates that it's supported if the CK buffer is present on the module.

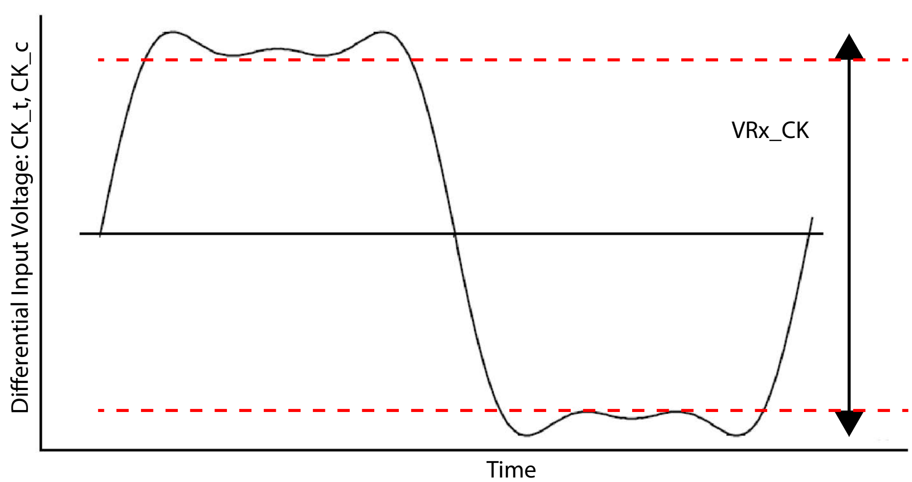


Figure 228 — VRx_CK

8.5.2 Differential Input Voltage Levels for Clock

Table 212 — Differential Clock (CK_t, CK_c) Input Levels for DDR5-3200 to DDR5-6400

From	Parameter [3]	DDR5 3200-6400	Note
$V_{IHdiff}CK$	Differential input high measurement level (CK_t, CK_c)	0.75 x $V_{diffpk-pk}$	1, 2
$V_{ILdiff}CK$	Differential input low measurement level (CK_t, CK_c)	0.25 x $V_{diffpk-pk}$	1, 2

NOTE 1 $V_{diffpk-pk}$ defined in Figure 229
NOTE 2 $V_{diffpk-pk}$ is the mean high voltage minus the mean low voltage over 8UI samples
NOTE 3 All parameters are defined over the entire clock common mode range

Table 213 — Differential Clock (CK_t, CK_c) Input Levels for DDR5-6800 to DDR5-8800

From	Parameter [3]	DDR5 6800-8800	Note
$V_{IHdiff}CK$	Differential input high measurement level (CK_t, CK_c)	0.75 x $V_{diffpk-pk}$	1, 2
$V_{ILdiff}CK$	Differential input low measurement level (CK_t, CK_c)	0.25 x $V_{diffpk-pk}$	1, 2

NOTE 1 $V_{diffpk-pk}$ defined in Figure 229
NOTE 2 $V_{diffpk-pk}$ is the mean high voltage minus the mean low voltage over 8UI samples
NOTE 3 All parameters are defined over the entire clock common mode range

8.5.3 Differential Input Slew Rate Definition for Clock (CK_t, CK_c)

Input slew rate for differential signals (CK_t, CK_c) are defined and measured as shown in Figure 229.

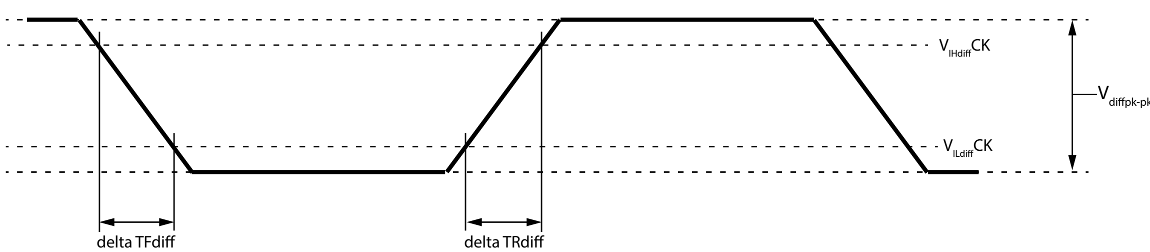


Figure 229 — Differential Input Slew Rate Definition for CK_t, CK_c

Table 214 — Differential Input Slew Rate Definition for CK_t, CK_c

Parameter	Measured		Defined by	Notes
	From	To		
Differential Input slew rate for rising edge (CK_t - CK_c)	$VIL_{diff}CK$	$VIH_{diff}CK$	$(VIH_{diff}CK - VIL_{diff}CK)$ /deltaTRdiff	
Differential Input slew rate for falling edge (CK_t - CK_c)	$VIH_{diff}CK$	$VIL_{diff}CK$	$(VIH_{diff}CK - VIL_{diff}CK)$ /deltaTFdiff	

8.5.3 Differential Input Slew Rate Definition for Clock (CK_t, CK_c) (cont'd)

Table 215 — Differential Input Slew Rate for CK_t, CK_c for DDR5-3200 to DDR5-4800

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Differential Input Slew Rate for CK_t, CK_c	SRIdiff_CK	2	14	2	14	2	14	2	14	2	14	V/ns	

Table 216 — Differential Input Slew Rate for CK_t, CK_c for DDR5-5200 to DDR5-6400

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
Differential Input Slew Rate for CK_t, CK_c	SRIdiff_CK	2	30	2	30	2	30	2	30	V/ns	

Table 217 — Differential Input Slew Rate for CK_t, CK_c for DDR5-6800 to DDR5-8800

Parameter	Symbol	DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Differential Input Slew Rate for CK_t, CK_c	SRIdiff_CK	2	30	2	30	2	30	2	30	2	30	2	30	V/ns	

8.6 Rx DQS Jitter Sensitivity

The receiver DQS jitter sensitivity test provides the methodology for testing the receiver's strobe sensitivity to an applied duty cycle distortion (DCD) and/or random jitter (Rj) at the forwarded strobe input without adding jitter, noise and ISI to the data. The receiver must pass the appropriate BER rate when no cross-talk nor ISI is applied, and must pass through the combination of applied DCD and Rj.

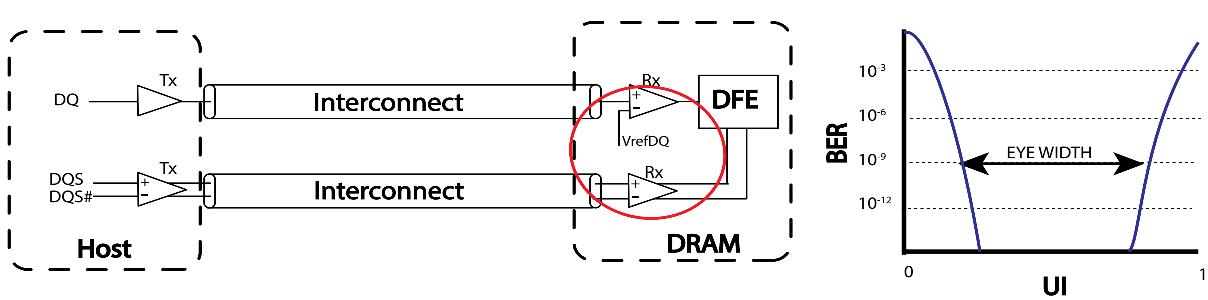


Figure 230 — SDRAM's Rx Forwarded Strobes for Jitter Sensitivity Testing

8.6.1 Rx DQS Jitter Sensitivity Specification

Table 218 provides Rx DQS Jitter Sensitivity Specification for the DDR5 DRAM receivers when operating at various possible transfer rates. These parameters are tested on the CTC2 card without Rx Equalization set. Additive DFE Gain Bias can be set.

Table 218 — Rx DQS Jitter Sensitivity Specification for DDR5-3200 to 4800

[BER = Bit Error Rate; DCD = Duty Cycle Distortion; Rj =Random Jitter]

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
DQ Timing Width	tRx_DQ_tMargin	0.900	-	0.875	-	0.825	-	0.825	-	0.825	-	UI	1, 2, 3, 8, 9, 10
Degradation of timing width compared to tRx_DQ_tMargin, with DCD injection in DQS	∆tRx_DQ_tMargin_DQS_DCD	-	0.06	-	0.06	-	0.06	-	0.06	-	0.06	UI	1, 4, 8, 9, 10
Degradation of timing width compared to tRx_DQ _tMargin, with Rj injection in DQS	∆tRx_DQ_tMargin_DQS_Rj	-	0.09	-	0.09	-	0.09	-	0.09	-	0.09	UI	1, 5, 8, 9, 10
Degradation of timing width compared to tRx_DQ_tMargin, with both DCD and Rj injection in DQS	∆tRx_DQ_tMargin_DQS_DCD_Rj	-	0.15	-	0.15	-	0.15	-	0.15	-	0.15	UI	1, 2, 6, 8, 9, 10
Delay of any data lane relative to the DQS_t/DQS_c crossing	tRx_DQS2DQ	114	937	114	833	114	750	114	738	114	729	ps	1, 7, 8, 9, 10
Delay of any data lane relative to any other data lane	tRX_DQ2DQ	-	50	-	50	-	50	-	50	-	50	ps	1, 7, 8, 9, 10, 11, 12

NOTE 1 Validation methodology will be defined in future ballots. 2UI is defined as 1tCK for this parameter

NOTE 2 Each of ∆tRx_DQ_tMargin_DQS_DCD, ∆tRx_DQ_tMargin_DQS_Rj, and ∆tRx_DQ_tMargin_DQS _DCD_Rj can be relaxed by up to 5% if tRx_DQ_tMargin exceeds the spec by 5% or more

NOTE 3 DQ Timing Width - timing width for any data lane using repetitive patterns (check note 4 for the pattern) measured at BER=E-9

NOTE 4 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD injection in forwarded strobe DQS compared to tRx_DQ_tMargin, measured at BER=E-9. The magnitude of DCD is specified under Test Conditions for Rx DQS Jitter Sensitivity Testing. Test using clock-like pattern of repeating 3 “1s” and 3 “0s”

NOTE 5 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with only Rj injection in forwarded strobe DQS measured at BER=E-9, compared to tRx_tMargin. The magnitude of Rj is specified under Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE 6 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD and Rj injection in forwarded strobe DQS measured at BER=E-9, compared to tRx_tMargin. The magnitudes of DCD and Rj are specified under Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE 7 Delay of any data lane relative to the strobe lane, as measured at the end of Tx+Channel. This parameter is a collective sum of effects of data clock mismatches in Tx and on the medium connecting Tx and Rx.

NOTE 8 All measurements at BER=E-9

NOTE 9 This test should be done after the DQS and DQ Voltage Sensitivity tests are completed and passing

NOTE 10 The user has the freedom to set the voltage swing and slew rates for strobe and DQ signals as long as they meet the specification. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

NOTE 11 DQ to DQ offset is skew between DQs within a nibble (x4) or a byte (x8, x16) at the DDR5 SDRAM balls.

NOTE 12 This parameter is relative to the per-pin trained DQS2DQ value, and only applies after per-pin DQS2DQ training is complete.

8.6.1 Rx DQS Jitter Sensitivity Specification (cont'd)

Table 219 — Rx DQS Jitter Sensitivity Specification for DDR5-5200 to 6400

[BER = Bit Error Rate; DCD = Duty Cycle Distortion; Rj =Random Jitter]

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
DQ Timing Width	tRx_DQ_tMargin	0.835	-	0.835	-	0.845	-	0.845	-	UI	1, 2, 3, 8, 9, 10
Degradation of timing width compared to tRx_DQ_tMargin, with DCD injection in DQS	∆tRx_DQ_tMargin_DQS_DCD	-	0.06	-	0.06	-	0.06	-	0.06	UI	1, 4, 8, 9, 10
Degradation of timing width compared to tRx_DQ _tMargin, with Rj injection in DQS	∆tRx_DQ_tMargin_DQS_Rj	-	0.09	-	0.09	-	0.09	-	0.09	UI	1, 5, 8, 9, 10
Degradation of timing width compared to tRx_DQ_tMargin, with both DCD and Rj injection in DQS	∆tRx_DQ_tMargin_DQS_DCD_Rj	-	0.15	-	0.15	-	0.15	-	0.15	UI	1, 2, 6, 8, 9, 10
Delay of any data lane relative to the DQS_t/DQS_c crossing	tRx_DQS2DQ	114	721	114	714	114	708	114	703	ps	1, 7, 8, 9, 10
Delay of any data lane relative to any other data lane	tRX_DQ2DQ	-	50	-	50	-	50	-	50	ps	1, 7, 8, 9, 10, 11, 12

NOTE 1 Validation methodology will be defined in future ballots. 2UI is defined as 1tCK for this parameter

NOTE 2 Each of ∆tRx_DQ_tMargin_DQS_DCD, ∆tRx_DQ_tMargin_DQS_Rj, and ∆tRx_DQ_tMargin_DQS _DCD_Rj can be relaxed by up to 5% if tRx_DQ_tMargin exceeds the spec by 5% or more

NOTE 3 DQ Timing Width - timing width for any data lane using repetitive patterns (check note 4 for the pattern) measured at BER=E-9

NOTE 4 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD injection in forwarded strobe DQS compared to tRx_DQ_tMargin, measured at BER=E-9. The magnitude of DCD is specified under Test Conditions for Rx DQS Jitter Sensitivity Testing. Test using clock-like pattern of repeating 3 "1s" and 3 "0s"

NOTE 5 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with only Rj injection in forwarded strobe DQS measured at BER=E-9, compared to tRx_tMargin. The magnitude of Rj is specified under Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE 6 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD and Rj injection in forwarded strobe DQS measured at BER=E-9, compared to tRx_tMargin. The magnitudes of DCD and Rj are specified under Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE 7 Delay of any data lane relative to the strobe lane, as measured at the end of Tx+Channel. This parameter is a collective sum of effects of data clock mismatches in Tx and on the medium connecting Tx and Rx.

NOTE 8 All measurements at BER=E-9

NOTE 9 This test should be done after the DQS and DQ Voltage Sensitivity tests are completed and passing

NOTE 10 The user has the freedom to set the voltage swing and slew rates for strobe and DQ signals as long as they meet the specification. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

NOTE 11 DQ to DQ offset is skew between DQs within a nibble (x4) or a byte (x8, x16) at the DDR5 SDRAM balls.

NOTE 12 This parameter is relative to the per-pin trained DQS2DQ value, and only applies after per-pin DQS2DQ training is complete.

8.6.1 Rx DQS Jitter Sensitivity Specification (cont'd)

Table 220 — Rx DQS Jitter Sensitivity Specification for DDR5-6800 to 8800

[BER = Bit Error Rate; DCD = Duty Cycle Distortion; Rj =Random Jitter]

Parameter	Symbol	DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
DQ Timing Width	tRx_DQ_tMargin	0.845	-	0.845	-	0.845	-	0.845	-	0.845	-	0.845	-	UI	1, 2, 3, 8, 9, 10
Degradation of timing width compared to tRx_DQ_tMargin, with DCD injection in DQS	∆tRx_DQ_tMargin _DQS_DCD	-	0.06	-	0.06	-	0.06	-	0.06	-	0.06	-	0.06	UI	1, 4, 8, 9, 10
Degradation of timing width compared to tRx_DQ _tMargin, with Rj injection in DQS	∆tRx_DQ_tMargin _DQS_Rj	-	0.09	-	0.09	-	0.09	-	0.09	-	0.09	-	0.09	UI	1, 5, 8, 9, 10
Degradation of timing width compared to tRx_DQ_tMargin, with both DCD and Rj injection in DQS	∆tRx_DQ_tMargin _DQS_DCD_Rj	-	0.15	-	0.15	-	0.15	-	0.15	-	0.15	-	0.15	UI	1, 2, 6, 8, 9, 10
Delay of any data lane relative to the DQS_t/ DQS_c crossing	tRx_DQS2DQ	114	703	114	703	114	703	114	703	114	703	114	703	ps	1, 7, 8, 9, 10
Delay of any data lane relative to any other data lane	tRX_DQ2DQ	-	50	-	50	-	50	-	50	-	50	-	50	ps	1, 7, 8, 9, 10, 11, 12

NOTE 1 Validation methodology will be defined in future ballots. 2UI is defined as 1tCK for this parameter

NOTE 2 Each of ∆tRx_DQ_tMargin_DQS_DCD, ∆tRx_DQ_tMargin_DQS_Rj, and ∆tRx_DQ_tMargin_DQS _DCD_Rj can be relaxed by up to 5% if tRx_DQ_tMargin exceeds the spec by 5% or more

NOTE 3 DQ Timing Width - timing width for any data lane using repetitive patterns (check note 4 for the pattern) measured at BER=E-9

NOTE 4 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD injection in forwarded strobe DQS compared to tRx_DQ_tMargin, measured at BER=E-9. The magnitude of DCD is specified under Test Conditions for Rx DQS Jitter Sensitivity Testing. Test using clock-like pattern of repeating 3 "1s" and 3 "0s"

NOTE 5 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with only Rj injection in forwarded strobe DQS measured at BER=E-9, compared to tRx_tMargin. The magnitude of Rj is specified under Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE 6 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD and Rj injection in forwarded strobe DQS measured at BER=E-9, compared to tRx_tMargin. The magnitudes of DCD and Rj are specified under Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE 7 Delay of any data lane relative to the strobe lane, as measured at the end of Tx+Channel. This parameter is a collective sum of effects of data clock mismatches in Tx and on the medium connecting Tx and Rx.

NOTE 8 All measurements at BER=E-9

NOTE 9 This test should be done after the DQS and DQ Voltage Sensitivity tests are completed and passing

NOTE 10 The user has the freedom to set the voltage swing and slew rates for strobe and DQ signals as long as they meet the specification. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

NOTE 11 DQ to DQ offset is skew between DQs within a nibble (x4) or a byte (x8, x16) at the DDR5 SDRAM balls.

NOTE 12 This parameter is relative to the per-pin trained DQS2DQ value, and only applies after per-pin DQS2DQ training is complete.

8.6.2 Test Conditions for Rx DQS Jitter Sensitivity Tests

Table 221 lists the amount of Duty Cycle Distortion (DCD) and/or Random Jitter (Rj) that must be applied to the forwarded strobe when measuring the Rx DQS Jitter Sensitivity parameters specified in Table 218 and Table 219.

Table 221 — Test Conditions for Rx DQS Jitter Sensitivity Testing for DDR5-3200 to 4800

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Applied DCD to the DQS	tRx_DQS_DCD	-	0.045	-	0.045	-	0.045	-	0.045	-	0.045	UI	1, 2, 3, 6, 7, 10
Applied Rj RMS to the DQS	tRx_DQS_Rj	-	0.0075	-	0.0075	-	0.0075	-	0.0075	-	0.0075	UI (RMS)	1, 2, 4, 6, 8, 10
Applied DCD and Rj RMS to the DQS	tRx_DQS_DCD_Rj	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	UI	1, 2, 5, 6, 7, 9, 10

NOTE 1 While imposing this spec, the strobe lane is stressed, but the data input is kept large amplitude and no jitter or ISI injection. The specified voltages are at the Rx input pin. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

NOTE 2 The jitter response of the forwarded strobe channel will depend on the input voltage, primarily due to bandwidth limitations of the clock receiver. For this revision, no separate specification of jitter as a function of input amplitude is specified, instead the response characterization done at the specified clock amplitude only. The specified voltages are at the Rx input pin

NOTE 3 Various DCD values should be tested, complying within the maximum limits

NOTE 4 Various Rj values should be tested, complying within the maximum limits

NOTE 5 Various combinations of DCD and Rj should be tested, complying within the maximum limits. The maximum timing margin degradation as a result of these injected jitter is specified in a separate table

NOTE 6 Although DDR5 has bursty traffic, current available BERTs that can be used for this test do not support burst traffic patterns. A continuous strobe and continuous DQ are used for this parameter. The clock like pattern repeating 3 “1s” and 3 “0s” is used for this test.

NOTE 7 Duty Cycle Distortion (in UI DCD) as applied to the input forwarded DQS from BERT (UI)

NOTE 8 RMS value of Rj (specified as Edge jitter) applied to the input forwarded DQS from BERT (values of the edge jitter RMS values specified as % of UI)

NOTE 9 Duty cycle distortion (specified as UI DCD) and rms values of Rj (specified as edge jitter) applied to the input forwarded DQS from BERT (values of the edge jitter RMS values specified as % of UI)

NOTE 10 The user has the freedom to set the voltage swing and slew rates for strobe and DQ signals as long as they meet the specification. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

8.6.2 Test Conditions for Rx DQS Jitter Sensitivity Tests (cont'd)

Table 222 — Test Conditions for Rx DQS Jitter Sensitivity Testing for DDR5-5200 to 6400

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
Applied DCD to the DQS	tRx_DQS_DCD	-	0.045	-	0.045	-	0.045	-	0.045	UI	1, 2, 3, 6, 7, 10
Applied Rj RMS to the DQS	tRx_DQS_Rj	-	0.0075	-	0.0075	-	0.0075	-	0.0075	UI (RMS)	1, 2, 4, 6, 8, 10
Applied DCD and Rj RMS to the DQS	tRx_DQS_DCD_Rj	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	UI	1, 2, 5, 6, 7, 9, 10

NOTE 1 While imposing this spec, the strobe lane is stressed, but the data input is kept large amplitude and no jitter or ISI injection. The specified voltages are at the Rx input pin. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

NOTE 2 The jitter response of the forwarded strobe channel will depend on the input voltage, primarily due to bandwidth limitations of the clock receiver. For this revision, no separate specification of jitter as a function of input amplitude is specified, instead the response characterization done at the specified clock amplitude only. The specified voltages are at the Rx input pin

NOTE 3 Various DCD values should be tested, complying within the maximum limits

NOTE 4 Various Rj values should be tested, complying within the maximum limits

NOTE 5 Various combinations of DCD and Rj should be tested, complying within the maximum limits. The maximum timing margin degradation as a result of these injected jitter is specified in a separate table

NOTE 6 Although DDR5 has bursty traffic, current available BERTs that can be used for this test do not support burst traffic patterns. A continuous strobe and continuous DQ are used for this parameter. The clock like pattern repeating 3 “1s” and 3 “0s” is used for this test.

NOTE 7 Duty Cycle Distortion (in UI DCD) as applied to the input forwarded DQS from BERT (UI)

NOTE 8 RMS value of Rj (specified as Edge jitter) applied to the input forwarded DQS from BERT (values of the edge jitter RMS values specified as % of UI)

NOTE 9 Duty cycle distortion (specified as UI DCD) and rms values of Rj (specified as edge jitter) applied to the input forwarded DQS from BERT (values of the edge jitter RMS values specified as % of UI)

NOTE 10 The user has the freedom to set the voltage swing and slew rates for strobe and DQ signals as long as they meet the specification. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

8.6.2 Test Conditions for Rx DQS Jitter Sensitivity Tests (cont'd)

Table 223 — Test Conditions for Rx DQS Jitter Sensitivity Testing for DDR5-6800 to 8800

Parameter	Symbol	DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Applied DCD to the DQS	tRx_DQS_DCD	-	0.045	-	0.045	-	0.045	-	0.045	-	0.045	-	0.045	UI	3, 6, 7, 10
Applied Rj RMS to the DQS	tRx_DQS_Rj	-	0.0075	-	0.0075	-	0.0075	-	0.0075	-	0.0075	-	0.0075	UI (RMS)	4, 6, 8, 10
Applied DCD and Rj RMS to the DQS	tRx_DQS_DCD_Rj	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	-	0.045UI DCD + 0.0075UI Rj RMS	UI	5, 6, 7, 9, 10

NOTE 1 While imposing this spec, the strobe lane is stressed, but the data input is kept large amplitude and no jitter or ISI injection. The specified voltages are at the Rx input pin. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

NOTE 2 The jitter response of the forwarded strobe channel will depend on the input voltage, primarily due to bandwidth limitations of the clock receiver. For this revision, no separate specification of jitter as a function of input amplitude is specified, instead the response characterization done at the specified clock amplitude only. The specified voltages are at the Rx input pin

NOTE 3 Various DCD values should be tested, complying within the maximum limits

NOTE 4 Various Rj values should be tested, complying within the maximum limits

NOTE 5 Various combinations of DCD and Rj should be tested, complying within the maximum limits. The maximum timing margin degradation as a result of these injected jitter is specified in a separate table

NOTE 6 Although DDR5 has bursty traffic, current available BERTs that can be used for this test do not support burst traffic patterns. A continuous strobe and continuous DQ are used for this parameter. The clock like pattern repeating 3 "1s" and 3 "0s" is used for this test.

NOTE 7 Duty Cycle Distortion (in UI DCD) as applied to the input forwarded DQS from BERT (UI)

NOTE 8 RMS value of Rj (specified as Edge jitter) applied to the input forwarded DQS from BERT (values of the edge jitter RMS values specified as % of UI)

NOTE 9 Duty cycle distortion (specified as UI DCD) and rms values of Rj (specified as edge jitter) applied to the input forwarded DQS from BERT (values of the edge jitter RMS values specified as % of UI)

NOTE 10 The user has the freedom to set the voltage swing and slew rates for strobe and DQ signals as long as they meet the specification. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

8.7 Rx DQS Voltage Sensitivity

8.7.1 Overview

The receiver DQS (strobe) input voltage sensitivity test provides the methodology for testing the receiver's sensitivity to varying input voltage in the absence of Inter-Symbol Interference (ISI), jitter (Rj, Dj, DCD) and crosstalk noise.

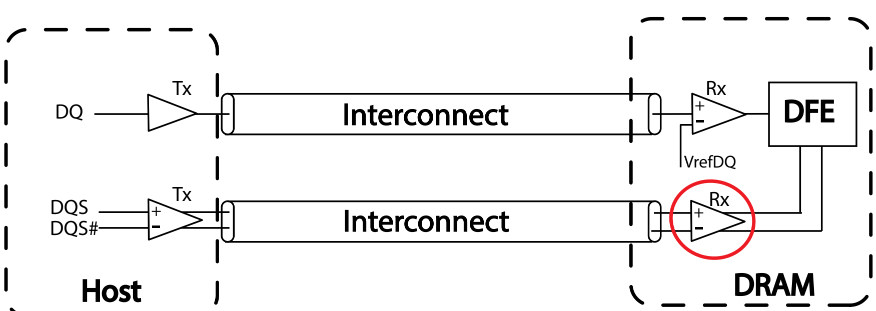


Figure 231 — Example of DDR5 Memory Interconnect

8.7.2 Receiver DQS Voltage Sensitivity Parameter

Input differential (DQS_t, DQS_c) VRx_DQS is defined and measured as shown in Table 224 through Table 226. The receiver must pass the minimum BER requirements for DDR5. These parameters are tested on the CTC2 card with neither additive gain nor Rx Equalization set.

Table 224 — Rx DQS Input Voltage Sensitivity Parameter for DDR5-3200 to 4800

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
DQS Rx Input Voltage Sensitivity (differential pp)	VRx_DQS	-	130	-	115	-	105	-	100	-	100	mV	1, 2, 3
NOTE 1 Refer to the minimum BER requirements for DDR5 NOTE 2 The validation methodology for this parameter will be covered in future ballot(s) NOTE 3 Test using clock like pattern of repeating 3 “1s” and 3 “0s”													

Table 225 — Rx DQS Input Voltage Sensitivity Parameter for DDR5-5200 to 6400

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		DDR5-6800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
DQS Rx Input Voltage Sensitivity (differential pp)	VRx_DQS	-	90	-	90	-	83	-	83	-	80	mV	1, 2, 3
NOTE 1 Refer to the minimum BER requirements for DDR5 NOTE 2 The validation methodology for this parameter will be covered in future ballot(s) NOTE 3 Test using clock like pattern of repeating 3 “1s” and 3 “0s”													

Table 226 — Rx DQS Input Voltage Sensitivity Parameter for DDR5-6800 to 8800

Parameter	Symbol	DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
DQS Rx Input Voltage Sensitivity (differential pp)	VRx_DQS	-	80	-	77.5	-	77.5	-	75	-	75	mV	1, 2, 3
NOTE 1 Refer to the minimum BER requirements for DDR5 NOTE 2 The validation methodology for this parameter will be covered in future ballot(s) NOTE 3 Test using clock like pattern of repeating 3 “1s” and 3 “0s”													

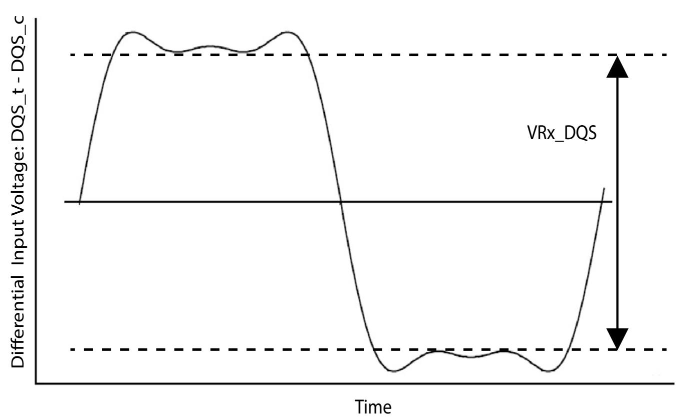


Figure 232 — VRx_DQS

8.8 Differential Strobe (DQS_t, DQS_c) Input Cross Point Voltage (VIX)

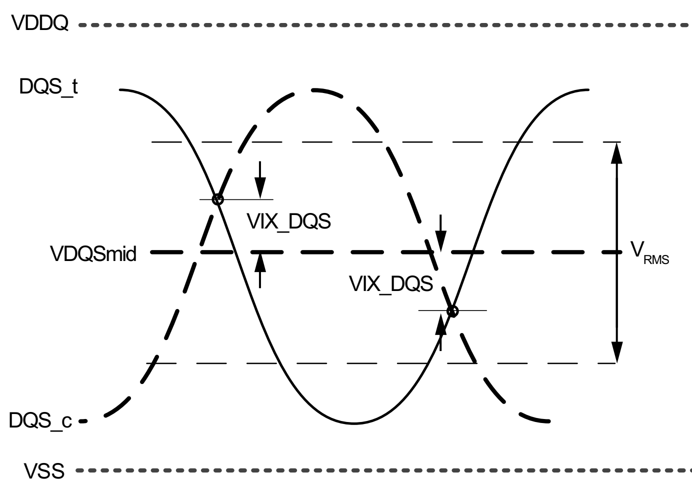


Figure 233 — VIX Definition (DQS)

Table 227 — Crosspoint Voltage (VIX) for DQS Differential Input Signals

Parameter	Symbol	DDR5-3200 - 8800		Unit	Notes
		Min	Max		
DQS differential input crosspoint voltage ratio	VIX_DQS_Ratio	-	50	%	1, 2, 3

NOTE 1 The VIX_DQS voltage is referenced to VDQSmid(mean) = (DQS_t voltage + DQS_c voltage) /2, where the mean is over 8 UI
NOTE 2 VIX_DQS_Ratio = (|VIX_DQS| / |V_{RMS}|)*100%, where V_{RMS} = RMS(DQS_t voltage - DQS_c voltage)
NOTE 3 Only applies when both DQS_t and DQS_c are transitioning (including preamble)

8.9 Rx DQ Voltage Sensitivity

8.9.1 Overview

The receiver data input voltage sensitivity test provides the methodology for testing the receiver’s sensitivity to varying input voltage in the absence of Inter-Symbol Interference (ISI), jitter (Rj, Dj, DCD) and crosstalk noise.

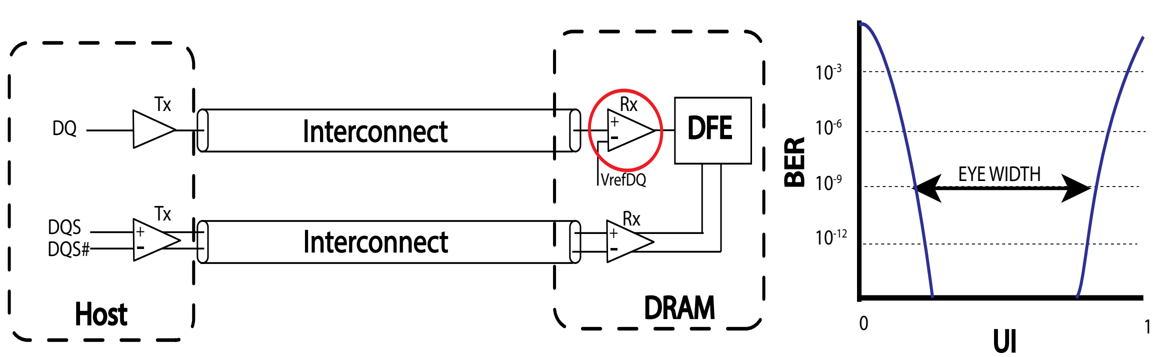


Figure 234 — Example of DDR5 Memory Interconnect

8.9.2 Receiver DQ Input Voltage Sensitivity Parameters

Input single-ended VRx_DQ is defined and measured as shown below. The receiver must pass the minimum BER requirements for DDR5. These parameters are tested on the CTC2 card with neither additive gain nor Rx Equalization set.

Table 228 — Rx DQ Input Voltage Sensitivity Parameters for DDR5-3200 to 4800

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Minimum DQ Rx input voltage sensitivity applied around Vref	VRx_DQ	-	85	-	75	-	70	-	65	-	65	mV	1, 2, 3

NOTE 1 Refer to the minimum BER requirements for DDR5
NOTE 2 The validation methodology for this parameter will be covered in future ballot(s)
NOTE 3 Test using clock like pattern of repeating 3 “1s” and 3 “0s”

Table 229 — Rx DQ Input Voltage Sensitivity Parameters for DDR5-5200 to 6400

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		DDR5-6800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Minimum DQ Rx input voltage sensitivity applied around Vref	VRx_DQ	-	60	-	60	-	55	-	55	-	50	mV	1, 2, 3

NOTE 1 Refer to the minimum BER requirements for DDR5
NOTE 2 The validation methodology for this parameter will be covered in future ballot(s)
NOTE 3 Test using clock like pattern of repeating 3 “1s” and 3 “0s”

Table 230 — Rx DQ Input Voltage Sensitivity Parameters for DDR5-6800 to 8800

Parameter	Symbol	DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Minimum DQ Rx input voltage sensitivity applied around Vref	VRx_DQ	-	50	-	50	-	50	-	47.5	-	47.5	mV	1, 2, 3

NOTE 1 Refer to the minimum BER requirements for DDR5
NOTE 2 The validation methodology for this parameter will be covered in future ballot(s)
NOTE 3 Test using clock like pattern of repeating 3 “1s” and 3 “0s”

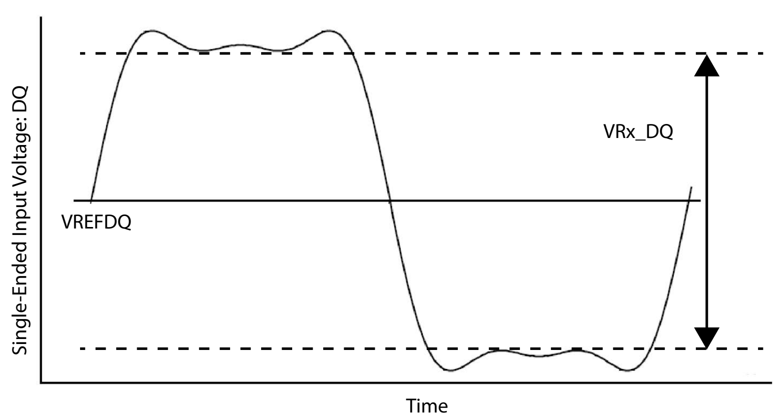


Figure 235 — VRx_DQ

8.9.3 Differential Input Levels for DQS

Table 231 — Differential Input Levels for DQS (DQS_t, DQS_c) for DDR5-3200 to DDR5-6400

From	Parameter	DDR5 3200-6400	Note
V_{IHdiff}DQS	Differential input high measurement level (DQS_t, DQS_c)	0.75 x $V_{diffpk-pk}$	1, 2, 3
V_{ILdiff}DQS	Differential input low measurement level (DQS_t, DQS_c)	0.25 x $V_{diffpk-pk}$	1, 2, 3
NOTE 1 $V_{diffpk-pk}$ defined in Figure 236 NOTE 2 $V_{diffpk-pk}$ is the mean high voltage minus the mean low voltage over 8UI samples NOTE 3 All parameters are defined over the entire clock common mode range			

Table 232 — Differential Input Levels for DQS (DQS_t, DQS_c) for DDR5-6800 to DDR5-8800

From	Parameter	DDR5 6800-8800	Note
V_{IHdiff}DQS	Differential input high measurement level (DQS_t, DQS_c)	0.75 x $V_{diffpk-pk}$	1, 2, 3
V_{ILdiff}DQS	Differential input low measurement level (DQS_t, DQS_c)	0.25 x $V_{diffpk-pk}$	1, 2, 3
NOTE 1 $V_{diffpk-pk}$ defined in Figure 236 NOTE 2 $V_{diffpk-pk}$ is the mean high voltage minus the mean low voltage over 8UI samples NOTE 3 All parameters are defined over the entire clock common mode range			

8.9.4 Differential Input Slew Rate for DQS_t, DQS_c

Input slew rate for differential signals are defined and measured as shown in Figure 236.

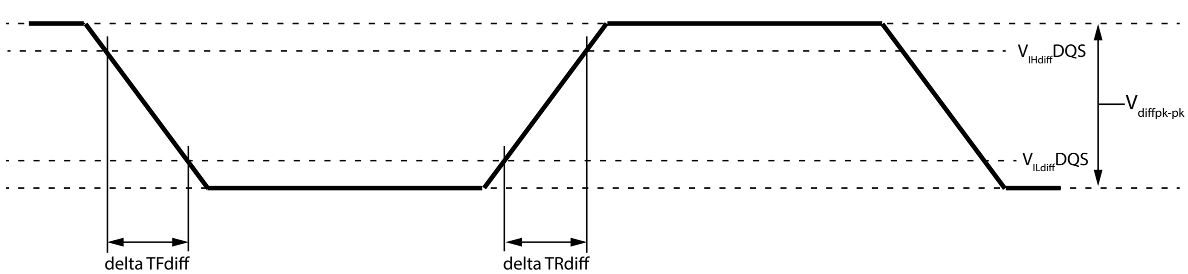


Figure 236 — Differential Input Slew Rate Definition for DQS_t, DQS_c

Table 233 — Differential Input Slew Rate Definition for DQS_t, DQS_c

Parameter	Measured		Defined by	Notes
	From	To		
Differential Input slew rate for rising edge (DQS_t, DQS_c)	V_{ILdiff}DQS	V_{IHdiff}DQS	(V_{IHdiff}DQS - V_{ILdiff}DQS) /deltaTRdiff	1, 2, 3
Differential Input slew rate for falling edge (DQS_t, DQS_c)	V_{IHdiff}DQS	V_{ILdiff}DQS	(V_{IHdiff}DQS - V_{ILdiff}DQS) /deltaTFdiff	1, 2, 3

8.9.4 Differential Input Slew Rate for DQS_t, DQS_c (cont'd)

Table 234 — Differential Input Slew Rate for DQS_t, DQS_c for DDR5-3200 to 4800

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Differential Input Slew Rate for DQS_t, DQS_c	SRIdiff_DQS	1.5	30	1.5	30	1.5	30	1.5	30	1.5	30	V/ns	1
NOTE 1 Only applies when both DQS_t and DQS_c are transitioning.													

Table 235 — Differential Input Slew Rate for DQS_t, DQS_c for DDR5-5200 to 6400

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
Differential Input Slew Rate for DQS_t, DQS_c	SRIdiff_DQS	2.0	30	2.0	30	2.0	30	2.0	30	V/ns	1
NOTE 1 Only applies when both DQS_t and DQS_c are transitioning.											

Table 236 — Differential Input Slew Rate for DQS_t, DQS_c for DDR5-6800 to 8800

Parameter	Symbol	DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Differential Input Slew Rate for DQS_t, DQS_c	SRIdiff_DQS	2.0	30	2.0	30	2.0	30	2.0	30	2.0	30	2.0	30	V/ns	1
NOTE 1 Only applies when both DQS_t and DQS_c are transitioning.															

8.10 Rx Stressed Eye

The stressed eye tests provide the methodology for creating the appropriate stress for the DRAM's receiver with the combination of ISI (both loss and reflective), jitter (Rj, Dj, DCD), and crosstalk noise. The receiver must pass the appropriate BER rate when the equivalent stressed eye is applied through the combination of ISI, jitter and crosstalk.

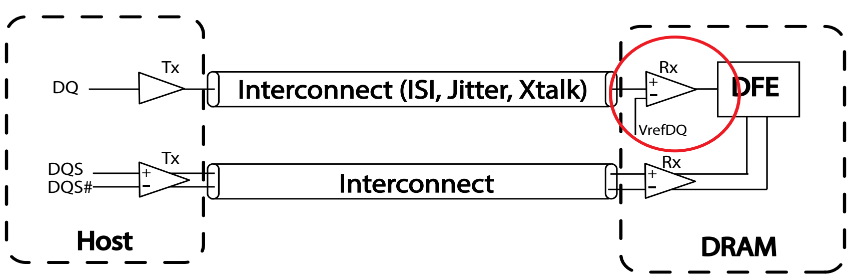


Figure 237 — Example of Rx Stressed Test Setup in the Presence of ISI, Jitter, and Crosstalk

8.10 Rx Stressed Eye (cont'd)

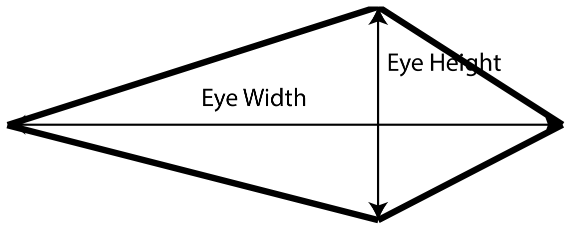


Figure 238 — Example of Rx Stressed Eye Height and Eye Width

8.10.1 Parameters for DDR5 Rx Stressed Eye Tests

Table 237 — Test Conditions for Rx Stressed Eye Tests for DDR5-3200 to 4800

[BER=Bit Error Rate; DCD=Duty Cycle Distortion; Rj=Random Jitter; Sj=Sinusoidal Jitter; p-p =peak to peak]

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Eye height of stressed eye for Golden Reference Channel 1	RxEH_Stressed_Eye_ Golden_Ref_Channel_1	95	-	85	-	80	-	75	-	70	-	mV	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Eye width of stressed eye Golden Reference Channel 1	RxEW_Stressed_Eye_ Golden_Ref_Channel_1	0.25	-	0.25	-	0.25	-	0.25	-	0.25	-	UI	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Vswing stress to meet above data eye	Vswing_Stressed_Eye_ Golden_Ref_Channel_1	-	600	-	600	-	600	-	600	-	600	mV	1. 2
Injected sinusoidal jitter at 200 MHz to meet above data eye	Sj_Stressed_Eye_Golden _Ref_Channel_1	0	0.45	0	0.45	0	0.45	0	0.45	0	0.45	UI p-p	1, 2
Injected Random wide band (10 MHz-1 GHz) Jitter to meet above data eye	Rj_Stressed_Eye_Golden _Ref_Channel_1	0	0.04	0	0.04	0	0.04	0	0.04	0	0.04	UI RMS	1, 2
Injected voltage noise as PRBS23, or Injected voltage noise at 2.1 GHz	Vnoise_Stressed_Eye_ Golden_Ref_Channel_1	0	125	0	125	0	125	0	125	0	125	mV p-p	1, 2
Golden Reference Channel 1 Characteristics as measured at TBD	Golden_Ref_Channel_1_ Characteristics	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	dB	3

NOTE 1 Must meet minimum BER of $1E^{-16}$ or better requirement with the stressed eye at the slice of the receiver (after equalization is applied in the summer). The eye shape is verified by measuring to BER E^{-9} and extrapolating to BER $E\text{-}^{16}$.

NOTE 2 These parameters are applied on the defined golden reference channel with parameters TBD.

NOTE 3 DFE Tap 1-4 Bias settings that give the best eye margin are used and referring to Table 131, Min/Max Ranges for the DFE Tap Coefficients. DFE tap range limits apply: sum of absolute values of Tap-2, Tap-3, and Tap-4 shall be less than 60 mV (|Tap-2| + |Tap-3| + |Tap-4| < 60 mV) after the tap multiflier is applied.

NOTE 4 Evaluated with no DC supply voltage drift.

NOTE 5 Evaluated with no temperature drift.

NOTE 6 Supply voltage noise limited according to DC bandwidth spec, see DC Operating Conditions

NOTE 7 The stressed eye is to be assumed to have a diamond shape

NOTE 8 The VREFDQ, DFE Gain Bias Step, and DFE Taps 1,2,3,4 Bias Step can be adjusted as needed, without exceeding the specifications, for this test, including the limits placed in Note 3.

NOTE 9 The stressed eye is defined as centered on the DQS_t/DQS_c crossing during the calibration. Measurement includes an optimal set of DQS_t/DQS_c location, VrefDQ, and DFE solution to give the best eye margin.

NOTE 10 The Rx stressed eye spec applies at 2933 MT/s and faster data rates.

NOTE 11 EH/EW are measured at the slicer of the receiver

LIGHT GREY - **All Light Grey text is defined as something that should be considered TBD. The content may be accurate or the same as previous technologies but has not yet been reviewed or determined to be the working assumption.**

8.10.1 Parameters for DDR5 Rx Stressed Eye Tests (cont'd)

Table 238 — Test Conditions for Rx Stressed Eye Tests for DDR5-5200 to 6400

[BER=Bit Error Rate; DCD=Duty Cycle Distortion; Rj=Random Jitter; Sj=Sinusoidal Jitter; p-p =peak to peak]

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
Eye height of stressed eye for Golden Reference Channel 1	RxEH_Stressed_Eye_ Golden_Ref_Channel_1	65	-	60	-	57.5	-	57.5	-	mV	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Eye width of stressed eye Golden Reference Channel 1	RxEW_Stressed_Eye_ Golden_Ref_Channel_1	0.235	-	0.235	-	0.230	-	0.230	-	UI	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Vswing stress to meet above data eye	Vswing_Stressed_Eye_ Golden_Ref_Channel_1	-	600	-	600	-	600	-	600	mV	1, 2
Injected sinusoidal jitter at 200 MHz to meet above data eye	Sj_Stressed_Eye_Golden _Ref_Channel_1	-	0.45	-	0.45	-	0.45	-	0.45	UI p-p	1, 2
Injected Random wide band (10 MHz-1 GHz) Jitter to meet above data eye	Rj_Stressed_Eye_Golden _Ref_Channel_1	0	0.04	0	0.04	0	0.04	0	0.04	UI RMS	1, 2
Injected voltage noise as PRBS23, or Injected voltage noise at 2.1 GHz	Vnoise_Stressed_Eye_ Golden_Ref_Channel_1	0	125	0	125	0	125	0	125	mV p-p	1, 2
Golden Reference Channel 1 Characteristics as measured at TBD	Golden_Ref_Channel_1_ Characteristics	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	dB	3

NOTE 1 Must meet minimum BER of $1E^{-16}$ or better requirement with the stressed eye at the slice of the receiver (after equalization is applied in the summer). The eye shape is verified by measuring to BER E^{-9} and extrapolating to BER $E\text{-}^{16}$.

NOTE 2 These parameters are applied on the defined golden reference channel with parameters TBD.

NOTE 3 DFE tap range limits apply: sum of absolute values of Tap-2, Tap-3, and Tap-4 shall be less than 6 0mV (|Tap-2| + |Tap-3| + |Tap-4| < 60 mV).

NOTE 4 Evaluated with no DC supply voltage drift.

NOTE 5 Evaluated with no temperature drift.

NOTE 6 Supply voltage noise limited according to DC bandwidth spec, see DC Operating Conditions

NOTE 7 The stressed eye is to be assumed to have a diamond shape

NOTE 8 The VREFDQ, DFE Gain Bias Step, and DFE Taps 1,2,3,4 Bias Step can be adjusted as needed, without exceeding the specifications, for this test, including the limits placed in Note 3.

NOTE 9 The stressed eye is defined as centered on the DQS_t/DQS_c crossing during the calibration. Measurement includes an optimal set of DQS_t/DQS_c location, VrefDQ, and DFE solution to give the best eye margin.

NOTE 10 EH/WE are measured at the slicer of the receiver.

8.10.1 Parameters for DDR5 Rx Stressed Eye Tests (cont'd)

Table 239 — Test Conditions for Rx Stressed Eye Tests for DDR5-6800 to 8800

[BER=Bit Error Rate; DCD=Duty Cycle Distortion; Rj=Random Jitter; Sj=Sinusoidal Jitter; p-p =peak to peak]

Parameter	Symbol	DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Eye height of stressed eye for Golden Reference Channel 1	RxEH_Stressed_Eye_ Golden_Ref_Channel_1	55	-	55	-	54	-	54	-	53	-	53	-	mV	1, 2, 3, 4, 5, 6, 7, 8, 9
Eye width of stressed eye Golden Reference Channel 1	RxEW_Stressed_Eye_ Golden_Ref_Channel_1	0.230	-	0.230	-	0.230	-	0.230	-	0.230	-	0.230	-	UI	1, 2, 3, 4, 5, 6, 7, 8, 9
Vswing stress to meet above data eye	Vswing_Stressed_Eye_ Golden_Ref_Channel_1	-	600	-	600	-	600	-	600	-	600	-	600	mV	1, 2
Injected sinusoidal jitter at 200 MHz to meet above data eye	Sj_Stressed_Eye_ Golden_Ref_Channel_1	-	0.45	-	0.45	-	0.45	-	0.45	-	0.45	-	0.45	UI p-p	1, 2
Injected Random wide band (10 MHz-1 GHz) Jitter to meet above data eye	Rj_Stressed_Eye_ Golden_Ref_Channel_1	0	0.04	0	0.04	0	0.04	0	0.04	-	0.04	-	0.04	UI RMS	1, 2
Injected voltage noise as PRBS23, or Injected voltage noise at 2.1 GHz	Vnoise_Stressed_Eye_ Golden_Ref_Channel_1	0	125	0	125	0	125	0	125	-	125	-	125	mV p-p	1, 2
Golden Reference Channel 1 Characteristics as measured at TBD	Golden_Ref_Channel_1 _Characteristics	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	dB	3

NOTE 1 Must meet minimum BER of $1E^{-16}$ or better requirement with the stressed eye at the slice of the receiver (after equalization is applied in the summer). The eye shape is verified by measuring to BER E^{-9} and extrapolating to BER E^{-16}.

NOTE 2 These parameters are applied on the defined golden reference channel with parameters TBD.

NOTE 3 DFE tap range limits apply: sum of absolute values of Tap-2, Tap-3, and Tap-4 shall be less than 60mV (|Tap-2| + |Tap-3| + |Tap-4| < 60 mV).

NOTE 4 Evaluated with no DC supply voltage drift.

NOTE 5 Evaluated with no temperature drift.

NOTE 6 Supply voltage noise limited according to DC bandwidth spec, see DC Operating Conditions.

NOTE 7 The stressed eye is to be assumed to have a diamond shape

NOTE 8 The VREFDQ, DFE Gain Bias Step, and DFE Taps 1,2,3,4 Bias Step can be adjusted as needed, without exceeding the specifications, for this test, including the limits placed in Note 3.

NOTE 9 The stressed eye is defined as centered on the DQS_t/DQS_c crossing during the calibration. Measurement includes an optimal set of DQS_t/DQS_c location, VrefDQ, and DFE solution to give the best eye margin.

8.11 Connectivity Test Mode - Input level and Timing Requirement

During CT Mode, input levels are defined below.
TEN pin: CMOS rail-to-rail with AC high and low at 80% and 20% of VDDQ
CS_n: CMOS rail-to-rail with AC high and low at 80% and 20% of VDDQ.
Test Input pins: CMOS rail-to-rail with AC high and low at 80% and 20% of VDDQ.
RESET_n: CMOS rail-to-rail with AC high and low at 80% and 20% of VDDQ.

Prior to the assertion of the TEN pin, all voltage supplies must be valid and stable.

Upon the assertion of the TEN pin, the CK_t and CK_c signals will be ignored and the DDR5 memory device will enter into the CT mode after time tCT_Enable. In the CT mode, no refresh activities in the memory arrays, initiated either externally (i.e., auto-refresh) or internally (i.e., self-refresh), will be maintained.

8.11.1 Connectivity Test Mode - Input level and Timing Requirement (cont'd)

The TEN pin may be asserted after the DRAM has completed power-on, after RESET_n has de-asserted, the wait time after the RESET_n de-assertion has elapsed, and prior to starting clocks (CK_t, CK_c).

The TEN pin may be de-asserted at any time in the CT mode. Upon exiting the CT mode, the states of the DDR5 memory device are unknown and the integrity of the original content of the memory array is not guaranteed; therefore, the reset initialization sequence is required.

All output signals at the test output pins will be stable within tCT_valid after the test inputs have been applied to the test input pins with TEN input and CS_n input maintained High and Low, respectively.

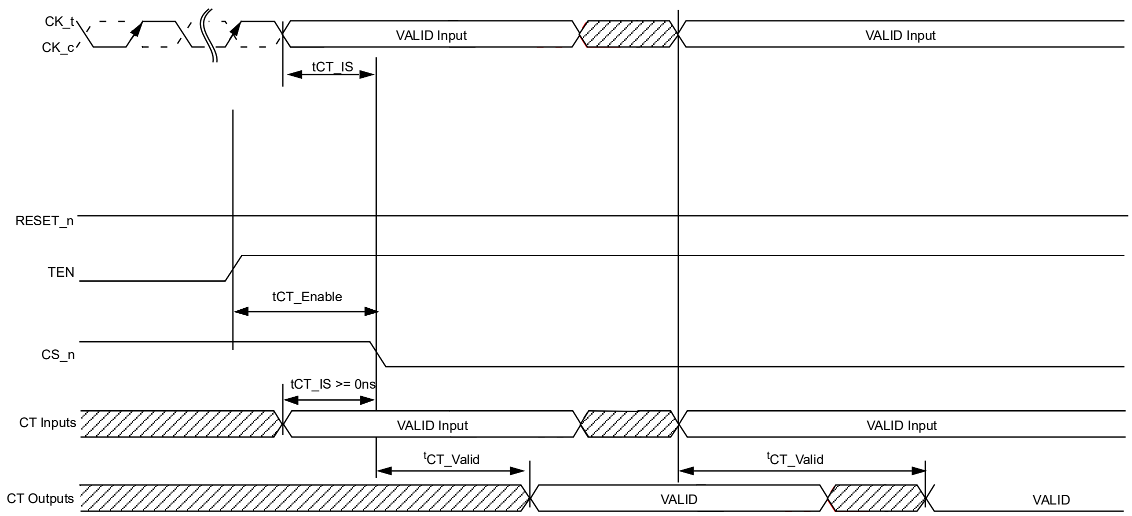


Figure 239 — Timing Diagram for Connectivity Test (CT) Mode

Table 240 — AC Parameters for Connectivity Test (CT) Mode

Symbol	Min	Max	Unit
tCT_IS	0	-	ns
tCT_Enable	200	-	ns
tCT_Valid	-	200	ns

8.11.1 Connectivity Test (CT) Mode Input Levels

Following input parameters will be applied for DDR5 SDRAM Input Signals during Connectivity Test Mode.

Table 241 — CMOS Rail to Rail Input Levels for TEN, CS_n, and Test inputs

Parameter	Symbol	Min	Max	Unit	Notes
TEN AC Input High Voltage	VIH(AC)_TEN	0.8 * VDDQ	VDDQ	V	1
TEN DC Input High Voltage	VIH(DC)_TEN	0.7 * VDDQ	VDDQ	V	
TEN DC Input Low Voltage	VIL(DC)_TEN	VSS	0.3 * VDDQ	V	
TEN AC Input Low Voltage	VIL(AC)_TEN	VSS	0.2 * VDDQ	V	2
TEN Input signal Falling time	TF_input_TEN	-	10	ns	
TEN Input signal Rising time	TR_input_TEN	-	10	ns	
NOTE 1 Overshoot might occur. It should be limited by the Absolute Maximum DC Ratings. NOTE 2 Undershoot might occur. It should be limited by Absolute Maximum DC Ratings.					

8.11.1 Connectivity Test (CT) Mode Input Levels (cont'd)

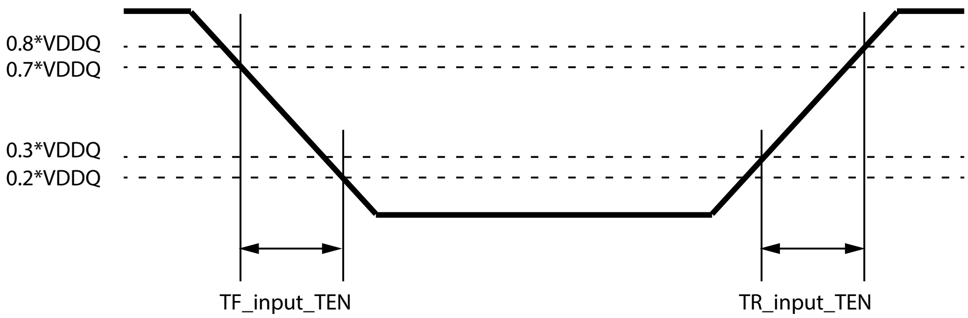


Figure 240 — TEN Input Slew Rate Definition

8.11.2 CMOS Rail to Rail Input Levels for RESET_n

Table 242 — CMOS Rail to Rail Input Levels for RESET_n

Parameter	Symbol	Min	Max	Unit	Note
AC Input High Voltage	VIH(AC)_RESET	0.8*VDDQ	VDDQ	V	5
DC Input High Voltage	VIH(DC)_RESET	0.7*VDDQ	VDDQ	V	2
DC Input Low Voltage	VIL(DC)_RESET	VSS	0.3*VDDQ	V	1
AC Input Low Voltage	VIL(AC)_RESET	VSS	0.2*VDDQ	V	6
Rising time	TR_RESET	-	1.0	μs	
RESET pulse width	tPW_RESET	1.0	-	μs	3, 4

NOTE 1 After RESET_n is registered LOW, RESET_n level shall be maintained below VIL(DC)_RESET during tPW_RESET, otherwise, SDRAM may not be reset.

NOTE 2 Once RESET_n is registered HIGH, RESET_n level must be maintained above VIH(DC)_RESET, otherwise, SDRAM operation will not be guaranteed until it is reset asserting RESET_n signal LOW.

NOTE 3 RESET is destructive to data contents.

NOTE 4 This definition is applied only for "Reset Procedure at Power Stable".

NOTE 5 Overshoot might occur. It should be limited by the Absolute Maximum DC Ratings.

NOTE 6 Undershoot might occur. It should be limited by Absolute Maximum DC Ratings

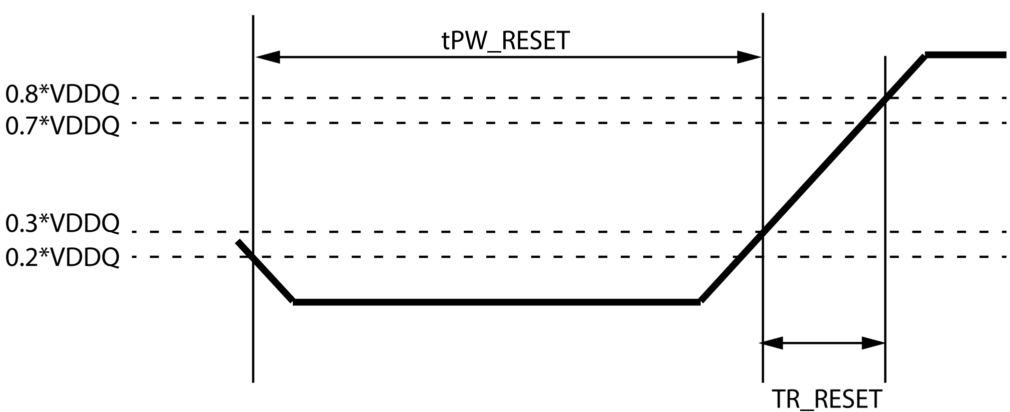


Figure 241 — RESET_n Input Slew Rate Definition

9 AC and DC Output Measurement Levels and Timing

9.1 Output Driver DC Electrical Characteristics for DQS and DQ

The DDR5 driver supports two different Ron values. These Ron values are referred as strong(low Ron) and weak mode(high Ron). A functional representation of the output buffer is shown in Figure 242.

Output driver impedance RON is defined as follows:

The individual pull-up and pull-down resistors (RON_{Pu} and RON_{Pd}) are defined as follows:

$$RON_{Pu} = \frac{VDDQ - Vout}{|I\,out|} \quad \text{under the condition that } RON_{Pd} \text{ is off}$$

$$RON_{Pd} = \frac{Vout}{|I\,out|} \quad \text{under the condition that } RON_{Pu} \text{ is off}$$

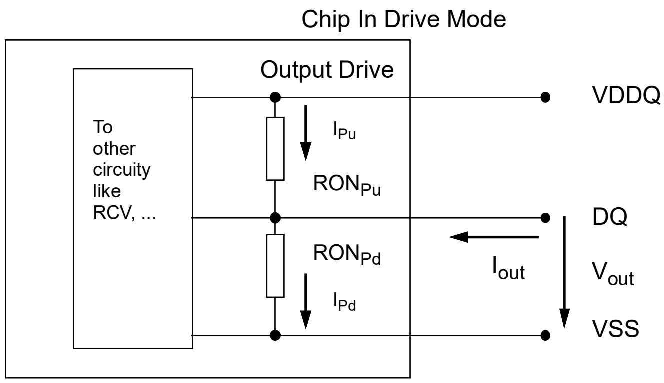


Figure 242 — Strong (Low Ron) and Weak Mode (High Ron) Output Buffer

9.1 Output Driver DC Electrical Characteristics for DQS and DQ (cont'd)

Table 243 — Output Driver DC Electrical Characteristics, Assuming RZQ = 240 Ohms; Entire Operating Temperature Range; after Proper ZQ Calibration

RON_{NOM}	Resistor	Vout	Min	Nom	Max	Unit	Notes
34Ω	RON34Pd	VOLdc= 0.5*VDDQ	0.73	1	1.1	RZQ/7	1, 2
		VOMdc= 0.8* VDDQ	0.83	1	1.1	RZQ/7	1, 2
		VOHdc= 0.95* VDDQ	0.83	1	1.25	RZQ/7	1, 2
	RON34Pu	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/7	1, 2
		VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/7	1, 2
		VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/7	1, 2
48Ω	RON48Pd	VOLdc= 0.5*VDDQ	0.73	1	1.1	RZQ/5	1, 2
		VOMdc= 0.8* VDDQ	0.83	1	1.1	RZQ/5	1, 2
		VOHdc= 0.95* VDDQ	0.83	1	1.25	RZQ/5	1, 2
	RON48Pu	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/5	1, 2
		VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/5	1, 2
		VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/5	1, 2
Mismatch between pull-up and pull-down, MMPuPd		VOMdc= 0.8* VDDQ	-10		10	%	1, 2, 3, 4
Mismatch DQ-DQ within byte variation pull-up, MMPudd		VOMdc= 0.8* VDDQ			10	%	1, 2, 4
Mismatch DQ-DQ within byte variation pull-dn, MMPddd		VOMdc= 0.8* VDDQ			10	%	1, 2, 4

NOTE 1 The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see "ZQ Calibration Commands" section.

NOTE 2 Pull-up and pull-dn output driver impedances are recommended to be calibrated at 0.8 * VDDQ. Other calibration schemes may be used to achieve the linearity spec shown above, e.g. calibration at 0.5 * VDDQ and 0.95 * VDDQ.

NOTE 3 Measurement definition for mismatch between pull-up and pull-down, MMPuPd : Measure RONPu and RONPD both at 0.8*VDD separately; Ronnom is the nominal Ron value

$$MMPuPd = \frac{RONPu - RONPd}{RONNOM} *100$$

NOTE 4 RON variance range ratio to RON Nominal value in a given component, including DQS_t and DQS_c.

$$MMPudd = \frac{RONPuMax - RONPuMin}{RONNOM} *100$$

$$MMPddd = \frac{RONPdMax - RONPdMin}{RONNOM} *100$$

NOTE 5 This parameter of x16 device is specified for Upper byte and Lower byte.

9.2 Output Driver DC Electrical Characteristics for Loopback Signals LBDQS, LBDQ

The DDR5 Loopback driver supports 34 ohms. A functional representation of the output buffer is shown in Figure 243.

$$RON_{Pu} = \frac{VDDQ - Vout}{|I\,out|}$$ under the condition that RONPd is off

$$RON_{Pd} = \frac{Vout}{|I\,out|}$$ under the condition that RONPu is off

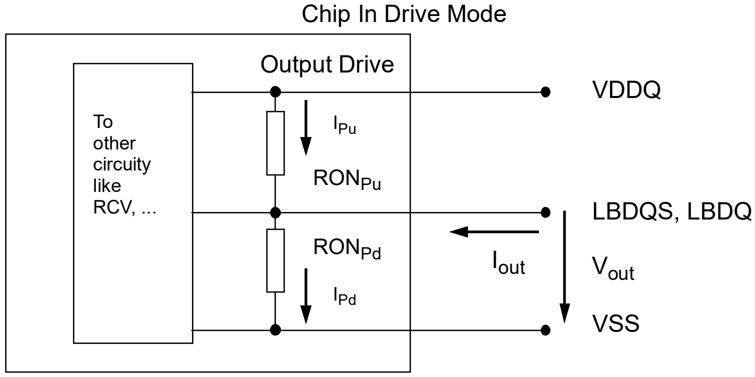


Figure 243 — Output Driver for Loopback Signals

Table 244 — Output Driver DC Electrical Characteristics, Assuming RZQ = 240 Ohms; Entire Operating Temperature Range; after Proper ZQ Calibration

RON_{NOM}	Resistor	Vout	Min	Nom	Max	Unit	Notes
34Ω	RON34Pd	VOLdc= 0.5*VDDQ	0.8	1	1.1	RZQ/7	1, 2
		VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/7	1, 2
		VOHdc= 0.95* VDDQ	0.9	1	1.25	RZQ/7	1, 2
	RON34Pu	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/7	1, 2
		VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/7	1, 2
		VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/7	1, 2
Mismatch between pull-up and pull-down, MMPuPd		VOMdc= 0.8* VDDQ	-10		10	%	1, 2, 3, 4
Mismatch LBDQS-LBDQ within device variation pull-up, MMPudd		VOMdc= 0.8* VDDQ			10	%	1, 2, 4
Mismatch LBDQS-LBDQ within device variation pull-dn, MMPddd		VOMdc= 0.8* VDDQ			10	%	1, 2, 4

NOTE 1 The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see “ZQ Calibration Commands” section.

NOTE 2 Pull-up and pull-dn output driver impedances are recommended to be calibrated at 0.8 * VDDQ. Other calibration schemes may be used to achieve the linearity spec shown above, e.g. calibration at 0.5 * VDDQ and 0.95 * VDDQ.

NOTE 3 Measurement definition for mismatch between pull-up and pull-down, MMPuPd : Measure RONPu and RONPD both at 0.8*VDD separately; Ronnom is the nominal Ron value

$$MMPuPd = \frac{RONPu - RONPd}{RONNOM} *100$$

NOTE 4 RON variance range ratio to RON Nominal value in a given component, including LBDQS and LBDQ.

$$MMPudd = \frac{RONPuMax - RONPuMin}{RONNOM} *100$$

$$MMPddd = \frac{RONPdMax - RONPdMin}{RONNOM} *100$$

9.3 Loopback Output Timing

Loopback strobe LBDQS to Loopback data LBDQ relationship is illustrated in Figure 244.

- tLBQSH describes the single-ended LBDQS strobe high pulse width
- tLBQSL describes the single-ended LBDQS strobe low pulse width
- tLBQ_Set describes the setup time of LBDQS and where LBDQ needs to remain stable
- tLBQ_Hld describes the hold time of LBDQS and where LBDQ needs to remain stable
- tLBDVW describes the data valid window per device per UI

Table 245 — Loopback Output Timing Parameters for DDR5-3200 to 4800

Speed		DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Units	Note
Parameter	Symbol	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Loopback Timing													
Loopback LBDQS Output Low Time	tLBQSL	0.7	-	0.7	-	0.7	-	0.7	-	0.7	-	tCK	1
Loopback LBDQS Output High Time	tLBQSH	0.7	-	0.7	-	0.7	-	0.7	-	0.7	-	tCK	1
Loopback Setup time for LBDQS	tLBQ_Set	0.5		0.5		0.5		0.5		0.5		tCK	1, 2
Loopback Hold time for LBDQS	tLBQ_Hld	0.5	-	0.5	-	0.5	-	0.5	-	0.5	-	tCK	1, 2
Loopback Data valid window of each UI per DRAM	tLBDVW	1.6	-	1.6	-	1.6	-	1.6	-	1.6	-	tCK	1

NOTE 1 Based on Loopback 4-way interleave setting (see MR53)
NOTE 2 tLBQ_Set is measured from LBDQ first transition to LBDQS falling edge and tLBQ_Hld is measured from LBDQS falling edge to LBDQ last transition.

Table 246 — Loopback Output Timing Parameters for DDR5-5200 to 6400

Speed		DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Units	Note
Parameter	Symbol	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Loopback Timing											
Loopback LBDQS Output Low Time	tLBQSL	0.7	-	0.7	-	0.75	-	0.75	-	tCK	1
Loopback LBDQS Output High Time	tLBQSH	0.7	-	0.7	-	0.75	-	0.75	-	tCK	1
Loopback Setup time for LBDQS	tLBQ_Set	0.5		0.5		0.5		0.5		tCK	1, 2
Loopback Hold time for LBDQS	tLBQ_Hld	0.5	-	0.5	-	0.5	-	0.5	-	tCK	1, 2
Loopback Data valid window of each UI per DRAM	tLBDVW	1.6	-	1.6	-	1.6	-	1.6	-	tCK	1

NOTE 1 Based on Loopback 4-way interleave setting (see MR53)
NOTE 2 tLBQ_Set is measured from LBDQ first transition to LBDQS falling edge and tLBQ_Hld is measured from LBDQS falling edge to LBDQ last transition.

Table 247 — Loopback Output Timing Parameters for DDR5-6800 to 8400

Speed		DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		Units	Note
Parameter	Symbol	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Loopback Timing													
Loopback LBDQS Output Low Time	tLBQSL	0.75	-	0.75	-	0.75	-	0.75	-	0.75	-	tCK	1
Loopback LBDQS Output High Time	tLBQSH	0.75	-	0.75	-	0.75	-	0.75	-	0.75	-	tCK	1
Loopback Setup time for LBDQS	tLBQ_Set	0.55	-	0.55	-	0.55	-	0.55	-	0.55	-	tCK	1, 2
Loopback Hold time for LBDQS	tLBQ_Hld	0.55	-	0.55	-	0.55	-	0.55	-	0.55	-	tCK	1, 2
Loopback Data valid window of each UI per DRAM	tLBDVW	1.6	-	1.6	-	1.6	-	1.6	-	1.6	-	tCK	1

NOTE 1 Based on Loopback 4-way interleave setting (see MR53)
NOTE 2 tLBQ_Set is measured from LBDQ first transition to LBDQS falling edge and tLBQ_Hld is measured from LBDQS falling edge to LBDQ last transition.

9.3.1 Loopback Output Timing (cont'd)

Table 248 — Loopback Output Timing Parameters for DDR5-8800

Speed		DDR5-8800		Units	NOTE
Parameter	Symbol	MIN	MAX		
Loopback Timing					
Loopback LBDQS Output Low Time	tLBQSL	0.75	-	tCK	1
Loopback LBDQS Output High Time	tLBQSH	0.75	-	tCK	1
Loopback Setup time for LBDQS	tLBQ_Set	0.55	-	tCK	1, 2
Loopback Hold time for LBDQS	tLBQ_Hld	0.55	-	tCK	1, 2
Loopback Data valid window of each UI per DRAM	tLBDVW	1.6	-	tCK	1
NOTE 1 Based on Loopback 4-way interleave setting (see MR53) NOTE 2 tLBQ_Set is measured from LBDQ first transition to LBDQS falling edge and tLBQ_Hld is measured from LBDQS falling edge to LBDQ last transition.					

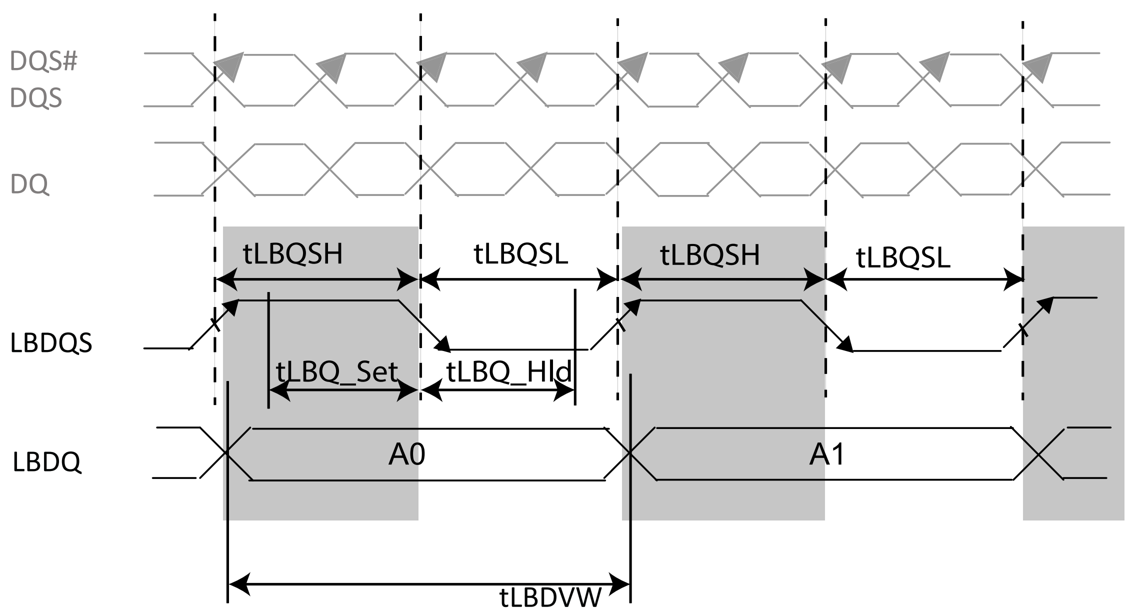


Figure 244 — Loopback Strobe to Data Relationship

9.3.1 ALERT_n Output Drive Characteristic

A functional representation of the output buffer is shown in Figure 245. Output driver impedance RON is defined as follows:

$$RON_{Pd} = \frac{Vout}{|\ Iout\ |} \text{ under the condition that } RON_{Pu} \text{ is off}$$

Alert Driver

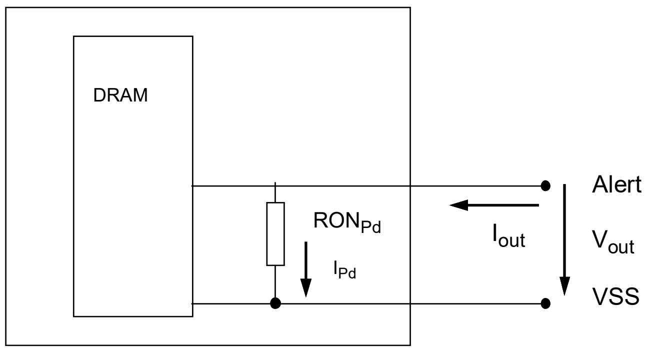


Figure 245 — Output Buffer Ron

Table 249 — Output Driver Impedance RON

Resistor	Vout	Min	Max	Unit	Note
RON_{Pd}	VOLdc= 0.1* VDDQ	0.3	1.1	$R_{ZQ}/7$	
	V_{OMdc} = 0.8* VDDQ	0.4	1.1	$R_{ZQ}/7$	
	V_{OHdc} = 0.95* VDDQ	0.4	1.25	$R_{ZQ}/7$	

9.3.2 Output Driver Characteristic of Connectivity Test (CT) Mode

Following Output driver impedance RON will be applied to the Test Output Pin during Connectivity Test (CT) Mode.

The individual pull-up and pull-down resistors (RONPu_CT and RONPd_CT) are defined as follows:

$$RON_{Pu\_CT} = \frac{V_{DDQ}-V_{OUT}}{|\,Iout\,|}$$

$$RON_{Pd\_CT} = \frac{V_{OUT}}{|\,Iout\,|}$$

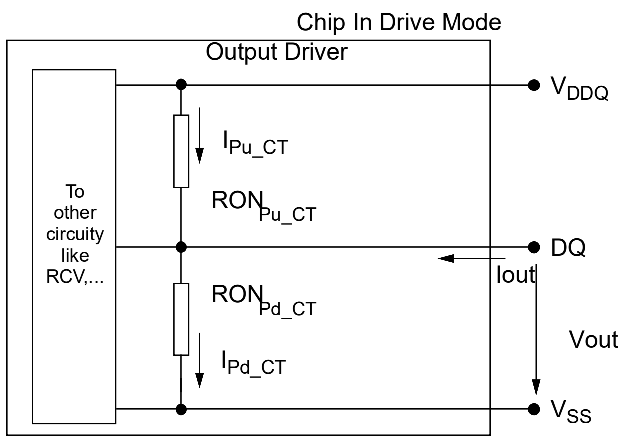


Figure 246 — Output Driver

Table 250 — Output Driver Characteristic of Connectivity Test (CT) Mode

RONNOM_CT	Resistor	Vout	Max	Units	Note
34 Ω	RONPd_CT	VOBdc = 0.2 x VDDQ	1.9	$R_{ZQ}/7$	1, 2
		VOLdc = 0.5 x VDDQ	2.0	$R_{ZQ}/7$	1, 2
		VOMdc = 0.8 x VDDQ	2.2	$R_{ZQ}/7$	1, 2
		VOHdc = 0.95 x VDDQ	2.5	$R_{ZQ}/7$	1, 2
	RONPu_CT	VOBdc = 0.2 x VDDQ	1.9	$R_{ZQ}/7$	1, 2
		VOLdc = 0.5 x VDDQ	2.0	$R_{ZQ}/7$	1, 2
		VOMdc = 0.8 x VDDQ	2.2	$R_{ZQ}/7$	1, 2
		VOHdc = 0.95 x VDDQ	2.5	$R_{ZQ}/7$	1, 2

NOTE 1 Connectivity test mode uses un-calibrated drivers, showing the full range over PVT. No mismatch between pull up and pull down is defined.
NOTE 2 Uncalibrated drive strength tolerance is specified at +/- 30%

9.4 Single-ended Output Levels - VOL/VOH

Table 251 — Single-ended Output Levels for DDR5-3200 to DDR5-6400

Symbol	Parameter	DDR5-3200-6400	Units	Notes
V_{OH}	Output high measurement level (for output SR)	0.75 x V_{pk-pk}	V	1
V_{OL}	Output low measurement level (for output SR)	0.25 x V_{pk-pk}	V	1
NOTE 1 V_{pk-pk} is the mean high voltage minus the mean low voltage over 8UI samples.				

Table 252 — Single-ended Output Levels for DDR5-6800 to DDR5-8800

Symbol	Parameter	DDR5-6800-8800	Units	Notes
VOH	Output high measurement level (for output SR)	0.75 x Vpk-pk	V	1
VOL	Output low measurement level (for output SR)	0.25 x Vpk-pk	V	1
NOTE 1 V_{pk-pk} is the mean high voltage minus the mean low voltage over 8UI samples.				

9.4.1 DDP Single-ended Output Levels - VOL/VOH

Table 253 — DDP Single-Ended Output Levels for DDR5 DDP 3200 to 6400

Symbol	Parameter	DDR5 DDP 3200-6400	Units	Notes
VOH	Output high measurement level (for output SR)	0.75 x V_{pk-pk}	V	1
VOL	Output low measurement level (for output SR)	0.25 x V_{pk-pk}	V	1
NOTE 1 V_{pk-pk} is the mean high voltage minus the mean low voltage over 8UI samples.				

9.5 Single-Ended Output Levels - VOL/VOH for Loopback Signals

Table 254 — Single-ended Output Levels for Loopback Signals DDR5-3200 to DDR5-6400

Symbol	Parameter	DDR5-3200-6400	Units	Notes
V_{OH}	Output high measurement level (for output SR)	0.75 x V_{pk-pk}	V	1
V_{OL}	Output low measurement level (for output SR)	0.25 x V_{pk-pk}	V	1
NOTE 1 V_{pk-pk} is the mean high voltage minus the mean low voltage over 8UI samples.				

Table 255 — Single-ended Output Levels for Loopback Signals DDR5-6800 to DDR5-8800

Symbol	Parameter	DDR5-6800-8800	Units	Notes
V_{OH}	Output high measurement level (for output SR)	0.75 x V_{pk-pk}	V	1
V_{OL}	Output low measurement level (for output SR)	0.25 x V_{pk-pk}	V	1
NOTE 1 V_{pk-pk} is the mean high voltage minus the mean low voltage over 8UI samples.				

9.5.1 DDP Single-Ended Output Levels - VOL/VOH for Loopback Signals

Table 256 — DDP Single-ended Output Levels for Loopback Signals DDR5 DDP 3200 to 6400

Symbol	Parameter	DDR5 DDP 3200-6400	Units	Notes
V_{OH}	Output high measurement level (for output SR)	0.75 x V_{pk-pk}	V	1
V_{OL}	Output low measurement level (for output SR)	0.25 x V_{pk-pk}	V	1
NOTE 1 V_{pk-pk} is the mean high voltage minus the mean low voltage over 8UI samples.				

9.6 Single-ended Output Slew Rate

With the reference load for timing measurements, output slew rate for falling and rising edges is defined and measured between V_{OL} and V_{OH} for single ended signals as shown in Table 257 and Figure 247.

Table 257 — Single-ended Output Slew Rate Definition

Description	Measured		Defined by
	From	To	
Single ended output slew rate for rising edge	V_{OL}	V_{OH}	$[V_{OH}-V_{OL}]$ / delta TRse
Single ended output slew rate for falling edge	V_{OH}	V_{OL}	$[V_{OH}-V_{OL}]$ / delta TFse
NOTE 1 Output slew rate is verified by design and characterization, and may not be subject to production test.			

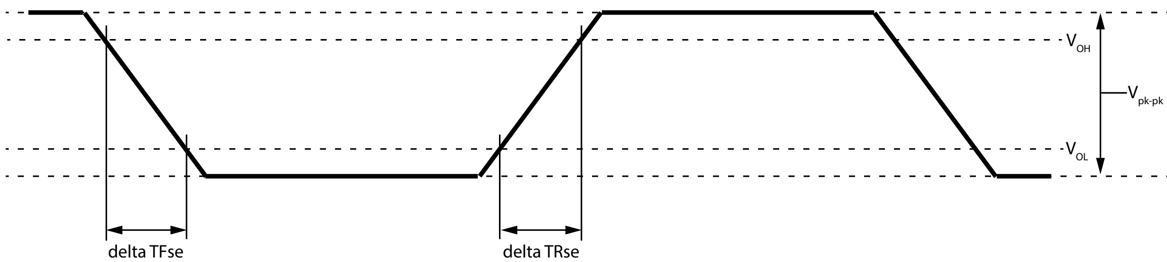


Figure 247 — Single-ended Output Slew Rate Definition

Table 258 — Single-ended Output Slew Rate for DDR5-3200 to DDR5-4800

Speed		DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Units	NOTE
Parameter	Symbol	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Single ended output slew rate	SRQse	8	24	8	24	8	24	8	24	8	24	V/ns	

Table 259 — Single-ended Output Slew Rate for DDR5-5200 to DDR5-6400

Speed		DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Units	NOTE
Parameter	Symbol	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Single ended output slew rate	SRQse	12	24	12	24	12	24	12	24	V/ns	

Table 260 — Single-ended Output Slew Rate for DDR5-6800 to DDR5-8800

Speed		DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Units	NOTE
Parameter	Symbol	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Single ended output slew rate	SRQse	12	24	12	24	12	24	12	24	12	24	12	24	V/ns	

9.6.1 DDP Single-Ended Output Slew Rate

With the reference load for timing measurements, output slew rate for falling and rising edges is defined and measured between V_{OL} and V_{OH} for single ended signals as shown in Table 261 and Figure 248.

Table 261 — DDP Single-ended Output Slew Rate Definition

Description	Measured		Defined by
	From	To	
Single ended output slew rate for rising edge	V_{OL}	V_{OH}	$[V_{OH}-V_{OL}]$ / delta TRse
Single ended output slew rate for falling edge	V_{OH}	V_{OL}	$[V_{OH}-V_{OL}]$ / delta TFse
NOTE 1 Output slew rate is verified by design and characterization, and may not be subject to production test.			

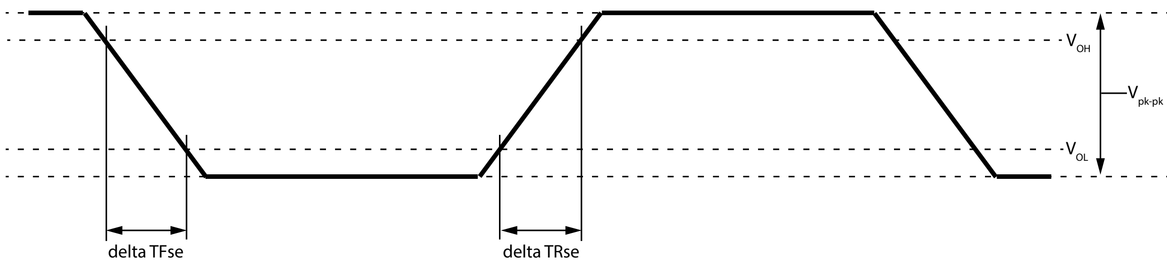


Figure 248 — Single-ended Output Slew Rate Definition

Table 262 — DDP Single-ended Output Slew Rate for DDR5-3200 to DDR5-6400

Speed		DDR5 DDP 3200-6400		Units	NOTE
Parameter	Symbol	MIN	MAX		
Single ended output slew rate	SRQse	4	15	V/ns	

9.7 Differential Output Levels

Table 263 — Differential Output levels for DDR5-3200 to DDR5-6400

Symbol	Parameter	DDR5-3200-6400	Units	Notes
V_{OHdiff}	Differential output high measurement level (for output SR)	0.75 x $V_{diffpk-pk}$	V	1
V_{OLdiff}	Differential output low measurement level (for output SR)	0.25 x $V_{diffpk-pk}$	V	1
NOTE 1 $V_{diffpk-pk}$ is the mean high voltage minus the mean low voltage over 8UI samples.				

Table 264 — Differential AC and DC Output Levels for DDR5-6800 to DDR5-8800

Symbol	Parameter	DDR5-6800-8800	Units	Notes
V_{OHdiff}	Differential output high measurement level (for output SR)	0.75 x $V_{diffpk-pk}$	V	1
V_{OLdiff}	Differential output low measurement level (for output SR)	0.25 x $V_{diffpk-pk}$	V	1
NOTE 1 $V_{diffpk-pk}$ is the mean high voltage minus the mean low voltage over 8UI samples.				

9.7.1 DDP Differential Output Levels

Table 265 — DDP Differential Output Levels for DDR5-3200 to DDR5-6400

Symbol	Parameter	DDR5-3200-6400	Units	Notes
V_{OHdiff}	Differential output high measurement level (for output SR)	0.75 x $V_{diffpk-pk}$	V	1
V_{OLdiff}	Differential output low measurement level (for output SR)	0.25 x $V_{diffpk-pk}$	V	1
NOTE 1 $V_{diffpk-pk}$ is the mean high voltage minus the mean low voltage over 8UI samples.				

9.8 Differential Output Slew Rate

With the reference load for timing measurements, output slew rate for falling and rising edges is defined and measured between $V_{OL\text{-}diff}$ and V_{OHdiff} for differential signals as shown in Table 266 and Figure 249

Table 266 — Differential Output Slew Rate Definition

Description	Measured		Defined by
	From	To	
Differential output slew rate for rising edge	V_{OLdiff}	V_{OHdiff}	$[V_{OHdiff}$-$V_{OLdiff}]$ / delta TRdiff
Differential output slew rate for falling edge	V_{OHdiff}	V_{OLdiff}	$[V_{OHdiff}$-$V_{OLdiff}]$ / delta TFdiff
NOTE 1 Output slew rate is verified by design and characterization, and may not be subject to production test.			

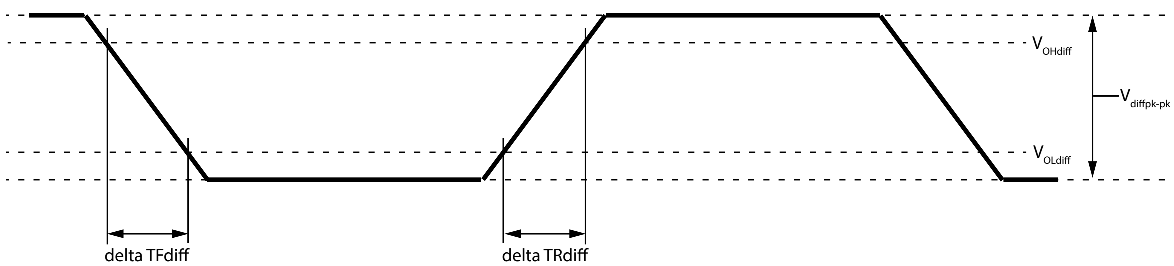


Figure 249 — Differential Output Slew Rate Definition

Table 267 — Differential Output Slew Rate for DDR5-3200 to DDR5-4800

Speed		DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Units	NOTE
Parameter	Symbol	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Differential output slew rate	SRQdiff	16	48	16	48	16	48	16	48	16	48	V/ns	

Table 268 — Differential Output Slew Rate for DDR5-5200 to DDR5-6400

Speed		DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Units	NOTE
Parameter	Symbol	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Differential output slew rate	SRQdiff	24	48	24	48	24	48	24	48	V/ns	

Table 269 — Differential Output Slew Rate for DDR5-6800 to DDR5-8800

Speed		DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Units	NOTE
Parameter	Symbol	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Differential output slew rate	SRQdiff	24	48	24	48	24	48	24	48	24	48	24	48	V/ns	

9.8.1 DDP Differential Output Slew Rate

With the reference load for timing measurements, output slew rate for falling and rising edges is defined and measured between V_{OLdiff} and V_{OHdiff} for differential signals as shown in Table 270 and Figure 250

Table 270 — DDP Differential output slew rate definition

Description	Measured		Defined by
	From	To	
Differential output slew rate for rising edge	V_{OLdiff}	V_{OHdiff}	[V_{OHdiff}-V_{OLdiff}] / delta TRdiff
Differential output slew rate for falling edge	V_{OHdiff}	V_{OLdiff}	[V_{OHdiff}-V_{OLdiff}] / delta TFdiff
NOTE 1 Output slew rate is verified by design and characterization, and may not be subject to production test.			

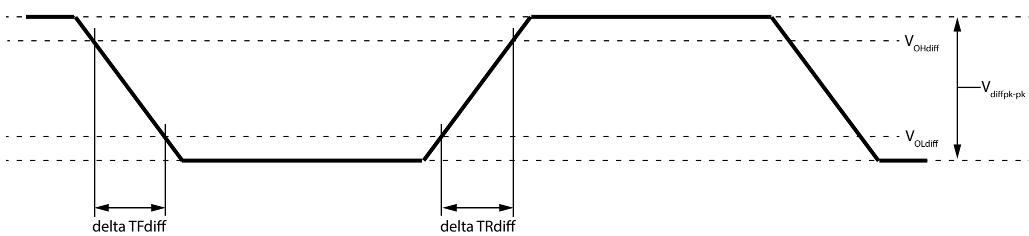


Figure 250 — Differential Output Slew Rate Definition

Table 271 — DDP Differential Output Slew Rate for DDR5-3200 to DDR5-6400

Speed		DDR5 DDP 3200-6400		Units	NOTE
Parameter	Symbol	MIN	MAX		
Differential output slew rate	SRQdiff	8	30	V/ns	

9.9 Tx DQS Jitter

The Random Jitter (Rj) specified is a random jitter meeting a Gaussian distribution. The Deterministic Jitter (Dj) specified is bounded. The DDR5 device output jitter must not exceed maximum values specified in Table 272.

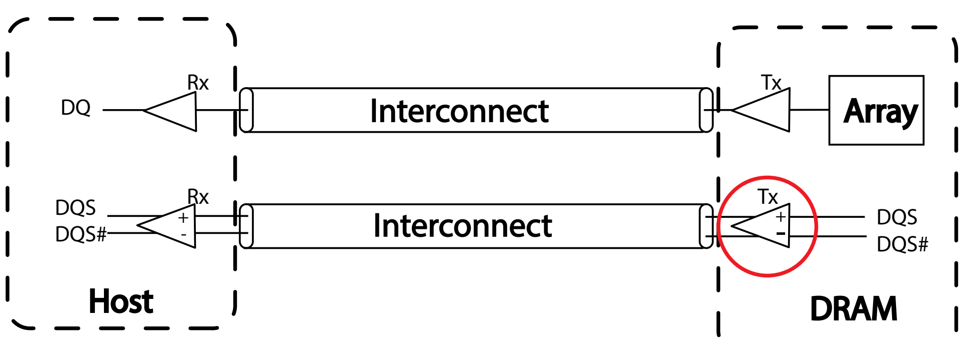


Figure 251 — Tx DQS Jitter

9.9 Tx DQS Jitter (cont'd)

Table 272 — Tx DQS Jitter Parameters for DDR5-3200 to 4800

[Dj=Deterministic Jitter; Rj=Random Jitter; DCD=Duty Cycle Distortion; BUJ=Bounded Uncorrelated Jitter; pp=Peak to Peak]

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Rj RMS Value of 1-UI Jitter without BUJ	tTx_DQS_1UI_Rj_NoBUJ	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	UI (RMS)	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
Dj pp Value of 1-UI Jitter without BUJ	tTx_DQS_1UI_Dj_NoBUJ	-	0.150	-	0.150	-	0.150	-	0.150	-	0.150	UI	1, 2, 3, 5, 6, 7, 8, 9, 10, 11
Rj RMS Value of N-UI jitter without BUJ, where 1<N< 4	tTx_DQS_NUI_Rj_NoBUJ	-	tCK_NUI_Rj_NoBUJ + 0.002	-	tCK_NUI_Rj_NoBUJ + 0.002	-	tCK_NUI_Rj_NoBUJ + 0.002	-	tCK_NUI_Rj_NoBUJ + 0.002	-	tCK_NUI_Rj_NoBUJ + 0.002	UI (RMS)	1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 12
Dj pp Value of N-UI Jitter without BUJ, where 1<N < 4	tTx_DQS_NUI_Dj_NoBUJ	-	0.150	-	0.150	-	0.150	-	0.150	-	0.150	UI	1, 2, 3, 5, 6, 7, 8, 9, 10, 11

NOTE 1 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 2 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases the contribution of BUJ is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers, so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 3 The validation methodology for these parameters will be covered in future ballots

NOTE 4 Rj RMS value of 1-UI jitter. Without BUJ, but on-die system like noise present. This extraction is to be done after software correction of DCD

NOTE 5 See Section 7.1 for details on the minimum BER requirements

NOTE 6 See Section 7.2 for details on UI, NUI and Jitter definitions

NOTE 7 Duty Cycle of the DQ pins must be adjusted as close to 50% as possible using the Global and Per Pin Duty Cycle Adjuster feature prior to running the Tx DQ Jitter test

NOTE 8 The Mode Registers for the Duty Cycle Adjuster are MR43 and MR44, and the Mode Registers for the Per Pin DCA of DQS are MR103 - MR110.

NOTE 9 Spread Spectrum Clocking (SSC) must be disabled while running the Tx DQ Jitter test

NOTE 10 These parameters are tested using the continuous clock pattern which are sent out from the dram device without the need for sending out continuous MRR commands. The MR25 OP[3] is set to "1" to enable this feature.

NOTE 11 Tested on the CTC2 card only

NOTE 12 The max value of tTx_DQS_Rj_1UI_NoBUJ and tTx_DQS_Rj_NUI_NoBUJ can be 6 mUI RMS

9.9 Tx DQS Jitter (cont'd)

Table 273 — Tx DQS Jitter Parameters for DDR5-5200 to 6400

[Dj=Deterministic Jitter; Rj=Random Jitter; DCD=Duty Cycle Distortion; BUJ=Bounded Uncorrelated Jitter; pp=Peak to Peak]

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
Rj RMS Value of 1-UI Jitter without BUJ	tTx_DQS_1UI_Rj_NoBUJ	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	UI (RMS)	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
Dj pp Value of 1-UI Jitter without BUJ	tTx_DQS_1UI_Dj_NoBUJ	-	0.130	-	0.130	-	0.130	-	0.130	UI	1, 2, 3, 5, 6, 7, 8, 9, 10, 11
Rj RMS Value of N-UI jitter without BUJ, where 1<N< 4	tTx_DQS_NUI_Rj_NoBUJ	-	tCK_NUI_Rj_NoBUJ + 0.002	-	tCK_NUI_Rj_NoBUJ + 0.002	-	-	-	-	UI (RMS)	1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 12
Dj pp Value of N-UI Jitter without BUJ, where 1<N < 4	tTx_DQS_NUI_Dj_NoBUJ	-	0.130	-	0.130	-	-	-	-	UI	1, 2, 3, 5, 6, 7, 8, 9, 10, 11
Rj RMS Value of N-UI jitter without BUJ, where 1<N< 5	tTx_DQS_NUI_Rj_NoBUJ	-	-	-	-	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	UI (RMS)	1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 12
Dj pp Value of N-UI Jitter without BUJ, where 1<N < 5	tTx_DQS_NUI_Dj_NoBUJ	-	-	-	-	-	0.130	-	0.130	UI	1, 2, 3, 5, 6, 7, 8, 9, 10, 11

NOTE 1 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 2 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases the contribution of BUJ is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers, so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 3 The validation methodology for these parameters will be covered in future ballots

NOTE 4 Rj RMS value of 1-UI jitter. Without BUJ, but on-die system like noise present. This extraction is to be done after software correction of DCD

NOTE 5 See Section 7.1 for details on the minimum BER requirements

NOTE 6 See Section 7.2 for details on UI, NUI and Jitter definitions

NOTE 7 Duty Cycle of the DQ pins must be adjusted as close to 50% as possible using the Global and Per Pin Duty Cycle Adjuster feature prior to running the Tx DQ Jitter test

NOTE 8 The Mode Registers for the Duty Cycle Adjuster are MR43 and MR44, and the Mode Registers for the Per Pin DCA of DQS are MR103 - MR110.

NOTE 9 Spread Spectrum Clocking (SSC) must be disabled while running the Tx DQ Jitter test

NOTE 10 These parameters are tested using the continuous clock pattern which are sent out from the dram device without the need for sending out continuous MRR commands. The MR25 OP[3] is set to "1" to enable this feature.

NOTE 11 Tested on the CTC2 card only

NOTE 12 The max value of tTx_DQS_Rj_1UI_NoBUJ and tTx_DQS_Rj_NUI_NoBUJ can be 6 mUI RMS

9.9 Tx DQS Jitter (cont'd)

Table 274 — Tx DQS Jitter Parameters for DDR5-6800 to 8800

[Dj=Deterministic Jitter; Rj=Random Jitter; DCD=Duty Cycle Distortion; BUJ=Bounded Uncorrelated Jitter; pp=Peak to Peak]

Parameter	Symbol	DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Rj RMS Value of 1-UI Jitter without BUJ	tTx_DQS_1UI_Rj_NoBUJ	-	0.006	-	0.006	-	0.006	-	0.006	-	0.006	-	0.006	UI (RMS)	1,2,3,4,5,6,7,8,9,10,11,12
Dj pp Value of 1-UI Jitter without BUJ	tTx_DQS_1UI_Dj_NoBUJ	-	0.130	-	0.130	-	0.130	-	0.130	-	0.130	-	0.130	UI	1,2,3,5,6,7,8,9,10,11
Rj RMS Value of N-UI jitter without BUJ, where 1<N< 6	tTx_DQS_NUI_Rj_NoBUJ	-	0.006	-	0.006	-	0.006	-	0.006	-	0.006	-	0.006	UI (RMS)	1,2,3,5,6,7,8,9,10,11,12
Dj pp Value of N-UI Jitter without BUJ, where 1<N < 6	tTx_DQS_NUI_Dj_NoBUJ	-	0.130	-	0.130	-	0.130	-	0.130	-	0.130	-	0.130	UI	1,2,3,5,6,7,8,9,10,11

NOTE 1 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 2 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases the contribution of BUJ is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers, so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 3 The validation methodology for these parameters will be covered in future ballots

NOTE 4 Rj RMS value of 1-UI jitter. Without BUJ, but on-die system like noise present. This extraction is to be done after software correction of DCD

NOTE 5 See Section 7.1 for details on the minimum BER requirements

NOTE 6 See Section 7.2 for details on UI, NUI and Jitter definitions

NOTE 7 Duty Cycle of the DQ pins must be adjusted as close to 50% as possible using the Global and Per Pin Duty Cycle Adjuster feature prior to running the Tx DQ Jitter test

NOTE 8 The Mode Registers for the Duty Cycle Adjuster are MR43 and MR44.

NOTE 9 Spread Spectrum Clocking (SSC) must be disabled while running the Tx DQ Jitter test

NOTE 10 These parameters are tested using the continuous clock pattern which are sent out from the dram device without the need for sending out continuous MRR commands. The MR25 OP[3] is set to "1" to enable this feature.

NOTE 11 Tested on the CTC2 card only

NOTE 12 The max value of tTx_DQS_Rj_1UI_NoBUJ and tTx_DQS_Rj_NUI_NoBUJ can be 6 mUI RMS

9.10 Tx DQ Jitter

9.10.1 Overview

The Random Jitter (Rj) specified is a random jitter meeting a Gaussian distribution. The Deterministic Jitter (Dj) specified is bounded. The DDR5 device output jitter must not exceed maximum values specified in **Table 275**.

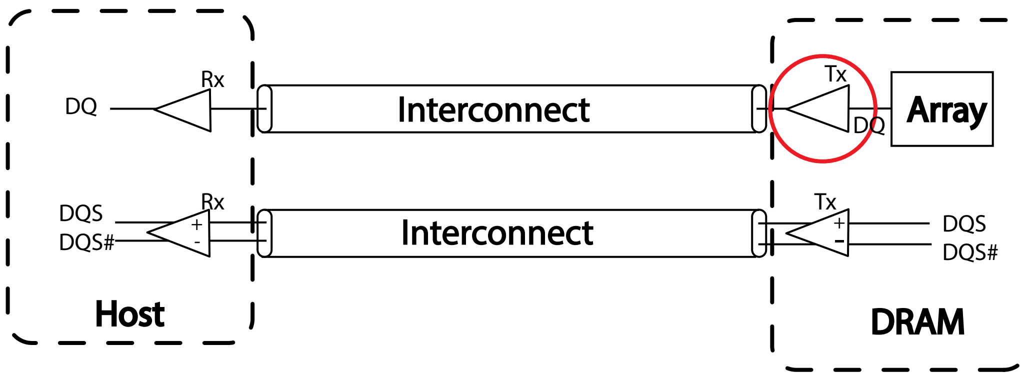


Figure 252 — Random Jitter R_j

9.10.2 Tx DQ Jitter Parameters

Table 275 — Tx DQ Jitter Parameters for DDR5-3200 to 4800

[Dj=Deterministic Jitter; Rj=Random Jitter; DCD=Duty Cycle Distortion; BUJ=Bounded Uncorrelated Jitter; pp=Peak to Peak]

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Rj RMS of 1-UI jitter without BUJ	tTx_DQ_1UI_Rj_NoBUJ	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	UI (RMS)	1, 3, 4, 5, 7, 8, 9, 10, 11, 12, 13, 14
Dj pp 1-UI jitter without BUJ	tTx_DQ_1UI_Dj_NoBUJ	-	0.150	-	0.150	-	0.150	-	0.150	-	0.150	UI	3, 5, 7, 8, 9, 10, 11, 12, 13
Rj RMS of N-UI jitter without BUJ, where 1<N<4	tTx_DQ_NUI_Rj_NoBUJ	-	tCK_NUI_Rj_NoBUJ + 0.002	-	tCK_NUI_Rj_NoBUJ + 0.002	-	tCK_NUI_Rj_NoBUJ + 0.002	-	tCK_NUI_Rj_NoBUJ + 0.002	-	tCK_NUI_Rj_NoBUJ + 0.002	UI (RMS)	3, 5, 7, 8, 9, 10, 11, 12, 13, 14
Dj pp N-UI jitter without BUJ, where 1<N<4	tTx_DQ_NUI_Dj_NoBUJ	-	0.150	-	0.150	-	0.150	-	0.150	-	0.150	UI	3, 6, 7, 8, 9, 10, 11, 12, 13
Delay of any data lane relative to strobe lane	tTx_DQS2DQ	-0.100	0.100	-0.100	0.100	-0.100	0.100	-0.100	0.100	-0.100	0.100	UI	3, 5, 6, 7, 9, 10, 11, 12, 13

NOTE 1 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility.

NOTE 2 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of BUJ is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 3 The validation methodology for these parameters will be covered in future ballots

NOTE 4 Rj RMS value of 1-UI jitter without BUJ, but on-die system like noise present. This extraction is to be done after software correction of DCD

NOTE 5 Delay of any data lane relative to strobe lane, as measured at Tx output

NOTE 6 Vref noise level to DQ jitter should be adjusted to minimize DCD

NOTE 7 See **Chapter 7** for details on the minimum BER requirements

NOTE 8 See **Chapter 7** for details on UI, NUI and Jitter definitions

NOTE 9 Duty Cycle of the DQ pins must be adjusted as close to 50% as possible using the Global and Per Pin Duty Cycle Adjuster feature prior to running this test

NOTE 10 The Mode Registers for the Duty Cycle Adjuster are MR43 and MR44. Also the Mode Registers for the Per Pin DCA of DQLx are MR(133+8x) and MR(134+8x), where $0 \le x \le 7$, and the Mode Registers for the Per Pin DCA of DQUy are MR(197+8y) and MR(198+8y), where $0 \le y \le 7$.

NOTE 11 Spread Spectrum Clocking (SSC) must be disabled while running this test

NOTE 12 These parameters are tested using the continuous clock pattern which are sent out from the dram device without the need for sending out continuous MRR commands. The MR25 OP[3] is set to “1” to enable this feature.

NOTE 13 Tested on the CTC2 card only

NOTE 14 The max value of tTx_DQ_Rj_1UI_NoBUJ and tTx_DQ_Rj_NUI_NoBUJ can be 6 mUI RMS

9.10.2 Tx DQ Jitter Parameters (cont'd)

Table 276 — Tx DQ Jitter Parameters for DDR5-5200 to 6400

[Dj=Deterministic Jitter; Rj=Random Jitter; DCD=Duty Cycle Distortion; BUJ=Bounded Uncorrelated Jitter; pp=Peak to Peak]

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
Rj RMS of 1-UI jitter without BUJ	tTx_DQ_1UI_Rj_NoBUJ	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	UI (RMS)	1, 3, 4, 5, 7, 8, 9, 10, 11, 12, 13, 14,
Dj pp 1-UI jitter without BUJ	tTx_DQ_1UI_Dj_NoBUJ	-	0.130	-	0.130	-	0.130	-	0.130	UI	3, 5, 7, 8, 9, 10, 11, 12, 13
Rj RMS of N-UI jitter without BUJ, where 1<N<4	tTx_DQ_NUI_Rj_NoBUJ	-	tCK_NUI_Rj_NoBUJ + 0.002	-	tCK_NUI_Rj_NoBUJ + 0.002	-	-	-	-	UI (RMS)	3, 5, 7, 8, 9, 10, 11, 12, 13, 14
Dj pp N-UI jitter without BUJ, where 1<N<4	tTx_DQ_NUI_Dj_NoBUJ	-	0.130	-	0.130	-	-	-	-	UI	3, 6, 7, 8, 9, 10, 11, 12, 13
Rj RMS of N-UI jitter without BUJ, where 1<N<5	tTx_DQ_NUI_Rj_NoBUJ	-	-	-	-	-	tCK_1UI_Rj_NoBUJ + 0.002	-	tCK_1UI_Rj_NoBUJ + 0.002	UI (RMS)	
Dj pp N-UI jitter without BUJ, where 1<N<5	tTx_DQ_NUI_Dj_NoBUJ	-	-	-	-	-	0.130		0.130	UI	
Delay of any data lane relative to strobe lane	tTx_DQS2DQ	-0.100	0.100	-0.100	0.100	-0.100	0.100	-0.100	0.100	UI	3, 5, 6, 7, 9, 10, 11, 12, 13

NOTE 1 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility.

NOTE 2 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of BUJ is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 3 The validation methodology for these parameters will be covered in future ballots

NOTE 4 Rj RMS value of 1-UI jitter without BUJ, but on-die system like noise present. This extraction is to be done after software correction of DCD

NOTE 5 Delay of any data lane relative to strobe lane, as measured at Tx output

NOTE 6 Vref noise level to DQ jitter should be adjusted to minimize DCD

NOTE 7 See **Chapter 7** for details on the minimum BER requirements

NOTE 8 See **Chapter 7** for details on UI, NUI and Jitter definitions

NOTE 9 Duty Cycle of the DQ pins must be adjusted as close to 50% as possible using the Global and Per Pin Duty Cycle Adjuster feature prior to running this test

NOTE 10 The Mode Registers for the Duty Cycle Adjuster are MR43 and MR44. Also the Mode Registers for the Per Pin DCA of DQLx are MR(133+8x) and MR(134+8x), where $0 \le x \le 7$, and the Mode Registers for the Per Pin DCA of DQUy are MR(197+8y) and MR(198+8y), where $0 \le y \le 7$.

NOTE 11 Spread Spectrum Clocking (SSC) must be disabled while running this test

NOTE 12 These parameters are tested using the continuous clock pattern which are sent out from the dram device without the need for sending out continuous MRR commands. The MR25 OP[3] is set to “1” to enable this feature.

NOTE 13 Tested on the CTC2 card only

NOTE 14 The max value of tTx_DQ_Rj_1UI_NoBUJ and tTx_DQ_Rj_NUI_NoBUJ can be 6mUI RMS

9.10.2 Tx DQ Jitter Parameters (cont'd)

Table 277 — Tx DQ Jitter Parameters for DDR5-6800 to 8800

[Dj=Deterministic Jitter; Rj=Random Jitter; DCD=Duty Cycle Distortion; BUJ=Bounded Uncorrelated Jitter; pp=Peak to Peak]

Parameter	Symbol	DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Rj RMS of 1-UI jitter without BUJ	tTx_DQ_1UI_Rj_NoBUJ	-	0.006	-	0.006	-	0.006	-	0.006	-	0.006	-	0.006	UI (RMS)	1, 3, 5, 9, 11, 12, 13, 14, 15, 16, 17, 18
Dj pp 1-UI jitter without BUJ	tTx_DQ_1UI_Dj_NoBUJ	-	0.130	-	0.130	-	0.130	-	0.130	-	0.130	-	0.130	UI	3, 6, 9, 11, 12, 13, 14, 15, 16, 17
Rj RMS of N-UI jitter without BUJ, where 1<N<6	tTx_DQ_NUI_Rj_NoBUJ	-	0.006	-	0.006	-	0.006	-	0.006	-	0.006	-	0.006	UI (RMS)	3, 7, 9, 11, 12, 13, 14, 15, 16, 17, 18
Dj pp N-UI jitter without BUJ, where 1<N<6	tTx_DQ_NUI_Dj_NoBUJ	-	0.130	-	0.130	-	0.130	-	0.130	-	0.130	-	0.130	UI	3, 8, 10, 11, 12, 13, 14, 15, 16, 17
Delay of any data lane relative to strobe lane	tTx_DQS2DQ	-0.100	0.100	-0.100	0.100	-0.100	0.100	-0.100	0.100	-0.100	0.100	-0.100	0.100	UI	3, 9, 10, 11, 13, 14, 15, 16, 17

NOTE 1 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility.

NOTE 2 On-die noise similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of BUJ is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining TX lanes send patterns to the corresponding RX receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility

NOTE 3 The validation methodology for these parameters will be covered in future ballots

NOTE 4 Rj RMS value of 1-UI jitter without BUJ, but on-die system like noise present. This extraction is to be done after software correction of DCD

NOTE 5 Delay of any data lane relative to strobe lane, as measured at Tx output

NOTE 6 Vref noise level to DQ jitter should be adjusted to minimize DCD

NOTE 7 See **Chapter 7** for details on the minimum BER requirements

NOTE 8 See **Chapter 7** for details on UI, NUI and Jitter definitions

NOTE 9 Duty Cycle of the DQ pins must be adjusted as close to 50% as possible using the Duty Cycle Adjuster feature prior to running this test

NOTE 10 The Mode Registers for the Duty Cycle Adjuster are MR43 and MR44

NOTE 11 Spread Spectrum Clocking (SSC) must be disabled while running this test

NOTE 12 These parameters are tested using the continuous clock pattern which are sent out from the dram device without the need for sending out continuous MRR commands. The MR25 OP[3] is set to "1" to enable this feature.

NOTE 13 Tested on the CTC2 card only

NOTE 14 The max value of tTx_DQ_Rj_1UI_NoBUJ and tTx_DQ_Rj_NU_NoBUJ can be 6mUI RMS

9.11 Tx DQ Stressed Eye

Tx DQ stressed eye height and eye width must meet minimum specification values at BER=E^{-9} and confidence level 99.5%. Tx DQ Stressed Eye shows the DQS to DQ skew for both Eye Width and Eye Height. In order to support different Host Receiver (Rx) designs, it is the responsibility of the Host to insure the advanced DQS edges are adjusted accordingly via the Read DQS Offset Timing mode register settings (MR40 OP[3:0]).

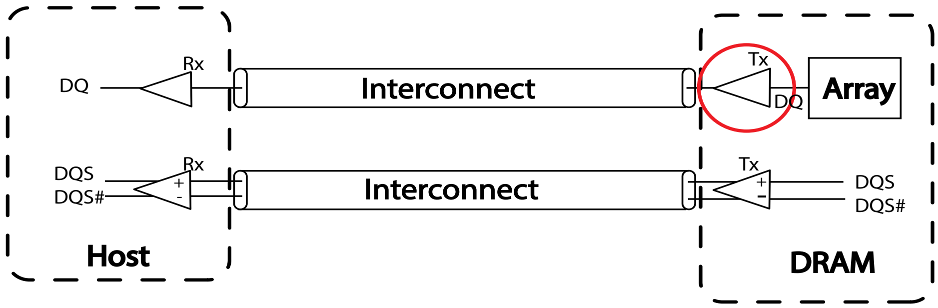


Figure 253 — Example of DDR5 Memory Interconnect

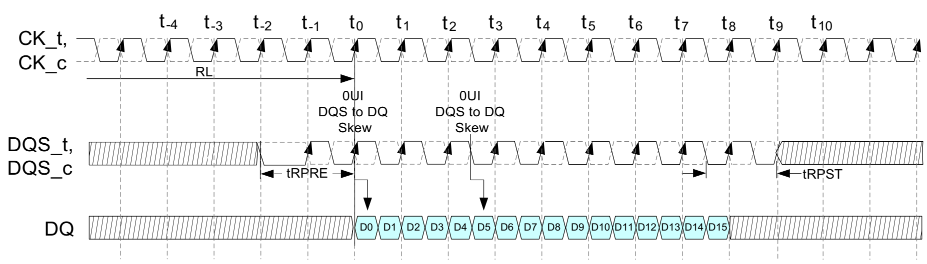


Figure 254 — Read Burst Example for Pin DQx Depicting Bit 0 and 5 Relative to the DQS Edge for 0 UI Skew

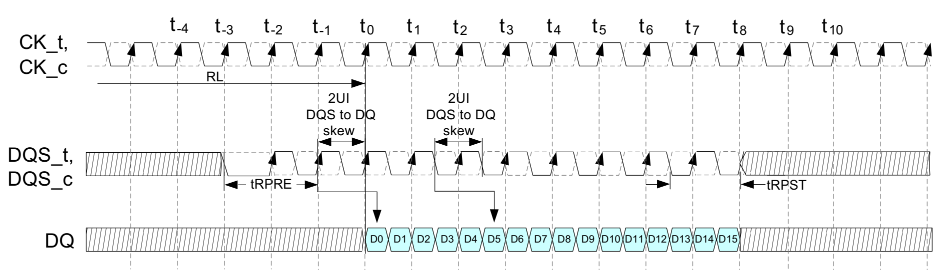


Figure 255 — Read Burst Example for Pin DQx Depicting Bit 0 and 5 Relative to the DQS Edge for 2 UI Skew with Read DQS Offset Timing Set to 1 Clock (2UI)

9.11.1 Tx DQ Stressed Eye Parameters

Table 278 — Tx DQ Stressed Eye Parameters for DDR5-3200 to 4800

[EH=Eye Height, EW=Eye Width; BER=Bit Error Rate, SES=Stressed Eye Skew]

Parameter	Symbol	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Eye Width specified at the transmitter with a skew between DQ and DQS of 1UI	TxEW_DQ_SES_1UI	0.72	-	0.72	-	0.72	-	0.72	-	0.72	-	UI	1, 2, 3, 4, 6, 7, 8, 9, 10, 11
Eye Width specified at the transmitter with a skew between DQ and DQS of 2UI	TxEW_DQ_SES_2UI	0.72	-	0.72	-	0.72	-	0.72	-	0.72	-	UI	1, 2, 3, 4, 6, 7, 8, 9, 10, 11
Eye Width specified at the transmitter with a skew between DQ and DQS of 3UI	TxEW_DQ_SES_3UI	0.72	-	0.72	-	0.72	-	0.72	-	0.72	-	UI	1, 2, 3, 4, 6, 7, 8, 9, 10, 11

NOTE 1 Minimum BER E^{-9} and Confidence Level of 99.5% per pin

NOTE 2 Refer to the minimum Bit Error Rate (BER) requirements for DDR5

NOTE 3 The validation methodology for these parameters will be covered in future ballot(s)

NOTE 4 Mismatch is defined as DQS to DQ mismatch, in UI increments

NOTE 5 The number of UI's accumulated will depend on the speed of the link. For higher speeds, higher UI accumulation may be specified. For lower speeds, N=4,5 UI may not be applicable

NOTE 6 Duty Cycle of the DQ pins must be adjusted as close to 50% as possible using the Global and Per Pin Duty Cycle Adjuster feature prior to running this test

NOTE 7 The Mode Registers for the Duty Cycle Adjuster are MR43 and MR44. Also the Mode Registers for the Per Pin DCA of DQS are MR103-MR110, the Mode Registers for the Per Pin DCA of DQLx are MR(133+8x) and MR(134+8x), where 0≤x≤7, and the Mode Registers for the Per Pin DCA of DQUy are MR(197+8y) and MR(198+8y), where 0≤y≤7.

NOTE 8 Spread Spectrum Clocking (SSC) must be disabled while running this test

NOTE 9 These parameters are tested using the continuous PRBS8 LFSR training pattern which are sent out on all DQ lanes off the dram device without the need for sending out continuous MRR commands. The MR25 OP[3] is set to "1" to enable this feature.

NOTE 10 Tested on the CTC2 card only

NOTE 11 Matched DQS to DQ would require the DQs to be adjusted by 0.5UI to place it in the center of the DQ eye. 1UI mismatch would require the DQS to be adjusted 1.5 UI. Generally, for XUI mismatch the DQ must be adjusted XUI + 0.5 UI to be placed in the center of the eye.

9.11.1 Tx DQ Stressed Eye Parameters (cont'd)

Table 279 — Tx DQ Stressed Eye Parameters for DDR5-5200 to 6400

[EH=Eye Height, EW=Eye Width; BER=Bit Error Rate]

Parameter	Symbol	DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
Eye Width specified at the transmitter with a skew between DQ and DQS of 1UI	TxEW_DQ_SES_1UI	0.74	-	0.74	-	0.75	-	0.75	-	UI	1, 2, 3, 4, 6, 7, 8, 9, 10, 11
Eye Width specified at the transmitter with a skew between DQ and DQS of 2UI	TxEW_DQ_SES_2UI	0.74	-	0.74	-	0.75	-	0.75	-	UI	1, 2, 3, 4, 6, 7, 8, 9, 10, 11
Eye Width specified at the transmitter with a skew between DQ and DQS of 3UI	TxEW_DQ_SES_3UI	0.74	-	0.74	-	0.75	-	0.75	-	UI	1, 2, 3, 4, 6, 7, 8, 9, 10, 11
Eye Width specified at the transmitter with a skew between DQ and DQS of 4UI	TxEW_DQ_SES_4UI	0.74	-	0.74	-	0.75	-	0.75	-	UI	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11
Eye Width specified at the transmitter with a skew between DQ and DQS of 5UI	TxEW_DQ_SES_5UI	-	-	-	-	0.75	-	0.75	-	UI	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

NOTE 1 Minimum BER E^{-9} and Confidence Level of 99.5% per pin

NOTE 2 Refer to the minimum Bit Error Rate (BER) requirements for DDR5

NOTE 3 The validation methodology for these parameters will be covered in future ballot(s)

NOTE 4 Mismatch is defined as DQS to DQ mismatch, in UI increments

NOTE 5 The number of UI's accumulated will depend on the speed of the link. For higher speeds, higher UI accumulation may be specified. For lower speeds, N=4,5 UI may not be applicable

NOTE 6 Duty Cycle of the DQ pins must be adjusted as close to 50% as possible using the Global and Per Pin Duty Cycle Adjuster feature prior to running this test

NOTE 7 The Mode Registers for the Duty Cycle Adjuster are MR43 and MR44. Also the Mode Registers for the Per Pin DCA of DQS are MR103-MR110, the Mode Registers for the Per Pin DCA of DQLx are MR(133+8x) and MR(134+8x), where $0 \le x \le 7$, and the Mode Registers for the Per Pin DCA of DQUy are MR(197+8y) and MR(198+8y), where $0 \le y \le 7$

NOTE 8 Spread Spectrum Clocking (SSC) must be disabled while running this test

NOTE 9 These parameters are tested using the continuous PRBS8 LFSR training pattern which are sent out on all DQ lanes off the dram device without the need for sending out continuous MRR commands. The MR25 OP[3] is set to "1" to enable this feature.

NOTE 10 Tested on the CTC2 card only

NOTE 11 Matched DQS to DQ would require the DQs to be adjusted by 0.5UI to place it in the center of the DQ eye. 1UI mismatch would require the DQS to be adjusted 1.5UI. Generally, for XUI mismatch the DQ must be adjusted XUI + 0.5UI to be placed in the center of the eye.

9.11.1 Tx DQ Stressed Eye Parameters (cont'd)

Table 280 — Tx DQ Stressed Eye Parameters for DDR5-6800 to 8400

[EH=Eye Height, EW=Eye Width; BER=Bit Error Rate]

Parameter	Symbol	DDR5-6800		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Eye Width specified at the transmitter with a skew between DQ and DQS of 1UI	TxEW_DQ_SES_1UI	0.76	-	0.76	-	0.76	-	0.76	-	0.76	-	0.76	-	UI	1, 2, 3, 4, 6, 7, 8, 9, 10, 11
Eye Width specified at the transmitter with a skew between DQ and DQS of 2UI	TxEW_DQ_SES_2UI	0.76	-	0.76	-	0.76	-	0.76	-	0.76	-	0.76	-	UI	1, 2, 3, 4, 6, 7, 8, 9, 10, 11
Eye Width specified at the transmitter with a skew between DQ and DQS of 3UI	TxEW_DQ_SES_3UI	0.76	-	0.76	-	0.76	-	0.76	-	0.76	-	0.76	-	UI	1, 2, 3, 4, 6, 7, 8, 9, 10, 11
Eye Width specified at the transmitter with a skew between DQ and DQS of 4UI	TxEW_DQ_SES_4UI	0.76	-	0.76	-	0.76	-	0.76	-	0.76	-	0.76	-	UI	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11
Eye Width specified at the transmitter with a skew between DQ and DQS of 5UI	TxEW_DQ_SES_5UI	0.76	-	0.76	-	0.76	-	0.76	-	0.76	-	0.76	-	UI	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

NOTE 1 Minimum BER E^{-9} and Confidence Level of 99.5% per pin

NOTE 2 Refer to the minimum Bit Error Rate (BER) requirements for DDR5

NOTE 3 The validation methodology for these parameters will be covered in future ballot(s)

NOTE 4 Mismatch is defined as DQS to DQ mismatch, in UI increments

NOTE 5 The number of UI's accumulated will depend on the speed of the link. For higher speeds, higher UI accumulation may be specified. For lower speeds, N=4,5 UI may not be applicable

NOTE 6 Duty Cycle of the DQ pins must be adjusted as close to 50% as possible using the Duty Cycle Adjuster feature prior to running this test

NOTE 7 The Mode Registers for the Duty Cycle Adjuster are MR43 and MR44

NOTE 8 Spread Spectrum Clocking (SSC) must be disabled while running this test

NOTE 9 These parameters are tested using the continuous PRBS8 LFSR training pattern which are sent out on all DQ lanes off the dram device without the need for sending out continuous MRR commands. The MR25 OP[3] is set to "1" to enable this feature.

NOTE 10 Tested on the CTC2 card only

NOTE 11 Matched DQS to DQ would require the DQs to be adjusted by 0.5UI to place it in the center of the DQ eye. 1UI mismatch would require the DQS to be adjusted 1.5UI. Generally, for XUI mismatch the DQ must be adjusted XUI + 0.5UI to be placed in the center of the eye.

10 Speed Bins

DDR5 Standard Speed Bins defined as:

3200 / 3600 / 4000 / 4400 / 4800 / 5200 / 5600 / 6000 / 6400 / 6800 / 7200 / 7600 / 8000 / 8400 / 8800

10.1 DDR5-3200 Speed Bins and Operations

Table 281 — DDR5-3200 Speed Bins and Operations

Speed Bin					DDR5-3200AN		DDR5-3200B		DDR5-3200BN		DDR5-3200C		Unit	Note
CL-nRCD-nRP					24-24-24		26-26-26		26-26-26		28-28-28			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	15.000	22.222	16.250	22.222	16.250	22.222	17.500	22.222	ns	12
Activate to Read or Write command delay time				tRCD	15.000	-	16.250	-	16.250	-	17.500	-	ns	7
Row Precharge time				tRP	15.000	-	16.250	-	16.250	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7,13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	47.000	-	48.250	-	48.250	-	49.500	-	ns	7,8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6,9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
Supported CL					22,24,26,28		22,26,28		22,26,28		22,28		nCK	12

10.2 DDR5-3600 Speed Bins and Operations

Table 282 — DDR5-3600 Speed Bins and Operations

Speed Bin					DDR5-3600AN		DDR5-3600B		DDR5-3600BN		DDR5-3600C		Unit	Note
CL-nRCD-nRP					26-26-26		30-30-30		30-30-30		32-32-32			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.444	22.222	16.250	22.222	16.666	22.222	17.500	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.444	-	16.250	-	16.666	-	17.500	-	ns	7
Row Precharge time				tRP	14.444	-	16.250	-	16.666	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7,13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.444	-	48.250	-	48.666	-	49.500	-	ns	7,8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6,9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
Supported CL					22,24,26,28,30,32		22,26,28,30,32		22,28,30,32		22,28,32		nCK	12

10.3 DDR5-4000 Speed Bins and Operations

Table 283 — DDR5-4000 Speed Bins and Operations

Speed Bin		DDR5-4000AN		DDR5-4000B		DDR5-4000BN		DDR5-4000C		Unit	Note
CL-nRCD-nRP		28-28-28		32-32-32		32-32-32		36-35-35			
Parameter	Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data	tAA	14.000	22.222	16.000	22.222	16.000	22.222	17.500	22.222	ns	12
Activate to Read or Write command delay time	tRCD	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Row Precharge time	tRP	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Activate to Precharge command period	tRAS	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7,13
Activate to Activate or Refresh command period	tRC (tRAS +tRP)	46.000	-	48.000	-	48.000	-	49.500	-	ns	7,8
CAS Write Latency	CWL	CL-2								nCK	12

Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6,9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	RESERVED		ns	
4000AN	14.000	14.000	28	tCK(AVG)	0.500	<0.555	RESERVED						ns	
Supported CL					22,24,26,28, 30,32,36		22,26,28, 30,32,36		22,26,28,30,32,36		22,28,32,36		nCK	12

10.4 DDR5-4400 Speed Bins and Operations

Table 284 — DDR5-4400 Speed Bins and Operations

Speed Bin					DDR5-4400AN		DDR5-4400B		DDR5-4400BN		DDR5-4400C		Unit	Note
CL-nRCD-nRP					32-32-32		36-36-36		36-36-36		40-39-39			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.545	22.222	16.000	22.222	16.363	22.222	17.500	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.545	-	16.000	-	16.363	-	17.500	-	ns	7
Row Precharge time				tRP	14.545	-	16.000	-	16.363	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 ~~x~~* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 ~~x~~* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 ~~x~~* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 ~~x~~* tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.545	-	48.000	-	48.363	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	RESERVED								ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	14.000	14.000	28	tCK(AVG)	RESERVED								ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	14.545	14.545	32	tCK(AVG)	0.454	<0.500	RESERVED						ns	
Supported CL					22,24,26,28,30,32,36,40		22,26,28,30,32,36,40		22,28,30,32,36,40		22,28,32,36,40		nCK	12

10.5 DDR5-4800 Speed Bins and Operations

Table 285 — DDR5-4800 Speed Bins and Operations

Speed Bin					DDR5-4800AN		DDR5-4800B		DDR5-4800BN		DDR5-4800C		Unit	Note
CL-nRCD-nRP					34-34-34		40-39-39		40-40-40		42-42-42			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.166	22.222	16.000	22.222	16.666	22.222	17.500	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.166	-	16.000	-	16.666	-	17.500	-	ns	7
Row Precharge time				tRP	14.166	-	16.000	-	16.666	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.166	-	48.000	-	48.666	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	14.000	14.000	28	tCK(AVG)	RESERVED								ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	RESERVED				ns	
4400AN	14.545	14.545	32	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	17.500	17.500	42	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	16.666	16.666	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	16.666	16.250	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	14.166	14.166	34	tCK(AVG)	0.416	<0.454	RESERVED						ns	
Supported CL					22,24,26,28,30,32,34, 36,40,42		22,26,28,30,32,36, 40,42		22,28,30,32,36,40,42		22,28,32,36,40,42		nCK	12

10.6 DDR5-5200 Speed Bins and Operations

Table 286 — DDR5-5200 Speed Bins and Operations

Speed Bin					DDR5-5200AN		DDR5-5200B		DDR5-5200BN		DDR5-5200C		Unit	Note
CL-nRCD-nRP					38-38-38		42-42-42		42-42-42		46-46-46			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.615	22.222	16.000	22.222	16.153	22.222	17.500	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.615	-	16.000	-	16.153	-	17.500	-	ns	7
Row Precharge time				tRP	14.615	-	16.000	-	16.153	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.615	-	48.000	-	48.153	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	RESERVED								ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	14.000	14.000	28	tCK(AVG)	RESERVED								ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	14.545	14.545	32	tCK(AVG)	RESERVED								ns	
4800C	17.500	17.500	42	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	16.666	16.666	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	16.666	16.250	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800AN	14.166	14.166	34	tCK(AVG)	RESERVED								ns	
5200C	17.692	17.692	46	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	16.153	16.153	42	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	RESERVED		ns	
5200AN	14.615	14.615	38	tCK(AVG)	0.384	<0.416	RESERVED						ns	
Supported CL					22,24,26,28,30,32,36, 38,40,42,46		22,26,28,30,32,36,40, 42,46		22,26,28,30,32,36,40, 42,46		22,28,32,36,40,42,46		nCK	12

10.7 DDR5-5600 Speed Bins and Operations

Table 287 — DDR5-5600 Speed Bins and Operations

Speed Bin					DDR5-5600AN		DDR5-5600B		DDR5-5600BN		DDR5-5600C		Unit	Note
CL-nRCD-nRP					40-40-40		46-45-45		46-46-46		50-49-49			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.285	22.222	16.000	22.222	16.428	22.222	17.500	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.285	-	16.000	-	16.428	-	17.500	-	ns	7
Row Precharge time				tRP	14.285	-	16.000	-	16.428	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 x* tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.285	-	48.000	-	48.428	-	49.500	-	ns	7 ,8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	14.000	14.000	28	tCK(AVG)	RESERVED								ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	RESERVED				ns	
4400AN	14.545	14.545	32	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	17.500	17.500	42	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	16.666	16.666	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	16.666	16.250	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	14.166	14.166	34	tCK(AVG)	RESERVED								ns	
5200C	17.692	17.692	46	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	16.153	16.153	42	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	14.615	14.615	38	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	17.857	17.500	50	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	16.428	16.428	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	16.428	16.071	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	14.285	14.285	40	tCK(AVG)	0.357	<0.384	RESERVED						ns	
Supported CL					22,24,26,28,30,32,36, 38,40,42,46,50		22,26,28,30,32,36,40, 42,46,50		22,28,30,32,36,40,42, 46,50		22,28,32,36,40,42, 46,50		nCK	12

10.8 DDR5-6000 Speed Bins and Operations

Table 288 — DDR5-6000 Speed Bins and Operations

Speed Bin					DDR5-6000AN		DDR5-6000B		DDR5-6000BN		DDR5-6000C		Unit	Note
CL-nRCD-nRP					42-42-42		48-48-48		48-48-48		54-53-53			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.000	22.222	16.000	22.222	16.000	22.222	17.500	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Row Precharge time				tRP	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7,13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.000	-	48.000	-	48.000	-	49.500	-	ns	7,8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6,9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	RESERVED		ns	
4000AN	14.000	14.000	28	tCK(AVG)	0.500	<0.555	RESERVED						ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	14.545	14.545	32	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	17.500	17.500	42	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	16.666	16.666	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	16.666	16.250	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800AN	14.166	14.166	34	tCK(AVG)	0.416	<0.454	RESERVED						ns	
5200C	17.692	17.692	46	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	16.153	16.153	42	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	RESERVED		ns	
5200AN	14.615	14.615	38	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	17.857	17.500	50	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	16.428	16.428	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	16.428	16.071	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600AN	14.285	14.285	40	tCK(AVG)	0.357	<0.384	RESERVED						ns	
6000C	18.000	17.666	54	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	16.000	16.000	48	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	RESERVED		ns	
6000AN	14.000	14.000	42	tCK(AVG)	0.333	<0.357	RESERVED						ns	
Supported CL					22,24,26,28,30,32,34, 36,38,40,42,46,48,50, 54		22,26,28,30,32,36,40, 42,46,48,50,54		22,26,28,30,32,36,40, 42,46,48,50,54		22,28,32,36,40,42,46, 50,54		nCK	12

10.9 DDR5-6400 Speed Bins and Operations

Table 289 — DDR5-6400 Speed Bins and Operations

Speed Bin					DDR5-6400AN		DDR5-6400B		DDR5-6400BN		DDR5-6400C		Unit	Note
CL-nRCD-nRP					46-46-46		52-52-52		52-52-52		56-56-56			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.375	22.222	16.000	22.222	16.250	22.222	17.500	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.375	-	16.000	-	16.250	-	17.500	-	ns	7
Row Precharge time				tRP	14.375	-	16.000	-	16.250	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 ~~x~~* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 ~~x~~* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 ~~x~~* tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 ~~x~~* tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.375	-	48.000	-	48.250	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	14.000	14.000	28	tCK(AVG)	RESERVED								ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	14.545	14.545	32	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	17.500	17.500	42	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	16.666	16.666	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	16.666	16.250	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800AN	14.166	14.166	34	tCK(AVG)	RESERVED								ns	
5200C	17.692	17.692	46	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	16.153	16.153	42	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	14.615	14.615	38	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	17.857	17.500	50	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	16.428	16.428	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	16.428	16.071	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	14.285	14.285	40	tCK(AVG)	RESERVED								ns	
6000C	18.000	17.666	54	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	16.000	16.000	48	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED		RESERVED		ns	
6000AN	14.000	14.000	42	tCK(AVG)	RESERVED								ns	
6400C	17.500	17.500	56	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	16.250	16.250	52	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	RESERVED		ns	
6400AN	14.375	14.375	46	tCK(AVG)	0.312	<0.333	RESERVED						ns	
Supported CL					22,24,26,28,30,32,36,38,40,42,46,48,50,52,54,56		22,26,28,30,32,36,40,42,46,48,50,52,54,56		22,26,28,30,32,36,40,42,46,50,52,54,56		22,28,32,36,40,42,46,50,54,56		nCK	12

10.10 DDR5-6800 Speed Bins and Operations

Table 290 — DDR5-6800 Speed Bins and Operations

Speed Bin					DDR5-6800AN		DDR5-6800B		DDR5-6800BN		DDR5-6800C		Unit	Note
CL-nRCD-nRP					48-48-48		56-55-55		56-56-56		60-60-60			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.117	-	16.000	-	16.470	-	17.500	-	ns	12
Activate to Read or Write command delay time				tRCD	14.117	-	16.000	-	16.470	-	17.500	-	ns	7
Row Precharge time				tRP	14.117	-	16.000	-	16.470	-	17.500		ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7,13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.117	-	48.000	-	48.000		49.500	-	ns	7,8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6,9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	14.000	14.000	28	tCK(AVG)	RESERVED								ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	RESERVED				ns	
4400AN	14.545	14.545	32	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	17.500	17.500	42	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	16.666	16.666	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	16.666	16.250	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	14.166	14.166	34	tCK(AVG)	0.416	<0.454	RESERVED						ns	
5200C	17.692	17.692	46	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	16.153	16.153	42	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	14.615	14.615	38	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	17.857	17.500	50	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	16.428	16.428	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600B	16.428	16.071	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	14.285	14.285	40	tCK(AVG)	0.357	<0.384	RESERVED						ns	
6000C	18.000	17.666	54	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	16.000	16.000	48	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED				ns	
6000AN	14.000	14.000	42	tCK(AVG)	RESERVED								ns	
6400C	17.500	17.500	56	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	16.250	16.250	52	tCK(AVG)	0.312	<0.333	0.312	<0.333	RESERVED				ns	
6400AN	14.375	14.375	46	tCK(AVG)	0.312	<0.333	RESERVED						ns	
6800C	17.647	17.647	60	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	16.470	16.470	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800B	16.470	16.176	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	RESERVED				ns	
6800AN	14.117	14.117	48	tCK(AVG)	0.294	<0.312	RESERVED						ns	
Supported CL					22,24,26,28,30,32,34,36,38,40,42,46,48,50,52,54,56,60		22,26,28,30,32,36,40,42,46,48,50,52,54,56,60		22,28,30,32,36,40,42,46,50,54,56,60		22,28,32,36,40,42,46,50,54,56,60		nCK	12

10.11 DDR5-7200 Speed Bins and Operations

Table 291 — DDR5-7200 Speed Bins and Operations

Speed Bin					DDR5-7200AN		DDR5-7200B		DDR5-7200BN		DDR5-7200C		Unit	Note
CL-nRCD-nRP					52-52-52		58-58-58		58-58-58		64-63-63			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.444	-	16.000	-	16.111	-	17.500	-	ns	12
Activate to Read or Write command delay time				tRCD	14.444	-	16.000	-	16.111	-	17.500	-	ns	7
Row Precharge time				tRP	14.444	-	16.000	-	16.111	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.444	-	48.000	-	48.111	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	14.000	14.000	28	tCK(AVG)	RESERVED								ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	14.545	14.545	32	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	17.500	17.500	42	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	16.666	16.666	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	16.666	16.250	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800AN	14.166	14.166	34	tCK(AVG)	RESERVED								ns	
5200C	17.692	17.692	46	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	16.153	16.153	42	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	RESERVED		ns	
5200AN	14.615	14.615	38	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	17.857	17.500	50	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	16.428	16.428	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	16.428	16.071	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600AN	14.285	14.285	40	tCK(AVG)	RESERVED								ns	
6000C	18.000	17.666	54	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	16.000	16.000	48	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED				ns	
6000AN	14.000	14.000	42	tCK(AVG)	RESERVED								ns	
6400C	17.500	17.500	56	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	16.250	16.250	52	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	RESERVED		ns	
6400AN	14.375	14.375	46	tCK(AVG)	RESERVED								ns	
6800C	17.647	17.647	60	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	16.470	16.470	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800B	16.470	16.176	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800AN	14.117	14.117	48	tCK(AVG)	RESERVED								ns	
7200C	17.777	17.500	64	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	0.277	<0.294	ns	
7200BN,B	16.111	16.111	58	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	RESERVED		ns	
7200AN	14.444	14.444	52	tCK(AVG)	0.277	<0.294	RESERVED						ns	
Supported CL					22,24,26,28,30,32,36,38,40,42,46,48,50,52,54,56,58,60,64		22,26,28,30,32,36,40,42,46,48,50,52,54,56,58,60,64		22,26,28,30,32,36,40,42,46,50,52,54,56,58,60,64		22,28,32,36,40,42,46,50,54,56,60,64		nCK	12

10.12 DDR5-7600 Speed Bins and Operations

Table 292 — DDR5-7600 Speed Bins and Operations

Speed Bin					DDR5-7600AN		DDR5-7600B		DDR5-7600BN		DDR5-7600C		Unit	Note
CL-nRCD-nRP					54-54-54		62-61-61		62-62-62		68-67-67			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.210	-	16.000	-	16.315	-	17.500	-	ns	12
Activate to Read or Write command delay time				tRCD	14.210	-	16.000	-	16.315	-	17.500	-	ns	7
Row Precharge time				tRP	14.210	-	16.000	-	16.315	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.210	-	48.000	-	48.315	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	14.000	14.000	28	tCK(AVG)	RESERVED								ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	14.545	14.545	32	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	17.500	17.500	42	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	16.666	16.666	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	16.666	16.250	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	14.166	14.166	34	tCK(AVG)	RESERVED								ns	
5200C	17.692	17.692	46	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	16.153	16.153	42	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	14.615	14.615	38	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	17.857	17.500	50	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	16.428	16.428	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	16.428	16.071	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	14.285	14.285	40	tCK(AVG)	0.357	<0.384	RESERVED						ns	
6000C	18.000	17.666	54	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	16.000	16.000	48	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED				ns	
6000AN	14.000	14.000	42	tCK(AVG)	RESERVED								ns	
6400C	17.500	17.500	56	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	16.250	16.250	52	tCK(AVG)	0.312	<0.333	0.312	<0.333	RESERVED				ns	
6400AN	14.375	14.375	46	tCK(AVG)	0.312	<0.333	RESERVED						ns	
6800C	17.647	17.647	60	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	16.470	16.470	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800B	16.470	16.176	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	RESERVED				ns	
6800AN	14.117	14.117	48	tCK(AVG)	RESERVED								ns	
7200C	17.777	17.500	64	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	0.277	<0.294	ns	
7200BN,B	16.111	16.111	58	tCK(AVG)	0.277	<0.294	0.277	<0.294	RESERVED				ns	
7200AN	14.444	14.444	52	tCK(AVG)	0.277	<0.294	RESERVED						ns	
7600C	17.894	17.631	68	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	0.263	<0.277	ns	
7600BN	16.315	16.315	62	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	RESERVED		ns	
7600B	16.315	16.052	62	tCK(AVG)	0.263	<0.277	0.263	<0.277	RESERVED				ns	
7600AN	14.210	14.210	54	tCK(AVG)	0.263	<0.277	RESERVED						ns	
Supported CL					22,24,26,28,30,32, 36,38,40,42,46,48, 50,52,54,56,58,60, 62,64,68		22,26,28,30,32,36, 40,42,46,48,50,52, 54,56,58,60,62,64, 68		22,28,30,32,36,40, 42,46,50,54,56,60, 62,64,68		22,28,32,36,40,42, 46,50,54,56,60,64, 68		nCK	12

10.13 DDR5-8000 Speed Bins and Operations

Table 293 — DDR5-8000 Speed Bins and Operations

Speed Bin					DDR5-8000AN		DDR5-8000B		DDR5-8000BN		DDR5-8000C		Unit	Note
CL-nRCD-nRP					56-56-56		64-64-64		64-64-64		70-70-70			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.000	-	16.000		16.000	-	17.500	-	ns	12
Activate to Read or Write command delay time				tRCD	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Row Precharge time				tRP	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.000	-	48.000	-	48.000	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	RESERVED		ns	
4000AN	14.000	14.000	28	tCK(AVG)	0.500	<0.555	RESERVED						ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	14.545	14.545	32	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	17.500	17.500	42	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	16.666	16.666	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	16.666	16.250	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800AN	14.166	14.166	34	tCK(AVG)	0.416	<0.454	RESERVED						ns	
5200C	17.692	17.692	46	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	16.153	16.153	42	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	RESERVED		ns	
5200AN	14.615	14.615	38	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	17.857	17.500	50	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	16.428	16.428	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	16.428	16.071	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600AN	14.285	14.285	40	tCK(AVG)	0.357	<0.384	RESERVED						ns	
6000C	18.000	17.666	54	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	16.000	16.000	48	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	RESERVED		ns	
6000AN	14.000	14.000	42	tCK(AVG)	0.333	<0.357	RESERVED						ns	
6400C	17.500	17.500	56	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	16.250	16.250	52	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	RESERVED		ns	
6400AN	14.375	14.375	46	tCK(AVG)	0.312	<0.333	RESERVED						ns	
6800C	17.647	17.647	60	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	16.470	16.470	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800B	16.470	16.176	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800AN	14.117	14.117	48	tCK(AVG)	0.294	<0.312	RESERVED						ns	
7200C	17.777	17.500	64	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	0.277	<0.294	ns	
7200BN,B	16.111	16.111	58	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	RESERVED		ns	
7200AN	14.444	14.444	52	tCK(AVG)	0.277	<0.294	RESERVED						ns	
7600C	17.894	17.631	68	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	0.263	<0.277	ns	
7600BN	16.315	16.315	62	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	RESERVED		ns	
7600B	16.315	16.052	62	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	RESERVED		ns	
7600AN	14.210	14.210	54	tCK(AVG)	0.263	<0.277	RESERVED						ns	
8000C	17.500	17.500	70	tCK(AVG)	0.250	<0.263	0.250	<0.263	0.250	<0.263	0.250	<0.263	ns	
8000BN,B	16.000	16.000	64	tCK(AVG)	0.250	<0.263	0.250	<0.263	0.250	<0.263	RESERVED		ns	
8000AN	14.000	14.000	56	tCK(AVG)	0.250	<0.263	RESERVED						ns	
Supported CL					22,24,26,28,30,32, 34,36,38,40,42,46, 48,50,52,54,56,58, 60,62,64,68,70		22,26,28,30,32,36, 40,42,46,48,50,52, 54,56,58,60,62,64, 68,70		22,26,28,30,32,36, 40,42,46,48,50,52, 54,56,58,60,62,64, 68,70		22,28,32,36,40,42, 46,50,54,56,60,64, 68,70		nCK	12

10.14 DDR5-8400 Speed Bins and Operations

Table 294 — DDR5-8400 Speed Bins and Operations

Speed Bin					DDR5-8400AN		DDR5-8400B		DDR5-8400BN		DDR5-8400C		Unit	Note
CL-nRCD-nRP					60-60-60		68-68-68		68-68-68		74-74-74			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.285	-	16.000	-	16.190	-	17.500	-	ns	12
Activate to Read or Write command delay time				tRCD	14.285	-	16.000	-	16.190	-	17.500	-	ns	7
Row Precharge time				tRP	14.285	-	16.000	-	16.190	-	17.500		ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.285	-	48.000	-	48.190	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	14.000	14.000	28	tCK(AVG)	RESERVED								ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	14.545	14.545	32	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	17.500	17.500	42	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	16.666	16.666	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	16.666	16.250	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800AN	14.166	14.166	34	tCK(AVG)	RESERVED								ns	
5200C	17.692	17.692	46	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	16.153	16.153	42	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	14.615	14.615	38	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	17.857	17.500	50	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	16.428	16.428	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	16.428	16.071	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	14.285	14.285	40	tCK(AVG)	0.357	<0.384	RESERVED						ns	
6000C	18.000	17.666	54	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	16.000	16.000	48	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED				ns	
6000AN	14.000	14.000	42	tCK(AVG)	RESERVED								ns	
6400C	17.500	17.500	56	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	16.250	16.250	52	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	RESERVED		ns	
6400AN	14.375	14.375	46	tCK(AVG)	0.312	<0.333	RESERVED						ns	
6800C	17.647	17.647	60	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	16.470	16.470	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800B	16.470	16.176	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	RESERVED				ns	
6800AN	14.117	14.117	48	tCK(AVG)	RESERVED								ns	
7200C	17.777	17.500	64	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	0.277	<0.294	ns	
7200BN,B	16.111	16.111	58	tCK(AVG)	0.277	<0.294	0.277	<0.294	RESERVED				ns	
7200AN	14.444	14.444	52	tCK(AVG)	0.277	<0.294	RESERVED						ns	
7600C	17.894	17.631	68	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	0.263	<0.277	ns	
7600BN	16.315	16.315	62	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	RESERVED		ns	
7600B	16.315	16.052	62	tCK(AVG)	0.263	<0.277	0.263	<0.277	RESERVED				ns	
7600AN	14.210	14.210	54	tCK(AVG)	RESERVED								ns	
8000C	17.500	17.500	70	tCK(AVG)	0.250	<0.263	0.250	<0.263	0.250	<0.263	0.250	<0.263	ns	
8000BN,B	16.000	16.000	64	tCK(AVG)	0.250	<0.263	0.250	<0.263	RESERVED				ns	
8000AN	14.000	14.000	56	tCK(AVG)	RESERVED								ns	
8400C	17.619	17.619	74	tCK(AVG)	0.238	<0.250	0.238	<0.250	0.238	<0.250	0.238	<0.250	ns	
8400BN,B	16.190	16.190	68	tCK(AVG)	0.238	<0.250	0.238	<0.250	0.238	<0.250	RESERVED		ns	
8400AN	14.285	14.285	60	tCK(AVG)	0.238	<0.250	RESERVED						ns	
Supported CL					22,24,26,28,30,32, 36,38,40,42,46,48, 50,52,54,56,58,60, 62,64,68,70,74		22,26,28,30,32,36, 40,42,46,48,50,52, 54,56,58,60,62,64, 68,70,74		22,26,28,30,32,36, 40,42,46,50,52,54, 56,60,62,64,68,70, 74		22,28,32,36,40,42, 46,50,54,56,60,64, 68,70,74		nCK	12

10.15 DDR5-8800 Speed Bins and Operations

Table 295 — DDR5-8800 Speed Bins and Operations

Speed Bin					DDR5-8800AN		DDR5-8800B		DDR5-8800BN		DDR5-8800C		Unit	Note
CL-nRCD-nRP					62-62-62		72-71-71		72-72-72		78-77-77			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	14.090	-	16.000	-	16.363	-	17.500	-	ns	12
Activate to Read or Write command delay time				tRCD	14.090	-	16.000	-	16.363	-	17.500	-	ns	7
Row Precharge time				tRP	14.090	-	16.000	-	16.363	-	17.500		ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.090	-	48.000	-	48.363	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	17.500	17.500	28	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	16.250	16.250	26	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	15.000	15.000	24	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	17.777	17.777	32	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	16.666	16.666	30	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	14.444	14.444	26	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	18.000	17.500	36	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	16.000	16.000	32	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	14.000	14.000	28	tCK(AVG)	RESERVED								ns	
4400C	18.181	17.727	40	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	16.363	16.363	36	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	14.545	14.545	32	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	17.500	17.500	42	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	16.666	16.666	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	16.666	16.250	40	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	14.166	14.166	34	tCK(AVG)	0.416	<0.454	RESERVED						ns	
5200C	17.692	17.692	46	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	16.153	16.153	42	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	14.615	14.615	38	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	17.857	17.500	50	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	16.428	16.428	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	16.428	16.071	46	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	14.285	14.285	40	tCK(AVG)	0.357	<0.384	RESERVED						ns	
6000C	18.000	17.666	54	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	16.000	16.000	48	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED				ns	
6000AN	14.000	14.000	42	tCK(AVG)	RESERVED								ns	
6400C	17.500	17.500	56	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	16.250	16.250	52	tCK(AVG)	0.312	<0.333	0.312	<0.333	RESERVED				ns	
6400AN	14.375	14.375	46	tCK(AVG)	0.312	<0.333	RESERVED						ns	
6800C	17.647	17.647	60	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	16.470	16.470	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800B	16.470	16.176	56	tCK(AVG)	0.294	<0.312	0.294	<0.312	RESERVED				ns	
6800AN	14.117	14.117	48	tCK(AVG)	0.294	<0.312	RESERVED						ns	
7200C	17.777	17.500	64	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	0.277	<0.294	ns	
7200BN,B	16.111	16.111	58	tCK(AVG)	0.277	<0.294	0.277	<0.294	RESERVED				ns	
7200AN	14.444	14.444	52	tCK(AVG)	0.277	<0.294	RESERVED						ns	
7600C	17.894	17.631	68	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	0.263	<0.277	ns	

Table 295 — DDR5-8800 Speed Bins and Operations (cont'd)

7600BN	16.315	16.315	62	tCK(AVG)	0.263	<0.277	0.263	<0.277	RESERVED				ns	
7600B	16.315	16.052	62	tCK(AVG)	0.263	<0.277	0.263	<0.277	RESERVED				ns	
7600AN	14.210	14.210	54	tCK(AVG)	0.263	<0.277	RESERVED						ns	
8000C	17.500	17.500	70	tCK(AVG)	0.250	<0.263	0.250	<0.263	0.250	<0.263	0.250	<0.263	ns	
8000BN,B	16.000	16.000	64	tCK(AVG)	0.250	<0.263	0.250	<0.263	RESERVED				ns	
8000AN	14.000	14.000	56	tCK(AVG)	RESERVED								ns	
8400C	17.619	17.619	74	tCK(AVG)	0.238	<0.250	0.238	<0.250	0.238	<0.250	0.238	<0.250	ns	
8400BN,B	16.190	16.190	68	tCK(AVG)	0.238	<0.250	0.238	<0.250	RESERVED				ns	
8400AN	14.285	14.285	60	tCK(AVG)	0.238	<0.250	RESERVED						ns	
8800C	17.727	17.500	78	tCK(AVG)	0.227	<0.238	0.227	<0.238	0.227	<0.238	0.227	<0.238	ns	
8800BN	16.363	16.363	72	tCK(AVG)	0.227	<0.238	0.227	<0.238	0.227	<0.238	RESERVED		ns	
8800B	16.363	16.136	72	tCK(AVG)	0.227	<0.238	0.227	<0.238	RESERVED				ns	
8800AN	14.090	14.090	62	tCK(AVG)	0.227	<0.238	RESERVED						ns	
Supported CL					22,24,26,28,30,32, 34,36,38,40,42,46, 48,50,52,54,56,58, 60,62,64,68,70,72, 74,78		22,26,28,30,32,36, 40,42,46,48,50,52, 54,56,58,60,62,64, 68,70,72,74,78		22,28,30,32,36,40, 42,46,50,54,56,60, 64,68,70,72,74,78		22,28,32,36,40,42, 46,50,54,56,60,64, 68,70,74,78		nCK	12

DDR5 Speed Bin Table Notes:

1. Minimum timing parameters are defined according to the rules in the Rounding Definitions and Algorithms section.
2. The translation of all timing parameters from ns values to nCK values shall follow the Rounding Algorithm. The translation of tAA to CL shall follow the explicit combinations listed in the Speed Bin Tables.
3. The CL setting and CWL setting result in tCK(avg).MIN and tCK(avg).MAX requirements. When selecting tCK(avg), requirements from the CL setting as well as requirements from the CWL setting shall be fulfilled.
4. 'Reserved' settings are not allowed. The user shall program a different value.
5. This column shows the intended native speed bin timings to be replaced and supported when down clocking. This column does not necessarily show the actual minimum speed bin timings allowed and supported when down clocking because the timings could be faster according to the Rounding Algorithm, depending on the specific speed bin and down clock frequency combination.
6. DDR5-3200 AC timings apply if the DRAM operates slower than the 2933 MT/s data rate. This is not limited to only the Speed Bin Table timings.
7. Parameters apply from tCK(avg)min to tCK(avg)max.
8. tRC(min) shall always be greater than or equal to tRAS(min) + tRP(min), and when using the appropriate rounding algorithms, nRC(min) shall always be greater than or equal to nRAS(min) + nRP(min).
9. tCK(avg).max of 1.010 ns (1980 MT/s data rate) is defined to allow for 1% SSC down-spreading at a data rate of 2000 MT/s according to JESD404-1.
10. Each speed bin lists the timing requirements that need to be supported in order for a given DRAM to be JEDEC standard. The JEDEC standard does not require support for all speed bins within a given speed. The JEDEC standard requires meeting the parameters for a least one of the listed speed bins.
11. Any speed bin also supports functional operation at slower frequencies as shown in the table which are not subject to Production Tests but are verified by Design/Characterization.
12. The CL Algorithm can be used to mathematically determine the valid CAS Latencies listed in the Speed Bin Tables. The CL Algorithm calculates supported CAS Latencies by rounding the operating frequency up to the next faster native speed bin (i.e., 3200 MT/s, 3600 MT/s...). Using the resulting tCK(AVG)min, and the bin target timings, the CL Algorithm then uses the Rounding Algorithm to calculate the valid CAS Latency. Because the DDR5 SDRAM specification only supports even CAS Latencies, odd CAS Latencies are rounded up to the next even CAS Latency. The 1980-2100 MT/s data rate always uses CL22. If tAA (corrected) or tRCDtRP(corrected) are violated, the CL Algorithm uses a slower combination of tAA(target) and tRCDtRP(target) to return slower valid CAS Latencies. The DDR5 SDRAM can support up to four valid CAS Latencies, CL(AN), CL(B), CL(BN), and CL(C), for a given frequency. tAA(corrected) and tRCDtRP(corrected) are calculated by reducing tAA(min), tRCD(min), and tRP(min) by the Rounding Algorithm correction factor. The proper setting of CL shall be determined by the memory controller, either by using the Speed Bin Tables, or by using the CL Algorithm, or by some other means. Refer to the Rounding Definitions and Algorithm section for more information. When Read CRC is enabled, CL is increased according to the Read CRC Latency Adder. When Write CRC is enabled, there is no Write CRC Latency Adder.

```
// Variables already defined in other areas of the DDR5 SDRAM specification
CorrFact       = 0.30                                        // (%)   Rounding Algorithm correction factor
ScaledCorrFact = 997                                         //       Scaled correction factor (1000*(1-0.30%))
tCKreal        =1011-952, 682-238                            // (ps)  Real application tCK(AVG) (1980-2100MT/s, 2933-8800MT/s)
tAAmin         MONO=14000-17500, 3DS=16000-20000             // (ps)  From Speed Bin Tables and DIMM SPD bytes 30-31
tRCDtRPmin     MONO=14000-17500, 3DS=14000-17500             // (ps)  From Speed Bin Tables and DIMM SPD bytes 32-33 (tRCD=tRP)
tAAcorr        = TRUNC(tAAmin*ScaledCorrFactor/1000)         // (ps)  Corrected tAA(min) per the Rounding Algorithm rules
tRCDtRPcorr    = TRUNC(tRCDtRPmin*ScaledCorrFactor/1000)     // (ps)  Corrected tRCD(min), tRP(min) per the Rounding Algorithm
FUNC[RA(targ)] = TRUNC((targ*ScaledCorrFact/tCKstd+1000)/1000) // (nCK) Use Rounding Algorithm to convert bin target timing to nCK
```

DDR5 Speed Bin Table Notes: (cont'd)

```
// Round tCKreal down to the next faster standard frequency (tCK in ps)
IF (TRUNC(2000000/(2000*99%))>=TRUNC(tCKreal)>=TRUNC(2000000/(2000*105%)))            // Check for  1980-2100 nominal data rates
   tCKstd=TRUNC(2000000/2000)                                                          // Assign standard 2000 tCK (ps)
ELSE IF (TRUNC(2000000/(2000+7*(133+1/3)))>=TRUNC(tCKreal)>=TRUNC(2000000/3200))       // Check for  2933-3200 nominal data rates
   tCKstd=TRUNC(2000000/3200)                                                          // Assign standard 3200 tCK (ps)
ELSE
   FOR (DataRateNom=3200; DataRateNom<=8400; DataRateNom=DataRateNom+400)              // Check for >3200-8800 nominal data rates
      IF (TRUNC(2000000/DataRateNom)>TRUNC(tCKreal)>=TRUNC(2000000/(DataRateNom+400)))
         tCKstd=TRUNC(2000000/(DataRateNom+400))                                       // Assign standard 3600-8800 tCK (ps)
      ELSE
         tCKstd=RESERVED                                                               // No valid data rate found

// Timing targets (ps) that have been used to define the Speed Bin Tables
// MONO targets             3DS targets
   BinAN_tAAtarg     = 14000 BinAN_tAAtarg     = 16000          // tAA target for AN bins
   BinB__tAAtarg     = 16000 BinB__tAAtarg     = 18500          // tAA target for AN, B bins
   BinBN_tAAtarg     = 16000 BinBN_tAAtarg     = 18500          // tAA target for AN, B, BN bins
   BinC__tAAtarg     = 17500 BinC__tAAtarg     = 20000          // tAA target for AN, B, BN, C bins
   BinAN_tRCDtRPtarg = 14000 BinAN_tRCDtRPtarg = 14000          // tRCD, tRP target for AN bins
   BinBN_tRCDtRPtarg = 16000 BinBN_tRCDtRPtarg = 16000          // tRCD, tRP target for AN, B, BN bins
   BinC__tRCDtRPtarg = 17500 BinC__tRCDtRPtarg = 17500          // tRCD, tRP target for AN, B, BN, C bins
IF (TRUNC(2000000/3600)>tCKstd)                                 // tRCD, tRP target for B bins is frequency dependent
   BinB__tRCDtRPtarg = 16000 BinB__tRCDtRPtarg = 16000          // tRCD, tRP target for AN, B bins data rates faster than 3600
ELSE
   // 16250=(2000000/3200)*EVEN(TRUNC((BinB__tRCDtRPtarg*ScaledCorrFact/(2000000/3200)+1000)/1000))
   BinB__tRCDtRPtarg = 16250 BinB__tRCDtRPtarg = 16250          // tRCD, tRP target for AN, B bins for data rates 3600 and slower

// CL Algorithm using variables defined above
// Up to four valid CL's can be returned for a specific freq: CL(AN), CL(B), CL(BN), CL(C), depending on tAAmin, tRCDmin, tRPmin
// The B and BN bins return the same CL
// Only even CL's (not odd CL's) are valid per the DDR5 SDRAM specification
// nRCD, nRP are only even at standard native frequencies for the AN, BN bins (can be even or odd at intermediate frequencies)
// nRCD, nRP may be even or odd at standard native frequencies for the B, C bins (can be even or odd at intermediate frequencies)
IF (TRUNC(2000000/2000)=tCKstd)                                 // CL22 is the only valid CL for 1980-2100 data rates
   CL(AN)=22                                                    // Valid even CL for AN bins
   CL(B )=22                                                    // Valid even CL for AN, B, bins
   CL(BN)=22                                                    // Valid even CL for AN, B, BN bins
   CL(C )=22                                                    // Valid even CL for AN, B, BN, C bins
ELSE IF (TRUNC(2000000/3200)>=tCKstd>=TRUNC(2000000/8800))      // Valid CL for 2933-8800 data rates
   IF      ((EVEN(RA(BinAN_tAAtarg))*tCKstd>=tAAcorr)AND(EVEN(RA(BinAN_tRCDtRPtarg))*tCKstd>=tRCDtRPcorr)) // nRCD, nRP even only
      CL(AN)=EVEN(RA(BinAN_tAAtarg))                            // Valid even CL for AN bins
      CL(B )=EVEN(RA(BinB__tAAtarg))                            // Valid even CL for AN, B bins
      CL(BN)=EVEN(RA(BinBN_tAAtarg))                            // Valid even CL for AN, B, BN bins
      CL(C )=EVEN(RA(BinC__tAAtarg))                            // Valid even CL for AN, B, BN, C bins
   ELSE IF ((EVEN(RA(BinB__tAAtarg))*tCKstd>=tAAcorr)AND(    (RA(BinB__tRCDtRPtarg))*tCKstd>=tRCDtRPcorr)) // nRCD, nRP even, odd
      CL(AN)=RESERVED                                           // Valid even CL for AN bins
      CL(B )=EVEN(RA(BinB__tAAtarg))                            // Valid even CL for AN, B bins
      CL(BN)=EVEN(RA(BinBN_tAAtarg))                            // Valid even CL for AN, B, BN bins
      CL(C )=EVEN(RA(BinC__tAAtarg))                            // Valid even CL for AN, B, BN, C bins
   ELSE IF ((EVEN(RA(BinBN_tAAtarg))*tCKstd>=tAAcorr)AND(EVEN(RA(BinBN_tRCDtRPtarg))*tCKstd>=tRCDtRPcorr)) // nRCD, nRP even only
      CL(AN)=RESERVED                                           // Valid even CL for AN bins
      CL(B )=RESERVED                                           // Valid even CL for AN, B bins
      CL(BN)=EVEN(RA(BinBN_tAAtarg))                            // Valid even CL for AN, B, BN bins
      CL(C )=EVEN(RA(BinC__tAAtarg))                            // Valid even CL for AN, B, BN, C bins
   ELSE IF ((EVEN(RA(BinC__tAAtarg))*tCKstd>=tAAcorr)AND(    (RA(BinC__tRCDtRPtarg))*tCKstd>=tRCDtRPcorr)) // nRCD, nRP even, odd
      CL(AN)=RESERVED                                           // Valid even CL for AN bins
      CL(B )=RESERVED                                           // Valid even CL for AN, B bins
      CL(BN)=RESERVED                                           // Valid even CL for AN, B, BN bins
      CL(C )=EVEN(RA(BinC__tAAtarg))                            // Valid even CL for AN, B, BN, C bins
   ELSE                                                         // No valid CL found (tAAmin, tRCDmin, tRPmin are too slow)
      CL(AN)=RESERVED                                           // Valid even CL for AN bins
      CL(B )=RESERVED                                           // Valid even CL for AN, B bins
      CL(BN)=RESERVED                                           // Valid even CL for AN, B, BN bins
      CL(C )=RESERVED                                           // Valid even CL for AN, B, BN, C bins
ELSE                                                            // No valid data rate found
   CL(AN)=RESERVED                                              // Valid even CL for AN bins
   CL(B )=RESERVED                                              // Valid even CL for AN, B bins
   CL(BN)=RESERVED                                              // Valid even CL for AN, B, BN bins
   CL(C )=RESERVED                                              // Valid even CL for AN, B, BN, C bins
```

13. tRAS(max) shall always be less than or equal to 5*tREFI1(max) during Normal Refresh Mode and less than or equal to 9*tREFI2(max) during Fine Granularity Refresh Mode, and when using the rounding algorithms, nRAS(max) shall always be less than or equal to 5*nREFI1(max) during Normal Refresh Mode and less than or equal to 9*nREFI2(max) during Fine Granularity Refresh Mode.

10.16 3DS DDR5-3200 Speed Bins and Operations

Table 296 — 3DS DDR5-3200 Speed Bins and Operations

Speed Bin					DDR5-3200AN 3DS		DDR5-3200B 3DS		DDR5-3200BN 3DS		DDR5-3200C 3DS		Unit	NOTE
CL-nRCD-nRP					26-24-24		30-26-26		30-26-26		32-28-28			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.250	22.222	18.750	22.222	18.750	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	15.000	-	16.250	-	16.250	-	17.500	-	ns	7
Row Precharge time				tRP	15.000	-	16.250	-	16.250	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	47.000	-	48.250	-	48.250	-	49.500	-	ns	7,8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6,9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	16.250	15.000	26	tCK(AVG)	0.625	0.681	RESERVED						ns	
Supported CL					22,26,30,32		22,30,32		22,30,32		22,32		nCK	12

10.17 3DS DDR5-3600 Speed Bins and Operations

Table 297 — 3DS DDR5-3600 Speed Bins and Operations

Speed Bin					DDR5-3600AN 3DS		DDR5-3600B 3DS		DDR5-3600BN 3DS		DDR5-3600C 3DS		Unit	NOTE
CL-nRCD-nRP					30-26-26		34-30-30		34-30-30		36-32-32			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.666	22.222	18.750	22.222	18.888	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.444	-	16.250	-	16.666	-	17.500	-	ns	7
Row Precharge time				tRP	14.444	-	16.250	-	16.666	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	ns	7
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.444	-	48.250	-	48.666	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	16.250	15.000	26	tCK(AVG)	RESERVED								ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
Supported CL					22,30,32,34,36		22,30,32,34,36		22,32,34,36		22,32,36		nCK	

10.18 3DS DDR5-4000 Speed Bins and Operations

Table 298 — 3DS DDR5-4000 Speed Bins and Operations

Speed Bin					DDR5-4000AN 3DS		DDR5-4000B 3DS		DDR5-4000BN 3DS		DDR5-4000C 3DS		Unit	NOTE
CL-nRCD-nRP					32-28-28		38-32-32		38-32-32		40-35-35			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.000	22.222	18.750	22.222	19.000	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Row Precharge time				tRP	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	ns	7
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.000	-	48.000	-	48.000	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	16.250	15.000	26	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	RESERVED				ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	RESERVED		ns	
4000AN	16.000	14.000	32	tCK(AVG)	0.500	<0.555	RESERVED						ns	
Supported CL					22,26,30,32,34,36,38, 40		22,30,32,34,36,38, 40		22,32,36,38,40		22,32,36,40		nCK	

10.19 3DS DDR5-4400 Speed Bins and Operations

Table 299 — 3DS DDR5-4400 Speed Bins and Operations

Speed Bin					DDR5-4400AN 3DS		DDR5-4400B 3DS		DDR5-4400BN 3DS		DDR5-4400C 3DS		Unit	NOTE
CL-nRCD-nRP					36-32-32		42-36-36		42-36-36		44-39-39			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.363	22.222	18.750	22.222	19.090	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.545	-	16.000	-	16.363	-	17.500	-	ns	7
Row Precharge time				tRP	14.545	-	16.000	-	16.363	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	ns	7
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.545	-	48.000	-	48.363	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	16.250	15.000	26	tCK(AVG)	RESERVED								ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	RESERVED				ns	
3600AN	16.666	14.444	30	tCK(AVG)	RESERVED								ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	16.000	14.000	32	tCK(AVG)	RESERVED								ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	16.363	14.545	36	tCK(AVG)	0.454	<0.500	RESERVED						ns	
Supported CL					22,30,32,34,36,38,40,42,44		22,30,32,34,36,38,40,42,44		22,32,36,40,42,44		22,32,36,40,44		nCK	

10.20 3DS DDR5-4800 Speed Bins and Operations

Table 300 — 3DS DDR5-4800 Speed Bins and Operations

Speed Bin					DDR5-4800AN 3DS		DDR5-4800B 3DS		DDR5-4800BN 3DS		DDR5-4800C 3DS		Unit	NOTE
CL-nRCD-nRP					40-34-34		46-39-39		46-40-40		48-42-42			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.666	22.222	18.750	22.222	19.166	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.166	-	16.000	-	16.666	-	17.500	-	ns	7
Row Precharge time				tRP	14.166	-	16.000	-	16.666	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	ns	7
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.166	-	48.000	-	48.666	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	16.250	15.000	26	tCK(AVG)	RESERVED								ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	RESERVED				ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	16.000	14.000	32	tCK(AVG)	RESERVED								ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	RESERVED				ns	
4400AN	16.363	14.545	36	tCK(AVG)	RESERVED								ns	
4800C	20.000	17.500	48	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	19.166	16.666	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	19.166	16.250	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	16.666	14.166	40	tCK(AVG)	0.416	<0.454	RESERVED						ns	
Supported CL					22,30,32,34,36,38,40, 42,44,46,48		22,30,32,34,36,38,40, 42,44,46,48		22,32,36,40,44,46,48		22,32,36,40,44,48		nCK	

10.21 3DS DDR5-5200 Speed Bins and Operations

Table 301 — 3DS DDR5-5200 Speed Bins and Operations

Speed Bin					DDR5-5200AN 3DS		DDR5-5200B 3DS		DDR5-5200BN 3DS		DDR5-5200C 3DS		Unit	NOTE
CL-nRCD-nRP					42-38-38		50-42-42		50-42-42		52-46-46			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.153	22.222	18.750	22.222	19.230	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.615	-	16.000	-	16.153	-	17.500	-	ns	7
Row Precharge time				tRP	14.615	-	16.000	-	16.153	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	ns	7
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.615	-	48.000	-	48.153	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	16.250	15.000	26	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	RESERVED				ns	
3600AN	16.666	14.444	30	tCK(AVG)	RESERVED								ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	16.000	14.000	32	tCK(AVG)	RESERVED								ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	RESERVED				ns	
4400AN	16.363	14.545	36	tCK(AVG)	RESERVED								ns	
4800C	20.000	17.500	48	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	19.166	16.666	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800B	19.166	16.250	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	16.666	14.166	40	tCK(AVG)	RESERVED								ns	
5200C	20.000	17.692	52	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	19.230	16.153	50	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	RESERVED		ns	
5200AN	16.153	14.615	42	tCK(AVG)	0.384	<0.416	RESERVED						ns	
Supported CL					22,26,30,32,34,36,38, 40,42,44,46,48,50,52		22,30,32,34,36,38, 40,42,44,46,48,50,52		22,32,36,40,44,48,50, 52		22,32,36,40,44,48,52		nCK	

10.22 3DS DDR5-5600 Speed Bins and Operations

Table 302 — 3DS DDR5-5600 Speed Bins and Operations

Speed Bin					DDR5-5600AN 3DS		DDR5-5600B 3DS		DDR5-5600BN 3DS		DDR5-5600C 3DS		Unit	NOTE
CL-nRCD-nRP					46-40-40		52-45-45		52-46-46		56-49-49			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.428	22.222	18.571	22.222	18.571	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.285	-	16.000	-	16.428	-	17.500	-	ns	7
Row Precharge time				tRP	14.285	-	16.000	-	16.4228	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	ns	7
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.285	-	48.000	-	48.428	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	16.250	15.000	26	tCK(AVG)	RESERVED								ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	16.000	14.000	32	tCK(AVG)	RESERVED								ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	RESERVED				ns	
4400AN	16.363	14.545	36	tCK(AVG)	RESERVED								ns	
4800C	20.000	17.500	48	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	19.166	16.666	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	19.166	16.250	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	16.666	14.166	40	tCK(AVG)	RESERVED								ns	
5200C	20.000	17.692	52	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	19.230	16.153	50	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	16.153	14.615	42	tCK(AVG)	RESERVED								ns	
5600C	20.000	17.500	56	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	18.571	16.428	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	18.571	16.071	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	16.428	14.285	46	tCK(AVG)	0.357	<0.384	RESERVED						ns	
Supported CL					22,30,32,34,36,38,40, 42,44,46,48,50,52,56		22,30,32,34,36,38,40, 42,44,46,48,50,52,56		22,32,34,36,40,44,46, 48,52,56		22,32,36,40,44,48,52, 56		nCK	

10.23 3DS DDR5-6000 Speed Bins and Operations

Table 303 — 3DS DDR5-6000 Speed Bins and Operations

Speed Bin					DDR5-6000AN 3DS		DDR5-6000B 3DS		DDR5-6000BN 3DS		DDR5-6000C 3DS		Unit	NOTE
CL-nRCD-nRP					48-42-42		56-48-48		56-48-48		60-53-53			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.000	22.222	18.571	22.222	18.666	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Row Precharge time				tRP	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	ns	7
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.000	-	48.000	-	48.000	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	<0.681	RESERVED		ns	
3200AN	16.250	15.000	26	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	RESERVED		ns	
4000AN	16.000	14.000	32	tCK(AVG)	0.500	<0.555	RESERVED						ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	16.363	14.545	36	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	20.000	17.500	48	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	19.166	16.666	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	19.166	16.250	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800AN	16.666	14.166	40	tCK(AVG)	0.416	<0.454	RESERVED						ns	
5200C	20.000	17.692	52	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	19.230	16.153	50	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	RESERVED		ns	
5200AN	16.153	14.615	42	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	20.000	17.500	56	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	18.571	16.428	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600B	18.571	16.071	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	16.428	14.285	46	tCK(AVG)	0.357	<0.384	RESERVED						ns	
6000C	20.000	17.666	60	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	18.666	16.000	56	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	RESERVED		ns	
6000AN	16.600	14.000	48	tCK(AVG)	0.333	<0.357	RESERVED						ns	
Supported CL					22,26,30,32,34,36,38, 40,42,44,46,48,50,52, 56,60		22,30,32,34,36,38,40, 42,44,46,48),50,52,56, 60		22,30,32,34,36,38,40, 42,44,46,48,50,52,56, 60		22,32,36,40,44,48,52, 56,60		nCK	

10.24 3DS DDR5-6400 Speed Bins and Operations

Table 304 — 3DS DDR5-6400 Speed Bins and Operations

Speed Bin					DDR5-6400AN 3DS		DDR5-6400B 3DS		DDR5-6400BN 3DS		DDR5-6400C 3DS		Unit	NOTE
CL-nRCD-nRP					52-46-46		60-52-52		60-52-52		64-56-56			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.250	22.222	18.571	22.222	18.750	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.375	-	16.000	-	16.250	-	17.500	-	ns	7
Row Precharge time				tRP	14.375	-	16.000	-	16.250	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	32.000	5 x tREFI1	ns	7
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.375	-	48.000	-	48.250	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Down Bins										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	16.250	15.000	26	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	16.000	14.000	32	tCK(AVG)	RESERVED								ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	16.363	14.545	36	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	20.000	17.500	48	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	19.166	16.666	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	19.166	16.250	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800AN	16.666	14.166	40	tCK(AVG)	RESERVED								ns	
5200C	20.000	17.692	52	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	19.230	16.153	50	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	16.153	14.615	42	tCK(AVG)	RESERVED								ns	
5600C	20.000	17.500	56	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	18.571	16.428	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600B	18.571	16.071	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	16.428	14.285	46	tCK(AVG)	RESERVED								ns	
6000C	20.000	17.666	60	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	18.666	16.000	56	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED				ns	
6000AN	16.600	14.000	48	tCK(AVG)	RESERVED								ns	
6400C	20.000	17.500	64	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	18.750	16.250	60	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	RESERVED		ns	
6400AN	16.250	14.375	52	tCK(AVG)	0.312	<0.333	RESERVED						ns	
Supported CL					22,26,30,32,34,36,38, 40,42,44,46,48,50,52, 56,60,64		22,30,32,34,36,38,40, 42,44,46,48,50,52,56, 60,64		22,30,32,34,36,40,42, 44,46,48,52,56,60,64		22,32,36,40,44,48,52, 56,60,64		nCK	

10.25 3DS DDR5-6800 Speed Bins and Operations

Table 305 — 3DS DDR5-6800 Speed Bins and Operations

Speed Bin					DDR5-6800AN 3DS		DDR5-6800B 3DS		DDR5-6800BN 3DS		DDR5-6800C 3DS		Unit	Note
CL-nRCD-nRP					56-48-48		64-55-55		64-56-56		68-60-60			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.470	22.222	18.571	22.222	18.823	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.117	-	16.000	-	16.470	-	17.500	-	ns	7
Row Precharge time				tRP	14.117	-	16.000	-	16.470	-	17.500		ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.117	-	48.000	-	48.000		49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	16.250	15.000	26	tCK(AVG)	RESERVED								ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	16.000	14.000	32	tCK(AVG)	RESERVED								ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	RESERVED				ns	
4400AN	16.363	14.545	36	tCK(AVG)	RESERVED								ns	
4800C	20.000	17.500	48	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	19.166	16.666	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	19.166	16.250	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	16.666	14.166	40	tCK(AVG)	0.416	<0.454	RESERVED						ns	
5200C	20.000	17.692	52	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	19.230	16.153	50	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	16.153	14.615	42	tCK(AVG)	RESERVED								ns	
5600C	20.000	17.500	56	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	18.571	16.428	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600B	18.571	16.071	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	16.428	14.285	46	tCK(AVG)	RESERVED								ns	
6000C	20.000	17.666	60	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	18.666	16.000	56	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED				ns	
6000AN	16.000	14.000	48	tCK(AVG)	RESERVED								ns	
6400C	20.000	17.500	64	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	18.750	16.250	60	tCK(AVG)	0.312	<0.333	0.312	<0.333	RESERVED				ns	
6400AN	16.250	14.375	52	tCK(AVG)	RESERVED								ns	
6800C	20.000	17.647	68	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	18.823	16.470	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800B	18.823	16.176	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	RESERVED				ns	
6800AN	16.470	14.117	56	tCK(AVG)	0.294	<0.312	RESERVED						ns	
Supported CL					22,30,32,34,36,38, 40,42,44,46,48,50, 52,56,60,64,68		22,30,32,34,36,38, 40,42,44,46,48,50, 52,56,60,64,68		22,32,34,36,40,44, 46,48,56,60,64, 68		22,32,36,40,44,48, 52,56,60,64,68		nCK	12

10.26 3DS DDR5-7200 Speed Bins and Operations

Table 306 — 3DS DDR5-7200 Speed Bins and Operations

Speed Bin					DDR5-7200AN 3DS		DDR5-7200B 3DS		DDR5-7200BN 3DS		DDR5-7200C 3DS		Unit	Note
CL-nRCD-nRP					58-52-52		68-58-58		68-58-58		72-63-63			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	~~M~~max		
Read command to first data				tAA	16.111	22.222	18.571	22.222	18.888	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.444	-	16.000	-	16.111	-	17.500	-	ns	7
Row Precharge ~~T~~time				tRP	14.444	-	16.000	-	16.111	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7,13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.444	-	48.000	-	48.111	-	49.500	-	ns	7,8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6,9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	16.250	15.000	26	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	16.000	14.000	32	tCK(AVG)	RESERVED								ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	16.363	14.545	36	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	20.000	17.500	48	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	19.166	16.666	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	19.166	16.250	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800AN	16.666	14.166	40	tCK(AVG)	RESERVED								ns	
5200C	20.000	17.692	52	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	19.230	16.153	50	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	RESERVED		ns	
5200AN	16.153	14.615	42	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	20.000	17.500	56	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	18.571	16.428	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600B	18.571	16.071	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	16.428	14.285	46	tCK(AVG)	RESERVED								ns	
6000C	20.000	17.666	60	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	18.666	16.000	56	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED				ns	
6000AN	16.000	14.000	48	tCK(AVG)	RESERVED								ns	
6400C	20.000	17.500	64	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	18.750	16.250	60	tCK(AVG)	0.312	<0.333	0.312	<0.333	RESERVED				ns	
6400AN	16.250	14.375	52	tCK(AVG)	RESERVED								ns	
6800C	20.000	17.647	68	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	18.823	16.470	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	RESERVED				ns	
6800B	18.823	16.176	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	RESERVED				ns	
6800AN	16.470	14.117	56	tCK(AVG)	RESERVED								ns	
7200C	20.000	17.500	72	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	0.277	<0.294	ns	
7200BN,B	18.888	16.111	68	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	RESERVED		ns	
7200AN	16.111	14.444	58	tCK(AVG)	0.277	<0.294	RESERVED						ns	
Supported CL					22,26,30,32,34,36, 38,40,42,44,46,48, 50,52,56,58,60,64, 68,72		22,30,32,34,36,38, 40,42,44,46,48,50, 52,56,60,64,68,72		22,32,34,36,40,42, 44,46,48,50,52,56, 60,64,68,72		22,32,36,40,44,48, 52,56,60,64,68,72		nCK	12

10.27 3DS DDR5-7600 Speed Bins and Operations

Table 307 — 3DS DDR5-7600 Speed Bins and Operations

Speed Bin					DDR5-7600AN 3DS		DDR5-7600B 3DS		DDR5-7600BN 3DS		DDR5-7600C 3DS		Unit	Note
CL-nRCD-nRP					62-54-54		72-61-61		72-62-62		76-67-67			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.315	22.222	18.571	22.222	18.947	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.210	-	16.000	-	16.315	-	17.500	-	ns	7
Row Precharge time				tRP	14.210	-	16.000	-	16.315	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.210	-	48.000	-	48.315	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	16.250	15.000	26	tCK(AVG)	RESERVED								ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	16.000	14.000	32	tCK(AVG)	RESERVED								ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	16.363	14.545	36	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	20.000	17.500	48	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	19.166	16.666	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	19.166	16.250	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	16.666	14.166	40	tCK(AVG)	RESERVED								ns	
5200C	20.000	17.692	52	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	19.230	16.153	50	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	16.153	14.615	42	tCK(AVG)	RESERVED								ns	
5600C	20.000	17.500	56	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	18.571	16.428	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	18.571	16.071	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	16.428	14.285	46	tCK(AVG)	0.357	<0.384	RESERVED						ns	
6000C	20.000	17.666	60	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	18.666	16.000	56	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED				ns	
6000AN	16.000	14.000	48	tCK(AVG)	RESERVED								ns	
6400C	20.000	17.500	64	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	18.750	16.250	60	tCK(AVG)	0.312	<0.333	0.312	<0.333	RESERVED				ns	
6400AN	16.250	14.375	52	tCK(AVG)	RESERVED								ns	
6800C	20.000	17.647	68	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	18.823	16.470	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800B	18.823	16.176	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	RESERVED				ns	
6800AN	16.470	14.117	56	tCK(AVG)	RESERVED								ns	
7200C	20.000	17.500	72	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	0.277	<0.294	ns	
7200BN,B	18.888	16.111	68	tCK(AVG)	0.277	<0.294	0.277	<0.294	RESERVED				ns	
7200AN	16.111	14.444	58	tCK(AVG)	RESERVED								ns	
7600C	20.000	17.631	76	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	0.263	<0.277	ns	
7600BN	18.347	16.315	72	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	RESERVED		ns	
7600B	18.347	16.052	72	tCK(AVG)	0.263	<0.277	0.263	<0.277	RESERVED				ns	
7600AN	16.315	14.210	62	tCK(AVG)	0.263	<0.277	RESERVED						ns	
Supported CL					22,30,32,34,36,38, 40,42,44,46,48,50, 52,56,60,62,64,68, 72,76		22,30,32,34,36,38, 40,42,44,46,48,50, 52,56,60,64,68,72, 76		22,32,34,36,40,42, 44,46,48,52,56,60, 64,68,72,76		22,32,36,40,44,48, 52,56,60,64,68,72, 76		nCK	12

10.28 3DS DDR5-8000 Speed Bins and Operations

Table 308 — 3DS DDR5-8000 Speed Bins and Operations

Speed Bin					DDR5-8000AN 3DS		DDR5-8000B 3DS		DDR5-8000BN 3DS		DDR5-8000C 3DS		Unit	Note
CL-nRCD-nRP					64-56-56		74-64-64		74-64-64		80-70-70			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.000	22.222	18.500	22.222	18.500	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Row Precharge time				tRP	14.000	-	16.000	-	16.000	-	17.500	-	ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7,13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.000	-	48.000	-	48.000	-	49.500	-	ns	7,8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6,9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	16.250	15.000	26	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	RESERVED		ns	
4000AN	16.000	14.000	32	tCK(AVG)	0.500	<0.555	RESERVED						ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	16.363	14.545	36	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	20.000	17.500	48	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	19.166	16.666	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	19.166	16.250	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800AN	16.666	14.166	40	tCK(AVG)	0.416	<0.454	RESERVED						ns	
5200C	20.000	17.692	52	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	19.230	16.153	50	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	RESERVED		ns	
5200AN	16.153	14.615	42	tCK(AVG)	0.384	<0.416	RESERVED						ns	
5600C	20.000	17.500	56	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	18.571	16.428	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	18.571	16.071	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600AN	16.428	14.285	46	tCK(AVG)	0.357	<0.384	RESERVED						ns	
6000C	20.000	17.666	60	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	18.666	16.000	56	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	RESERVED		ns	
6000AN	16.000	14.000	48	tCK(AVG)	0.333	<0.357	RESERVED						ns	
6400C	20.000	17.500	64	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	18.750	16.250	60	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	RESERVED		ns	
6400AN	16.250	14.375	52	tCK(AVG)	0.312	<0.333	RESERVED						ns	
6800C	20.000	17.647	68	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	18.823	16.470	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800B	18.823	16.176	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800AN	16.470	14.117	56	tCK(AVG)	0.294	<0.312	RESERVED						ns	
7200C	20.000	17.500	72	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	0.277	<0.294	ns	
7200BN,B	18.888	16.111	68	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	RESERVED		ns	
7200AN	16.111	14.444	58	tCK(AVG)	0.277	<0.294	RESERVED						ns	
7600C	20.000	17.631	76	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	0.263	<0.277	ns	
7600BN	18.347	16.315	72	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	RESERVED		ns	
7600B	18.347	16.052	72	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	RESERVED		ns	
7600AN	16.315	14.210	62	tCK(AVG)	0.263	<0.277	RESERVED						ns	
8000C	20.000	17.500	80	tCK(AVG)	0.250	<0.263	0.250	<0.263	0.250	<0.263	0.250	<0.263	ns	
8000BN,B	18.500	16.000	74	tCK(AVG)	0.250	<0.263	0.250	<0.263	0.250	<0.263	RESERVED		ns	
8000AN	16.000	14.000	64	tCK(AVG)	0.250	<0.263	RESERVED						ns	
Supported CL					22,26,30,32,34,36, 38,40,42,44,46,48, 50,52,56,58,60,62, 64,68,72,74,76,80		22,30,32,34,36,38, 40,42,44,46,48,50, 52,56,60,64,68,72, 74,76,80		22,30,32,34,36,38, 40,42,44,46,48,50, 52,56,60,64,68,72, 74,76,80		22,32,36,40,44,48, 52,56,60,64,68,72, 76,80		nCK	12

10.29 3DS DDR5-8400 Speed Bins and Operations

Table 309 — 3DS DDR5-8400 Speed Bins and Operations

Speed Bin					DDR5-8400AN 3DS		DDR5-8400B 3DS		DDR5-8400BN 3DS		DDR5-8400C 3DS		Unit	Note
CL-nRCD-nRP					68-60-60		78-68-68		78-68-68		84-74-74			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.190	22.222	18.500	22.222	18.571	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.285	-	16.000	-	16.190	-	17.500	-	ns	7
Row Precharge time				tRP	14.285	-	16.000	-	16.190	-	17.500		ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.285	-	48.000	-	48.190	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	RESERVED		ns	
3200AN	16.250	15.000	26	tCK(AVG)	0.625	0.681	RESERVED						ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	16.000	14.000	32	tCK(AVG)	RESERVED								ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	16.363	14.545	36	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	20.000	17.500	48	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	19.166	16.666	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	19.166	16.250	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	16.666	14.166	40	tCK(AVG)	RESERVED								ns	
5200C	20.000	17.692	52	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	19.230	16.153	50	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	16.153	14.615	42	tCK(AVG)	RESERVED								ns	
5600C	20.000	17.500	56	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	18.571	16.428	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	RESERVED		ns	
5600B	18.571	16.071	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	16.428	14.285	46	tCK(AVG)	0.357	<0.384	RESERVED						ns	
6000C	20.000	17.666	60	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	18.666	16.000	56	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED				ns	
6000AN	16.000	14.000	48	tCK(AVG)	RESERVED								ns	
6400C	20.000	17.500	64	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	18.750	16.250	60	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	RESERVED		ns	
6400AN	16.250	14.375	52	tCK(AVG)	0.312	<0.333	RESERVED						ns	
6800C	20.000	17.647	68	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	18.823	16.470	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800B	18.823	16.176	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	RESERVED				ns	
6800AN	16.470	14.117	56	tCK(AVG)	RESERVED								ns	
7200C	20.000	17.500	72	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	0.277	<0.294	ns	
7200BN,B	18.888	16.111	68	tCK(AVG)	0.277	<0.294	0.277	<0.294	RESERVED				ns	
7200AN	16.111	14.444	58	tCK(AVG)	RESERVED								ns	
7600C	20.000	17.631	76	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	0.263	<0.277	ns	
7600BN	18.347	16.315	72	tCK(AVG)	0.263	<0.277	0.263	<0.277	RESERVED				ns	
7600B	18.347	16.052	72	tCK(AVG)	0.263	<0.277	0.263	<0.277	RESERVED				ns	
7600AN	16.315	14.210	62	tCK(AVG)	RESERVED								ns	
8000C	20.000	17.500	80	tCK(AVG)	0.250	<0.263	0.250	<0.263	0.250	<0.263	0.250	<0.263	ns	
8000BN,B	18.500	16.000	74	tCK(AVG)	0.250	<0.263	0.250	<0.263	RESERVED				ns	
8000AN	16.000	14.000	64	tCK(AVG)	RESERVED								ns	
8400C	20.000	17.619	84	tCK(AVG)	0.238	<0.250	0.238	<0.250	0.238	<0.250	0.238	<0.250	ns	
8400BN,B	18.571	16.190	78	tCK(AVG)	0.238	<0.250	0.238	<0.250	0.238	<0.250	RESERVED		ns	
8400AN	16.190	14.285	68	tCK(AVG)	0.238	<0.250	RESERVED						ns	
Supported CL					22,26,30,32,34,36,38,40,42,44,46,48,50,52,56,60,64,68,72,74,76,78,80,84		22,30,32,34,36,38,40,42,44,46,48,50,52,56,60,64,68,72,74,76,78,80,84		22,30,32,34,36,40,42,44,46,48,52,56,60,64,68,72,76,78,80,84		22,32,36,40,44,48,52,56,60,64,68,72,76,80,84		nCK	12

10.30 3DS DDR5-8800 Speed Bins and Operations

Table 310 — 3DS DDR5-8800 Speed Bins and Operations

Speed Bin					DDR5-8800AN 3DS		DDR5-8800B 3DS		DDR5-8800BN 3DS		DDR5-8800C 3DS		Unit	Note
CL-nRCD-nRP					72-62-62		82-71-71		82-72-72		88-77-77			
Parameter				Symbol	Min	Max	Min	Max	Min	Max	Min	Max		
Read command to first data				tAA	16.363	22.222	18.500	22.222	18.636	22.222	20.000	22.222	ns	12
Activate to Read or Write command delay time				tRCD	14.090	-	16.000	-	16.363	-	17.500	-	ns	7
Row Precharge time				tRP	14.090	-	16.000	-	16.363	-	17.500		ns	7
Activate to Precharge command period				tRAS	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	32.000	5 * tREFI1 (Norm) 9 * tREFI2 (FGR)	ns	7, 13
Activate to Activate or Refresh command period				tRC (tRAS +tRP)	46.090	-	48.000	-	48.363	-	49.500	-	ns	7, 8
CAS Write Latency				CWL	CL-2								nCK	12
Speed Bin[5]	tAAmin (ns)[5]	tRCDmin tRPmin (ns)[5]	Read CL[12]	Supported Frequency Table										
-	20.952	-	22	tCK(AVG)	0.952	1.010	0.952	1.010	0.952	1.010	0.952	1.010	ns	6, 9
3200C	20.000	17.500	32	tCK(AVG)	0.625	0.681	0.625	0.681	0.625	0.681	0.625	0.681	ns	
3200BN,B	18.750	16.250	30	tCK(AVG)	0.625	0.681	0.625	0.681	RESERVED				ns	
3200AN	16.250	15.000	26	tCK(AVG)	RESERVED								ns	
3600C	20.000	17.777	36	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	0.555	<0.625	ns	
3600BN,B	18.888	16.666	34	tCK(AVG)	0.555	<0.625	0.555	<0.625	0.555	<0.625	RESERVED		ns	
3600AN	16.666	14.444	30	tCK(AVG)	0.555	<0.625	RESERVED						ns	
4000C	20.000	17.500	40	tCK(AVG)	0.500	<0.555	0.500	<0.555	0.500	<0.555	0.500	<0.555	ns	
4000BN,B	19.000	16.000	38	tCK(AVG)	0.500	<0.555	0.500	<0.555	RESERVED				ns	
4000AN	16.000	14.000	32	tCK(AVG)	RESERVED								ns	
4400C	20.000	17.727	44	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	0.454	<0.500	ns	
4400BN,B	19.090	16.363	42	tCK(AVG)	0.454	<0.500	0.454	<0.500	0.454	<0.500	RESERVED		ns	
4400AN	16.363	14.545	36	tCK(AVG)	0.454	<0.500	RESERVED						ns	
4800C	20.000	17.500	48	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	0.416	<0.454	ns	
4800BN	19.166	16.666	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	0.416	<0.454	RESERVED		ns	
4800B	19.166	16.250	46	tCK(AVG)	0.416	<0.454	0.416	<0.454	RESERVED				ns	
4800AN	16.666	14.166	40	tCK(AVG)	0.416	<0.454	RESERVED						ns	
5200C	20.000	17.692	52	tCK(AVG)	0.384	<0.416	0.384	<0.416	0.384	<0.416	0.384	<0.416	ns	
5200BN,B	19.230	16.153	50	tCK(AVG)	0.384	<0.416	0.384	<0.416	RESERVED				ns	
5200AN	16.153	14.615	42	tCK(AVG)			RESERVED						ns	
5600C	20.000	17.500	56	tCK(AVG)	0.357	<0.384	0.357	<0.384	0.357	<0.384	0.357	<0.384	ns	
5600BN	18.571	16.428	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600B	18.571	16.071	52	tCK(AVG)	0.357	<0.384	0.357	<0.384	RESERVED				ns	
5600AN	16.428	14.285	46	tCK(AVG)	0.357	<0.384	RESERVED						ns	
6000C	20.000	17.666	60	tCK(AVG)	0.333	<0.357	0.333	<0.357	0.333	<0.357	0.333	<0.357	ns	
6000BN,B	18.666	16.000	56	tCK(AVG)	0.333	<0.357	0.333	<0.357	RESERVED				ns	
6000AN	16.000	14.000	48	tCK(AVG)	RESERVED								ns	
6400C	20.000	17.500	64	tCK(AVG)	0.312	<0.333	0.312	<0.333	0.312	<0.333	0.312	<0.333	ns	
6400BN,B	18.750	16.250	60	tCK(AVG)	0.312	<0.333	0.312	<0.333	RESERVED				ns	
6400AN	16.250	14.375	52	tCK(AVG)	RESERVED								ns	
6800C	20.000	17.647	68	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	0.294	<0.312	ns	
6800BN	18.823	16.470	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	0.294	<0.312	RESERVED		ns	
6800B	18.823	16.176	64	tCK(AVG)	0.294	<0.312	0.294	<0.312	RESERVED				ns	
6800AN	16.470	14.117	56	tCK(AVG)	0.294	<0.312	RESERVED						ns	
7200C	20.000	17.500	72	tCK(AVG)	0.277	<0.294	0.277	<0.294	0.277	<0.294	0.277	<0.294	ns	
7200BN,B	18.888	16.111	68	tCK(AVG)	0.277	<0.294	0.277	<0.294	RESERVED				ns	
7200AN	16.111	14.444	58	tCK(AVG)	RESERVED								ns	

Table 310 — 3DS DDR5-8800 Speed Bins and Operations (cont'd)

7600C	20.000	17.631	76	tCK(AVG)	0.263	<0.277	0.263	<0.277	0.263	<0.277	0.263	<0.277	ns	
7600BN	18.347	16.315	72	tCK(AVG)	0.263	<0.277	0.263	<0.277	RESERVED				ns	
7600B	18.347	16.052	72	tCK(AVG)	0.263	<0.277	0.263	<0.277	RESERVED				ns	
7600AN	16.315	14.210	62	tCK(AVG)	RESERVED								ns	
8000C	20.000	17.500	80	tCK(AVG)	0.250	<0.263	0.250	<0.263	0.250	<0.263	0.250	<0.263	ns	
8000BN,B	18.500	16.000	74	tCK(AVG)	0.250	<0.263	0.250	<0.263	RESERVED				ns	
8000AN	16.000	14.000	64	tCK(AVG)	RESERVED								ns	
8400C	20.000	17.619	84	tCK(AVG)	0.238	<0.250	0.238	<0.250	0.238	<0.250	0.238	<0.250	ns	
8400BN,B	18.571	16.190	78	tCK(AVG)	0.238	<0.250	0.238	<0.250	RESERVED				ns	
8400AN	16.190	14.285	68	tCK(AVG)	RESERVED								ns	
8800C	20.000	17.500	88	tCK(AVG)	0.227	<0.238	0.227	<0.238	0.227	<0.238	0.227	<0.238	ns	
8800BN	18.636	16.363	82	tCK(AVG)	0.227	<0.238	0.227	<0.238	0.227	<0.238	RESERVED		ns	
8800B	18.636	16.136	82	tCK(AVG)	0.227	<0.238	0.227	<0.238	RESERVED				ns	
8800AN	16.363	14.090	72	tCK(AVG)	0.227	<0.238	RESERVED						ns	
Supported CL					22,30,32,34,36,38,40,42,44,46,48,50,52,56,60,64,68,72,74,76,78,80,82,84,88		22,30,32,34,36,38,40,42,44,46,48,50,52,56,60,64,68,72,74,76,78,80,82,84,88		22,32,34,36,40,42,44,46,48,52,56,60,64,68,72,76,80,82,84,88		22,32,36,40,44,48,52,56,60,64,68,72,76,80,84,88		nCK	12

DDR5 Speed Bin Table Notes:

1. Minimum timing parameters are defined according to the rules in the Rounding Definitions and Algorithms section.
2. The translation of all timing parameters from ns values to nCK values shall follow the Rounding Algorithm. The translation of tAA to CL shall follow the explicit combinations listed in the Speed Bin Tables.
3. The CL setting and CWL setting result in tCK(avg).MIN and tCK(avg).MAX requirements. When selecting tCK(avg), requirements from the CL setting as well as requirements from the CWL setting shall be fulfilled.
4. 'Reserved' settings are not allowed. The user shall program a different value.
5. This column shows the intended native speed bin timings to be replaced and supported when down clocking. This column does not necessarily show the actual minimum speed bin timings allowed and supported when down clocking because the timings could be faster according to the Rounding Algorithm, depending on the specific speed bin and down clock frequency combination.
6. DDR5-3200 AC timings apply if the DRAM operates slower than the 2933 MT/s data rate. This is not limited to only the Speed Bin Table timings.
7. Parameters apply from tCK(avg)min to tCK(avg)max.
8. tRC(min) shall always be greater than or equal to tRAS(min) + tRP(min), and when using the appropriate rounding algorithms, nRC(min) shall always be greater than or equal to nRAS(min) + nRP(min).
9. tCK(avg).max of 1.010 ns (1980 MT/s data rate) is defined to allow for 1% SSC down-spreading at a data rate of 2000 MT/s according to JESD404-1.
10. Each speed bin lists the timing requirements that need to be supported in order for a given DRAM to be JEDEC standard. The JEDEC standard does not require support for all speed bins within a given speed. The JEDEC standard requires meeting the parameters for a least one of the listed speed bins.
11. Any speed bin also supports functional operation at slower frequencies as shown in the table which are not subject to Production Tests but are verified by Design/Characterization.
12. The CL Algorithm can be used to mathematically determine the valid CAS Latencies listed in the Speed Bin Tables. The CL Algorithm calculates supported CAS Latencies by rounding the operating frequency up to the next faster native speed bin (i.e., 3200 MT/s, 3600 MT/s...). Using the resulting tCK(AVG)min, and the bin target timings, the CL Algorithm then uses the Rounding Algorithm to calculate the valid CAS Latency. Because the DDR5 SDRAM specification only supports even CAS Latencies, odd CAS Latencies are rounded up to the next even CAS Latency. The 1980-2100 MT/s data rate always uses CL22. If tAA(corrected) or tRCDtRP(corrected) are violated, the CL Algorithm uses a slower combination of tAA(target) and tRCDtRP(target) to return slower valid CAS Latencies. The DDR5 SDRAM can support up to four valid CAS Latencies, CL(AN), CL(B), CL(BN), and CL(C), for a given frequency. tAA(corrected) and tRCDtRP(corrected) are calculated by reducing tAA(min), tRCD(min), and tRP(min) by the Rounding Algorithm correction factor. The proper setting of CL shall be determined by the memory controller, either by using the Speed Bin Tables, or by using the CL Algorithm, or by some other means. Refer to the Rounding Definitions and Algorithm section for more information. When Read CRC is enabled, CL is increased according to the Read CRC Latency Adder. When Write CRC is enabled, there is no Write CRC Latency Adder.

DDR5 Speed Bin Table Notes (cont'd):

```
// Variables already defined in other areas of the DDR5 SDRAM specification
CorrFact       = 0.30                                       // (%)   Rounding Algorithm correction factor
ScaledCorrFact = 997                                        //       Scaled correction factor (1000*(1-0.30%))
tCKreal        =1011-952, 682-238                           // (ps)  Real application tCK(AVG) (1980-2100MT/s, 2933-8800MT/s)
tAAmin         MONO=14000-17500, 3DS=16000-20000            // (ps)  From Speed Bin Tables and DIMM SPD bytes 30-31
tRCDtRPmin     MONO=14000-17500, 3DS=14000-17500            // (ps)  From Speed Bin Tables and DIMM SPD bytes 32-33 (tRCD=tRP)
tAAcorr        = TRUNC(tAAmin*ScaledCorrFactor/1000)        // (ps)  Corrected tAA(min) per the Rounding Algorithm rules
tRCDtRPcorr    = TRUNC(tRCDtRPmin*ScaledCorrFactor/1000)    // (ps)  Corrected tRCD(min), tRP(min) per the Rounding Algorithm
FUNC[RA(targ)] = TRUNC((targ*ScaledCorrFact/tCKstd+1000)/1000) // (nCK) Use Rounding Algorithm to convert bin target timing to nCK
// Round tCKreal down to the next faster standard frequency (tCK in ps)
IF (TRUNC(2000000/(2000*99%))>=TRUNC(tCKreal)>=TRUNC(2000000/(2000*105%)))            // Check for  1980-2100 nominal data rates
   tCKstd=TRUNC(2000000/2000)                                                          // Assign standard 2000 tCK (ps)
ELSE IF (TRUNC(2000000/(2000+7*(133+1/3)))>=TRUNC(tCKreal)>=TRUNC(2000000/3200))       // Check for  2933-3200 nominal data rates
   tCKstd=TRUNC(2000000/3200)                                                          // Assign standard 3200 tCK (ps)
ELSE
   FOR (DataRateNom=3200; DataRateNom<=8400; DataRateNom=DataRateNom+400)              // Check for >3200-8800 nominal data rates
      IF (TRUNC(2000000/DataRateNom)>TRUNC(tCKreal)>=TRUNC(2000000/(DataRateNom+400)))
         tCKstd=TRUNC(2000000/(DataRateNom+400))                                       // Assign standard 3600-8800 tCK (ps)
      ELSE
         tCKstd=RESERVED                                                               // No valid data rate found

// Timing targets (ps) that have been used to define the Speed Bin Tables
// MONO targets              3DS targets
   BinAN_tAAtarg     = 14000 BinAN_tAAtarg     = 16000        // tAA target for AN bins
   BinB__tAAtarg     = 16000 BinB__tAAtarg     = 18500        // tAA target for AN, B bins
   BinBN_tAAtarg     = 16000 BinBN_tAAtarg     = 18500        // tAA target for AN, B, BN bins
   BinC__tAAtarg     = 17500 BinC__tAAtarg     = 20000        // tAA target for AN, B, BN, C bins
   BinAN_tRCDtRPtarg = 14000 BinAN_tRCDtRPtarg = 14000        // tRCD, tRP target for AN bins
   BinBN_tRCDtRPtarg = 16000 BinBN_tRCDtRPtarg = 16000        // tRCD, tRP target for AN, B, BN bins
   BinC__tRCDtRPtarg = 17500 BinC__tRCDtRPtarg = 17500        // tRCD, tRP target for AN, B, BN, C bins
IF (TRUNC(2000000/3600)>tCKstd)                               // tRCD, tRP target for B bins is frequency dependent
   BinB__tRCDtRPtarg = 16000 BinB__tRCDtRPtarg = 16000        // tRCD, tRP target for AN, B bins data rates faster than 3600
ELSE
   // 16250=(2000000/3200)*EVEN(TRUNC((BinB__tRCDtRPtarg*ScaledCorrFact/(2000000/3200)+1000)/1000))
   BinB__tRCDtRPtarg = 16250 BinB__tRCDtRPtarg = 16250        // tRCD, tRP target for AN, B bins for data rates 3600 and slower

// CL Algorithm using variables defined above
// Up to four valid CL's can be returned for a specific freq: CL(AN), CL(B), CL(BN), CL(C), depending on tAAmin, tRCDmin, tRPmin
// The B and BN bins return the same CL
// Only even CL's (not odd CL's) are valid per the DDR5 SDRAM specification
// nRCD, nRP are only even at standard native frequencies for the AN, BN bins (can be even or odd at intermediate frequencies)
// nRCD, nRP may be even or odd at standard native frequencies for the B, C bins (can be even or odd at intermediate frequencies)
IF (TRUNC(2000000/2000)=tCKstd)                               // CL22 is the only valid CL for 1980-2100 data rates
   CL(AN)=22                                                  // Valid even CL for AN bins
   CL(B )=22                                                  // Valid even CL for AN, B, bins
   CL(BN)=22                                                  // Valid even CL for AN, B, BN bins
   CL(C )=22                                                  // Valid even CL for AN, B, BN, C bins
ELSE IF (TRUNC(2000000/3200)>=tCKstd>=TRUNC(2000000/8800))    // Valid CL for 2933-8800 data rates
   IF      ((EVEN(RA(BinAN_tAAtarg))*tCKstd>=tAAcorr)AND(EVEN(RA(BinAN_tRCDtRPtarg))*tCKstd>=tRCDtRPcorr)) // nRCD, nRP even only
      CL(AN)=EVEN(RA(BinAN_tAAtarg))                          // Valid even CL for AN bins
      CL(B )=EVEN(RA(BinB__tAAtarg))                          // Valid even CL for AN, B bins
      CL(BN)=EVEN(RA(BinBN_tAAtarg))                          // Valid even CL for AN, B, BN bins
      CL(C )=EVEN(RA(BinC__tAAtarg))                          // Valid even CL for AN, B, BN, C bins
   ELSE IF ((EVEN(RA(BinB__tAAtarg))*tCKstd>=tAAcorr)AND(     (RA(BinB__tRCDtRPtarg))*tCKstd>=tRCDtRPcorr)) // nRCD, nRP even, odd
      CL(AN)=RESERVED                                         // Valid even CL for AN bins
      CL(B )=EVEN(RA(BinB__tAAtarg))                          // Valid even CL for AN, B bins
      CL(BN)=EVEN(RA(BinBN_tAAtarg))                          // Valid even CL for AN, B, BN bins
      CL(C )=EVEN(RA(BinC__tAAtarg))                          // Valid even CL for AN, B, BN, C bins
   ELSE IF ((EVEN(RA(BinBN_tAAtarg))*tCKstd>=tAAcorr)AND(EVEN(RA(BinBN_tRCDtRPtarg))*tCKstd>=tRCDtRPcorr)) // nRCD, nRP even only
      CL(AN)=RESERVED                                         // Valid even CL for AN bins
      CL(B )=RESERVED                                         // Valid even CL for AN, B bins
      CL(BN)=EVEN(RA(BinBN_tAAtarg))                          // Valid even CL for AN, B, BN bins
      CL(C )=EVEN(RA(BinC__tAAtarg))                          // Valid even CL for AN, B, BN, C bins
   ELSE IF ((EVEN(RA(BinC__tAAtarg))*tCKstd>=tAAcorr)AND(     (RA(BinC__tRCDtRPtarg))*tCKstd>=tRCDtRPcorr)) // nRCD, nRP even, odd
      CL(AN)=RESERVED                                         // Valid even CL for AN bins
      CL(B )=RESERVED                                         // Valid even CL for AN, B bins
      CL(BN)=RESERVED                                         // Valid even CL for AN, B, BN bins
      CL(C )=EVEN(RA(BinC__tAAtarg))                          // Valid even CL for AN, B, BN, C bins
   ELSE                                                       // No valid CL found (tAAmin, tRCDmin, tRPmin are too slow)
      CL(AN)=RESERVED                                         // Valid even CL for AN bins
      CL(B )=RESERVED                                         // Valid even CL for AN, B bins
      CL(BN)=RESERVED                                         // Valid even CL for AN, B, BN bins
      CL(C )=RESERVED                                         // Valid even CL for AN, B, BN, C bins
ELSE                                                          // No valid data rate found
   CL(AN)=RESERVED                                            // Valid even CL for AN bins
   CL(B )=RESERVED                                            // Valid even CL for AN, B bins
   CL(BN)=RESERVED                                            // Valid even CL for AN, B, BN bins
   CL(C )=RESERVED                                            // Valid even CL for AN, B, BN, C bins
```

13. tRAS(max) shall always be less than or equal to 5*tREFI1(max) during Normal Refresh Mode and less than or equal to 9*tREFI2(max) during Fine Granularity Refresh Mode, and when using the rounding algorithms, nRAS(max) shall always be less than or equal to 5*nREFI1(max) during Normal Refresh Mode and less than or equal to 9*nREFI2(max) during Fine Granularity Refresh Mode.

11 IDD, IDDQ, and IPP Specification Parameters and Test conditions

11.1 IDD, IPP, and IDDQ Measurement Conditions

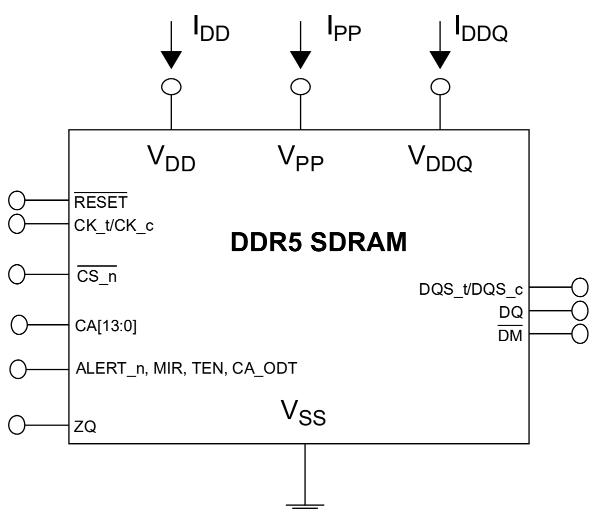
In this chapter, IDD, IPP, and IDDQ measurement conditions such as test load and patterns are defined. Figure 256 shows the setup and test load for IDD, IPP, and IDDQ measurements.

- IDD currents (such as IDD0, IDDQ0, IPP0, IDD0F, IDDQ0F, IPP0F, IDD2N, IDDQ2N, IPP2N, IDD2NT, IDDQ2NT, IPP2NT, IDD2P, IDDQ2P, IPP2P, IDD3N, IDDQ3N, IPP3N, IDD3P, IDDQ3P, IPP3P, IDD4R, IDDQ4R, IPP4R, IDD4RC, IDD4W, IDDQ4W, IPP4W, IDD4WC, IDD5F, IDDQ5F, IPP5F, IDD5B, IDDQ5B, IPP5B, IDD5C, IDDQ5C, IPP5C, IDD6N, IDDQ6N, IPP6N, IDD6E, IDDQ6E, IPP6E, IDD7, IDDQ7, IPP7, IDD8, IDDQ8, IPP8 and IDD9, IDDQ9, IPP9) are measured as time-averaged currents with all VDD balls of the DDR5 SDRAM under test tied together. Any IDDQ or IPP current is not included in IDD currents.
- IDDQ currents are measured as time-averaged currents with all VDDQ balls of the DDR5 SDRAM under test tied together. Any IDD or IPP current is not included in IDDQ currents.
- IPP currents are measured as time-averaged currents with all VPP balls of the DDR5 SDRAM under test tied together. Any IDD or IDDQ current is not included in IPP currents.

Attention: IDDQ values cannot be directly used to calculate IO power of the DDR5 SDRAM. They can be used to support correlation of simulated IO power to actual IO power as outlined in Figure 257.

For IDD, IPP, and IDDQ measurements, the following definitions apply:

- “0” and “LOW” is defined as VIN <= VILAC(max).
- “1” and “HIGH” is defined as VIN >= VIHAC(min).
- “MID-LEVEL” is defined as inputs are VREF = 0.75 * VDDQ.
- Basic IDD, IPP and IDDQ Measurement Conditions are described in Table 311.
- Detailed IDD, IPP and IDDQ Measurement-Loop Patterns are described in Chapter 11.2 through Chapter 11.11.
- IDD Measurements are done after properly initializing and training the DDR5 SDRAM. This includes but is not limited to setting
  TDQS_t disabled in MR5;
  CRC disabled in MR50;
  DM disabled in MR5;
  1N mode enabled and set CS assertion duration (MR2:OP[4]) as 1_B in MR2, unless otherwise specified in the IDD, IDDQ, and IPP patterns’ conditions definitions;
- Attention: The IDD, IPP and IDDQ Measurement-Loop Patterns need to be executed at least one time before actual IDD, IDDQ or IPP measurement is started, with the exception of IDD9 which can be measured any time after the DRAM has entered MBIST mode.
- T_{CASE} defined as 0 - 95 °C, unless stated in the specific condition definition Table 311.
- For all IDD, IDDQ and IPP measurement loop timing parameters, refer to the timing parameters defined in the spec to calculate the nCK required.



NOTES:
1. DIMM level Output test load condition may be different from above

Figure 256 — Measurement Setup and Test Load for IDD, IPP, and IDDQ Measurements

11.1 IDD, IPP, and IDDQ Measurement Conditions (cont'd)

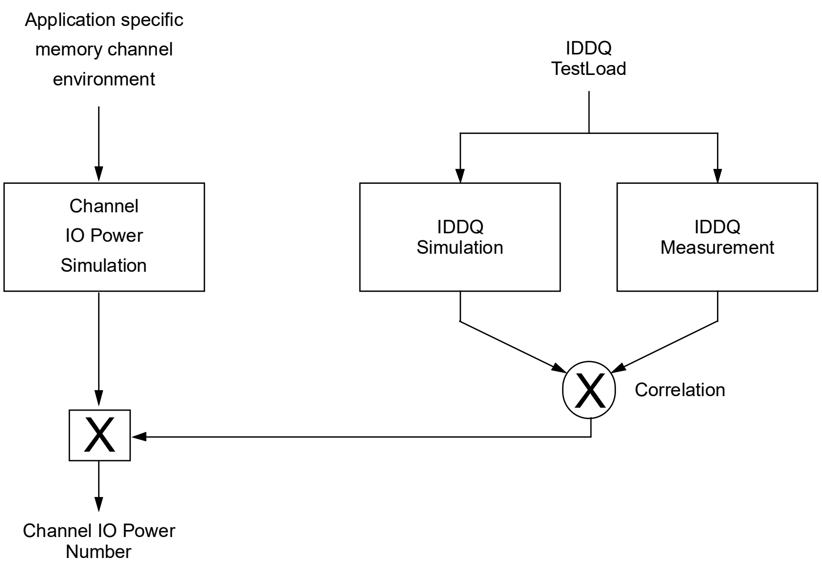


Figure 257 — Correlation from Simulated Channel IO Power to Actual Channel IO Power Supported by IDDQ Measurement

Table 311 — Basic IDD, IDDQ, and IPP Measurement Conditions

Symbol	Description
IDD0	**Operating One Bank Active-Precharge Current** **External clock:** On; **tCK, nRC, nRAS, nRP, nRRD**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **BL:** 16[1]; **CS_n:** High between ACT and PRE; **CA Inputs:** partially toggling according to Table 312; **Data IO:** VDDQ; **DM_n:** stable at 1; **Bank Activity:** Cycling with one bank active at a time: 0,0,1,1,2,2,... (see Table 312); **Output Buffer and RTT:** Enabled in Mode Registers[2]; **Pattern Details:** see Table 312
IDDQ0	**Operating One Bank Active-Precharge IDDQ Current** **Same condition with IDD0,** however measuring IDDQ current instead of IDD current
IPP0	**Operating One Bank Active-Precharge IPP Current** **Same condition with IDD0,** however measuring IPP current instead of IDD current
IDD0F	**Operating Four Bank Active-Precharge Current** **External clock:** On; **tCK, nRC, nRAS, nRP, nRRD**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **BL:** 16[1]; **CS_n:** High between ACT and PRE; **CA Inputs:** partially toggling according to Table 313; **Data IO:** VDDQ; **DM_n:** stable at 1; **Bank Activity:** Cycling with four bank active at a time: (see Table 313); **Output Buffer and RTT:** Enabled in Mode Registers[2]; **Pattern Details:** see Table 313
IDDQ0F	**Operating Four Bank Active-Precharge IDDQ Current** **Same condition with IDD0F,** however measuring IDDQ current instead of IDD current
IPP0F	**Operating Four Bank Active-Precharge IPP Current** **Same condition with IDD0F,** however measuring IPP current instead of IDD current
IDD2N	**Precharge Standby Current** **External clock:** On; **tCK**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **CS_n:** stable at 1; **CA Inputs:** partially toggling according to Table 314; **Data IO:** VDDQ; **DM_n:** stable at 1; **Bank Activity:** all banks closed; **Output Buffer and RTT:** Enabled in Mode Registers[2]; **Pattern Details:** see Table 314
IDDQ2N	**Precharge Standby IDDQ Current** **Same condition with IDD2N,** however measuring IDDQ current instead of IDD current
IPP2N	**Precharge Standby IPP Current** **Same condition with IDD2N,** however measuring IPP current instead of IDD current
IDD2NT	**Precharge Standby Non-Target Command Current** **External clock:** On; **tCK**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **BL:** 16[1]; **CS_n:** High between WRITE commands; **CS_n, CA Inputs:** partially toggling according to Table 315; **Data IO:** VDDQ; **DM_n:** stable at 1; **Bank Activity:** all banks closed; **Output Buffer and RTT:** Enabled in Mode Registers[2]; **Pattern Details:** see Table 315

Table 311 — Basic IDD, IDDQ, and IPP Measurement Conditions (cont'd)

Symbol	Description
IDDQ2NT (Optional)	**Precharge Standby Non-Target Command IDDQ Current** Same condition with IDD2NT, however measuring IDDQ current instead of IDD current
IPP2NT (Optional)	**Precharge Standby Non-Target Command IPP Current** Same condition with IDD2NT, however measuring IPP current instead of IDD current
IDD2P	**Precharge Power-Down** **Device in Precharge Power-Down, External clock:** On; **tCK**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **CS_n:** stable at 1 after Power Down Entry command; **CA Inputs:** stable at 1; CA11=H during the PDE command; **Data IO:** VDDQ; **DM_n:** stable at 1; **Bank Activity:** all banks closed; **Output Buffer and RTT:** Enabled in Mode Registers[2];
IDDQ2P	**Precharge Power-Down** **Same condition with IDD2P,** however measuring IDDQ current instead of IDD current
IPP2P	**Precharge Power-Down** **Same condition with IDD2P,** however measuring IPP current instead of IDD current
IDD3N	**Active Standby Current** **External clock:** On; **tCK**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **CS_n:** stable at 1; **CA Inputs:** partially toggling according to Table 314; **Data IO:** VDDQ; **DM_n:** stable at 1; **Bank Activity:** all banks open; **Output Buffer and RTT:** Enabled in Mode Registers[2]; **Pattern Details:** see Table 314
IDDQ3N	**Active Standby IDDQ Current** **Same condition with IDD3N,** however measuring IDDQ current instead of IDD current
IPP3N	**Active Standby IPP Current** **Same condition with IDD3N,** however measuring IPP current instead of IDD current
IDD3P	**Active Power-Down Current** **Device in Active Power-Down, External clock:** On; **tCK**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **CS_n:** stable at 1 after Power Down Entry command; **CA Inputs:** stable at 1; CA11=H during the PDE command; **Data IO:** VDDQ; **DM_n:** stable at 1; **Bank Activity:** all banks open; **Output Buffer and RTT:** Enabled in Mode Registers[2];
IDDQ3P	**Active Power-Down IDDQ Current** **Same condition with IDD3P,** however measuring IDDQ current instead of IDD current
IPP3P	**Active Power-Down IPP Current** **Same condition with IDD3P,** however measuring IPP current instead of IDD current
IDD4R	**Operating Burst Read Current** **External clock:** On; **tCK, nCCD_S, CL**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **BL:** 16[1]; **CS_n:** High between RD; **CA Inputs:** partially toggling according to Table 316; **Data IO:** seamless read data burst with different data between one burst and the next one according to Table 316; **DM_n:** stable at 1; **Bank Activity:** all banks open, RD commands cycling through banks: 0,0,1,1,2,2,... (see Table 316); **Output Buffer and RTT:** Enabled in Mode Registers[2]; **Pattern Details:** see Table 316
IDD4RC	**Operating Burst Read Current with Read CRC** **Read CRC enabled[4]. Other conditions:** see IDD4R
IDDQ4R	**Operating Burst Read IDDQ Current** Same definition like for IDD4R, however measuring IDDQ current instead of IDD current
IPP4R	**Operating Burst Read IPP Current** **Same condition with IDD4R,** however measuring IPP current instead of IDD current
IDD4W	**Operating Burst Write Current** **External clock:** On; **tCK, nCCD_S_WR, CL**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **BL:** 16[1]; **CS_n:** High between WR; **CA Inputs:** partially toggling according to Table 317; **Data IO:** seamless write data burst with different data between one burst and the next one according to Table 317; **DM_n:** stable at 1; **Bank Activity:** all banks open, WR commands cycling through banks: 0,0,1,1,2,2,... (see Table 317); **Output Buffer and RTT:** Enabled in Mode Registers[2]; **Pattern Details:** see Table 317
IDD4WC	**Operating Burst Write Current with Write CRC** **Write CRC enabled[3], Other conditions:** see IDD4W
IDDQ4W	**Operating Burst Write IDDQ Current** **Same condition with IDD4W,** however measuring IDDQ current instead of IDD current
IPP4W	**Operating Burst Write IPP Current** **Same condition with IDD4W,** however measuring IPP current instead of IDD current
IDD5B	**Burst Refresh Current (Normal Refresh Mode)** **External clock:** On; **tCK, nRFC**1: refer to Chapter 13.3 Timing Parameters by Speed Grade; **BL:** 16[1]; **CS_n**: High between REF; **CA Inputs:** partially toggling according to Table 318; **Data IO:** VDDQ; **DM_n:** stable at 1; **Bank Activity:** REF command every nRFC1 (see Table 318); **Output Buffer and RTT:** Enabled in Mode Registers[2]; **Pattern Details:** see Table 318

Table 311 — Basic IDD, IDDQ, and IPP Measurement Conditions (cont'd)

Symbol	Description
IDDQ5B	**Burst Refresh IDDQ Current (Normal Refresh Mode)** **Same condition with IDD5B,** however measuring IDDQ current instead of IDD current
IPP5B	**Burst Refresh IPP Current (Normal Refresh Mode)** **Same condition with IDD5B,** however measuring IPP current instead of IDD current
IDD5F	**Burst Refresh Current (Fine Granularity Refresh Mode)** **tRFC=tRFC2, Other conditions:** see IDD5B
IDDQ5F	**Burst Refresh IDDQ Current (Fine Granularity Refresh Mode)** **Same condition with IDD5F,** however measuring IDDQ current instead of IDD current
IPP5F	**Burst Refresh IPP Current (Fine Granularity Refresh Mode)** **Same condition with IDD5F,** however measuring IPP current instead of IDD current
IDD5C	**Burst Refresh Current (Same Bank Refresh Mode)** **External clock:** On; **tCK, nRFCsb**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **BL:** 16[1]; **CS_n**: High between REF; **CA Inputs:** partially toggling according to Table 319; **Data IO:** VDDQ; **DM_n:** stable at 1; **Bank Activity:** REF command every nRFCsb (see Table 319); **Output Buffer and RTT:** Enabled in Mode Registers[2]; **Pattern Details:** see Table 319
IDDQ5C	**Burst Refresh IDDQPP Current (Same Bank Refresh Mode)** **Same condition with IDD5C,** however measuring IDDQ current instead of IDD current
IPP5C	**Burst Refresh IPP Current (Same Bank Refresh Mode)** **Same condition with IDD5C,** however measuring IPP current instead of IDD current
IDD6N	**Self Refresh Current: Normal Temperature Range** ***T*CASE:** 0 - 85 °C; **External clock:** Off; **CK_t and CK_c**: HIGH; **tCK, nCPDED**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **BL:** 16[1]; **CS_n#:** low; **CA, Data IO:** High; **DM_n:** stable at 1; **Bank Activity:** Self-Refresh operation; **Output Buffer and RTT:** Disabled by the DRAM upon entry into Self-Refresh
IDDQ6N	**Self Refresh IDDQ Current: Normal Temperature Range** **Same condition with IDD6N,** however measuring IDDQ current instead of IDD current
IPP6N	**Self Refresh IPP Current: Normal Temperature Range** **Same condition with IDD6N,** however measuring IPP current instead of IDD current
IDD6E	**Self Refresh Current: Extended Temperature Range**) **T_{CASE}:** 85 - 95 °C; Extended[4]; **External clock:** Off; **CK_t and CK_c**: HIGH; **tCK, nCPDED**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **BL:** 16[1]; **CS_n:** low; **CA, Data IO:** High; **DM_n**:stable at 1; **Bank Activity:** Self-Refresh operation; **Output Buffer and RTT:** Disabled by the DRAM upon entry into Self-Refresh
IDDQ6E	**Self Refresh IDDQ Current: Extended Temperature Range** **Same condition with IDD6E,** however measuring IDDQ current instead of IDD current
IPP6E	**Self Refresh IPP Current: Extended Temperature Range** **Same condition with IDD6E,** however measuring IPP current instead of IDD current
IDD7	**Operating Bank Interleave Read Current** **External clock:** On; **tCK, nRC, nRAS, nRCD, nRRD_S, nFAW, tCCD_S CL**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **BL:** 16[1]; **CS_n**: High between ACT and RDA; **CA Inputs:** partially toggling according to Table 321; **Data IO:** read data bursts with different data between one burst and the next one according to Table 321; **DM_n:** stable at 1; **Bank Activity:** two times interleaved cycling through banks (0, 1, ...7) with different addressing, see Table 321; **Output Buffer and RTT:** Enabled in Mode Registers[2]; **Pattern Details:** see Table 321
IDDQ7	**Operating Bank Interleave Read IDDQ Current** **Same condition with IDD7,** however measuring IDDQ current instead of IDD current
IPP7	**Operating Bank Interleave Read IPP Current** **Same condition with IDD7,** however measuring IPP current instead of IDD current
IDD8	**Maximum Power Saving Deep Power Down Current** **External clock:** Off; **CK_t and CK_c**: HIGH; **tCK, nCPDED**: refer to Chapter 13.3 Timing Parameters by Speed Grade; **BL:** 16[1]; **CS_n#:** low; **CA:**High**, DM_n:** stable at 1; **Bank Activity:** All banks closed and device in MPSM deep power down mode5; **Output Buffer and RTT:** Enabled in Mode Registers[2]; **Patterns Details**: same as IDD6N but MPSM is enabled in mode register.
IDDQ8	**Maximum Power Saving Deep Power Down IDDQ Current** **Same condition with IDD8,** however measuring IDDQ current instead of IDD current
IPP8	**Maximum Power Saving Deep Power Down IPP Current** **Same condition with IDD8,** however measuring IPP current instead of IDD current
IDD9 (Optional)	**MBIST Current** **Device in MBIST mode, External clock:** On; **CS_n**: Stable at 1 after MBIST entry; **CA Inputs:** stable at 1; **Data IO:** VDDQ; **Bank Activity:** MBIST operation; **Output Buffer and RTT:** Enabled in Mode Registers[2];

Table 311 — Basic IDD, IDDQ, and IPP Measurement Conditions (cont'd)

Symbol	Description
IDDQ9 (Optional)	**MBIST IDDQ Current** **Same condition with IDD9,** however measuring IDDQ current instead of IDD current
IPP9 (Optional)	**MBIST IPP Current** **Same condition with IDD9,** however measuring IPP current instead of IDD current

NOTE 1 Burst Length: BL16 fixed by MR0 OP[1:0]=00.

NOTE 2 Output Buffer Enable
- set MR5 OP[0] = 0] : Qoff = Output buffer enabled
- set MR5 OP[2:1] = 00]: Pull-Up Output Driver Impedance Control = RZQ/7
- set MR5 OP[7:6] = 00]: Pull-Down Output Driver Impedance Control = RZQ/7

RTT_Nom enable
- set MR35 OP[5:0] = 110110: RTT_NOM_WR = RTT_NOM_RD = RZQ/6

RTT_WR enable
- set MR34 OP[5:3] = 010 RTT_WR = RZQ/2

CA/CS/CK ODT, DQS_RTT_PART, and RTT_PARK disable
- set MR32 OP[5:0] = 000000
- set/MR33 OP[5:0] = 000000
- set MR34 OP[2:0] = 000

NOTE 3 WRITE CRC enabled
- set MR50 OP[2:1] = 11

NOTE 4 Read CRC enabled
- set MR50:OP[0]=1

NOTE 5 MPSM Deep Power Down Mode
- set MR2:OP[3]=1 if PDA Enumerate ID not equal to 15
- set MR2:OP[5]=1 if PDA Enumerate ID equal to 15

11.2 IDD0, IDDQ0, IPP0 Pattern

Executes Active and PreCharge commands with tightest timing possible while exercising all Bank and Bank Group addresses. Note 2 applies to the entire table.

Table 312 — IDD0, IDDQ0, IPP0

Sub-Loop	Sequence	Command	CS_n	C/A [13:0]	Row Address [17:0]	BA [1:0]	BG [2:0]	CID [2:0]	Special Instructions
0	0	ACT	L H	-	0x00000	0x0	0x00	0x0	
	1	DES	H	Toggling[1]					Repeat sequence to satisfy tRAS(min), truncate if required
	2	PREpb	L	-		0x0	0x00	0x0	
	3	DES	H	Toggling[1]					Repeat sequence to satisfy tRP(min), truncate if required
	4	ACT	L H	-	0x03FFF	0x0	0x00	0x0	
	5	DES	H	Toggling[1]					Repeat sequence to satisfy tRAS(min), truncate if required
	6	PREpb	L	-		0x0	0x00	0x0	
	7	DES	H	Toggling[1]					Repeat sequence to satisfy tRP(min), truncate if required
1	8-15	Repeat sub-loop 0, use BG[2:0]=0x1 instead							
2	16-23	Repeat sub-loop 0, use BG[2:0]=0x2 instead							
3	24-31	Repeat sub-loop 0, use BG[2:0]=0x3 instead							
4	32-39	Repeat sub-loop 0, use BG[2:0]=0x4 instead							skip for x16
5	40-47	Repeat sub-loop 0, use BG[2:0]=0x5 instead							skip for x16
6	48-55	Repeat sub-loop 0, use BG[2:0]=0x6 instead							skip for x16
7	56-63	Repeat sub-loop 0, use BG[2:0]=0x7 instead							skip for x16
8-15	64-127	Repeat sub loops 0-7, use BA[1:0]=0x1 instead							
16-23	128-191	Repeat sub loops 0-7, use BA[1:0]=0x2 instead							
24-31	192-255	Repeat sub loops 0-7, use BA[1:0]=0x3 instead							
...	...	Repeat sub loops 0-31 for each 3DS logical rank, if applicable							CID[2:0]=0x1-0x7

NOTE 1 Utilize DESELECTs between commands while toggling all C/A bits per the 4-cycle sequence defined in the IDD2N, IDD3N pattern.
NOTE 2 For 3DS, all banks of all “non-target” logical ranks are Idd2N condition.

11.3 IDD0F, IDDQ0F, IPP0F Pattern

Executes a rolling four bank group Active and PreCharge commands per tRC time while exercising all Bank, Bank, Group and CID addresses. Note 2 applies to the entire table.

Table 313 — IDD0F, IDDQ0F, IPP0F

Sub-Loop	Sequence	Command	CS	C/A [13:0]	Row Address [17:0]	BA [1:0]	BG [2:0]	CID [2:0]	Special Instructions
0	0	ACT	L H		0x00000	0x0	0x00	0x0	
	1	DES	H	Toggling[1]					Repeat to satisfy tRRD(min) (6 DES to meet 8nCK)
	2	ACT	L H		0x00000	0x0	0x01	0x0	
	3	DES	H	Toggling[1]					Repeat to satisfy tRRD(min) (6 DES to meet 8nCK)
	4	ACT	L H		0x00000	0x0	0x02	0x0	
	5	DES	H	Toggling[1]					Repeat to satisfy tRRD(min) (6 DES to meet 8nCK)
	6	ACT	L H		0x00000	0x0	0x03	0x0	
	7	DES	H	Toggling[1]					Repeat to satisfy tRAS(min) from Sequence 0
	8	PREpb	L			0x0	0x00	0x0	
	9	DES	H	Toggling[1]					Repeat for tRRD(min) (7 DES to meet 8nCK) This allows for next PRE to meet tRAS(min)
	10	PREpb	L			0x0	0x01	0x0	
	11	DES	H	Toggling[1]					Repeat for tRRD(min) (7 DES to meet 8nCK) This allows for next PRE to meet tRAS(min)
	12	PREpb	L			0x0	0x02	0x0	
	13	DES	H	Toggling[1]					Repeat for tRRD(min) (7 DES to meet 8nCK) This allows for next PRE to meet tRAS(min)
	14	PREpb	L			0x0	0x03	0x0	
	15	DES	H	Toggling[1]					Repeat for tRRD(min) (7 DES to meet 8nCK) This allows for next PRE to meet tRAS(min)
	16	DES	H	Toggling[1]					Repeat for tRC(min) from Sequence 0 first ACTIVATE. This will be zero DESELECTS for speed 4000MT/s and slower.
1	17-33	Repeat sub-loop 0, use Row Address = 0x03FFF for the ACT instead							
2-3	34-67	Repeat sub-loop 0-1, use BG[2:0]=0x4,0x5,0x6,0x7 instead of 0x0,0x1,0x2,0x3							skip for x16
4-7	68-101	Repeat sub-loops 0-3, use BA[1:0]=0x1 instead							
8-11	102-135	Repeat sub-loops 0-3, use BA[1:0]=0x2 instead							
12-15	136-169	Repeat sub-loops 0-3, use BA[1:0]=0x3 instead							
...	...	Repeat sub loops 0-15 for each 3DS logical rank, if applicable							CID[2:0]=0x1-0x7

NOTE 1 Utilize DESELECTs between commands while toggling C/A bits per the 4-cycle sequence defined in the IDD2N, IDD3N pattern.
NOTE 2 For 3DS, all banks of all “non-target” logical ranks are Idd2N condition.

11.4 IDD2N, IDD3N Pattern

Executes DESELECT commands while exercising all command/address pins in a predefined pattern. All notes apply to entire table.

Table 314 — IDD2N, IDDQ2N, IPP2N, IDD3N, IDDQ3N, IPP3N

Sequence	Command	CS	C/A [13:0]
0	DES	H	0x0000
1	DES	H	0x3FFF
2	DES	H	0x3FFF
3	DES	H	0x3FFF

NOTE 1 Data is pulled to VDDQ
NOTE 2 DQS_t and DQS_c are pulled to VDDQ
NOTE 3 Command / Address ODT is disabled
NOTE 4 Repeat sequence 0 through 3.
NOTE 5 All banks of all logical ranks mimic the same test condition.

11.5 IDD2NT, IDDQ2NT, IPP2NT Pattern

Executes Non-Target WRITE commands simulating Rank to Rank timing while exercising all C/A bits. Notes 3-6 apply to entire table.

Table 315 — IDD2NT, IDDQ2NT, IPP2NT

Sequence	Command	CS_n	C/A [13:0]	Special Instructions
0	WRITE [1]	L	0x002D	All valid C/A inputs to VSS
		L	0x0000	
1	DES	H	Toggling [2]	Repeat sequence to meet 1*tCCD_S_WR (min), truncate if required
2	WRITE[1]	L	0x3FED	All valid C/A inputs to VDDQ
		L	0x3FFF	
3	DES	H	Toggling [2]	Repeat sequence to meet 1*tCCD_S_WR (min), truncate if required

NOTE 1 WRITE with CS_n=L on both cycles indicated a non-target WRITE.
NOTE 2 Utilize DESELECTs between commands while toggling C/A bits per the 4-cycle sequence defined in the IDD2N, IDD3N pattern.
NOTE 3 Time between Non-Target WRITEs reflect 1 * tCCD_S_WR (min) for one ranks.
NOTE 4 DQ signals are VDDQ.
NOTE 5 DQS_t, DQS_c are VDDQ.
NOTE 6 Repeat 0 through 3.

11.6 IDD4R, IDDQ4R, IPP4R Pattern

Executes READ commands with tightest timing possible while exercising all Bank, Bank Group and CID addresses. Notes 2-9 apply to entire table

Table 316 — IDD4R, IDDQ4R, IPP4R

Sub-Loop	Sequence	Command	CS_n	C/A [13:0]	Column Address [10:0]	BA [1:0]	BG [2:0]	CID [3:0]	Data Burst (BL=16)	Special Instructions
0	0	READ	L H	-	0x000	0x00	0x0	0x0	Pattern A	All “Valid” inputs = VDDQ
	1	DES	H	Toggling[1]	-					Repeat sequence to satisfy tCCD_S(min), truncate if required
1	2	READ	L H	-	0x3F0	0x00	0x1	0x0	Pattern B	All “Valid” inputs = VDDQ
	3	DES	H	Toggling[1]	-					Repeat sequence to satisfy tCCD_S(min), truncate if required
2	4-5	Repeat sub-loop 0, use BG[2:0]=0x2 instead								
3	6-7	Repeat sub-loop 1, use BG[2:0]=0x3 instead								
4	8-9	Repeat sub-loop 0, use BG[2:0]=0x4 instead								skip for x16
5	10-11	Repeat sub-loop 1, use BG[2:0]=0x5 instead								skip for x16
6	12-13	Repeat sub-loop 0, use BG[2:0]=0x6 instead								skip for x16
7	14-15	Repeat sub-loop 1, use BG[2:0]=0x7 instead								skip for x16
8	16	READ	L H	-	0x3F0	0x00	0x0	0x0	Pattern B	All “Valid” inputs = VDDQ
	17	DES	H	Toggling[1]						Repeat sequence to satisfy tCCD_S(min), truncate if required
9	18	READ	L H	-	0x000	0x00	0x1	0x0	Pattern A	All “Valid” inputs = VDDQ
	19	DES	H	Toggling[1]						Repeat sequence to satisfy tCCD_S(min), truncate if required
10	20-21	Repeat sub-loop 8, use BG[2:0]=0x2 instead								
11	22-23	Repeat sub-loop 9, use BG[2:0]=0x3 instead								
12	24-25	Repeat sub-loop 8, use BG[2:0]=0x4 instead								skip for x16
13	26-27	Repeat sub-loop 9, use BG[2:0]=0x5 instead								skip for x16
14	28-29	Repeat sub-loop 8, use BG[2:0]=0x6 instead								skip for x16
15	30-31	Repeat sub-loop 9, use BG[2:0]=0x7 instead								skip for x16
16-31	32-33	Repeat sub-loops 0-15, use BA[1:0]=0x1 instead								
32-47	34-35	Repeat sub-loops 0-15, use BA[1:0]=0x2 instead								
48-63	36-37	Repeat sub-loops 0-15, use BA[1:0]=0x3 instead								
...	...	Repeat sub-loops 0-63 for each 3DS logical rank, if applicable								CID[2:0]=0x1-0x7

NOTE 1 Utilize DESELECTs between commands while toggling all C/A bits per the 4-cycle sequence defined in the IDD2N, IDD3N pattern.
NOTE 2 READs performed with Auto Precharge = H, Burst Chop = H.
NOTE 3 Row address is set to 0x0000
NOTE 4 Data reflects burst length of 16.
NOTE 5 Data Pattern A for x4: 0x0, 0xF, 0xF, 0x0, 0x0, 0xF, 0x0, 0xF, 0xF, 0x0, 0x0, 0xF, 0x0, 0xF, 0x0, 0xF.
NOTE 6 Data Pattern B for x4: 0xF, 0x0, 0x0, 0xF, 0xF, 0x0, 0xF, 0x0, 0x0, 0xF, 0xF, 0x0, 0xF, 0x0, 0xF, 0x0.
NOTE 7 Data Pattern for x8 each beat will reflect two like nibbles (Data Pattern A = 0x00, 0xFF, 0xFF...).
NOTE 8 Data Pattern for x16 each beat will reflect two like bytes (Data Pattern A = 0x0000, 0xFFFF, 0xFFFF...).
NOTE 9 Where C/A column is not populated, refer to command truth table, column address, BA, BG, and CID for the C/A state

11.7 IDD4W, IDDQ4W, IPP4W Pattern

Executes WRITE commands with tightest timing possible while exercising all Bank, Bank Group and ~~CDI~~ CID addresses. Notes 2-6 apply to entire table.

Table 317 — IDD4W, IDDQ4W, IPP4W

Sub-Loop	Sequence	Command	CS_n	C/A [13:0]	Column Address [10:0]	BA [1:0]	BG [2:0]	CID [3:0]	Data Burst (BL=16)	Special Instructions
0	0	WRITE	L H	-	0x000	0x00	0x0	0x0	Pattern A	All "Valid" inputs = VDDQ
	1	DES	H	Toggling[1]	-					Repeat sequence to satisfy tCCD_S_WR (min), truncate if required
1	2	WRITE	L H	-	0x3F0	0x00	0x1	0x0	Pattern B	All "Valid" inputs = VDDQ
	3	DES	H	Toggling[1]	-					Repeat sequence to satisfy tCCD_S_WR (min), truncate if required
2	4-5	Repeat sub-loop 0, use BG[2:0]=0x2 instead								
3	6-7	Repeat sub-loop 1, use BG[2:0]=0x3 instead								
4	8-9	Repeat sub-loop 0, use BG[2:0]=0x4 instead								skip for x16
5	10-11	Repeat sub-loop 1, use BG[2:0]=0x5 instead								skip for x16
6	12-13	Repeat sub-loop 0, use BG[2:0]=0x6 instead								skip for x16
7	14-15	Repeat sub-loop 1, use BG[2:0]=0x7 instead								skip for x16
8	16	WRITE	L H	-	0x3F0	0x00	0x0	0x0	Pattern B	All "Valid" inputs = VDDQ
	17	DES	H	Toggling[1]						Repeat sequence to satisfy tCCD_S_WR (min), truncate if required
9	18	WRITE	L H	-	0x000	0x00	0x1	0x0	Pattern A	All "Valid" inputs = VDDQ
	19	DES	H	Toggling[1]						Repeat sequence to satisfy tCCD_S_WR (min), truncate if required
10	20-21	Repeat sub-loop 8, use BG[2:0]=0x2 instead								
11	22-23	Repeat sub-loop 9, use BG[2:0]=0x3 instead								
12	24-25	Repeat sub-loop 8, use BG[2:0]=0x4 instead								skip for x16
13	26-27	Repeat sub-loop 9, use BG[2:0]=0x5 instead								skip for x16
14	28-29	Repeat sub-loop 8, use BG[2:0]=0x6 instead								skip for x16
15	30-31	Repeat sub-loop 9, use BG[2:0]=0x7 instead								skip for x16
16-31	32-33	Repeat sub-loops 0-15, use BA[1:0]=0x1 instead								
32-47	34-35	Repeat sub-loops 0-15, use BA[1:0]=0x2 instead								
48-63	36-37	Repeat sub-loops 0-15, use BA[1:0]=0x3 instead								
...	...	Repeat sub-loops 0-63 for each 3DS logical rank, if applicable								CID[2:0]=0x1-0x7

NOTE 1 Utilize DESELECTs between commands per the 4-cycle sequence defined in the IDD2N, IDD3N pattern.
NOTE 2 WRITEs performed with Auto Precharge = H, Burst Chop = H.
NOTE 3 Row address is set to 0x0000.
NOTE 4 Data reflects burst length of 16.
NOTE 5 Refer to IDD4R measurement loop table for data pattern definition.
NOTE 6 Where C/A column is not populated, refer to command truth table, column address, BA, BG, and CID for the C/A state.

11.8 IDD5B, IDDQ5B, IPP5B, IDD5F, IDDQ5F, IPP5F Pattern

Executes Refresh (all Banks) command at minimum tRFC. Notes 3-6 apply to entire table.

Table 318 — IDD5B, IDD5B, IDDQ5B, IPP5B, IDD5F, IDDQ5F, IPP5F

Sequence	Command	CS	C/A [13:0]	CA[9:8]	CID [2:0]	Special Instructions
0	REFab	L	-	H	0x0	All “valid” inputs = VDDQ
1	DES	H	Toggling[1]	-	-	Repeat sequence to satisfy tRFC(min)2, truncate if required
2	REFab	L	-	H	0x0	All “valid” inputs = VDDQ
3	DES	H	Toggling[1]	-	-	Repeat sequence to satisfy tRFC(min)2, truncate if required
...	Repeat sequence 0-3 for each 3DS logical rank, if applicable					CID[2:0]=0x1-0x7

NOTE 1 Utilize DESELECTs between commands per the 4-cycle sequence defined in the IDD2N, IDD3N pattern.
NOTE 2 For IDD5B, use tRFC1(min). For IDD5F, use tRFC2(min).
NOTE 3 DQ signals are VDDQ.
NOTE 4 All banks of all “non-target” logical ranks are Idd2N condition.
NOTE 5 Where C/A[13:0] column is not populated, refer to command truth table, CA[9:8], and CID columns for the C/A state.
NOTE 6 Must set CA8=H on REFab commands to indicate 1X refresh rate on devices that support RIR.

11.9 IDD5C, IDDQ5C and IPP5C Patterns

Executes Refresh (Same Bank) command at minimum tRFCsb. Notes 2-5 apply to entire table.

Table 319 — IDD5C, IDDQ5C, IPP5C

Sequence	Command	CS	C/A [13:0]	CA[9:8]	BA [1:0]	CID [2:0]	Special Instructions
0	REFsb	L	-	H	0x0	0x0	
1	DES	H	Toggling[1]	-			Repeat sequence to satisfy tRFCsb(min), truncate if required
2	REFsb	L	-	H	0x1	0x0	
3	DES	H	Toggling[1]	-			Repeat sequence to satisfy tRFCsb(min), truncate if required
4	REFsb	L	-	H	0x2	0x0	
5	DES	H	Toggling[1]	-			Repeat sequence to satisfy tRFCsb(min), truncate if required
6	REFsb	L	-	H	0x3	0x0	
7	DES	H	Toggling[1]	-			Repeat sequence to satisfy tRFCsb(min), truncate if required
...	Repeat sequence 0-7 for each 3DS logical rank, if applicable						CID[2:0]=0x1-0x7

NOTE 1 Utilize DESELECTs between commands per the 4-cycle sequence defined in the IDD2N, IDD3N pattern.
NOTE 2 DQ signals are VDDQ.
NOTE 3 All banks of all “non-target” logical ranks are Idd2N condition.
NOTE 4 Where C/A[13:0] column is not populated, refer to command truth table, CA[9:8], and CID columns for the C/A state.
NOTE 5 All banks of all “non-target” logical ranks are Idd2N condition.

11.10 IDD6N, IDDQ6N, IPP6N, IDD6E, IDDQ6E, IPP6E, IDD6R, IDDQ6R, IPP6R Pattern

All notes apply to entire table.

Table 320 — IDD6N, IDDQ6N, IPP6N, IDD6E, IDDQ6E, IPP6E

Sequence	Command	Clock	CS	C/A [13:0]	Special Instructions
0	SRE	Valid	L	0x3BF7	Clocks must be valid tCKLCS(min) time
1	DES	Valid	H	0x3FFF	Repeat sequence to satisfy tCPDED(min), truncate if required
2	All C/A=H	Valid	L	0x3FFF	
3	All C/A = H	CK_t = CK_c = H	L	0x3FFF	Repeat sequence indefinitely
NOTE 1 Data is pulled to VDDQ NOTE 2 DQS_t and DQS_c are pulled to VDDQ NOTE 3 For 3DS, all banks of all logical ranks mimic the same test condition.					

11.11 IDD7, IDDQ7 and IPP7 Patterns

Executes ACTVATE, READ/A commands with tightest timing possible while exercising all Bank, Bank Group and CID addresses. Notes 2-6 apply to entire table.

Table 321 — IDD7, IDD7Q, IPP7

Sub-Loop	Sequence	Command	CS	C/A [13:0]	Row Address [17:0]	Column Address [10:0]	BA [1:0]	BG [2:0]	CID [2:0]	Data Burst (BL=16)	Special Instructions
0	0	ACT	L H	-	0x00000	-	0x0	0x0	0x0	-	
	1	DES	H	Toggling[1]							Repeat sequence to satisfy tRRD_S(min), tFAW(min), and tRCD(min). Truncate if required.
1	2	ACT	L H	-	0x03FFF	-	0x0	0x1	0x0	-	
	3	DES	H	Toggling[1]							Repeat sequence to satisfy tRRD_S(min), tFAW(min), and tRCD(min). Truncate if required.
2	4-5	Repeat sub-loop 0, use BG[2:0]=0x2 instead									
3	6-7	Repeat sub-loop 1, use BG[2:0]=0x3 instead									
4	8-9	Repeat sub-loop 0, use BG[2:0]=0x4 instead									skip for x16
5	10-11	Repeat sub-loop 1, use BG[2:0]=0x5 instead									skip for x16
6	12-13	Repeat sub-loop 0, use BG[2:0]=0x6 instead									skip for x16
7	14-15	Repeat sub-loop 1, use BG[2:0]=0x7 instead									skip for x16
8	16	RDA	L H	-	-	0x3F0	0x0	0x0	0x0	Pattern A	
	17	ACT	L H	-	0x00000	-	0x1	0x0	0x0	-	
	18	DES	H	Toggling[1]							Repeat sequence to satisfy tCCD_S(min), tFAW(min), and tRCD(min). Truncate if required.
9	19	RDA	L H	-	-	0x000	0x0	0x1	0x0	Pattern B	
	20	ACT	L H	-	0x03FFF	-	0x1	0x1	0x0	-	
	21	DES	H	Toggling[1]							Repeat sequence to satisfy tCCD_S(min), tFAW(min), and tRCD(min). Truncate if required.
10	22-24	Repeat sub-loop 8, use BG[2:0]=0x2 instead									
11	25-27	Repeat sub-loop 9, use BG[2:0]=0x3 instead									
12	28-30	Repeat sub-loop 8, use BG[2:0]=0x4 instead									skip for x16
13	31-33	Repeat sub-loop 9, use BG[2:0]=0x5 instead									skip for x16
14	34-36	Repeat sub-loop 8, use BG[2:0]=0x6 instead									skip for x16
15	37-39	Repeat sub-loop 9, use BG[2:0]=0x7 instead									skip for x16
16-23	40-64	Repeat sub-loops 8-15, use BA[1:0]=0x1 for the RDA and BA[1:0]=0x2 for the ACT									
24-31	65-89	Repeat sub-loops 8-15, use BA[1:0]=0x2 for the RDA and BA[1:0]=0x3 for the ACT									
32-39	90-114	Repeat sub-loops 8-15, use BA[1:0]=0x3 for the RDA and BA[1:0]=0x0 for the ACT									
...	...	Repeat sub-loops 0-18 for each 3DS logical rank, if applicable									CID[2:0]=0x1-0x7

NOTE 1 Utilize DESELECTs between commands per the 4-cycle sequence defined in the IDD2N, IDD3N pattern.
NOTE 2 READs performed with Auto Precharge = L, Burst Chop = H.
NOTE 3 x8 or x16 may have different Bank or Bank Group Address.
NOTE 4 Data reflects burst length of 16.
NOTE 5 Refer to IDD4R measurement loop table for data pattern definition
NOTE 6 For 3DS, all banks of all “non-target” logical ranks are Idd2N condition

12 Input/Output Capacitance

Table 322 — Silicon Pad I/O Capacitance DDR5-3200 to DDR5-6400

Symbol	Parameter	DDR5-3200/ 3600/4000		DDR5-4400/ 4800		DDR5-5200/5600		DDR5-6000/6400		Unit	NOTE
		Min	Max	Min	Max	Min	Max	Min	Max		
C_{IO}	Input/output capacitance (DQ, DM_n, DQS_t, DQS_c, TDQS_t, TDQS_c)	0.3	0.9	0.3	0.9	0.3	0.85	0.3	0.8	pF	1, 2
C_{DIO}	Input/output capacitance delta (DQ, DM_c)	-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1	pF	1, 2, 8
C_{DDQS}	Input/output capacitance delta (DQS_t and DQS_c)		0.04		0.04		0.04		0.04	pF	1, 2, 4
C_{CK}	Input capacitance (CK_t and CK_c)	0.2	0.7	0.2	0.6	0.2	0.55	0.2	0.5	pF	1, 2
C_{DCK}	Input capacitance delta (CK_t and CK_c)		0.05		0.05		0.05		0.05	pF	1, 2, 3
C_I	Input capacitance (CS_n & CA[13:0] pins only)	0.2	0.7	0.2	0.6	0.2	(C_I Option1) 0.55 (C_I Option2) 0.5	0.2	0.5	pF	1, 2, 5, 12
$C_{DI\_CS\_n}$	Input capacitance delta (CS_n pin only)	-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1	pF	1, 2, 6
C_{DI_CA}	Input capacitance delta (CA[13:0] pins only)	-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1	pF	1, 2, 7
C_{ALERT}	Input/output capacitance of ALERT	0.3	1.5	0.3	1.5	0.3	1.5	0.3	1.5	pF	1, 2
$C_{Loopback}$	Input/output capacitance of Loopback (LBDQ, LBDQS)	0.3	1.0	0.3	1.0	0.3	1.0	0.3	1.0	pF	1, 2
CTEN	Input capacitance of TEN	0.2	2.3	0.2	2.3	0.2	2.3	0.2	2.3	pF	1, 2, 9
C_{ZQ}	Input capacitance of ZQ	-	5	-	5	-	5	-	5	pF	1, 2, 11
C_{STRAP}	Input capacitance of MIR, CAI, CA_ODT pins	-	10	-	10	-	10	-	10	pF	1, 2, 10

NOTE 1 This parameter is not subject to production test. This parameter is measured by using vendor specific measurement methodology.
NOTE 2 This parameter applies to monolithic devices only; stacked/dual-die devices are not covered here
NOTE 3 Absolute value CIO(CK_t)-CIO(CK_c)
NOTE 4 Absolute value of CIO(DQS_t)-CIO(DQS_c)
NOTE 5 CI applies to CS_n and CA[13:0]
NOTE 6 CDI_CS_n = CI(CS_n)-0.5*(CI(CK_t)+CI(CK_c))
NOTE 7 CDI_CA = CI(CA[13:0])-0.5*(CI(CK_t)+CI(CK_c))
NOTE 8 CDIO = CIO(DQ,DM)-Avg(CIO(DQ,DM))
NOTE 9 TEN pin may be DRAM internally pulled low through a weak pull-down resistor to VSS. In this case CTEN might not be valid and system shall verify TEN signal with Vendor specific information.
NOTE 10 MIR, CAI, and CA_ODT are strap pins used to configure module or point to point use cases depending on power, signal integrity, and termination requirements. No active AC signaling requirements defined for these pins.
NOTE 11 Maximum external load capacitance on ZQ pin: 25 pF. The ZQ functionality / accuracy with the max capacitive load is characterized.
NOTE 12 CI Options are incorporated VcIVW in Table 199 in Section 8.2.

12 Input/Output Capacitance (cont'd)

Table 323 — Silicon Pad I/O Capacitance DDR5-6800 to DDR5-8800

Symbol	Parameter	DDR5-6800/ 7200		DDR5-7600/8000		DDR5-8400/8800		Unit	NOTE
		Min	Max	Min	Max	Min	Max		
C_{IO}	Input/output capacitance (DQ, DM_n, DQS_t, DQS_c, TDQS_t, TDQS_c)	0.3	0.8	0.3	0.75	0.3	0.70	pF	1, 2
C_{DIO}	Input/output capacitance delta (DQ, DM_c)	-0.1	0.1	-0.1	0.1	-0.1	0.1	pF	1, 2, 8
C_{DDQS}	Input/output capacitance delta (DQS_t and DQS_c)		0.04		0.04		0.04	pF	1, 2, 4
C_{CK}	Input capacitance (CK_t and CK_c)	0.2	0.5	0.2	0.45	0.2	0.40	pF	1, 2
C_{DCK}	Input capacitance delta (CK_t and CK_c)		0.05		0.05		0.05	pF	1, 2, 3
C_I	Input capacitance (CS_n & CA[13:0] pins only)	0.2	0.5	0.2	0.45	0.2	0.40	pF	1, 2, 5
$C_{DI\_CS\_n}$	Input capacitance delta (CS_n pin only)	-0.1	0.1	-0.1	0.1	-0.1	0.1	pF	1, 2, 6
C_{DI_CA}	Input capacitance delta (CA[13:0] pins only)	-0.1	0.1	-0.1	0.1	-0.1	0.1	pF	1, 2, 7
C_{ALERT}	Input/output capacitance of ALERT	0.3	1.5	0.3	1.5	0.3	1.5	pF	1, 2
$C_{Loopback}$	Input/output capacitance of Loop-back (LBDQ, LBDQS)	0.3	1.0	0.3	1.0	0.3	1.0	pF	1, 2
CTEN	Input capacitance of TEN	0.2	2.3	0.2	2.3	0.2	2.3	pF	1, 2, 9
C_{ZQ}	Input capacitance of ZQ	-	5	-	5	-	5	pF	1, 2, 11
C_{STRAP}	Input capacitance of MIR, CAI, CA_ODT pins	-	10	-	10	-	10	pF	1, 2, 10

NOTE 1 This parameter is not subject to production test. This parameter is measured by using vendor specific measurement methodology.
NOTE 2 This parameter applies to monolithic devices only; stacked/dual-die devices are not covered here
NOTE 3 Absolute value CIO(CK_t)-CIO(CK_c)
NOTE 4 Absolute value of CIO(DQS_t)-CIO(DQS_c)
NOTE 5 CI applies to CS_n and CA[13:0]
NOTE 6 CDI_CS_n = CI(CS_n)-0.5*(CI(CK_t)+CI(CK_c))
NOTE 7 CDI_CA = CI(CA[13:0])-0.5*(CI(CK_t)+CI(CK_c))
NOTE 8 CDIO = CIO(DQ,DM)-Avg(CIO(DQ,DM))
NOTE 9 TEN pin may be DRAM internally pulled low through a weak pull-down resistor to VSS. In this case CTEN might not be valid and system shall verify TEN signal with Vendor specific information.
NOTE 10 MIR, CAI, and CA_ODT are strap pins used to configure module or point to point use cases depending on power, signal integrity, and termination requirements. No active AC signaling requirements defined for these pins.
NOTE 11 Maximum external load capacitance on ZQ pin: 25 pF. The ZQ functionality / accuracy with the max capacitive load is characterized.

12 Input/Output Capacitance (cont'd)

Table 324 — Silicon Pad I/O Capacitance DDR5 DDP 3200 to 6400

Symbol	Parameter	DDR5-3200/ 3600/4000 DDP		DDR5-4400/ 4800 DDP		DDR5-5200/5600 DDP		DDR5- 6000/6400 DDP		Unit	NOTE
		Min	Max	Min	Max	Min	Max	Min	Max		
C_{IO}	Input/output capacitance (DQ, DM_n, DQS_t, DQS_c, TDQS_t, TDQS_c)	0.6	1.8	0.6	1.8	0.6	1.7	0.6	1.6	pF	1, 2
C_{DIO}	Input/output capacitance delta (DQ, DM_c)	-0.2	0.2	-0.2	0.2	-0.2	0.2	-0.2	0.2	pF	1, 2, 9
C_{DDQS}	Input/output capacitance delta (DQS_t and DQS_c)		0.08		0.08		0.08		0.08	pF	1, 2, 4
C_{CK}	Input capacitance (CK_t and CK_c)	0.4	1.4	0.4	1.2	0.4	1.1	0.4	1.0	pF	1, 2
C_{DCK}	Input capacitance delta (CK_t and CK_c)		0.1		0.1		0.1		0.1	pF	1, 2, 3
C_{I_CA}	Input capacitance (CA[13:0] pins)	0.4	1.4	0.4	1.2	0.4	1.1	0.4	1.0	pF	1, 2, 5, 12
$C_{I\_CS\_n}$	Input capacitance (CS_n pin)	0.2	0.7	0.2	0.6	0.2	0.55	0.2	0.5	pF	1, 2, 6, 12
$C_{DI\_CS\_n}$	Input capacitance delta (CS_n pin only)	-0.2	0.2	-0.2	0.2	-0.2	0.2	-0.2	0.2	pF	1, 2, 7
C_{DI_CA}	Input capacitance delta (CA[13:0] pins only)	-0.2	0.2	-0.2	0.2	-0.2	0.2	-0.2	0.2	pF	1, 2, 8
C_{ALERT}	Input/output capacitance of ALERT	0.6	3.0	0.6	3.0	0.6	3.0	0.6	3.0	pF	1, 2
$C_{Loopback}$	Input/output capacitance of Loopback (LBDQ, LBDQS)	0.6	2.0	0.6	2.0	0.6	2.0	0.6	2.0	pF	1, 2
C_{ZQ}	Input capacitance of ZQ	-	5	-	5	-	5	-	5	pF	1, 2, 11, 13
C_{STRAP}	Input capacitance of MIR, CAI, CA_ODT pins	-	20	-	20	-	20	-	20	pF	1, 2, 10

NOTE 1 This parameter is not subject to production test. This parameter is measured by using vendor specific measurement methodology.

NOTE 2 This parameter applies to DDP only, but does not include RDL layer portion. RDL layer portion is covered in package electrical specification.

NOTE 3 Absolute value C_{CK}(CK_t)-C_{CK}(CK_c).

NOTE 4 Absolute value of C_{IO}(DQS_t)-C_{IO}(DQS_c).

NOTE 5 C_{I_CA} applies to CA[13:0] pins.

NOTE 6 $C_{I\_CS\_n}$ applies to CS_n pin and CS1_n pin individually.

NOTE 7 $C_{DI\_CS\_n}$ = 2*$C_{I\_CS\_n}$-0.5*(C_{CK}(CK_t)+C_{CK}(CK_c)).

NOTE 8 C_{DI_CA} = $C_{I\_CA[13:0]}$-0.5*(C_{CK}(CK_t)+C_{CK}(CK_c)).

NOTE 9 C_{DIO} = C_{IO}(DQ,DM)-Avg(C_{IO}(DQ,DM)).

NOTE 10 MIR and CA_ODT are strap pins used to configure module or point to point use cases depending on power, signal integrity, and termination requirements. No active AC signaling requirements defined for these pins.

NOTE 11 Maximum external load capacitance on ZQ pin: 25 pF. The ZQ functionality / accuracy with the max capacitive load is characterized.

NOTE 12 C_{I_CA}, $C_{I\_CS\_n}$ Options are incorporated VcIVW in Section 8.2.

NOTE 13 C_{ZQ} applies to ZQ pin and ZQ1 pin individually.

12 Input/Output Capacitance (cont'd)

Table 325 — DRAM Package Electrical Specifications (x4/x8)

Parameter	Symbol	DDR5-3200 to DDR5-4800		DDR5-5200 to DDR5-6400		DDR5-6800 to 8800		Unit	NOTE
		Min	Max	Min	Max	Min	Max		
Input/output Zpkg	Z_{pkg_DQ}	45	75	50	75	50	75	Ω	1, 2, 4, 5, 10
Input/output Pkg Delay	$T_{pkg\_delay\_DQ}$	10	35	10	32	10	32	ps	1, 3, 4, 5, 10
DQS_t, DQS_c Zpkg	Z_{pkg_DQS}	45	75	50	75	50	75	Ω	1, 2, 5,10, 12
DQS_t, DQS_c Pkg Delay	$T_{pkg\_delay\_DQS}$	10	35	10	32	10	32	ps	1, 3, 5, 10, 12
Delta Zpkg DQS_t, DQS_c	DZ_{pkg_DQS}	-	5	-	5	-	5	Ω	1, 2, 5, 7, 10
Delta Delay DQS_t, DQS_c	$DT_{pkg\_delay\_DQS}$	-	2	-	2	-	2	ps	1, 3, 5, 7, 10
Input- CTRL pins Zpkg	Z_{pkg_CTRL}	45	75	45	75	45	75	Ω	1, 2, 5, 9, 10
Input- CTRL pins Pkg Delay	$T_{pkg\_delay\_CTRL}$	10	35	10	28	10	28	ps	1, 3, 5, 9, 10
Input- CMD ADD pins Zpkg	Z_{pkg_CA}	45	75	45	75	45	75	Ω	1, 2, 5, 8, 10
Input- CMD ADD pins Pkg Delay	$T_{pkg\_delay\_CA}$	10	35	10	(Option 1) 35 (Option 2) 32 (Option 3) 29	10	(Option 1) 35 (Option 2) 32 (Option 3) 29	ps	1, 3, 5, 8, 10
CK_t & CK_c Zpkg	Z_{pkg_CK}	45	75	45	75	45	75	Ω	1, 2, 5, 10
CK_t & CK_c Pkg Delay	$T_{pkg\_delay\_CK}$	10	30	10	25	10	25	ps	1, 3, 5, 10
Delta Zpkg CK_t & CK_c	$DZ_{pkg\_delay\_CK}$	-	5	-	5	-	5	Ω	1, 2, 5, 6, 10
Delta Delay CK_t & CK_c	$DT_{pkg\_delay\_CK}$	-	2	-	2	-	2	ps	1, 3, 5, 6, 10
ALERT Zpkg	Z_{pkg_ALERT}	45	75	45	75	45	75	Ω	1, 2, 5, 10
ALERT Delay	$T_{pkg\_delay\_ALERT}$	10	60	10	60	10	60	ps	1, 3, 5, 10
Loopback Zpkg	$Z_{pkg\_Loopback}$	45	75	45	75	45	75	Ω	1, 2, 5, 10, 11
Loopback Delay	$T_{pkg\_delay\_Loopback}$	10	60	10	60	10	60	ps	1, 3, 5, 10, 11

NOTE 1 This parameter is not subject to production test.
NOTE 2 This parameter is measured by using vendor specific measurement methodology to calculate the average Z_{pkg_xx} over the interval $T_{pkg\_delay\_xx}$
NOTE 3 This parameter is measured by using vendor specific measurement methodology.
NOTE 4 Z_{pkg_DQ} and $T_{pkg\_delay\_DQ}$ applies to DQ, DM
NOTE 5 This parameter applies to monolithic devices only; stacked/dual-die devices are not covered here
NOTE 6 Absolute value of $Z_{pkg\_CK\_t}$ - $Z_{pkg\_CK\_c}$ for impedance(Z) or absolute value of $T_{pkg\_delay\_CK\_t}$ - $T_{pkg\_delay\_CK\_c}$ for delay (T_{pkg_delay}).
NOTE 7 Absolute value of Z_{pkg}(DQS_t)-Z_{pkg}(DQS_c) for impedance(Z) or absolute value of $T_{pkg\_delay\_DQS\_t}$ - $T_{pkg\_delay\_DQS\_c}$ for delay (T_{pkg_delay})
NOTE 8 Z_{pkg_CA} and $T_{pkg\_delay\_CA}$ applies to CA[13:0]
NOTE 9 Z_{pkg_CTRL} and $T_{pkg\_delay\_CTRL}$ applies to CS_n
NOTE 10 Package implementations shall meet spec if the designed Z_{pkg} and T_{pkg_delay} fall within the ranges shown.
NOTE 11 $Z_{pkg\_Loopback}$ and $T_{pkg\_delay\_Loopback}$ applies to LBDQ and LBDQS.
NOTE 12 Z_{pkg_DQS} and $T_{pkg\_delay\_DQS}$ applies to DQS_t & DQS_c, TDQS_t and TDQS_c.

12 Input/Output Capacitance (cont'd)

Table 326 — DRAM Package Electrical Specifications (x16)

Parameter	Symbol	DDR5-3200 to DDR5-4800		DDR5-5200 to DDR5-8800		Unit	NOTE
		Min	Max	Min	Max		
Input/output Zpkg	Z_{pkg_DQ}	45	75	45	75	Ω	1, 2, 4, 5, 10
Input/output Pkg Delay	$T_{pkg\_delay\_DQ}$	10	40	10	38	ps	1, 3, 4, 5, 10
DQS_t, DQS_c Zpkg	Z_{pkg_DQS}	45	75	45	75	Ω	1, 2, 5, 10, 12
DQS_t, DQS_c Pkg Delay	$T_{pkg\_delay\_DQS}$	10	40	10	38	ps	1, 3, 5, 10, 12
Delta Zpkg DQS_t, DQS_c	DZ_{pkg_DQS}	-	5	-	5	Ω	1, 2, 5, 7, 10
Delta Delay DQS_t, DQS_c	$DT_{pkg\_delay\_DQS}$	-	2	-	2	ps	1, 3, 5, 7, 10
Input- CTRL pins Zpkg	Z_{pkg_CTRL}	45	75	45	75	Ω	1, 2, 5, 9, 10
Input- CTRL pins Pkg Delay	$T_{pkg\_delay\_CTRL}$	10	40	10	40	ps	1, 3, 5, 9, 10
Input- CMD ADD pins Zpkg	Z_{pkg_CA}	45	75	45	75	Ω	1, 2, 5, 8, 10
Input- CMD ADD pins Pkg Delay	$T_{pkg\_delay\_CA}$	10	45	10	44	ps	1, 3, 5, 8, 10
CK_t & CK_c Zpkg	Z_{pkg_CK}	45	75	45	75	Ω	1, 2, 5, 10
CK_t & CK_c Pkg Delay	$T_{pkg\_delay\_CK}$	10	45	10	43	ps	1, 3, 5, 10
Delta Zpkg CK_t & CK_c	$DZ_{pkg\_delay\_CK}$	-	5	-	5	Ω	1, 2, 5, 6, 10
Delta Delay CK_t & CK_c	$DT_{pkg\_delay\_CK}$	-	2	-	2	ps	1, 3, 5, 6, 10
ALERT Zpkg	Z_{pkg_ALERT}	45	75	45	75	Ω	1, 2, 5, 10
ALERT Delay	$T_{pkg\_delay\_ALERT}$	10	60	10	60	ps	1, 3, 5, 10
Loopback Zpkg	$Z_{pkg\_Loopback}$	45	75	45	75	Ω	1, 2, 5, 10, 11
Loopback Delay	$T_{pkg\_delay\_Loopback}$	10	60	10	60	ps	1, 3, 5, 10, 11

NOTE 1 This parameter is not subject to production test.
NOTE 2 This parameter is measured by using vendor specific measurement methodology.
NOTE 3 This parameter is measured by using vendor specific measurement methodology.
NOTE 4 Z_{pkg_DQ} and $T_{pkg\_delay\_DQ}$ applies to DQ, DM.
NOTE 5 This parameter applies to monolithic devices only; stacked/dual-die devices are not covered here
NOTE 6 Absolute value of $Z_{pkg\_CK\_t}$ - $Z_{pkg\_CK\_c}$ for impedance(Z) or absolute value of $T_{pkg\_delay\_CK\_t}$ - $T_{pkg\_delay\_CK\_c}$ for delay (T_{pkg_delay}).
NOTE 7 Absolute value of Z_{pkg}(DQS_t)-Z_{pkg}(DQS_c) for impedance(Z) or absolute value of $T_{pkg\_delay\_DQS\_t}$ - $T_{pkg\_delay\_DQS\_c}$ for delay (T_{pkg_delay})
NOTE 8 Z_{pkg_CA} and $T_{pkg\_delay\_CA}$ applies to CA[13:0]
NOTE 9 Z_{pkg_CA} and $T_{pkg\_delay\_CTRL}$ applies to CS_n
NOTE 10 Package implementations shall meet spec if the designed Z_{pkg} and T_{pkg_delay} fall within the ranges shown.
NOTE 11 $Z_{pkg\_Loopback}$ and $T_{pkg\_delay\_Loopback}$ applies to LBDQ and LBDQS.
NOTE 12 Z_{pkg_DQS} and $T_{pkg\_delay\_DQS}$ applies to DQS_t & DQS_c, TDQS_t & TDQS_c.

12 Input/Output Capacitance (cont'd)

Table 327 — DRAM DDP Package Electrical Specifications (x4/x8)

Parameter	Symbol	DDR5-3200 to DDR5-6400		Unit	NOTE
		Min	Max		
Input/output Zpkg	Z_{pkg_DQ}	20	75	Ω	1, 2, 3, 4, 9
Input/output Pkg Delay	$T_{pkg\_delay\_DQ}$	25	100	ps	1, 2, 3, 4, 9
DQS_t, DQS_c Zpkg	Z_{pkg_DQS}	20	75	Ω	1, 2, 4, 9, 11
DQS_t, DQS_c Pkg Delay	$T_{pkg\_delay\_DQS}$	25	100	ps	1, 2, 4, 9, 11
Delta Zpkg DQS_t, DQS_c	DZ_{pkg_DQS}	-	5	Ω	1, 2, 4, 6, 9
Delta Delay DQS_t, DQS_c	$DT_{pkg\_delay\_DQS}$	-	2	ps	1, 2, 4, 6, 9
Input- CTRL pins Zpkg	Z_{pkg_CTRL}	20	75	Ω	1, 2, 4, 8, 9
Input- CTRL pins Pkg Delay	$T_{pkg\_delay\_CTRL}$	25	100	ps	1, 2, 4, 8, 9
Input- CMD ADD pins Zpkg	Z_{pkg_CA}	20	75	Ω	1, 2, 4, 7, 9
Input- CMD ADD pins Pkg Delay	$T_{pkg\_delay\_CA}$	25	100	ps	1, 2, 4, 7, 9
CK_t & CK_c Zpkg	Z_{pkg_CK}	20	75	Ω	1, 2, 4, 9
CK_t & CK_c Pkg Delay	$T_{pkg\_delay\_CK}$	25	100	ps	1, 2, 4, 9
Delta Zpkg CK_t & CK_c	$DZ_{pkg\_delay\_CK}$	-	5	Ω	1, 2, 4, 5, 9
Delta Delay CK_t & CK_c	$DT_{pkg\_delay\_CK}$	-	2	ps	1, 2, 4, 5, 9
ALERT Zpkg	Z_{pkg_ALERT}	20	75	Ω	1, 2, 4, 9
ALERT Delay	$T_{pkg\_delay\_ALERT}$	25	100	ps	1, 2, 4, 9
Loopback Zpkg	$Z_{pkg\_Loopback}$	20	75	Ω	1, 2, 4, 9, 10
Loopback Delay	$T_{pkg\_delay\_Loopback}$	25	100	ps	1, 2, 4, 9, 10

NOTE 1 This parameter is not subject to production test.
NOTE 2 This parameter is verified by design.
NOTE 3 Z_{pkg_DQ} and $T_{pkg\_delay\_DQ}$ applies to DQ, DM
NOTE 4 This parameter applies to DDPonly.
NOTE 5 Absolute value of $Z_{pkg\_CK\_t}$ - $Z_{pkg\_CK\_c}$ for impedance(Z) or absolute value of $T_{pkg\_delay\_CK\_t}$ - $T_{pkg\_delay\_CK\_c}$ for delay (T_{pkg_delay}).
NOTE 6 Absolute value of Z_{pkg}(DQS_t)-Z_{pkg}(DQS_c) for impedance(Z) or absolute value of $T_{pkg\_delay\_DQS\_t}$ - $T_{pkg\_delay\_DQS\_c}$ for delay (T_{pkg_delay})
NOTE 7 Z_{pkg_CA} and $T_{pkg\_delay\_CA}$ applies to CA[13:0]
NOTE 8 Z_{pkg_CTRL} and $T_{pkg\_delay\_CTRL}$ applies to CS_n and CS1_n
NOTE 9 Package implementations shall meet spec if the designed Z_{pkg} and T_{pkg_delay} fall within the ranges shown.
NOTE 10 $Z_{pkg\_Loopback}$ and $T_{pkg\_delay\_Loopback}$ applies to LBDQ and LBDQS.
NOTE 11 Z_{pkg_DQS} and $T_{pkg\_delay\_DQS}$ applies to DQS_t & DQS_c, TDQS_t & TDQS_c.

12.1 Electrostatic Discharge Sensitivity Characteristics

Table 328 — Electrostatic Discharge Sensitivity Characteristics

Parameter [1]	Symbol	Min	Max	Unit	Notes
Human body model (HBM)	ESDHBM	1000	-	V	2
Charged-device model (CDM)	ESD_{CDM}	250	-	V	3

NOTE 1 State-of-the-art basic ESD control measures have to be in place when handling devices
NOTE 2 Refer to ESDA / JEDEC Joint Standard JS-001 for measurement procedures.
NOTE 3 Refer to ESDA / JEDEC Joint Standard JS-002 f for measurement procedures

13 Electrical Characteristics and AC Timing

13.1 Reference Load for AC Timing and Output Slew Rate - No Ballot

Figure 258 represents the effective reference load of 50 ohms used in defining the relevant AC timing parameters of the device as well as output slew rate measurements.

Ron nominal of DQ, DQS_t and DQS_c drivers uses 34 ohms to specify the relevant AC timing parameter values of the device.

The maximum DC High level of Output signal = 1.0 * VDDQ,

The minimum DC Low level of Output signal = { 34 /(34 + 50) } *VDDQ = 0.4* VDDQ

The nominal reference level of an Output signal can be approximated by the following:

The center of maximum DC High and minimum DC Low = { (1 + 0.4) / 2 } * VDDQ = 0.7 * VDDQ

The actual reference level of Output signal might vary with driver Ron and reference load tolerances. Thus, the actual reference level or midpoint of an output signal is at the widest part of the output signal's eye. Prior to measuring AC parameters, the reference level of the verification tool should be set to an appropriate level.

It is not intended as a precise representation of any particular system environment or a depiction of the actual load presented by a production tester. System designers should use IBIS or other simulation tools to correlate the timing reference load to a system environment. Manufacturers correlate to their production test conditions, generally one or more coaxial transmission lines terminated at the tester electronics.

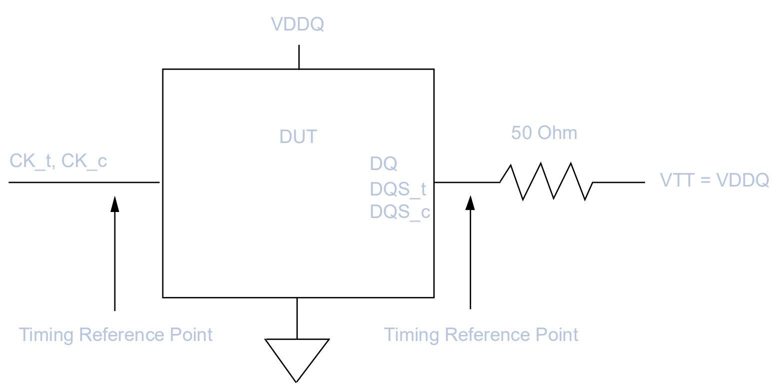


Figure 258 — Reference Load for AC Timing and Output Slew Rate

13.2 Rounding Definitions and Algorithms

Software algorithms for calculation of timing parameters are subject to rounding errors from many sources. For example, a system may use a memory clock with a nominal frequency of 2200 MHz (4400 MT/s) for the DDR5-4400 speed bin, which mathematically yields a nominal clock period tCK(AVG) of 0.454545... ns. In most cases, it is impossible to express all digits after the decimal point exactly, and rounding must be used. The DDR5 SDRAM specification establishes a minimum granularity for timing parameters of 1 ps.

Rules for rounding must be defined to allow optimization of device performance without violating device parameters. All timing parameters specified in the time domain (ns, ps, etc.) which must then be converted to the clock domain (nCK units) shall be defined to align with these rules. The key point is, minimum and maximum timing parameter values shall generally use the same rounding rules used to define tCK(AVG)min and tCK(AVG)max. The resulting rounding algorithms rely on results that are within correction factors of device testing and specification to avoid losing performance due to rounding errors.

These rules are:

- DEFINING TIMING PARAMETER VALUES: Minimum and maximum timing parameter values, including tCK(AVG)min and tCK(AVG)max, are rounded down and to be defined to 1 ps of accuracy in the DDR5 SDRAM specification based on the non-rounded nominal tCK(AVG) for a given speed bin. If the nominal timing parameter values require more than 1 ps of accuracy, they can be rounded down (faster) to the next 1 ps according to the rounding algorithms, and the DDR5 SDRAM is responsible for absorbing the resulting small parameter extensions. In other words, the DDR5 SDRAM specification only lists the nominal parameter values rounded down to the next 1 ps. For example, this extends the DDR5-4400 tCK(AVG)min definition to be exactly 0.454 ns which is slightly smaller (faster) than the nominal memory clock period of 0.454545... ns by less than 1 ps.
- CALCULATING THE REAL MINIMUM TIMING PARAMETER VALUES: For minimum timing parameters, other than tCK(AVG)min, to avoid losing performance due to additional erroneous nCKs and to calculate the true real minimum values, their nominal values listed in the DDR5 SDRAM specification must be reduced (faster) by the maximum %error (correction factor) used to define tCK(AVG)min. The DDR5 SDRAM is responsible for absorbing the resulting small parameter extensions. For example, tWRmin has a nominal value of 30.000 ns, however, applying the 0.30% correction factor allows a more aggressive timing (for example, 29.910 ns) to be supported, which allows the intended smaller (faster) nCK value to be maintained when rounding tCK(AVG)min down to the next 1 ps. Note, parameter values defined to be 0 ps do not need to be reduced by a correction factor, and therefore don't require these rounding algorithms.
- CALCULATING THE REAL MAXIMUM TIMING PARAMETER VALUES: For maximum timing parameters, including tCK(AVG)max, to avoid losing performance due to excluding erroneous nCKs and to calculate the true real maximum values, their nominal values listed in the DDR5 SDRAM specification must be increased (slower) by the maximum %error caused by rounding tCK(AVG) down to the next 1 ps. The DDR5 SDRAM is responsible for absorbing the resulting small parameter extensions. For example, tREFImax has a nominal value of 3.9 µs. And the DDR5-8400 speed bin mathematically yields a nominal clock period tCK(AVG) of 0.238095... ns, resulting in a theoretical maximum %error of 0.42% (0.239 ns/0.238 ns-100%). So the true real tREFImax value is 3.916386... µs (3.9 µs*0.239 ns/ 0.238 ns) for the DDR5-8400 speed bin.
- ROUNDING ALGORITHM FOR MINIMUM TIMING PARAMETER VALUES: Round down only integer number math is commonly used in the industry to calculate nCK values. This rounding algorithm for minimum timing parameters uses scaling by 1000 to allow use of integer math. Nominal minimum timing parameters like tWRmin, tRCDmin, etc. which are programmed in systems in numbers of clocks (nCK) but expressed in units of time(ps), are rounded down to the next 1ps, multiplied by the scaled inverse correction factor (1000-3=997) prior to division by the application memory clock period rounded down to the next 1ps, and adding 1 scaled by 1000 to that result effectively adds 1nCK. Division by 1000 undoes the scaling effects, resulting in a number of clocks as the final answer which has been effectively rounded up to the next 1 nCK by adding 1nCK in the previous step and then rounding down to the next 1 nCK. The caveat is, effectively adding 1 prior to rounding down is mostly equivalent to rounding up except when the result is equal to an integer, in which case the result won't be rounded down as intended, and therefore performance would be lost. To address this, the maximum 0.28% correction factor needed for 3600 MHz (0.277ns/0.2777777...ns-100%) operation has been increased slightly to 0.30% in this rounding algorithms. This accounts for all integer boundary conditions, except for the specific case when the nominal minimum timing parameter value is defined to be 0 ps. No rounding is required for parameter values equal to 0 ps. This rounding algorithm is not optimized for parameter values that are less than or equal to 0ps, and may result in unintended lost performance if used for parameter values that are less than or equal to 0 ps.

$$nCK = truncate\left[\frac{\left(\dfrac{truncate[\text{parameter\_nominal\_in\_ps}] \text{ x } 997}{truncate[\text{tCK(AVG)\_real\_in\_ps}]}\right) + 1000}{1000}\right]$$

13.2 Rounding Definitions and Algorithms (cont'd)

- ROUNDING ALGORITHM FOR MAXIMUM TIMING PARAMETER VALUES: Round down only integer number math is commonly used in the industry to calculate nCK values. This rounding algorithm is used for maximum timing parameters. Nominal maximum timing parameters like tREFImax, etc. which programmed in systems in numbers of clocks (nCK) but expressed in units of time (ps), are rounded down to the next 1 ps prior to division by the application memory clock period rounded down to the next 1 ps, resulting in a number of clocks as the final answer which is rounded down to the next 1 nCK. No rounding is required for parameter values equal to 0 ps, but this rounding algorithm can still be used for parameter values equal to 0ps and no performance will be lost. This rounding algorithm is not optimized for parameter values that are less than 0 ps (negative parameter values), and may result in violating the parameter specification if used for parameter values that are less than 0 ps.

$$nCK = truncate\left[\frac{truncate[\text{parameter\_nominal\_in\_ps}]}{truncate[tCKAVG)]}\right]$$

- CL ALGORITHM FOR STANDARD SPEED BINS: The math rounding algorithms shall be used for all timing parameters when converting from the time domain (ns, ps, etc.) to the clock domain (nCK units), except for when converting tAA to CL. If these rounding algorithms are used to convert tAA to CL, they'll return invalid CL's for some cases when down clocking (and the DIMM SPD CL Mask doesn't protect against all of these cases). The proper setting of CL shall be determined by the memory controller, either by using the speed bin tables, or by using the CL algorithm, or by some other means. Refer to the Speed Bins and Operations section for more information. Note, the CL algorithm replaces the need to use the DIMM SPD CL Mask.
- CL ALGORITHM FOR CUSTOM SPEED BINS: If the DDR5 SDRAM supports non-standard tCK, tAA, tRCD, and tRP speed bin timings, the CL algorithm will still only return valid CL's as defined in the speed bin tables, which may not be the intended CL's for non-standard speed bins. In these cases, the rounding algorithms may need to be used to convert tAA to CL, instead of the CL algorithm. The CL returned by the rounding algorithms shall be incremented up to the next supported CL according to the DIMM SPD CL Mask. Consult the memory vendor for more information.

13.2.1 Example 1, Using Integer Math to Convert $t_{WR(min)}$ from ns to nCK

```
// This algorithm reduces the nominal minimum timing parameter value by a 0.30% correction factor,
// rounds tCK(AVG) down, calculates nCK, adds 1 nCK, and rounds nCK down to the next integer value

int TwrMin, Correction, ClockPeriod, TempTwr, TempNck, TwrInNck;

TwrMin      = 30000;                         // tWRmin in ps
Correction  = 3                              // 0.30% per the rounding algorithm
ClockPeriod = ApplicationTck;                // Clock period in ps is application specific
TempTwr     = TwrMin * (1000 - Correction);  // Apply correction factor, scaled by 1000
TempNck     = TempTwr / ClockPeriod ;        // Initial nCK calculation, scaled by 1000
TempNck     = TempNck + 1000;                // Add 1, scaled by 1000, to effectively round up
TwrInNck    = (int)(TempNck / 1000);         // Round down to next integer
```

Table 329 — Example 1, Using Round Down only Integer Number Math

DDR5 Device Operating at Standard Application Frequencies Timing Parameter: tWR(min) = 30.000 ns = 30000 ps					
Application Speed Grade	**Device tWR**	**Application tCK**	**Device tWR / Application tCK**	**Device tWR * (1000 - Correction) / Application tCK + 1000**	**Truncate Corrected nCK / 1000**
MT/s	**ps**	**ps**	**nCK (real)**	**scaled nCK (corrected)**	**nCK (integer)**
3200	30000	625	48.00	48856	48
3600	30000	555	54.05	54891	54
4000	30000	500	60.00	60820	60
4400	30000	454	66.08	66881	66
4800	30000	416	72.12	72899	72

13.3 Timing Parameters by Speed Grade

The analog timing parameters in this section have been defined based on nominal tCA(avg)min according to the rounding rules which can be found in the Rounding Definitions and Algorithms section.

13.3.1 Timing Parameters for DDR5-3200 to DDR5-4000

Table 330 — Timing Parameters for DDR5-3200 to DDR5-4000

Speed		DDR5-3200		DDR5-3600		DDR5-4000		Units	Note
Parameter	Symbol	Min	Max	Min	Max	Min	Max		
Clock Timing									
Average Clock Period	tCK(avg)	0.625	-	0.555	-	0.500	-	ns	1
Command and Address Timing									
Read to Read command delay for same bank in same bank group	tCCD_L	Max(RBL/2, 5ns)						nCK,ns	8
Write to Write command delay for same bank in same bank group	tCCD_L_WR	Max(32nCK, 20ns)						nCK,ns	8
Write to Write command delay for same bank group, second write not RMW	tCCD_L_WR2	Max(16nCK, 10ns)						nCK,ns	8
Read to Write command delay for same bank group	tCCD_L_RTW	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 8
Write to Read command delay for same bank in same bank group	tCCD_L_WTR	CWL + WBL/2 + Max(16nCK,10ns)						nCK,ns	4, 6, 8
Read to Read command delay for different bank in same bank group	tCCD_M	tCCD_L						nCK,ns	8
Write to Write command delay for different bank in same bank group	tCCD_M_WR	tCCD_L_WR						nCK,ns	8
Write to Read command delay for different bank in same bank group	tCCD_M_WTR	tCCD_L_WTR						nCK,ns	4, 6, 8
Read to Read command delay for different bank group	tCCD_S	RBL/2						nCK	8
Write to Write command delay for different bank group	tCCD_S_WR	WBL/2						nCK	8
Read to Write command delay for different bank group	tCCD_S_RTW	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 8
Write to Read command delay for different bank group	tCCD_S_WTR	CWL + WBL/2 + Max(4nCK,2.5ns)						nCK,ns	4, 6, 8
Write to Read with Auto Precharge command delay for same bank	tCCD_WTRA	CWL + WBL/2 + tWR - tRTP						nCK,ns	2, 4, 6, 8
Activate to Activate command delay to same bank group for 1KB page size	tRRD_L(1K)	Max(8nCK, 5ns)						nCK,ns	8
Activate to Activate command delay to same bank group for 2KB page size	tRRD_L(2K)	Max(8nCK, 5ns)						nCK,ns	8
Activate to Activate command delay to different bank group for 1KB page size	tRRD_S(1K)	8						nCK	8
Activate to Activate command delay to different bank group for 2 KB page size	tRRD_S(2K)	8						nCK	8
Four activate window for 1 KB page size	tFAW (1K)	Max(32nCK, 20.000ns)	-	Max(32nCK, 17.777ns)	-	Max(32nCK, 16.000ns)	-	nCK, ns	
Four activate window for 2 KB page size	tFAW (2K)	Max(40nCK, 25.000ns)	-	Max(40nCK, 22.222ns)	-	Max(40nCK, 20.000ns)	-	nCK, ns	
Read to Precharge command delay	tRTP	Max(12nCK, 7.5ns)						nCK,ns	8
Precharge to Precharge command delay	tPPD	2						nCK	7, 8
Write recovery time	tWR	30						ns	8

13.3.2 Timing Parameters for DDR-4400 to DDR5-5200

The analog timing parameters in this section have been defined based on nominal tCK(avg)min according to the rounding rules which can be found in the Rounding Definitions and Algorithms section.

Table 331 — Timing Parameters for DDR5-4400 to DDR5-5200

Speed		DDR5-4400		DDR5-4800		DDR5-5200		Units	NOTE
Parameter	Symbol	Min	Max	Min	Max	Min	Max		
Clock Timing									
Average Clock Period	tCK(avg)	0.454	-	0.416	-	0.384	-	ns	1
Command and Address Timing									
Read to Read command delay for same bank in same bank group	tCCD_L	Max(RBL/2, 5ns)						nCK,ns	8
Write to Write command delay for same bank in same bank group	tCCD_L_WR	Max(32nCK, 20ns)						nCK,ns	8
Write to Write command delay for same bank group, second write not RMW	tCCD_L_WR2	Max(16nCK, 10ns)						nCK,ns	8
Read to Write command delay for same bank group	tCCD_L_RTW	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 8
Write to Read command delay for same bank in same bank group	tCCD_L_WTR	CWL + WBL/2 + Max(16nCK,10ns)						nCK,ns	4, 6, 8
Read to Read command delay for different bank in same bank group	tCCD_M	tCCD_L						nCK,ns	8
Write to Write command delay for different bank in same bank group	tCCD_M_WR	tCCD_L_WR						nCK,ns	8
Write to Read command delay for different bank in same bank group	tCCD_M_WTR	tCCD_L_WTR						nCK,ns	4, 6, 8
Read to Read command delay for different bank group	tCCD_S	RBL/2						nCK	8
Write to Write command delay for different bank group	tCCD_S_WR	WBL/2						nCK	8
Read to Write command delay for different bank group	tCCD_S_RTW	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 8
Write to Read command delay for different bank group	tCCD_S_WTR	CWL + WBL/2 + Max(4nCK,2.5ns)						nCK,ns	4, 6, 8
Write to Read with Auto Precharge command delay for same bank	tCCD_WTRA	CWL + WBL/2 + tWR - tRTP						nCK,ns	2, 4, 6, 8
Activate to Activate command delay to same bank group for 1KB page size	tRRD_L(1K)	Max(8nCK, 5ns)						nCK,ns	8
Activate to Activate command delay to same bank group for 2KB page size	tRRD_L(2K)	Max(8nCK, 5ns)						nCK,ns	8
Activate to Activate command delay to different bank group for 1KB page size	tRRD_S(1K)	8						nCK	8
Activate to Activate command delay to different bank group for 2KB page size	tRRD_S(2K)	8						nCK	8
Four activate window for 1 KB page size	tFAW (1K)	Max(32nCK, 14.545ns)	-	Max(32nCK, 13.333ns)	-	Max(32nCK, 12.307ns)	-	nCK, ns	
Four activate window for 2 KB page size	tFAW (2K)	Max(40nCK, 18.181ns)	-	Max(40nCK, 16.666ns)	-	Max(40nCK, 15.384ns)	-	nCK, ns	
Read to Precharge command delay	tRTP	Max(12nCK, 7.5ns)						nCK,ns	8
Precharge to Precharge command delay	tPPD	2						nCK	7, 8
Write recovery time	tWR	30						ns	8

13.3.3 Timing Parameters for DDR-5600 to DDR5-6400

Analog timing parameters in this section have been defined on nominal tCK(avg)min according to the rounding rules which can be found in the Rounding Definitions and Algorithms section.

Table 332 — Timing Parameters for DDR5-5600 to DDR5-6400

Speed		DDR5-5600		DDR5-6000		DDR5-6400		Units	NOTE
Parameter	Symbol	Min	Max	Min	Max	Min	Max		
Clock Timing									
Average Clock Period	tCK(avg)	0.357	-	0.333	-	0.312	-	ns	1
Command and Address Timing									
Read to Read command delay for same bank in same bank group	tCCD_L	Max(RBL/2, 5ns)						nCK,ns	8
Write to Write command delay for same bank in same bank group	tCCD_L_WR	Max(32nCK, 20ns)						nCK,ns	8
Write to Write command delay for same bank group, second write not RMW	tCCD_L_WR2	Max(16nCK, 10ns)						nCK,ns	8
Read to Write command delay for same bank group	tC-CD_L_RTW	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 8
Write to Read command delay for same bank in same bank group	tCCD_L_WTR	CWL + WBL/2 + Max(16nCK,10ns)						nCK,ns	4, 6, 8
Read to Read command delay for different bank in same bank group	tCCD_M	tCCD_L						nCK,ns	8
Write to Write command delay for different bank in same bank group	tCCD_M_WR	tCCD_L_WR						nCK,ns	8
Write to Read command delay for different bank in same bank group	tC-CD_M_WTR	tCCD_L_WTR						nCK,ns	4, 6, 8
Read to Read command delay for different bank group	tCCD_S	RBL/2						nCK	8
Write to Write command delay for different bank group	tCCD_S_WR	WBL/2						nCK	8
Read to Write command delay for different bank group	tC-CD_S_RTW	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 8
Write to Read command delay for different bank group	tCCD_S_WTR	CWL + WBL/2 + Max(4nCK,2.5ns)						nCK,ns	4, 6, 8
Write to Read with Auto Precharge command delay for same bank	tCCD_WTRA	CWL + WBL/2 + tWR - tRTP						nCK,ns	2, 4, 6, 8
Activate to Activate command delay to same bank group for 1KB page size	tRRD_L(1K)	Max(8nCK, 5ns)						nCK,ns	8
Activate to Activate command delay to same bank group for 2KB page size	tRRD_L(2K)	Max(8nCK, 5ns)						nCK,ns	8
Activate to Activate command delay to different bank group for 1KB page size	tRRD_S(1K)	8						nCK	8
Activate to Activate command delay to different bank group for 2KB page size	tRRD_S(2K)	8						nCK	8
Four activate window for 1KB page size	tFAW (1K)	Max(32nCK, 11.428ns)	-	Max(32nCK, 10.666ns)	-	Max(32nCK, 10.000ns)	-	nCK, ns	
Four activate window for 2KB page size	tFAW (2K)	Max(40nCK, 14.285ns)	-	Max(40nCK, 13.333ns)	-	Max(40nCK, 12.500ns)	-	nCK, ns	
Read to Precharge command delay	tRTP	Max(12nCK, 7.5ns)						nCK,ns	8
Precharge to Precharge command delay	tPPD	2						nCK	7, 8
Write recovery time	tWR	30						ns	8

13.3.4 Timing Parameters for DDR-6800 to DDR5-7600

The analog timing parameters in this section have been defined based on nominal tCK(avg)min according to the rounding rules which can be found in the Rounding Definitions and Algorithms section.

Table 333 — Timing Parameters for DDR5-6800 to DDR5-7600

Speed		DDR5-6800		DDR5-7200		DDR5-7600		Units	NOTE
Parameter	Symbol	Min	Max	Min	Max	Min	Max		
Average Clock Period	tCK(avg)	0.294	-	0.277	-	0.263	-	ns	1
Command and Address Timing									
Read to Read command delay for same bank in same bank group	tCCD_L	max(RBL/2,5ns)				max(RBL/2,5ns)		nCK,ns	8
Write to Write command delay for same bank in same bank group	tCCD_L_WR	max(32nCK,20ns)				max(32nCK,20ns)		nCK,ns	8
Write to Write command delay for same bank group, second write not RMW	tCCD_L_WR2	Max(16nCK,10ns)				Max(16nCK,10ns)		nCK,ns	8
Read to Write command delay for same bank group	tCCD_L_RTW	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE				CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE		nCK,ns	3, 5, 6, 8
Write to Read command delay for same bank in same bank group	tCCD_L_WTR	CWL + WBL/2 + Max(16nCK,10ns)				CWL + WBL/2 + Max(16nCK,10ns)		nCK,ns	4, 6, 8
Read to Read command delay for different bank in same bank group	tCCD_M	max(8nCK, 4.705ns)	-	max(8nCK, 4.444ns)	-	max(8nCK, 4.210ns)	-	nCK,ns	
Write to Write command delay for different bank in same bank group	tCCD_M_WR	max(32nCK, 18.823ns)	-	max(32nCK, 17.777ns)	-	max(32nCK, 16.842ns)	-	nCK,ns	
Write to Read command delay for different bank in same bank group	tCCD_M_WTR	CWL+WBL/2 +Max(16nCK, 9.411ns)	-	CWL+WBL/2 +Max(16nCK, 8.888ns)	-	CWL+WBL/2 +Max(16nCK, 8.421ns)	-	nCK,ns	4, 6
Read to Read command delay for different bank group	tCCD_S	RBL/2						nCK	8
Write to Write command delay for different bank group	tCCD_S_WR	WBL/2						nCK	8
Read to Write command delay for different bank group	tCCD_S_RTW	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE				CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE		nCK,ns	3, 5, 6, 8
Write to Read command delay for different bank group	tCCD_S_WTR	CWL+WBL/2 +Max(4nCK, 2.352ns)	-	CWL+WBL/2 +Max(4nCK, 2.222ns)	-	CWL+WBL/2 +Max(4nCK, 2.105ns)	-	nCK,ns	4, 6
Write to Read with Auto Precharge command delay for same bank	tCCD_WTRA	CWL + WBL/2 + tWR - tRTP				CWL + WBL/2 + tWR - tRTP		nCK,ns	2, 4, 6, 8
Activate to Activate command delay to same bank group for 1KB page size	tRRD_L(1K)	max(8nCK, 4.705ns)	-	max(8nCK, 4.444ns)	-	max(8nCK, 4.210ns)	-	nCK,ns	
Activate to Activate command delay to same bank group for 2KB page size	tRRD_L(2K)	max(8nCK, 4.705ns)	-	max(8nCK, 4.444ns)	-	max(8nCK, 4.210ns)	-	nCK,ns	
Activate to Activate command delay to different bank group for 1KB page size	tRRD_S(1K)	8				8		nCK	8
Activate to Activate command delay to different bank group for 2KB page size	tRRD_S(2K)	8				8		nCK	8
Four activate window for 1KB page size	tFAW(1K)	Max(32nCK, 9.411ns)	-	Max(32nCK, 8.888ns)	-	Max(32nCK, 8.421ns)	-	nCK,ns	
Four activate window for 2KB page size	tFAW(2K)	Max(40nCK, 11.764ns)	-	Max(40nCK, 11.111ns)	-	Max(40nCK, 10.526ns)	-	nCK,ns	
Read to Precharge command delay	tRTP	Max(12nCK,7.5ns)				Max(12nCK,7.5ns)		nCK,ns	8
Precharge to Precharge command delay	tPPD	2				4		nCK	7, 8
Write recovery time	tWR	30				30		ns	8

13.3.5 Timing Parameters for DDR-8000 to DDR5-8800

Table 334 — Timing Parameters for DDR5-8000 to DDR5-8800

Speed		DDR5-8000		DDR5-8400		DDR5-8800		Units	NOTE
Parameter	Symbol	Min	Max	Min	Max	Min	Max		
Average Clock Period	**tCK(avg)**	**0.250**	**-**	**0.238**	**-**	**0.227**	**-**	ns	1
Command and Address Timing									
Read to Read command delay for same bank in same bank group	tCCD_L	Max(8nCK, 5ns)						nCK,ns	8
Write to Write command delay for same bank in same bank group	tCCD_L_WR	Max(32nCK, 20ns)						nCK,ns	8
Write to Write command delay for same bank group, second write not RMW	tCCD_L_WR2	Max(16nCK, 10ns)						nCK,ns	8
Read to Write command delay for same bank group	tCCD_L_RTW	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 8
Write to Read command delay for same bank in same bank group	tCCD_L_WTR	CWL + WBL/2 + Max(16nCK,10ns)						nCK,ns	4, 6, 8
Read to Read command delay for different bank in same bank group	tCCD_M	max(8nCK, 4.000ns)	-	max(8nCK, 4.000ns)	-	max(8nCK, 3.863ns)	-	nCK,ns	
Write to Write command delay for different bank in same bank group	tCCD_M_WR	max(32nCK, 16.000ns)	-	max(32nCK, 15.238ns)	-	max(32nCK, 14.545ns)	-	nCK,ns	
Write to Read command delay for different bank in same bank group	tCCD_M_WTR	CWL+WBL/ 2+Max(16nCK, 8.000ns)	-	CWL+WBL/ 2+Max(16nCK, 7.619ns)	-	CWL+WBL/ 2+Max(16nCK, 7.272ns)	-	nCK,ns	4, 6, 8
Read to Read command delay for different bank group	tCCD_S	RBL/2						nCK	8
Write to Write command delay for different bank group	tCCD_S_WR	WBL/2						nCK	8
Read to Write command delay for different bank group	tCCD_S_RTW	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 8
Write to Read command delay for different bank group	tCCD_S_WTR	CWL + WBL/2 + Max(4nCK, 2.000ns)	-	CWL + WBL/2 + Max(4nCK, 1.904ns)	-	CWL + WBL/2 + Max(4nCK, 1.818ns)	-	nCK,ns	4, 6, 8
Write to Read with Auto Precharge command delay for same bank	tCCD_WTRA	CWL + WBL/2 + tWR - tRTP						nCK,ns	2, 4, 6, 8
Activate to Activate command delay to same bank group for 1KB page size	tRRD_L(1K)	Max(8nCK, 4.000ns)	-	max(8nCK, 4.000ns)	-	max(8nCK, 3.863ns)	-	nCK,ns	
Activate to Activate command delay to same bank group for 2KB page size	tRRD_L(2K)	Max(8nCK, 4.000ns)	-	max(8nCK, 4.000ns)	-	max(8nCK, 3.863ns)	-	nCK,ns	
Activate to Activate command delay to different bank group for 1KB page size	tRRD_S(1K)	8						nCK	8
Activate to Activate command delay to different bank group for 2KB page size	tRRD_S(2K)	8						nCK	8
Four activate window for 1KB page size	tFAW (1K)	Max(32nCK, 8.000ns)	-	Max(32nCK, 7.619ns)	-	Max(32nCK, 7.272ns)	-	nCK, ns	
Four activate window for 2KB page size	tFAW (2K)	Max(40nCK, 10.000ns)	-	Max(40nCK, 9.523ns)	-	Max(40nCK, 9.090ns)	-	nCK, ns	
Read to Precharge command delay	tRTP	Max(12nCK, 7.5ns)						nCK,ns	8
Precharge to Precharge command delay	tPPD	4						nCK	7, 8
Write recovery time	tWR	30						ns	8

Timing Parameters Table Notes for DDR5-3200 to DDR5-8800:

1. tCK(avg)min listed for reference only, refer to the Speed Bins and Operations section which lists all valid tCK(avg) values.
2. tCCD_WTRA(min) shall always be greater than or equal to CWL + WBL/2 + tWR(min) - tRTP(min), and when using the appropriate rounding algorithms, nCCD_WTRA(min) shall always be greater than or equal to CWL + WBL/2 + nWR(min) - nRTP(min).
3. RBL: Read burst length associated with Read command:
   RBL = 32 (36 w/ RCRC on) for fixed BL32 and BL32 in BL32 OTF mode
   RBL = 16 (18 w/ RCRC on) for fixed BL16 and BL16 in BL32 OTF mode
   RBL = 16 (18 w/ RCRC on) for BL16 in BC8 OTF mode and BC8 in BC8 OTF mode
4. WBL: Write burst length associated with Write command:
   WBL = 32 (36 w/ WCRC on) for fixed BL32 and BL32 in BL32 OTF mode
   WBL = 16 (18 w/ WCRC on) for fixed BL16 and BL16 in BL32 OTF mode
   WBL = 16 (18 w/ WCRC on) for BL16 in BC8 OTF mode and BC8 in BC8 OTF mode
5. The following is considered for tRTW equation:
   1tCK needs to be added due to tDQS2CK
   Read DQS offset timing can pull in the tRTW timing
   1tCK needs to be added when 1.5tCK postamble
6. CWL=CL-2
7. tPPD applies to any combination of precharge commands (PREab, PREsb, PREpb). tPPD also applies to any combination of precharge commands to a different die in a 3DS DDR5 SDRAM.
8. This parameter only specifies minimum values (there is no maximum value). The maximum value cells have been merged in the table to improve legibility.

13.3.6 Timing Parameters for 3DS-DDR5-3200 to 3DS-DDR5-4000 x4 2H and 4H

Analog timing parameters defined in this section are to be rounded to 1 ps of accuracy. Parameter min values which scale with tCKmin are to be defined using the tCKmin in the associated data rate.

Table 335 — Timing Parameters for x4 2H and 4H 3DS-DDR5-3200 to 3DS-DDR5-4000

Speed		DDR5-3200		DDR5-3600		DDR5-4000		Units	Notes
Parameter	Symbol	Min	Max	Min	Max	Min	Max		
Clock Timing									
Average Clock Period	tCK(avg)	0.625	-	0.555	-	0.500	-	ns	1
Command and Address Timing for 3DS									
Read to Read command delay for same bank group in same logical rank	tCCD_L_slr	Max(8nCK, 5ns)						nCK,ns	15
Write to Write command delay for same bank group in same logical rank	tCCD_L_ WR_slr	Max(32nCK, 20ns)						nCK,ns	15
Write to Write command delay for same bank group in same logical rank, second write not RMW	tCCD_L_ WR2_slr	Max(16nCK, 10ns)						nCK,ns	15
Read to Write command delay for same bank group in same logical rank	tCCD_L_ RTW_slr	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6
Write to Read command delay for same bank group in same logical rank	tCCD_L_ WTR_slr	CWL + WBL/2 + Max(16nCK,10ns)						nCK,ns	4, 6
Read to Read command delay for different bank in same bank group	tCCD_M_slr	tCCD_L_slr						nCK,ns	15
Write to Write command delay for different bank in same bank group	tCCD_M_WR_slr	tCCD_L_WR_slr						nCK,ns	15
Write to Read command delay for different bank in same bank group	tCCD_M_WTR_slr	tCCD_L_WTR_slr						nCK,ns	4, 6, 15
Read to Read command delay for different bank group in same logical rank	tCCD_S_slr	RBL/2						nCK	15
Write to Write command delay for different bank group in same logical rank	tCCD_S_ WR_slr	WBL/2						nCK	15
Read to Write command delay for different bank group in same logical rank	tCCD_S_ RTW_slr	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6
Write to Read command delay for different bank group in same logical rank	tCCD_S_ WTR_slr	CWL + WBL/2 + Max(4nCK,2.5ns)						nCK,ns	4, 6
Write to Read with Auto Precharge command for same bank in same logic rank	tCCD_ WTRA_slr	CWL + WBL/2 + tWR_slr - tRTP_slr						nCK,ns	2, 4, 6
Activate to Activate command delay to same bank group in the same logical rank	tRRD_L_ slr(1K)	Max(8nCK, 5ns)						nCK,ns	15
Activate to Activate command delay to different bank group in the same logical rank	tRRD_S_ slr(1K)	8						nCK	15
Four activate window to the same logical rank	tFAW_ slr(1K)	max(32nCK, 20.000ns)	-	max(32nCK, 17.777ns)	-	max(32nCK, 16.000ns)	-	nCK, ns	9
Read command to Precharge command delay in same logical rank	tRTP_slr	Max(12nCK, 7.5ns)						nCK,ns	15
Precharge to Precharge delay in same logical rank	tPPD_slr	2						nCK	7, 15
Write recovery time in same logical rank	tWR_slr	30						ns	15
Read to Read command delay in different logical ranks	tCCD_dlr	Max(8nCK, 5.000ns)	-	Max(8nCK, 4.444ns)	-	Max(8nCK, 4.000ns)	-	nCK,ns	
Write to Write command delay in different logical ranks	tCCD_ WR_dlr	Max(8nCK, 5.000ns)	-	Max(8nCK, 4.444ns)	-	Max(8nCK, 4.000ns)	-	nCK,ns	12
Read to Write command delay in different logical ranks	tCCD_ RTW_dlr	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6
Write to Read command delay in different logical ranks	tCCD_ WTR_dlr	CWL + WBL/2 + Max(4nCK,2.5ns)						nCK,ns	4, 6
Activate to Activate command delay to different logical ranks	tRRD_dlr	Max(4nCK, 2.500ns)	-	Max(4nCK, 2.222ns)	-	Max(4nCK, 2.000ns)	-	nCK,ns	
Four activate window to different logical ranks	tFAW_dlr	Max(16nCK, 10.000ns)	-	Max(16nCK, 8.888ns)	-	Max(16nCK, 8.000ns)	-	nCK,ns	
Precharge to Precharge delay in different logical rank	tPPD_dlr	2						nCK	7, 15
Minimum Write to Write command delay in different 3DS or DDP physical ranks	tCCD_ WR_dpr	8						nCK	12, 13, 15
Activate window by DIMM channel	tDCAW	128						nCK	10, 11, 13, 14, 15
DIMM Channel Activate Command Count in tDCAW	nDCAC	32						ACT	10, 11, 13, 14, 16

13.3.7 Timing Parameters for 3DS-DDR5-4400 to 3DS-DDR5-5200 x4 2H and 4H

Analog timing parameters defined in this section are to be rounded to 1 ps of accuracy. Parameter min values which scale with tCKmin are to be defined using the tCKmin in the associated data rate.

Table 336 — Timing Parameters for x4 2H and 4H 3DS-DDR5-4400 to 3DS-DDR5-5200

Speed		DDR5-4400		DDR5-4800		DDR5-5200		Units	Notes
Parameter	Symbol	Min	Max	Min	Max	Min	Max		
Clock Timing									
Average Clock Period	tCK(avg)	0.454	-	0.416	-	0.384	-	ns	1
Command and Address Timing for 3DS									
Read to Read command delay for same bank group in same logical rank	tCCD_L_slr	Max(8nCK, 5ns)						nCK,ns	15
Write to Write command delay for same bank group in same logical rank	tCCD_L_ WR_slr	Max(32nCK, 20ns)						nCK,ns	15
Write to Write command delay for same bank group in same logical rank, second write not RMW	tCCD_L_ WR2_slr	Max(16nCK, 10ns)						nCK,ns	15
Read to Write command delay for same bank group in same logical rank	tCCD_L_ RTW_slr	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6
Write to Read command delay for same bank group in same logical rank	tCCD_L_ WTR_slr	CWL + WBL/2 + Max(16nCK,10ns)						nCK,ns	4, 6
Read to Read command delay for different bank in same bank group	tCCD_M_slr	tCCD_L_slr						nCK,ns	15
Write to Write command delay for different bank in same bank group	tCCD_M_WR_slr	tCCD_L_WR_slr						nCK,ns	15
Write to Read command delay for different bank in same bank group	tCCD_M_WTR_slr	tCCD_L_WTR_slr						nCK,ns	4, 6, 15
Read to Read command delay for different bank group in same logical rank	tCCD_S_slr	RBL/2						nCK	15
Write to Write command delay for different bank group in same logical rank	tCCD_S_ WR_slr	WBL/2						nCK	15
Read to Write command delay for different bank group in same logical rank	tCCD_S_ RTW_slr	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6
Write to Read command delay for different bank group in same logical rank	tCCD_S_ WTR_slr	CWL + WBL/2 + Max(4nCK,2.5ns)						nCK,ns	4, 6
Write to Read with Auto Precharge command for same bank in same logic rank	tCCD_ WTRA_slr	CWL + WBL/2 + tWR_slr - tRTP_slr						nCK,ns	2, 4, 6
Activate to Activate command delay to same bank group in the same logical rank	tRRD_L_ slr(1K)	Max(8nCK, 5ns)						nCK,ns	15
Activate to Activate command delay to different bank group in the same logical rank	tRRD_S_ slr(1K)	8						nCK	15
Four activate window to the same logical rank	tFAW_ slr(1K)	max(32nCK, 14.545ns)	-	max(32nCK, 13.333ns)	-	max(32nCK, 12.307ns)	-	nCK, ns	9
Read command to Precharge command delay in same logical rank	tRTP_slr	Max(12nCK, 7.5ns)						nCK,ns	15
Precharge to Precharge delay in same logical rank	tPPD_slr	2						nCK	7, 15
Write recovery time in same logical rank	tWR_slr	30						ns	15
Read to Read command delay in different logical ranks	tCCD_dlr	Max(8nCK, 3.636ns)	-	Max(8nCK, 3.333ns)	-	Max(8nCK, 3.333ns)	-	nCK,ns	
Write to Write command delay in different logical ranks	tCCD_ WR_dlr	Max(8nCK, 3.636ns)	-	Max(8nCK, 3.333ns)	-	Max(8nCK, 3.333ns)	-	nCK,ns	12
Read to Write command delay in different logical ranks	tCCD_ RTW_dlr	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6
Write to Read command delay in different logical ranks	tCCD_ WTR_dlr	CWL + WBL/2 + Max(4nCK,2.5ns)						nCK,ns	4, 6
Activate to Activate command delay to different logical ranks	tRRD_dlr	Max(4nCK, 1.818ns)	-	Max(4nCK, 1.666ns)	-	Max(4nCK, 1.666ns)	-	nCK,ns	
Four activate window to different logical ranks	tFAW_dlr	Max(16nCK, 7.272ns)	-	Max(16nCK, 6.666ns)	-	Max(16nCK, 6.666ns)	-	nCK,ns	
Precharge to Precharge delay in different logical rank	tPPD_dlr	2						nCK	7, 15
Minimum Write to Write command delay in different 3DS or DDP physical ranks	tCCD_ WR_dpr	8						nCK	12, 13, 15
Activate window by DIMM channel	tDCAW	128						nCK	10, 11, 13, 14, 15
DIMM Channel Activate Command Count in tDCAW	nDCAC	32						ACT	10, 11, 13, 14, 16

13.3.8 Timing Parameters for 3DS-DDR5-5600 to 3DS-DDR5-6400 x4 2H and 4H

Analog timing parameters defined in this section are to be rounded to 1 ps of accuracy. Parameter min values which scale with tCKmin are to be defined using the tCKmin in the associated data rate.

Table 337 — Timing Parameters for x4 2H and 4H 3DS-DDR5-5600 to 3DS-DDR5-6400

Speed		DDR5-5600		DDR5-6000		DDR5-6400		Units	Notes
Parameter	Symbol	Min	Max	Min	Max	Min	Max		
Clock Timing									
Average Clock Period	tCK(avg)	0.357	-	0.333	-	0.312	-	ns	1
Command and Address Timing for 3DS									
Read to Read command delay for same bank group in same logical rank	tCCD_L_slr	Max(8nCK, 5ns)						nCK,ns	15
Write to Write command delay for same bank group in same logical rank	tCCD_L_ WR_slr	Max(32nCK, 20ns)						nCK,ns	15
Write to Write command delay for same bank group in same logical rank, second write not RMW	tCCD_L_ WR2_slr	Max(16nCK, 10ns)						nCK,ns	15
Read to Write command delay for same bank group in same logical rank	tCCD_L_ RTW_slr	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6
Write to Read command delay for same bank group in same logical rank	tCCD_L_ WTR_slr	CWL + WBL/2 + Max(16nCK,10ns)						nCK,ns	4, 6
Read to Read command delay for different bank in same bank group	tCCD_M_slr	tCCD_L_slr						nCK,ns	15
Write to Write command delay for different bank in same bank group	tCCD_M_WR_slr	tCCD_L_WR_slr						nCK,ns	15
Write to Read command delay for different bank in same bank group	tCCD_M_WTR_slr	tCCD_L_WTR_slr						nCK,ns	4, 6, 15
Read to Read command delay for different bank group in same logical rank	tCCD_S_slr	RBL/2						nCK	15
Write to Write command delay for different bank group in same logical rank	tCCD_S_ WR_slr	WBL/2						nCK	15
Read to Write command delay for different bank group in same logical rank	tCCD_S_RTW_slr	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6
Write to Read command delay for different bank group in same logical rank	tCCD_S_WTR_slr	CWL + WBL/2 + Max(4nCK,2.5ns)						nCK,ns	4, 6
Write to Read with Auto Precharge command for same bank in same logic rank	tCCD_WTRA_slr	CWL + WBL/2 + tWR_slr - tRTP_slr						nCK,ns	2, 4, 6
Activate to Activate command delay to same bank group in the same logical rank	tRRD_L_slr(1K)	Max(8nCK, 5ns)						nCK,ns	15
Activate to Activate command delay to different bank group in the same logical rank	tRRD_S_slr(1K)	8						nCK	15
Four activate window to the same logical rank	tFAW_slr(1K)	max(32nCK, 11.428ns)	-	max(32nCK, 10.666ns)	-	max(32nCK, 10.000ns)	-	nCK, ns	9
Read command to Precharge command delay in same logical rank	tRTP_slr	Max(12nCK, 7.5ns)						nCK,ns	15
Precharge to Precharge delay in same logical rank	tPPD_slr	2						nCK	7, 15
Write recovery time in same logical rank	tWR_slr	30						ns	15
Read to Read command delay in different logical ranks	tCCD_dlr	Max(8nCK, 3.214ns)	-	Max(8nCK, 3.214ns)	-	Max(8nCK, 3.125ns)	-	nCK,ns	
Write to Write command delay in different logical ranks	tCCD_ WR_dlr	Max(8nCK, 3.214ns)	-	Max(8nCK, 3.214ns)	-	Max(8nCK, 3.125ns)	-	nCK,ns	12
Read to Write command delay in different logical ranks	tCCD_RTW_dlr	CL - CWL + RBL/2 + 2tCK - (Read DQS offset) + (tRPST - 0.5tCK) + tWPRE						nCK,ns	3, 5, 6
Write to Read command delay in different logical ranks	tCCD_WTR_dlr	CWL + WBL/2 + Max(4nCK,2.5ns)						nCK,ns	4, 6
Activate to Activate command delay to different logical ranks	tRRD_dlr	Max(4nCK, 1.666ns)	-	Max(4nCK, 1.666ns)	-	Max(4nCK, 1.666ns)	-	nCK,ns	
Four activate window to different logical ranks	tFAW_dlr	Max(16nCK, 6.666ns)	-	Max(16nCK, 6.666ns)	-	Max(16nCK, 6.666ns)	-	nCK,ns	
Precharge to Precharge delay in different logical rank	tPPD_dlr	2						nCK	7, 15
Minimum Write to Write command delay in different 3DS or DDP physical ranks	tCCD_ WR_dpr	8						nCK	12, 13, 15
Activate window by DIMM channel	tDCAW	128						nCK	10, 11, 13, 14, 15
DIMM Channel Activate Command Count in tDCAW	nDCAC	32						ACT	10, 11, 13, 14, 16

13.3.9 Timing Parameters for 3DS-DDR5-6800 to 3DS-DDR5-7600 x4 2H and 4H

Analog timing parameters defined in this section are to be rounded to 1 ps of accuracy. Parameter min values which scale with tCKmin are to be defined using the tCKmin in the associated data rate

Table 338 — Timing Parameters for x4 2H and 4H 3DS-DDR5-6800 to 3DS-DDR5-7600

Speed		DDR5-6800		DDR5-7200		DDR5-7600		Units	Notes
Parameter	Symbol	Min	Max	Min	Max	Min	Max		
Clock Timing									
Average Clock Period	tCK(avg)	0.294	-	0.277	-	0.263	-	ns	1
Command and Address Timing for 3DS									
Read to Read command delay for same bank group in same logical rank	tCCD_L_slr	Max(8nCK, 5ns)						nCK,ns	15
Write to Write command delay for same bank group in same logical rank	tCCD_L_ WR_slr	Max(32nCK, 20ns)						nCK,ns	15
Write to Write command delay for same bank group in same logical rank, second write not RMW	tCCD_L_ WR2_slr	Max(16nCK, 10ns)						nCK,ns	15
Read to Write command delay for same bank group in same logical rank	tCCD_L_ RTW_slr	CL — CWL + RBL/2 + 2tCK — (Read DQS Offset) + (tRPST—0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 15
Write to Read command delay for same bank group in same logical rank	tCCD_L_ WTR_slr	CWL + WBL/2 + Max(16nCK,10ns)						nCK,ns	4, 6, 15
Read to Read command delay for different bank in same bank group	tCCD_M_slr	Max(8nCK, 4.705ns)	-	Max(8nCK, 4.444ns)	-	Max(8nCK, 4.210ns)	-	nCK,ns	
Write to Write command delay for different bank in same bank group	tCCD_M_WR_slr	Max(32nCK, 18.823ns)	-	Max(32nCK, 17.777ns)	-	Max(32nCK, 16.842ns)	-	nCK,ns	
Write to Read command delay for different bank in same bank group	tCCD_M_WTR_slr	CWL+WBL/2 +Max(16nCK, 9.411ns)	-	CWL+WBL/2 +Max(16nCK,8. 888ns)	-	CWL+WBL/2 +Max(16nCK, 8.421ns)	-	nCK,ns	4, 6
Read to Read command delay for different bank group in same logical rank	tCCD_S_slr	RBL/2						nCK	15
Write to Write command delay for different bank group in same logical rank	tCCD_S_ WR_slr	WBL/2						nCK	15
Read to Write command delay for different bank group in same logical rank	tCCD_S_RTW_slr	CL — CWL + RBL/2 + 2tCK — (Read DQS Offset) + (tRPST—0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 15
Write to Read command delay for different bank group in same logical rank	tCCD_S_WTR_slr	CWL+WBL/2 +Max(4nCK, 2.352ns)	-	CWL+WBL/2 +Max(4nCK, 2.222ns)	-	CWL+WBL/2 +Max(4nCK, 2.105ns)	-	nCK,ns	4, 6
Write to Read with Auto Precharge command for same bank in same logic rank	tCCD_WTRA_slr	CWL + WBL/2 + tWR_slr — tRTP_slr						nCK,ns	2, 4, 6, 15
Activate to Activate command delay to same bank group in the same logical rank	tRRD_L_slr(1K)	Max(8nCK, 4.705ns)	-	Max(8nCK, 4.444ns)	-	Max(8nCK, 4.210ns)	-	nCK,ns	
Activate to Activate command delay to different bank group in the same logical rank	tRRD_S_slr(1K)	8						nCK	15
Four activate window to the same logical rank	tFAW_slr(1K)	Max(32nCK, 9.411ns)	-	Max(32nCK, 8.888ns)	-	Max(32nCK, 8.421ns)	-	nCK, ns	9
Read command to Precharge command delay in same logical rank	tRTP_slr	Max(12nCK,7.5ns)						nCK,ns	15
Precharge to Precharge delay in same logical rank	tPPD_slr	2				4		nCK	7, 15
Write recovery time in same logical rank	tWR_slr	30						ns	15
Read to Read command delay in different logical ranks	tCCD_dlr	Max(8nCK, 3.125ns)	-	Max(8nCK, 3.0560ns)	-	Max(8nCK, 3.056ns)	-	nCK,ns	
Write to Write command delay in different logical ranks	tCCD_ WR_dlr	Max(8nCK, 3.125ns)	-	Max(8nCK, 3.0560ns)	-	Max(8nCK, 3.056ns)	-	nCK,ns	12
Read to Write command delay in different logical ranks	tCCD_RTW_dlr	CL — CWL + RBL/2 + 2tCK — (Read DQS Offset) + (tRPST—0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 15
Write to Read command delay in different logical ranks	tCCD_WTR_dlr	CWL + WBL/2 + Max(4nCK,2.5ns)						nCK,ns	4, 6, 15
Activate to Activate command delay to different logical ranks	tRRD_dlr	Max(4nCK, 1.666ns)						nCK,ns	15
Four activate window to different logical ranks	tFAW_dlr	Max(16nCK, 6.666ns)						nCK,ns	15
Precharge to Precharge delay in different logical rank	tPPD_dlr	2				4		nCK	7, 15
Minimum Write to Write command delay in different 3DS or DDP physical ranks	tCCD_ WR_dpr	8						nCK	12, 13, 15
Activate window by DIMM channel	tDCAW	128						nCK	10, 11, 13, 14, 15
DIMM Channel Activate Command Count in tDCAW	nDCAC	32						ACT	10, 11, 13, 14, 15, 16

13.3.10 Timing Parameters for 3DS-DDR5-8000 to 3DS-DDR5-8800 x4 2H and 4H

Analog timing parameters defined in this section are to be rounded to 1 ps of accuracy. Parameter min values which scale with tCKmin are to be defined using the tCKmin in the associated data rate**.**

Table 339 — Timing Parameters for x4 2H and 4H 3DS-DDR5-8000 to 3DS-DDR5-8800

Speed		DDR5-8000		DDR5-8400		DDR5-8800		Units	Notes
Parameter	Symbol	Min	Max	Min	Max	Min	Max		
Clock Timing									
Average Clock Period	tCK(avg)	0.250	-	0.238	-	0.227	-	ns	1
Command and Address Timing for 3DS									
Read to Read command delay for same bank group in same logical rank	tCCD_L_slr	Max(8nCK, 5ns)						nCK,ns	15
Write to Write command delay for same bank group in same logical rank	tCCD_L_ WR_slr	Max(32nCK, 20ns)						nCK,ns	15
Write to Write command delay for same bank group in same logical rank, second write not RMW	tCCD_L_ WR2_slr	Max(16nCK, 10ns)						nCK,ns	15
Read to Write command delay for same bank group in same logical rank	tCCD_L_ RTW_slr	CL — CWL + RBL/2 + 2tCK — (Read DQS Offset) + (tRPST—0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 15
Write to Read command delay for same bank group in same logical rank	tCCD_L_ WTR_slr	CWL + WBL/2 + Max(16nCK,10ns)						nCK,ns	4, 6, 15
Read to Read command delay for different bank in same bank group	tCCD_M_slr	Max(8nCK, 4.000ns)	-	Max(8nCK, 4.000ns)	-	Max(8nCK, 3.863ns)	-	nCK,ns	
Write to Write command delay for different bank in same bank group	tCCD_M_WR_slr	Max(32nCK, 16.000ns)	-	Max(32nCK, 15.238ns)	-	Max(32nCK, 14.545ns)	-	nCK,ns	
Write to Read command delay for different bank in same bank group	tCCD_M_WTR_slr	CWL+WBL/2 +Max(16nCK, 8.000ns)	-	CWL+WBL/2 +Max(16nCK,7 .619ns)	-	CWL+WBL/2 +Max(16nCK, 7.272ns)	-	nCK,ns	4, 6
Read to Read command delay for different bank group in same logical rank	tCCD_S_slr	RBL/2						nCK	15
Write to Write command delay for different bank group in same logical rank	tCCD_S_ WR_slr	WBL/2						nCK	15
Read to Write command delay for different bank group in same logical rank	tCCD_S_RTW_slr	CL — CWL + RBL/2 + 2tCK — (Read DQS Offset) + (tRPST—0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 15
Write to Read command delay for different bank group in same logical rank	tCCD_S_WTR_slr	CWL+WBL/2 +Max(4nCK, 2.000ns)	-	CWL+WBL/2 +Max(4nCK, 1.904ns)	-	CWL+WBL/2 +Max(4nCK, 1.818ns)	-	nCK,ns	4, 6
Write to Read with Auto Precharge command for same bank in same logic rank	tCCD_WTRA_slr	CWL + WBL/2 + tWR_slr — tRTP_slr						nCK,ns	2, 4, 6, 15
Activate to Activate command delay to same bank group in the same logical rank	tRRD_L_slr(1K)	Max(8nCK, 4.000ns)	-	Max(8nCK, 4.000ns)	-	Max(8nCK, 3.863ns)	-	nCK,ns	
Activate to Activate command delay to different bank group in the same logical rank	tRRD_S_slr(1K)	8						nCK	15
Four activate window to the same logical rank	tFAW_slr(1K)	Max(32nCK, 8.000ns)	-	Max(32nCK, 7.619ns)	-	Max(32nCK, 7.272ns)	-	nCK, ns	9
Read command to Precharge command delay in same logical rank	tRTP_slr	Max(12nCK,7.5ns)						nCK,ns	15
Precharge to Precharge delay in same logical rank	tPPD_slr	4						nCK	7, 15
Write recovery time in same logical rank	tWR_slr	30						ns	15
Read to Read command delay in different logical ranks	tCCD_dlr	Max(8nCK, 3.000ns)	-	Max(8nCK, 3.0000ns)	-	Max(8nCK, 2.955ns)	-	nCK,ns	
Write to Write command delay in different logical ranks	tCCD_ WR_dlr	Max(8nCK, 3.000ns)	-	Max(8nCK, 3.000ns)	-	Max(8nCK, 2.955ns)	-	nCK,ns	12
Read to Write command delay in different logical ranks	tCCD_RTW_dlr	CL — CWL + RBL/2 + 2tCK — (Read DQS Offset) + (tRPST—0.5tCK) + tWPRE						nCK,ns	3, 5, 6, 15
Write to Read command delay in different logical ranks	tCCD_WTR_dlr	CWL + WBL/2 + Max(4nCK,2.5ns)						nCK,ns	4, 6, 15
Activate to Activate command delay to different logical ranks	tRRD_dlr	Max(4nCK, 1.666ns)						nCK,ns	15
Four activate window to different logical ranks	tFAW_dlr	Max(16nCK, 6.666ns)						nCK,ns	15
Precharge to Precharge delay in different logical rank	tPPD_dlr	4						nCK	7, 15
Minimum Write to Write command delay in different 3DS or DDP physical ranks	tCCD_ WR_dpr	8						nCK	12, 13, 15
Activate window by DIMM channel	tDCAW	128						nCK	10, 11, 13, 14, 15
DIMM Channel Activate Command Count in tDCAW	nDCAC	32						ACT	10, 11, 13, 14, 15, 16

Timing Parameters Table Notes for 3DS-DDR5-3200 to 3DS-DDR5-8800 x4 2H and 4H:

1. tCK(avg)min listed for reference only, refer to the Speed Bins and Operations section which lists all valid tCK(avg) values.
2. tCCD_WTRA_slr(min) shall always be greater than or equal to CWL + WBL/2 + tWR_slr(min) - tRTP_slr(min), and when using the appropriate rounding algorithms, nCCD_WTRA_slr(min) shall always be greater than or equal to CWL + WBL/2 + nWR_slr(min) - nRTP_slr(min).
3. RBL: Read burst length associated with Read command

    RBL = 32 for fixed BL32 and BL32 in BL32 OTF mode

    RBL = 16 for fixed BL16 and BL16 in BL32 OTF mode

    RBL = 16 for BL16 in BC8 OTF mode and BC8 in BC8 OTF mode
4. WBL: Write burst length associated with Write command

    WBL = 32 for fixed BL32 and BL32 in BL32 OTF mode

    WBL = 16 for fixed BL16 and BL16 in BL32 OTF mode

    WBL = 16 for BL16 in BC8 OTF mode and BC8 in BC8 OTF mode
5. The following is considered for tRTW equation

    1tCK needs to be added due to tDQS2CK

    Read DQS offset timing can pull in the tRTW timing

    1tCK needs to be added when 1.5tCK postamble
6. CWL=CL-2
7. tPPD applies to any combination of precharge commands (PREab, PREsb, PREpb). tPPD also applies to any combination of precharge commands to a different die in a 3DS DDR5 SDRAM.
8. These timings contained in this table are for x4 2H and 4H 3Ds device
9. For x4 devices only.
10. Activate commands to different channels on the same DIMM may be issued on the same cycle, not requiring any stagger.
11. Activate commands to the same channel on a DIMM are subject to tDCAW. No more than nDCAC activate commands may be issued to the same channel on a DIMM within tDCAW.
12. tCCD_ WR_dlr and tCCD_ WR_dpr also apply to the WRITE PATTERN command.
13. Parameter applies to all DDP, or dual-physical-rank (36 and 40 placement) 3DS-based DIMMs built with JEDEC PMICXXXX, but may not apply to DIMMs built with higher current capacity PMICs.
14. Activate commands to a DIMM channel include all Activate commands to the same logical rank (SLR), all Activate commands to different logical ranks (DLR), and all Activate commands to different physical ranks (DPR).
15. This parameter only specifies minimum values (there is no maximum value). The maximum value cells have been merged in the table to improve legibility.
16. This parameter only specifies maximum values (there is no minimum value). The minimum value cells have been merged in the table to improve legibility.

14 DDR5 Module Rank and Channel Timings

14.1 Module Rank and Channel Limitations for DDR5 DIMMs

To achieve efficient module power supply design for JEDEC-standard DDR5 DIMMs, minimum timings as well as limitations in the number of DRAMs are provided for Refresh, and Write operations occurring on a single module. As well, since these modules are organized as two independent 36-bit or 40-bit channels (32 bits for non-ECC DIMMs), additional restrictions apply in order to limit localized power delivery noise on the module. To provide best performance, the different channels may initiate commands on the same cycle provided the rank to rank timings are met, the maximum number of DRAMs in a given activity is not exceeded, and the applicable component timings shown elsewhere in this specification are met. The timing and operational relationships for DDR5 DIMMs are shown in Table 340.

Table 340 — DDR5 Module Rank and Channel Timings for DDR5 DIMMs

DIMM Configuration	Maximum Number of DRAM Die in Simultaneous or Overlapping Activity			
	Refresh (All-Bank Refresh)		Write, Write-Pattern	
	Die per Physical Rank	Die per DIMM	Die per Channel	Die per DIMM
SR x16	No restriction			
DR x16	No restriction		2	4
SR x8	No restriction			
DR x8	No restriction		5	10
SR x4	No restriction			
DR x4	No restriction		10	20
DR x4 (2H 3DS)	No restriction	40	10	20
DR x4 (4H 3DS)	30	40	10	20
DR x4 (8H 3DS)	30	40	10	20
Notes	1, 2, 3, 4, 7, 8, 9		1, 5, 6, 7, 9	

NOTE 1 Any combination of commands with up to the maximum of die per channel and per DIMM, per condition is allowed.
NOTE 2 Refresh commands to different channels do not require stagger.
NOTE 3 tRFC_dlr must be met for refresh commands to different logical ranks within a package rank on the same channel.
NOTE 4 Any DRAM is considered to be in Refresh mode until tRFC time has been met.
NOTE 5 tCCD parameters must be met for Write and Write-Pattern commands to different logical ranks or physical ranks within the same channel; no overlapping write data bus activity is allowed on two physical or logical ranks within the same channel.
NOTE 6 Write and Write-Pattern commands to different channels do not require stagger.
NOTE 7 Each rank consists of one group of DRAMs making up a 36 or 40 bit channel (32 bits for non-ECC DIMMs).
NOTE 8 These restrictions only apply to explicit all-banks refresh commands (REFab) and not to self-refresh entry or exit
NOTE 9 Restrictions apply to DIMMs built with JEDEC PMICXXXX, but may not apply to DIMMs built with higher current capacity PMICs

15 DDR5 System RAS Improvement

15.1 Design Guidelines for DDR5 Bounded Fault RAS Improvement

15.1.1 DDR5 Reliability Design Guidelines Overview

These DDR5 reliability design guidelines aim to bound bits impacted by certain DRAM failures. This limits the number of failure patterns seen by the memory controller such that correction of many failures can be reliably performed in DIMMs with one ECC device.

Many internal DRAM failures may impact only a portion of the data from a fetch. The likelihood of a specific component failing may also vary between process generations and DRAM vendors. These guidelines can be used by DDR5 DRAM's to bound failures from the components most likely to fail. These design guidelines can only address failures that impact a portion of the data from a 128-bit fetch. Failures that impact all the data in a 128-bit fetch from a device (e.g., device failure, bank failure) cannot be bounded as described here.

15.1.2 Reliability Design Guidelines

In a x8 device bounded failures are defined by the following qualities:

- A bounded fault will not impact more than 32 bits of data in a 128-bit fetch
- The data bits impacted by a bounded failure will be confined to at most 2 DQs (i.e. the failures will be DQ aligned)
- The DQs transmitting data impacted by the failure will both be in either the first nibble or the second nibble of a burst. That is, the impacted DQs will both be in the set of the first 4 DQs (DQ0, DQ1, DQ2, DQ3) or both in the set of the last four DQs (DQ4, DQ5, DQ6, DQ7)

Figure 259 shows examples of fault boundaries for x8 devices. Device on the left has a bounded fault in lanes DQ0 and DQ1 in the first nibble. Device on the right has a bounded fault in lanes DQ4 and DQ6 in the second nibble.

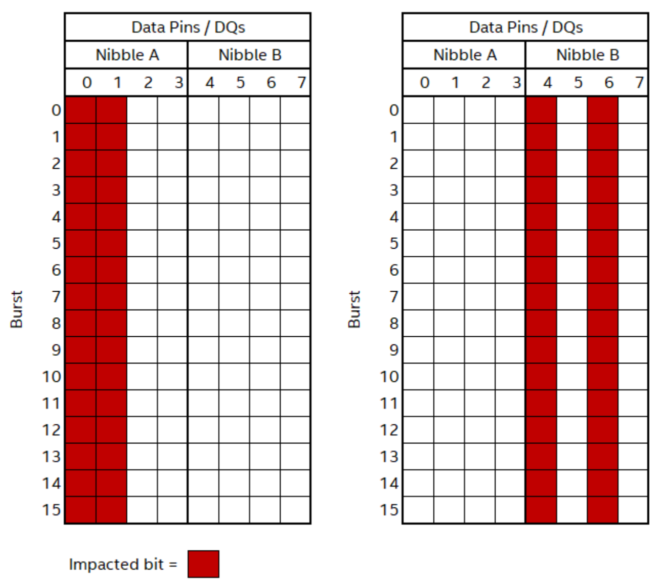


Figure 259 — Examples of x8 Fault Boundaries

An x4 device has similarly defined qualities for the failure boundaries, with the noted exception of the number of bits impacted by a bounded failure. In a DRAM device in a 9x4 configuration bounded failures should not impact more than 16 bits. However, in a 10x4 device, the bounded failure may impact up to 32 bits.

15.1.2 Reliability Design Guidelines (cont'd)

In x4 devices bounded failures are defined by the following qualities:

- A bounded failure in a DRAM device in a 9x4 configuration shall not impact more than 16 bits of data in a 128-bit prefetch.
- A bounded failure in a DRAM device in a 10x4 configuration shall not impact more than 32 bits of data in a 128-bit prefetch.
- For a device in a 9x4 configuration, the data bits impacted by a bounded failure will be confined to one DQ.
- For a device in a 10x4 configuration, the data bits impacted by a bounded failure will be confined to at most two DQs.

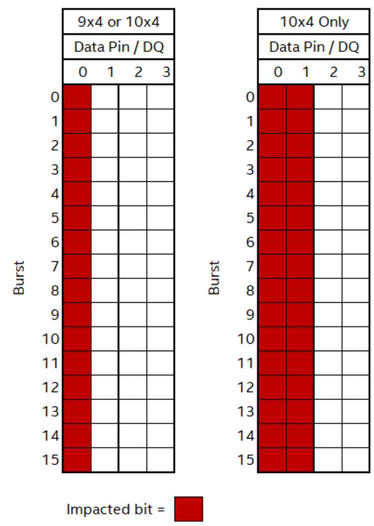


Figure 260 — Example of x4 Fault Boundary

15.2 Single Error Correction (SEC) Code Properties

To maintain the bounded fault design guidelines, miscorrections by the on-die ECC must be restricted when a bounded fault causes a multi-bit error. In devices with bounded fault it is suggested that the DRAM vendor uses an on-die ECC code that maintains the fault boundaries. That is, if an error is contained in one boundary, then the on-die ECC should not spread the error into a second boundary by miscorrection. Note that if faults are not bounded in the device, then the on-die ECC does not need to have these properties.

To restrict miscorrection in devices following bounded fault design guidelines, the data may be divided into blocks aligned with the bounded fault and miscorrection should be restricted in the case an error is contained in a single data block. The 128 data bits used to compute a set of 8 check bits may be divided into data blocks up to 32 bits in size. These data blocks are to be determined by the memory vendor to best align with internal DRAM faults. For example, a common component fault may impact 32 bits per 128 data bits, then the vendor may choose to divide the 128 data bits into four 32-bit blocks each of which correspond to the bits impacted by one of the components failing.

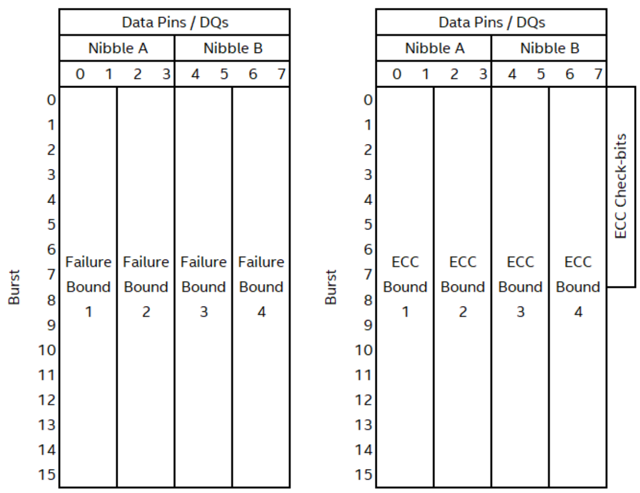


Figure 261 — Example of Fault Boundaries versus On-die ECC Data Blocks

If a multi-bit error occurs on a read and is limited to one data block, then the on-die ECC SEC code should do one of the following:

- Miscorrect a bit in the data block containing the error
- Miscorrect a bit in the on-die ECC check bits
- Detect the error (a SEC code may detect some multi-bit errors, but detection of these errors is not guaranteed)

The selection of block size and data blocks is determined by the vendor to best suit the device architecture and possible modes of fault.

15.3 Writeback of Data During ECS and x4 RMW Operations

The DRAM device may optionally support the suppressing of the writeback for ECS operations and x4 RMW Read-Modify-Write (RMW) operations. DRAM may implement the feature in MR9 or MR15. DDR5 SPD Byte 14 Bits[2:1] indicates if feature is supported and will also indicate whether to use MR9 or MR15 for enabling the modes.

If MR9 OP1=1 or MR15 OP7=1, then x4 DRAM's on writes will perform an internal 'read-modify-write' operation for BL16. BL32 mode does not requires an internal 'read modify write' operation. The DRAM will correct any single bit errors that result from the internal read of 128 bit data before merging the incoming 64 bit data and then re-compute 8 ECC Check bits. Note that ECC check bits are computed after merging of the incoming data with the corrected data from the array. DRAM will then write the incoming 64b data to the array along with recomputed 8 ECC Check bits. DRAM will suppress the writeback of 64 bit data that was fetched from the array irrespective of whether the 64 bit data needed any correction or not. Suppression of writeback (where applicable due to BC8 or DM usage) is not supported on x8/x16 devices; MR9:OP[1] or MR15:OP[7] must be set to "0_B". Suppression of writeback may or may not occur when BC8 mode is used on x4 devices.

If MR9 OP0=1 or MR15 OP6=1, then DRAM will suppress writeback of data and ECC Check bits during ECS operation for x4, x8, and x16 DDR5. DRAM will continue to count errors to provide transparency.

Table 341 — MR9 or MR15 Register Information

Function	Register Type	Operand	Data	Notes
x4 Writes	R/W	MR9:OP[1] or MR15:OP[7]	0B: Do not suppress writeback of Data during RMW (Default) 1B: Suppress writeback of Data during RMW (Optional)	1
ECS Write-back	R/W	MR9:OP[0] or MR15:OP[6]	0B: Do not suppress writeback of Data and ECC Check Bits (Default) 1B: Suppress writeback of Data and ECC Check Bits (Optional)	1
NOTE 1 DDR5 SPD Byte 14 Bits[2:1] indicates if feature is supported and will also indicate whether to use MR9 or MR15 for enabling the modes.				

DDR5 SPD Byte 14 Definition (Referenced from JESD400-5)

Byte 14 (0x00E): SDRAM Fault Handling and Temperature Sense

This byte defines support for SDRAM fault handling features and for wide on-die temperature sensing range (see MR4). This value comes from the DDR5 SDRAM specification, JESD79-5.

Table 342 — SDRAM Fault Handling and Temperature Sense

SDRAM Fault Handling Features and Temperature Sense			
Bit 7	**Bit 6**	**Bit 5**	**Bit 4**
Reserved; must be coded as 0000			
Bit 3	**Bit 2**	**Bit 1**	**Bit 0**
Wide Temperature Sense	**x4 RMW/ECS Writeback Suppression**	**x4 RMW/ECS Writeback Suppression MR Selector**	**Bounded Fault**
0: Wide temperature sense and reporting not supported 1: Wide temperature sense and reporting supported	0: Writeback suppression not supported 1: Writeback suppression supported	0: Writeback suppression control in MR9 1: Writeback suppression control in MR15	0: Bounded Fault not supported 1: Bounded Fault supported
NOTES: ECS = Error Check and Scrub RMW = Read-Modify-Write MR = Mode Register			

16 DDR5 Per Row Activation Counting

16.1 Introduction

Deterministic counting of row activations is an optional feature on DDR5 16Gb and higher density devices, with support indicated by MR70:OP[0]. This feature detects row activity that may adversely affect the data stored in cells on physically adjacent rows within the DRAM. As one or more rows' activation counts approach or reach a maximum activation threshold after a Refresh (REF) or Refresh Management (RFM) command, the host and DRAM may need to take action to prevent data in affected cells from flipping states.

The intended implementation for Per Row Activation Counting (PRAC) is to add Activation Counter bits to every row in the DRAM. These bits store a count associated with the number of activations received by a row since the last time it was refreshed. Activations to a row include the ACT, REF and RFM commands, and the MPC Manual ECS function.

A DDR5 design that supports PRAC shall continue to support "legacy" methods for maintaining data integrity in DRAM cells for backward compatibility. Designs supporting PRAC shall continue to support all features and functions as defined in this specification when running with PRAC disabled.

When PRAC is enabled, (MR70:OP[1]=1_B), the following modes are disabled by the DRAM, with MR58 and MR59 retaining their access types and values and those MRs will reflect the values of any subsequent MRWs:

- RFM Required, as indicated by MR58:OP[0], is disabled internally by the DRAM and not required by the host regardless of value set in the MR
- DRFM Enable, as configured by MR59:OP[0], is disabled internally by the DRAM regardless of host write value

Functionality of the PRAC RFM command is enabled when MR70:OP[1]=1_B and replaces the command usage described in Section 4.42 Refresh Management (RFM). If Refresh Management is not required, MR58:OP[0]=0_B, and PRAC is disabled, MR70:OP[1]=0B, CA9 is only required to be valid ("V") for a REF command, and the DRAM will treat a RFM command as a REF command. If Refresh Management is required, MR58:OP[0]=1_B, or PRAC is enabled, MR70:OP[1]=1B, a REF command requires CA9=H.

DRFM functionality of the WRPA, WRA, RDA, PREpb and DRFM commands is disabled regardless of MR59:OP[0] host write value when MR70:OP[1]=1_B. The DRFM address sampling operation is disabled for the WRPA, WRA, RDA, and PREpb commands. CA9 is only required to be valid ("V") for WRPA, WRA, and RDA commands. When CID3 is not used, CA5 is required to be Valid ("V") for RFM and PREpb commands.

The requirements for enabling and using PRAC are described in Sections 16.2-16.9. Updates to usage of the PRAC RFM command are described in Sections 16.4 and 16.5.

On-the-fly switching between modes is not permitted. Enabling PRAC requires initialization, as described below. Disabling PRAC (re-enabling legacy) requires a DRAM Reset.

When PRAC is supported (MR70:OP[0]=1_B) but disabled (MR70:OP[1]=0_B), all associated Mode Registers can be written to and read from, however no PRAC-related behaviors will be executed until PRAC is enabled (MR70:OP[1]=1_B). Activation Counter Initialization, PRAC Testing, and Alert Verification shall be disabled prior to enabling PRAC.

16.2 Activation Counter Initialization

Upon power-up or any time that DRAM refresh requirements are violated, the Activation Counter bits may be in an unknown state, requiring an initialization to put the bits into a known state.

The default state for DDR5 Per Row Activation Counting (PRAC) is disabled (MR70:OP[1]=0_B), as this is an optional feature. Prior to initialization of the activation counter bits, the PRAC shall be enabled (MR70:OP[1]=1_B) by the host. Once the PRAC is enabled, a full array Activation Counter Initialization (ACI) shall be performed. The DRAM will not track activation counts, nor issue an Alert Back-Off (ABO), until the Activation Counter bits are initialized. Legacy mode for maintaining of data integrity will not be performed after PRAC is enabled.

To initialize the Activation Counter bits, the host sets MR70:OP[2]=1_B to indicate to the DRAM that a full array refresh will take place. Only REF and MRR commands are permitted during initialization. The REF commands during the initialization period may be issued by the host up to two times (2x) the normal refresh rate (Normal mode: 0.5*tREFI1; FGR mode: 0.5*tREFI2). Upon completion of a full refresh cycle, the DRAM shall indicate completion by setting MR70:OP[3]=1_B, which the host shall then follow by setting MR70:OP[2]=0_B. The DRAM shall start counting activations from that point forward and issue the ABO as necessary.

During ACI, the DRAM does not need to refresh the main array, and any data previously written may be corrupted.

If the host reenters Activation Counter Initialization by setting MR70:OP[2]=1_B, the DRAM will reset MR70:OP[3]=0_B until the initialization has been completed. Likewise, a system reset to disable PRAC (MR70:OP[1]=0_B) will also reset MR70:OP[3]=0_B.

Like normal DRAM cells, the Activation Counter bits require refresh to maintain the stored values. Any time refresh is violated during ACI or after ACI in modes like MPSM or other idle periods, the ACI shall be performed by the host to set the Activation Counter bits to a known state. Since array data is also corrupted by refresh violations, previous Activation Counter values become irrelevant.

The MBIST operation impacts the Activation Counter bit values, requiring an ACI to be performed prior to rewriting data following the MBIST/mPPR operation.

16.3 Per Row Activation Counting Core Timing Parameters

Per Row Activation Counting (PRAC) requires additional counter cells per row in the DRAM to store Activation (ACT) command counts. Updating the activation counter cells requires a read-modify-write to the activation counter cells, known as Activation Counter Update (ACU), resulting in a change in some of the core timing parameters, when Per Row Activation Counting is enabled (timings not noted here remain the same as when PRAC is disabled):

Table 343 — Per Row Activation Counting (PRAC) Core Timing Parameters

Parameter	Symbol	Min	Max	Unit	Notes
ACT to PRE	tRAS	16	3 * tREFI1 (Normal), 5 * tREFI2 (FGR)	nS	1
Row Precharge	tRP	36	-	nS	2
ACT to ACT / REF	tRC	52	-	nS	
Read to Precharge	tRTP	5	-	nS	3
Write Recovery	tWR	10	-	nS	3
NOTE 1 tRASmax is based on the maximum allowance for postponed REF commands NOTE 2 Clock-based nRP timings are defined in the Speed Bin tables. NOTE 3 Clock-based nRTP and nWR timings are defined in MR6.					

Examples of the timings affected by the updating of the activation counter bits are shown in Figure 262 through Figure 264. The “i” references DRAM internal commands, not external commands issued by the Host.

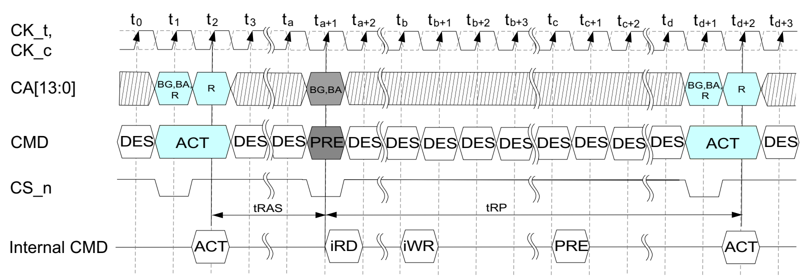


Figure 262 — Example ACT-PRE with ACU

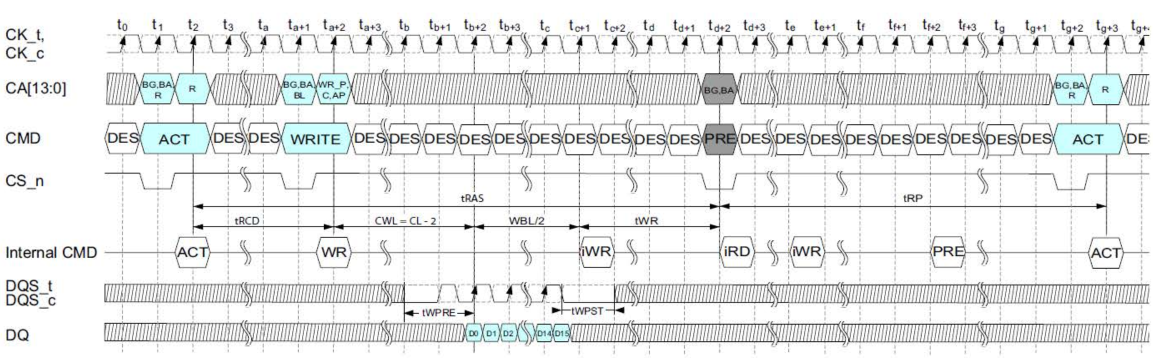


Figure 263 — Example ACT-WR-PRE with ACU

16.3 Per Row Activation Counting Core Timing Parameters (cont'd)

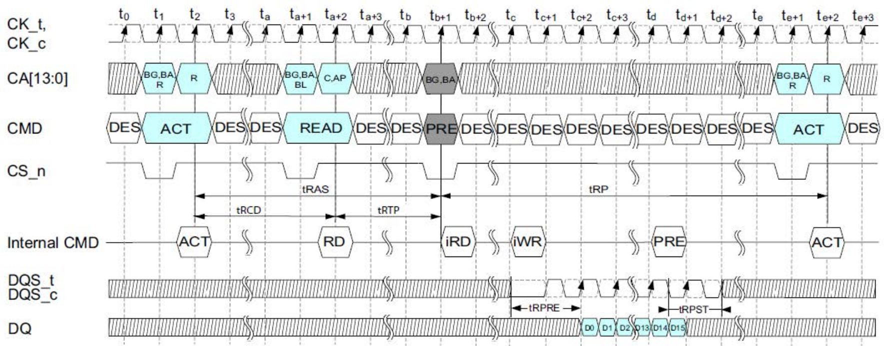


Figure 264 — Example ACT-RD-PRE with ACU

16.3.1 Refresh Operation Scheduling Flexibility

Refresh Operation Scheduling Flexibility is reduced when PRAC is enabled to allow targeted refreshes to occur at an improved rate.

In Normal Refresh mode, a maximum of 2 REFab command can be postponed. In the case that 2 REFab command is postponed, the resulting maximum interval between the surrounding REFab commands is limited to 3 × tREFI1. At any given time, a maximum of 2 REFab commands can be issued within 1 x tREFI1 window.

In FGR Mode, the maximum REFab commands that may be postponed is 4, with the resulting maximum interval between the surrounding REFab commands limited to 5 x tREFI2. At any given time, a maximum of 5 REFab commands can be issued within 1 x tREFI2 window.

16.3.2 Precharge Timing

Unlike the traditional Precharge (PRE) command, the PRAC PRE initiates an internal read followed by an internal write to increment the Activation Counting bits for all open rows affected by the PRE command. To allow for adequate power distribution network (PDN) recovery, the Row Precharge time, tRP, shall be scaled by the number of rows being closed.

Table 344 — Per Row Activation Counting (PRAC) Precharge Timing

Parameter	Number of Rows	tRP Required	Notes
Per Bank Precharge (PREpb)	1	tRP,min	
Same Bank Precharge (PREsb)	1 - 8	tRP,min	
All Bank Precharge (PREab)	1 - 8 9 - 32	tRP,min tRP,min + 4 ns	

16.4 Per Row Activation Counting Response

Since the DRAM is counting ACT commands on a per row basis, action may be required upon one or more rows reaching a counter threshold level or levels, or when a queue holding row addresses for targeted refreshing reaches or passes a critical level. This action is a request by the DRAM for the host to pause normal activity for a recovery period where refresh management (RFM commands) or other functions may be performed. This action is a “back-off”. The back-off protocol is described in the next section.

The counter and/or queue threshold level(s) may be changed (reduced) by setting the Adaptive Per Row Activation Counting level in MR71:OP[3:2] to something other than the default MR71:OP[3:2]=00_B. These alternate levels are based on the default starting value. The threshold level(s) reductions are approximately Level A = default - 10%, Level B = default - 20%, and Level C = default - 30%. Changing levels requires an ACI or full array refresh to set/reset all counters to a known state below critical threshold levels and to empty the queue.

In addition to the back-off which is required to keep activation counts from exceeding the threshold level, the DRAM can proactively perform targeted refreshes during Refresh (REF) and/or Refresh Management (RFM) commands to reduce and/or prevent the occurrence of the Alert Back-Off being triggered.

16.4.1 Targeted Refresh

For targeted refreshing during Refresh commands, the DRAM will adjust the normal refreshes accordingly to allow time for the additional row refreshing to occur. The back-off protocol will exist for cases where the DRAM cannot keep up during normal Refresh commands. There is no change to tRFC when PRAC is enabled.

16.4.2 Targeted RFM

DDR5 does not require issuing RFM commands as proactive mitigation when PRAC is enabled.

However, as a guideline for a host that chooses to have the DRAM proactively perform targeted refreshing during Refresh Management commands, the host may track the level of activity that occurs on a per-bank or sub-bank basis. When the Bank Activation Threshold (BAT) is reached (example BAT shown in Table 345), the RFMsb or RFMab command is issued. The duration of the RFM command (tRFM) is sufficient for the Row Refresh Span of +/-1 and +/-2 physically adjacent rows and the target row to be refreshed. This tRFM duration is directly related to the time required to refresh the rows. Due to a single row being refreshed on multiple banks corresponding to the RFMab or RFMsb command being issued, the per row refresh duration is tRRF.

Table 345 — Example Bank Activation Threshold (BAT)

Parameters	Symbol	Value	Units	Notes
Bank Activation Threshold	BAT	75	Activations	1
NOTE 1 The example BAT is derived from tREFI1=3.9 µs / tRC,min=52 ns.				

Table 346 — Per Row Refresh Timing

Parameters	Symbol	Duration	Units	Notes
All bank per row refresh	tRRFab(min)	70	ns	
Same bank per row refresh	tRRFsb(min)	60	ns	

Table 347 — Refresh Management Command Timing

Parameters	Symbol	Duration	Units	Notes
RFM All Bank (RFMab)	tRFMab	(4+1) * tRRFab	ns	
RFM Same Bank (RFMsb)	tRFMsb	(4+1) * tRRFsb	ns	

16.5 Back-off Protocol

DDR5 DRAM devices may support the optional Per Row Activation Counting (PRAC) for monitoring per row activity to ensure all rows are refreshed as needed to maintain data integrity. To facilitate the refresh activity, the DRAM may request the host pause normal activity for a maintenance period where refresh management (RFM commands) or other functions may be performed. This is called a “back-off”.

16.5.1 Alert Back-Off Protocol

The DDR5 back-off protocol uses the ALERT_n signal and mode register settings to provide a handshake between the DRAM and the host. This is the Alert Back-Off (ABO).

16.5.1.1 PRAC Triggered Alert Back-off

The Alert Back-off (ABO) protocol is enabled when PRAC is enabled by setting MR70:OP[1]=1_B. Upon reaching critical threshold levels by one or more rows, or when a queue holding addresses for targeted refreshing reaches or passes a critical level, the DRAM asserts ALERT_n and sets the ABO Flag at MR70:OP[5]=1_B, indicating that ALERT_n was asserted due to additional refresh management (RFM) being required. All commands received by the DRAM are executed while ALERT_n is asserted to maintain synchronization between DRAMs in the same rank on a DIMM.

Commands that may trigger the ABO include PREab/PREsb/PREpb, WRPA, WRA, RDA, REFab/REFsb and RFMab/RFMab and the MPC function Manual ECS. ALERT_n will be asserted within tABO_Assert, which occurs between issuing the trigger command and before the max(5ns,10nCK) following the completion of the respective command duration shown in Table 348.

Table 348 — Duration to ALERT_n Assertion

Symbol	Description	Command	Max
tABO_Assert	Duration to ALERT_n assertion	PREab/PREsb/PREpb	tRP+max(5ns, 10nCK)
		WRPA/WRA	CWL+WBL/2+tWR+tRP+max(5ns,10nCK)
		RDA	tRTP+tRP+max(5ns,10nCK)
		REFab (Normal)	tRFC1+max(5ns,10nCK)
		REFab (FGR)	tRFC2+max(5ns,10nCK)
		REFsb	tRFCsb+max(5ns,10nCK)
		RFMab	tRFMab+max(5ns,10nCK)
		RFMsb	tRFMsb+max(5ns,10nCK)
		MPC function Manual ECS	tECSc+max(5ns,10nCK)

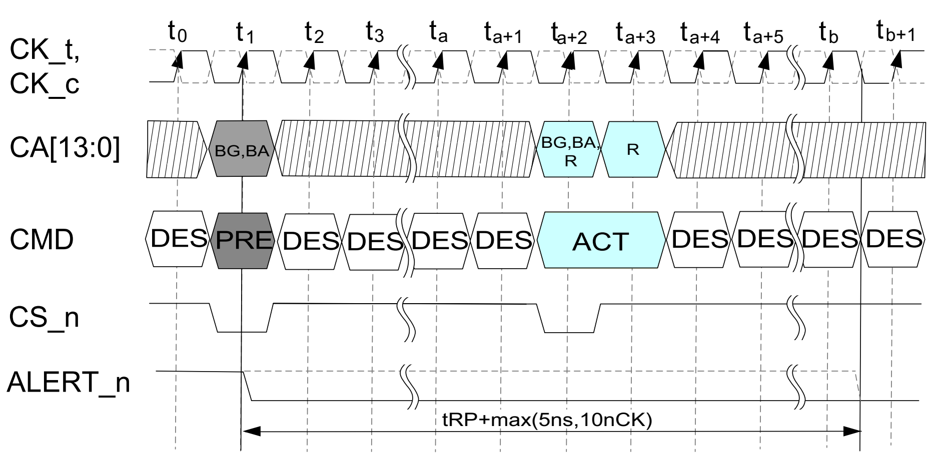


Figure 265 — Example Duration to ALERT_n Assertion for PRE Command

Since the PREpb required during hPPR/sPPR and the PREab during mPPR have different functions than closing a row, no counter bits will be incremented associated with the repaired row, therefore no ABO will be triggered in this instance.

16.5.1.2 Alert Back-off

To instantiate the ABO, ALERT_n is asserted by the DRAM with a fixed duration of clock cycles as defined by tABO_PW.

Table 349 — Alert Back-off Timing Parameters

Symbol	Description	Min	Typ	Max	Unit	Note
tABO_PW	Alert Back-Off pulse width	-	640	-	nCK	
tABO_ACT	Normal traffic to the DRAM allowed	-	-	180	ns	

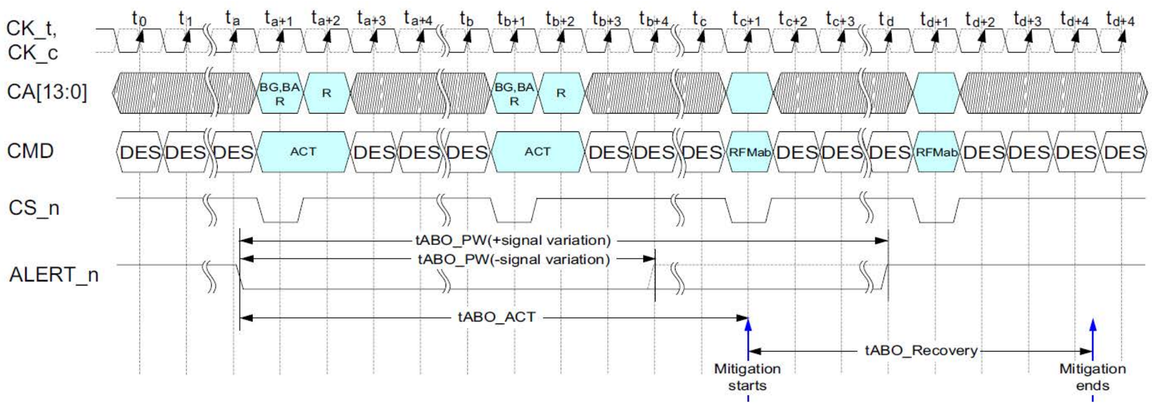


Figure 266 — Alert Back-Off Timing Diagram

After the ALERT_n is asserted, a bounded number of Activate (ACT) commands are permitted before the host starts the RFM-based recovery period. Issuing ACT commands along with normal traffic is allowed within the tABO_ACT window.

At the end of the tABO_ACT window, the host begins issuing Refresh Management (RFM) commands. The only RFM commands allow during the tABO_Recovery period are RFMab. The duration of the RFMab command is described in the previous section (Per Row Activation Counting Response). The number of RFMab commands required to be issued per ABO maintenance period are defined by mode register ABO_RFM bits, MR71:OP[1:0]. Systems that have stricter requirements for isochronous bandwidth may limit the number of RFMab commands by setting MR71:OP[1:0] to a lower value.

The recovery period begins when host issues the first RFMab command in response to ALERT_n. The DRAM self-clears the ABO flag by setting MR70:OP[5]=0_B by the end of the tABO_Recovery period, which ends tRFMab after the last RFMab command.

If additional RFM commands are needed, the Alert Back-Off protocol is restarted following the Alert Back-Off delay, as described below.

During the ABO_Recovery period, a host may issue only RFMab, REFab/REFsb, PDE, PDX, SRE, PREab and MRR commands. Refresh commands issued are to ensure that DRAM refresh requirements are not violated. The Refresh commands do not perform any of the refresh management required.

If Power Down is entered after ABO has been asserted but before the ALERT_n pulse has completed, the DRAM continues to assert ALERT_n for the expected tABO_PW duration, however the Host only resumes issuing RFM commands after exiting Power Down.

If Power Down is entered after ABO has been asserted and the tABO_PW duration has completed, the Host is responsible for issuing all of the expected RFM commands as of combination of commands before entering and after exiting Power Down.

In Self Refresh, the DRAM de-asserts ALERT_n. Upon exiting Self Refresh, the DRAM will reevaluate and issue a new ALERT_n if ABO is still needed.

16.5.1.3 Alert Back-Off Delays

ABO_Delay defines an interval in which DRAM shall service a minimum number of Activate commands before it can reassert ALERT_n for another Alert Back-Off. ALERT_n asserted for other operation such as Write DQ CRC errors is allowed with no timing restrictions.

ABO_Delay is specified by the number of Activate commands in MR71:OP[1:0].Commands that activate one or more rows in a DRAM include ACT, REFab/REFsb and RFMab/RFMsb and the MPC function Manual ECS, and any that occur are included in the total ABO_Delay Activates count allowed per bank.

Following ABO_Delay, any of the commands included in section 16.5.1.1 may trigger the ABO, as well as an ACT command.

Table 350 — Alert Back-Off Delay Parameter

Symbol	Description	Value	Unit	Note
ABO_Delay	Minimum number of Activate commands allowed	MR71:OP[1:0]	Activates	

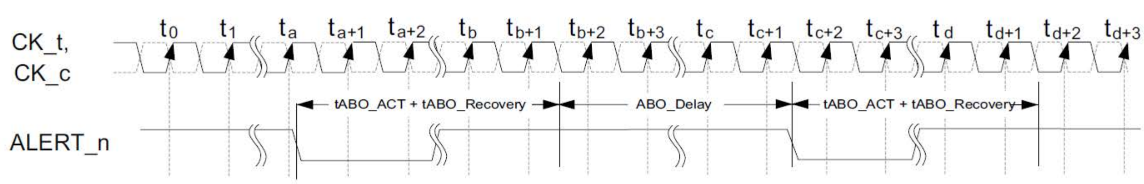


Figure 267 — Back to Back Alert Back-Off Delay

16.6 ALERT_n Priorities

At times, both Write CRC and Alert Back-Off may occur simultaneously resulting in an overlap of ALERT_n assertions, or the occurrence of the two events back-to-back (non-overlapped) may result in a short “glitch” on ALERT_n. For all events, the DRAM shall treat the ALERT_n as a “NOR” function where ALERT_n is issued whenever an event occurs, and the DRAM sets the corresponding Write CRC (MR50:OP[3]) or Alert Back-Off (MR70:OP[5] flag indicating which type of event, or possibly both, occurred.

16.7 Activation Counter Errors

The DRAM has the capability to detect errors that occur within the counter bits, and the following protocol defines how the DRAM will report the associated failing row address to the host.

If a counter bit error is detected, the DRAM sets the Counter Error Flag, MR72:OP[0]=1_B.

While the Counter Error Flag is set, the counter function for the specific row with the error is disabled. The DRAM will not assert the Alert Back-Off to request RFM commands for errors associated with that row. All other rows without counter errors continue to count activity and assert ALERT_n, if necessary.

During ECS (Manual or Auto mode), if the Counter Error Flag is set and the DRAM logs a failing row address, the DRAM will update MR73-MR75 with the failing row address and set the Counter Address Report Flag, MR72:OP[1]=1_B, when the ECS reaches the failing row. Only the first failing row will be reported. Additional rows will be reported in a subsequent scrub if they continue to fail after the Counter Address Report Flag is cleared by the host.

A counter RAS error that occurs during the ECS may still trigger the Counter Error Flag, but may not set the Counter Address Report Flag nor report the failing row address.

With the failing row address information, the host can repair the bad row with hPPR/sPPR or off-line that row.

The host clears the Counter Error Flag and Counter Address Report Flag, once the necessary information about the failing row has been collected and mitigative actions taken.

16.8 Per Row Activation Counting Testing

An optional test mode for PRAC may be included to check the ability of the DRAM to reach a threshold per row or fill the row address queue and issue the ABO. Test mode support is indicated by MR71:OP[4]=1_B.

PRAC Testing is enabled by setting MR71:OP[5]=1_B. Once enabled, the counter for target row to be tested is initialized by the host setting MR71:OP[6]=1_B, followed by an ACT/PRE to the target row. The host then sets MR71:OP[6]=0_B and issues 32 ACT/PRE or fewer to the target row or rows to trigger the ABO (multiple increments to the counter bits may occur depending on tRAS duration, mimicking normal operation). ABO is required as describe previously before testing subsequent rows.

Depending on counter bit starting values, all DRAMs on a DIMM may not trigger the Alert Back-Off at the same time (detected by MR70:OP[5]), however all DRAMs will trigger within 32 Activations.

After PRAC Testing has been complete, the test mode is cleared by setting MR71:OP[5]=0_B. Since the counter cells will be in an unknown state following testing, a full Activation Counter Initialization (ACI) is required before resuming normal operations.

Refresh commands are not allowed while PRAC Testing is enabled. Data in normal cells is not maintained during in PRAC Testing.

16.9 ALERT_n Verification

Support for an optional mode to verify ALERT_n is indicated by MR70:OP[6]=1_B.

ALERT_n Verification is triggered by setting MR70:OP[7]=1_B. The DRAM asserts ALERT_n and sets the ABO Flag at MR70:OP[5]=1_B within tMRD after the MRW and will keep ALERT_n asserted for tABO_PW. The host shall follow all normal ABO procedures for clearing the ALERT_n. MR70:OP[7] and MR70:OP[5] will be self-cleared by the DRAM prior to the end of the tABO_Recovery period.

Data in normal cells is maintained during ALERT_n Verification, as long as refresh is maintained. No per row counter values nor thresholds are altered during this mode.

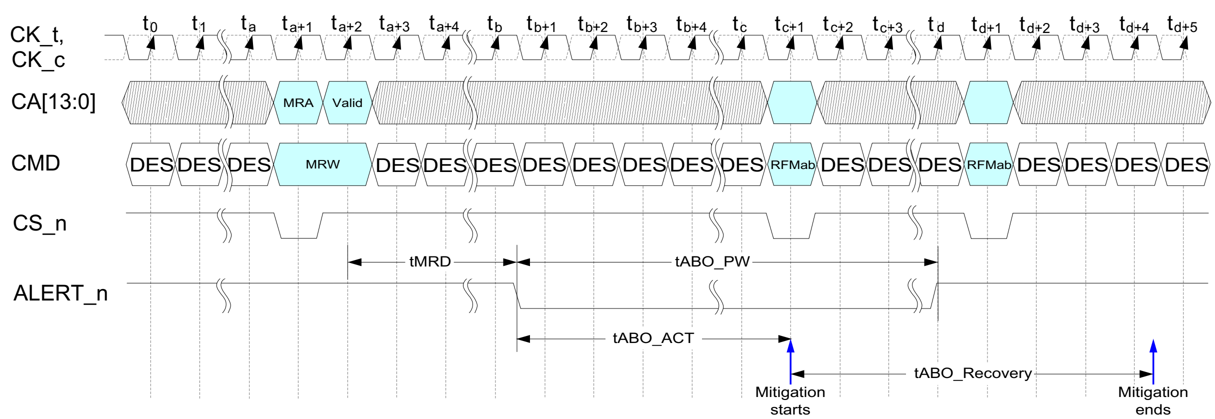


Figure 268 — Alert Verification Mode

16.10 PRAC and ABO Mode Register Definition

Table 351 — MR6 Register Information

Function	Register Type	Operand	Description	Notes
Write Recovery Time Legacy / PRAC	R/W	OP[3:0]	0000_B: 48nCK (Legacy) / 16nCK (PRAC) 0001_B: 54nCK (Legacy) / 18nCK (PRAC) 0010_B: 60nCK (Legacy) / 20nCK (PRAC) 0011_B: 66nCK (Legacy) / 22nCK (PRAC) 0100_B: 72nCK (Legacy) / 24nCK (PRAC) 0101_B: 78nCK (Legacy) / 26nCK (PRAC) 0110_B: 84nCK (Legacy) / 28nCK (PRAC) 0111_B: 90nCK (Legacy) / 30nCK (PRAC) 1000_B: 96nCK (Legacy) / 32nCK (PRAC) 1001_B: 102nCK (Legacy) / 34nCK (PRAC) 1010_B: 108nCK (Legacy) / 36nCK (PRAC) 1011_B: 114nCK (Legacy) / 38nCK (PRAC) 1110_B: 120nCK (Legacy) / 40nCK (PRAC) 1101_B: 126nCK (Legacy) / 42nCK (PRAC) 1110_B: 132nCK (Legacy) / 44nCK (PRAC) All other encodings reserved	1
tRTP Legacy / PRAC	R/W	OP[7:4]	0000_B: 12nCK (Legacy) / 8nCK (PRAC) 0001_B: 14nCK (Legacy) / 9nCK (PRAC) 0010_B: 15nCK (Legacy) / 10nCK (PRAC) 0011_B: 17nCK (Legacy) / 11nCK (PRAC) 0100_B: 18nCK (Legacy) / 12nCK (PRAC) 0101_B: 20nCK (Legacy) / 13nCK (PRAC) 0110_B: 21nCK (Legacy) / 14nCK (PRAC) 0111_B: 23nCK (Legacy) / 15nCK (PRAC) 1000_B: 24nCK (Legacy) / 16nCK (PRAC) 1001_B: 26nCK (Legacy) / 17nCK (PRAC) 1010_B: 27nCK (Legacy) / 18nCK (PRAC) 1011_B: 29nCK (Legacy) / 19nCK (PRAC) 1110_B: 30nCK (Legacy) / 20nCK (PRAC) 1101_B: 32nCK (Legacy) / 21nCK (PRAC) 1110_B: 33nCK (Legacy) / 22nCK (PRAC) All other encodings reserved	2

NOTE 1 tWR,min is defined in the “Timing Parameters” tables. Host must operate with MR settings resulting in tCK * MR6:OP[3:0] >= tWR,min.
NOTE 2 tRTP, min is defined in the “Timing Parameters” tables. Host must operate with MR settings resulting in tCK * MR6:OP[7:4]>= tRTP,min.
NOTE 3 All nCK conversions require rounding algorithm consideration.

16.10 PRAC and ABO Mode Register Definition (cont'd)

Table 352 — MR70 Register Information

Function	Register Type	Operand	Description	Notes
Per Row Activation Counting and Alert Back-Off Support (Optional)	R	OP[0]	0_B: PRAC and ABO not supported 1_B: PRAC and ABO supported	
Per Row Activation Counting and Alert Back-Off Enable/Disable	R/W	OP[1]	0_B: Disable PRAC and ABO (Default) 1_B: Enable PRAC and ABO	2
Activation Counter Initialization	R/W	OP[2]	0_B: Normal operating mode (Default) 1_B: Initialization mode	2
Activation Counter Initialization Complete	R	OP[3]	0_B: Initialization not completed 1_B: Initialization completed	2
RFU	RFU	OP[4]	RFU	
Alert Back-Off Flag	R	OP[5]	0_B: Clear 1_B: Source of ALERT_n	1
ALERT_n Verification Support (Optional)	R	OP[6]	0_B: ALERT_n Verification not supported 1_B: ALERT_n Verification supported	
ALERT_n Verification	R/W	OP[7]	0_B: ALERT_n Verification Disabled (Default) 1_B: ALERT_n Verification Trigger	

NOTE 1 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS-DDR5 stack is referenced in the MR70:OP[5] Alert Back-Off Flag source

NOTE 2 When PRAC is disabled, MR70:OP[1]=0_B, both MR70:OP[2] and MR70:OP[3] are reset to the default 0_B by the DRAM. Likewise when ACI is enabled, MR70:OP[2]=1_B, MR70:OP[3] is reset to 0_B by the DRAM.

Table 353 — MR71 Register Information

Functions	Type	Operand	Description	Notes
Min RFMab Commands during Recovery Period (ABO_RFM) and min ACT commands during Alert Back-Off Delay (ABO_ Delay)	R/W	OP[1:0]	00_B: 4 (Default) 01_B: 2 10_B: 1 11_B: RFU	
Adaptive Per Row Activation Counting	R/W	OP[3:2]	00_B: Default Mitigation Threshold (Default) 01_B: Mitigation Threshold Level A 10_B: Mitigation Threshold Level B 11_B: Mitigation Threshold Level C	
PRAC Testing Support (Optional)	R	OP[4]	0_B: PRAC Testing not supported 1_B: PRAC Testing supported	
PRAC Testing Enable/Disable	R/W	OP[5]	0_B: Disable PRAC Testing (Default) 1_B: Enable PRAC Testing	
PRAC Testing Initialization	R/W	OP[6]	0_B: Disable PRAC Testing Initialization (Default) 1_B: Enable PRAC Testing Initialization	
RFU	RFU	OP[7]	RFU	

16.10 PRAC and ABO Mode Register Definition (cont'd)

Table 354 — MR72 Register Information

Function	Register Type	Operand	Data	Notes
Counter Error Flag	R/W	OP[0]	0_B: Clear 1_B: Counter error detected	
Counter Address Report Flag	R/W	OP[1]	0_B: Clear 1_B: Failing row address available in MR73-MR75	
RFU	RFU	OP[7:2]	RFU	

Table 355 — MR73 Register Information

Function	Register Type	Operand	Data	Notes
8 bits of the row address	R	OP[7:0]	Activation Counter Error Row Address R[7:0]	1
NOTE 1 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS-DDR5 stack is referenced in the MR73 through MR75 Activation Counter Error address data				

Table 356 — MR74 Register Information

Function	Register Type	Operand	Data	Notes
8 bits of the row address	R	OP[7:0]	Activation Counter Error Row Address R[15:8]	1
NOTE 1 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS-DDR5 stack is referenced in the MR73 through MR75 Activation Counter Error address data				

Table 357 — MR75 Register Information

Function	Register Type	Operand	Data	Notes
2 bits of the row address	R	OP[1:0]	Activation Counter Error Row Address R[17:16]	1
2 bits of the bank address	R	OP[3:2]	Activation Counter Error Row Bank Address BA[1:0]	1
3 bits of the bank group	R	OP[6:4]	Activation Counter Error Row Bank Group BG[2:0]	1
RFU	RFU	OP[7]	RFU	
NOTE 1 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS-DDR5 stack is referenced in the MR73 through MR75 Activation Counter Error address data				

ANNEX A - (Informative) Differences between JESD79-5A and JESD79-5

This annex briefly describes most of the changes made to this standard, JESD79-5A, compared to its predecessor, JESD79-5 (July 2020). Some editorial changes are not included.

Changes Made in JESD79-5A

Section	Description of Change
2.2	Updated Section
2.6	Updated Table 3
2.7	Updated Table 6
3.1	Updated Section
3.3	Updated Table 9
3.3.1	Updated Section
3.4.1	Updated Section
3.5	Updated Section
3.5.1	Updated Table 24
3.5.6	Updated Section
3.5.8	Updated Section
3.5.9	Updated Section
3.5.11	Updated Section
3.5.14	Updated Section
3.5.15	Updated Section Updated Table 28
3.5.16	Updated Section
3.5.17	Updated Section
3.5.21	Updated Section
3.5.23	Updated Section
3.5.24	Updated Section
3.5.25	Updated Section
3.5.44	Updated Section
3.5.51	Updated Section
3.5.59	Updated Section
3.5.60	Updated Section
3.5.61	Updated Section
3.5.65	Added Section
3.5.66	Added Section
3.5.67	Added Section
3.5.68	Added Section
3.5.69	Added Section
3.5.70	Added Section
3.5.80	Updated Section